



Hardware for Artificial Intelligence

From Desktop PCs to
Enterprise Workstations

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Chapter 1 - The CPU in Consumer Desktop PCs

Note on Data Updates. This chapter is updated as of **July 2026**. Architectural concepts and protocols (PCIe, sockets, cache, IPC) are stable over time; what changes rapidly are **prices, the latest available generations, and availability**. In 2026, in particular, there is an anomalous market condition: a **severe shortage of DDR5 memory**, with RAM prices having risen by 5–8 times compared to normal. This heavily distorts the "total platform cost" that we will discuss, to the point that often a kit of fast RAM costs more than the CPU itself. All indicated prices should therefore be taken as an indicative order of magnitude and verified at the time of purchase.

1. What the CPU Does

1.1 Definition and Role in the System

The acronym **CPU** stands for *Central Processing Unit*. It is the component that executes program instructions: it is, literally, the "brain" that coordinates and controls the rest of the computer. Everything the PC does — opening a file, calculating a formula, moving the cursor, deciding what to send to the video card — passes in some form through the CPU, which orchestrates its flow.

To understand *how* a CPU works, one must start from the fundamental cycle it repeats billions of times per second, called **fetch-decode-execute**. In the *fetch* phase, the CPU retrieves the next instruction to execute from memory; in the *decode* phase, it interprets it, understanding what operation it is and what data it involves; in the *execute* phase, it actually performs it, for example, adding two numbers or writing a value to memory. Once execution is complete, the result is written (the *write-back* phase) and the cycle restarts with the next instruction. This process, trivial when described in words, is what happens billions of times per second inside each core, and is the reason why frequency ("GHz") has such a direct impact on performance: more cycles per second, more instructions processed.

The CPU does not work alone: it constantly communicates with other system components, and understanding this communication is essential for making sensible purchases. With **RAM** (*Random Access Memory*, the volatile working memory), the CPU exchanges the data it is currently using: RAM is fast but empties when powered off, and the CPU critically depends on it because that's where it

keeps the "hot" data of the running program. With **storage** (the storage unit, typically an NVMe SSD today), the CPU interacts to read and write permanent data, but storage is enormously slower than RAM, so data is loaded from storage to RAM before being processed. With the **GPU** (*Graphics Processing Unit*, the graphics processing unit, i.e., the video card), the CPU cooperates in graphical loads: the CPU prepares the scene, calculates game logic, physics, enemy artificial intelligence, and drawing instructions, then "hands over" this work to the GPU which transforms it into pixels on the screen. In a video game, therefore, the CPU and GPU continuously pass the baton, and if one of them is too slow compared to the other, the phenomenon of *bottleneck* occurs, which we will discuss in the section dedicated to selection.

1.2 CPU vs GPU: Two Computing Paradigms (Serial vs. Parallel)

One of the most common confusions for those new to hardware is thinking that CPU and GPU are "the same thing, one more powerful than the other." In reality, they are two profoundly different computing philosophies, and this difference explains why both are needed.

The CPU is optimized for **complex serial computation**: it has a few very sophisticated cores (typically 4 to 24 in consumer desktops), each capable of performing a wide variety of operations and making complex decisions quickly, managing conditional jumps, unexpected events, and instruction dependencies. It's like having a few brilliant engineers, each capable of solving difficult and diverse problems, but one at a time with great care.

The GPU, on the other hand, is optimized for **massive parallel computation**: it has thousands of much simpler computing units, each individually not very intelligent, but all together capable of performing the same operation on huge amounts of data simultaneously. It's like having an army of thousands of workers, each only capable of performing a simple task, but who together can paint a huge wall in a few seconds because they all work in parallel. This is exactly what is needed for graphics (calculating the color of millions of pixels simultaneously) and, not coincidentally, also for training artificial intelligence neural networks, which is parallel computation on matrices.

The practical consequence is clear: *sequential* and decision-rich tasks (game logic, code compilation, opening a spreadsheet) run better on the CPU; *massively parallel and repetitive* tasks (3D rendering, graphic computation, deep learning)

run much better on the GPU. Neither replaces the other, which is why a balanced PC needs both properly sized.

2. Internal CPU Architecture

2.1 Physical Cores: What They Are and Why They Matter

A **core** is a complete and independent processing unit within the CPU: it is, in practice, a small autonomous processor, capable of executing its own fetch-decode-execute cycle independently. A CPU with 8 cores contains 8 of these computing engines side-by-side on the same piece of silicon.

The number of cores matters because it determines how many *truly independent* tasks the CPU can perform simultaneously. With a single core, the computer can execute one instruction at a time (and simulates multitasking by rapidly switching between programs); with multiple cores, it can truly execute multiple workflows in parallel. Be aware, however, of a common misconception: *not all programs can use multiple cores*. Software is called "multi-threaded" when it is written to divide its work into parallel portions; many games, especially older ones, and many everyday applications, however, remain predominantly "single-threaded," meaning they only effectively utilize one or two cores. This is why, for gaming, "more cores" beyond a certain threshold (today 6–8 cores are more than sufficient for the vast majority of titles) offers diminishing returns, while for video rendering, compilation, virtualization, and scientific computing, additional cores are invaluable.

2.2 Threads and SMT/Hyper-Threading: Difference Between Cores and Threads

A **thread** is a single, ordered stream of instructions. The confusion between "core" and "thread" arises from a technology called **SMT** (Simultaneous Multi-Threading), which Intel markets under the name **Hyper-Threading**. Thanks to SMT, a single physical core can handle *two threads simultaneously*, presenting itself to the operating system as if it were two "logical" cores.

The trick works because a core, while executing a thread, often has inactive internal units: perhaps it's waiting for data from memory, and in the meantime, some of its computational circuits remain idle. SMT fills those "gaps" with the work of a second thread, making better use of existing resources. The typical gain is on the order of 15–30% more throughput in well-parallelized workloads: it

doesn't double performance (two threads on one core are not equivalent to two cores), but it's a free increase in efficiency.

When do threads help and when do they not? They help in heavily multi-threaded and parallel workloads (rendering, compression, virtualization, heavy editing), where keeping the core's units constantly occupied provides a concrete advantage. They help little or not at all in many games and single-thread workloads, where what matters is the speed of the single thread. In some extreme cases, in the past, SMT could even marginally reduce game performance due to resource contention, and it's one of the reasons why it was sometimes recommended to disable it for specific e-sports competitions. A notable and recent fact: **Intel, with the Arrow Lake architecture (Core Ultra series 2, from 2024), has abandoned Hyper-Threading on desktops**, focusing instead on a larger number of efficient cores. AMD, on the other hand, maintains SMT across its entire Ryzen range. Consequently, today, counting "threads" to directly compare Intel and AMD has become misleading: one must look at the overall architecture and benchmarks.

2.3 Base Frequency vs. Boost Frequency

Clock frequency measures how many cycles per second the core completes, expressed in gigahertz (GHz, billions of cycles per second). Modern CPUs do not have just one frequency but at least two declared values. The **base frequency** is the one guaranteed under prolonged load conditions, within standard power consumption and temperature limits: it's the "bare minimum" that the CPU always maintains. The **boost frequency** (or *turbo*) is the maximum that the CPU opportunistically reaches when it has thermal and power headroom, to accelerate work peaks.

It is crucial to distinguish between two types of boost. **Single-core boost** is the maximum frequency achievable when only one core (or very few) is engaged: it is high because heat is concentrated in one spot and there is ample headroom, and it is the value prominently advertised on the box. **All-core boost**, on the other hand, is the sustained frequency when *all* cores are working together: it is always lower than the single-core peak, because the heat produced by all cores simultaneously is much greater and needs to be dissipated. This distinction has important practical consequences: for a game, which often stresses few cores, single-core boost matters; for a rendering that loads all cores to 100%, all-core boost and the cooler's ability to maintain it matter. A common mistake is to be captivated by the

"5.7 GHz" written on the package without understanding that this value applies only to one core, for a few moments, and not under the intended workload.

2.4 L1, L2, L3 Caches and the Case of 3D V-Cache

Cache is a very small but very fast memory, integrated within the CPU, which stores frequently used data and instructions to avoid having to request them from RAM every time, which is much slower. The speed difference is enormous: accessing the cache costs the CPU a few cycles, accessing RAM costs hundreds. Every time the sought-after data is already in cache (a situation called *cache hit*), the CPU saves a huge wait; when it's not (*cache miss*), it has to fetch it from RAM, wasting precious time. Optimizing the cache is therefore one of the most effective ways to increase real performance.

Caches are organized in levels, in increasing order of size and decreasing order of speed. **L1 cache** is the smallest (a few tens of kilobytes per core) and fastest, private to each core. **L2 cache** is larger (from hundreds of kilobytes to a few megabytes per core) and slightly slower, also typically private to the core. **L3 cache** is the largest (from several megabytes up to over 100 MB) and is normally *shared* among all cores: it acts as a large common reserve, reducing RAM accesses for the entire processor.

Cache Level	Typical Size	Speed	Scope
L1	32–80 KB per core	Maximum	Core-private
L2	512 KB – 3 MB per core	High	Core-private
L3	16 – 128 MB and above	Medium	Shared among cores

The reason why a large L3 cache matters so much is most strikingly exemplified by AMD's **3D V-Cache** technology. In essence, AMD vertically "stacks" an additional block of SRAM cache memory *on top of* (or, in more recent versions, below) the compute die, increasing the L3 cache to enormous values — for example, 96 MB on a Ryzen 7 with 3D V-Cache. This has proven to be a tremendous advantage, especially in gaming, because many games have data sets that, if they all fit in cache, avoid continuous and very costly trips to RAM, with frame rate gains that can be double-digit percentages. This is why AMD models with the **X3D** suffix (like the famous Ryzen 7 9800X3D and the more recent 9850X3D from 2026) are considered among the best gaming CPUs ever. It should be noted, however, that 3D V-Cache does not help *every* game equally, and the

benefit tends to decrease as the resolution increases (at 4K, the bottleneck shifts to the GPU). Furthermore, historically, the first X3D generations had to slightly reduce their frequency to manage the heat of the stacked cache, penalizing productivity workloads; with more recent generations, this compromise has been greatly mitigated.

2.5 IPC: why frequency alone is not enough

IPC stands for *Instructions Per Clock*: it measures how much useful work a core can accomplish in a single cycle. It's the concept that debunks the myth that "more GHz = faster." The intuitive formula for a core's performance is indeed $performance \approx IPC \times frequency$: two multiplied factors, not just one.

This means that a modern CPU at 4.5 GHz can easily beat a ten-year-old CPU at 5.0 GHz, because the former executes many more instructions per cycle thanks to a more efficient architecture (better branch prediction, more execution units, larger caches, optimized pipelines). IPC is the reason why you cannot compare GHz across generations or between different brands: these numbers mean different things depending on the underlying architecture. Every new CPU generation typically boasts a "+X% IPC" over the previous one, and it's that gain — not the GHz — that makes the real difference. The practical lesson for the buyer is clear: never compare CPUs by GHz, but by benchmarks, which capture the real-world $IPC \times frequency$ product.

2.6 TDP and real consumption: PL1/PL2 and PPT

TDP (*Thermal Design Power*) is a value expressed in watts that, originally, indicated how much heat the cooler must be able to dissipate for the CPU to operate within specifications. In modern practice, it has become an ambiguous and often misleading figure, because it **does not coincide with the CPU's actual maximum consumption** under heavy load.

Intel manages power consumption through two limits called **PL1** and **PL2** (*Power Limit 1* and *Power Limit 2*). PL1 is the long-term sustained power limit and roughly corresponds to the nominal TDP. PL2 is the much higher peak limit that the CPU can draw for short intervals (governed by a timer called *Tau*) during boost. This is why a CPU with "TDP 125 W" can actually draw 250 W or more at peak: that number on the box describes the sustained regime, not the peak. Many motherboards, moreover, default to "unlocked" limits that let the CPU consume as much as it wants, further altering the picture.

AMD uses a different and more honest metric called **PPT** (*Package Power Tracking*), which indicates the effective power ceiling that the entire CPU package can draw from the socket. PPT is typically about 1.35 times the nominal TDP: an AMD CPU with "TDP 105 W" generally has a PPT around 142 W, which represents the real maximum consumption under full load.

The practical lesson is twofold. First: TDP is not enough to size the power supply or the cooler; you need to look at the real peak consumption (PL2 or PPT). Second, and crucial for the buyer: buying a powerful CPU and then pairing it with an inadequate cooler means throwing away performance, because the CPU, unable to dissipate heat, automatically reduces its frequency (a phenomenon called *thermal throttling*) and never maintains the declared boost.

2.7 Intel Hybrid Architectures: P-cores and E-cores

Since the twelfth generation, Intel has introduced a **hybrid** architecture, which mixes two types of cores on the same processor. **P-cores** (*Performance cores*) are large, powerful, and power-hungry cores, designed for heavy and latency-sensitive tasks, such as the main thread of a game. **E-cores** (*Efficient cores*) are smaller, more frugal cores, designed to efficiently handle many background tasks and highly parallel workloads, occupying little space and little energy: many can be placed in the space of one P-core.

The idea is to entrust the operating system (via a hardware component called *Thread Director* that suggests how to distribute the work) with assigning the right tasks to the right cores: critical tasks to P-cores, secondary and massive tasks to E-cores. The result, when it works well, is an excellent balance between peak performance and multi-thread efficiency. However, practical implications exist. The first is that the number of Intel cores must be read carefully: a "Core Ultra 9 285K" with "24 cores" actually has 8 P-cores and 16 E-cores, with very different performance characteristics. The second is that, in the early days, some old software (particularly certain game anti-cheat systems and old programs) got confused in thread assignment, leading to performance drops; the problem has been largely resolved with Windows updates and drivers, but it's useful to know. AMD, for its part, adopts a different approach: in consumer desktops, it uses all "full-size" identical cores, possibly combining multiple groups of cores (the so-called CCDs, the compute chiplets), which simplifies scheduling but introduces other subtleties in dual-chiplet models.

2.8 Integrated iGPU: when it's needed and when it's not

Many CPUs integrate an **iGPU** (*integrated GPU*): a small video card built into the processor itself. It doesn't have the power of a dedicated video card, but it's sufficient for displaying the desktop, playing videos, browsing, and playing light games. Its presence or absence is directly encoded in the commercial suffixes, and is a source of costly purchasing errors.

For **Intel**, the suffix **F** (for example, "Core Ultra 5 245KF") indicates a CPU *without a functional iGPU*: it costs a few tens of euros less, but absolutely requires a dedicated video card to function, otherwise nothing will be displayed on screen. For **AMD**, the reasoning is reversed and more insidious: most "normal" desktop Ryzen CPUs only have a minimal iGPU, just good enough for the desktop; models truly designed for gaming without a dedicated video card are **APUs** marked with the suffix **G** (like the Ryzen 7 8700G), which integrate much more powerful Radeon graphics, capable of running several games decently at 1080p with reduced details.

When is an iGPU (or an APU) needed? It's needed for an office, a media center, a secondary PC, or an inexpensive PC to use while waiting to buy a video card, and it serves as a "safety net" for diagnosing dedicated GPU failures. It's not needed, and it's useless to pay for it, if a powerful dedicated video card will be installed for gaming anyway: in that case, choosing a model with an F suffix (Intel) can save some money. The classic mistake, which we will discuss at the end, is buying an "F" CPU thinking of saving money and then realizing you don't have any video card to turn on the PC.

3. The socket: compatibility between CPU and motherboard

3.1 What a socket physically is

The **socket** is the physical connector on the motherboard where the CPU is inserted: it is the mechanical and electrical interface that connects the thousand-plus contacts of the processor to the traces on the board. The socket rigidly determines *which* CPUs can be mounted on a given motherboard: a CPU can only work on a socket for which it was designed, period. It is the first and most absolute compatibility rule of the entire system.

There are two main design philosophies. In **LGA** (*Land Grid Array*), the pins — i.e., the metal contacts — are *on the motherboard socket*, while the CPU only has flat

pads (the *lands*) that rest on them. In **PGA** (*Pin Grid Array*), it's the opposite: the pins are *on the CPU*, and the motherboard socket has corresponding holes. The difference has practical consequences for fragility: in PGA, if a pin bends, the expensive CPU is damaged; in LGA, a bent pin is on the motherboard socket (still difficult to repair, but often it's the mobo that's at risk). Intel has used the LGA format for years; AMD long used PGA on the AM4 platform but switched to LGA with the new AM5 platform.

3.2 Current and recent AMD sockets: AM4 and AM5

AMD has built an enormous reputation on **platform longevity**, meaning its ability to support many generations of CPUs on the same socket, allowing upgrades without changing the motherboard. The **AM4** socket (PGA format) is the prime example: introduced in 2016, it supported an impressive range of Ryzen generations, from the first Ryzen 1000 up to the Ryzen 5000 (and AMD continued to launch new models on it even years later, celebrating its "10 years" in 2026 with special editions). Those who bought a good AM4 motherboard in 2017 could, years later, install a much more powerful CPU with a simple firmware update.

The **AM5** socket (LGA format, since 2022) is AMD's current platform for Ryzen 7000 and 9000 series (and their X3D models). It introduces mandatory support for **DDR5** memory and **PCIe 5.0**. Here too, AMD has promised and confirmed long longevity: **AM5 support is guaranteed until at least 2029**, which makes an AM5 motherboard purchased today a reasonably future-proof investment, capable of hosting even upcoming generations. It's worth noting that next-generation CPUs based on the Zen 6 architecture (the "Ryzen 10000" family, codenamed "Olympic Ridge") have been, according to mid-2026 rumors, **postponed to 2027**, and will still remain on the AM5 socket: those who buy AM5 today will most likely be upgradeable to that generation as well.

3.3 Current and Recent Intel Sockets: LGA1700 and LGA1851

Intel has historically had a policy opposite to AMD's: it **changes sockets much more frequently**, typically every one or two generations, often forcing those upgrading their CPU to also change their motherboard. The number in the socket name (all in LGA format) indicates the number of contacts. The **LGA1700** socket hosted Intel generations from the twelfth to the fourteenth (12th, 13th, and 14th gen Core), with support for both DDR4 and DDR5 depending on the motherboard.

The current socket is **LGA1851**, introduced with the **Core Ultra 2 series "Arrow Lake"** (2024) and with their update **"Arrow Lake Refresh" / Core Ultra 200S Plus**

in March 2026 (models like the 270K Plus and 250K Plus). LGA1851 exclusively supports DDR5. Here, however, the classic warning about Intel's policy kicks in: **LGA1851 is already end-of-life**. The 800 series chipsets are the last generation for this socket, and Intel will introduce a new socket (referred to as **LGA1954**) for the next "Nova Lake" generation in the second half of 2026. In practical terms, buying an Intel LGA1851 platform today means acquiring, in all likelihood, a "dead end" in terms of future CPU upgrades: you will only be able to change CPUs within the current family. This is a decisive factor when choosing between Intel and AMD for those planning upgrades years down the line.

Socket	Manufacturer	Type	CPU Generations	Memory	PCIe (from CPU)	Status (Jul. 2026)
AM4	AMD	PGA	Ryzen 1000–5000	DDR4	4.0	Legacy, but still sold (budget segment)
AM5	AMD	LGA	Ryzen 7000–9000 (and future)	DDR5	5.0	Current , supported until at least 2029
LGA1700	Intel	LGA	Core 12th–14th gen	DDR4/DDR5	5.0	Previous, still available
LGA1851	Intel	LGA	Core Ultra 200S / 200S Plus	DDR5	5.0	Current , but end-of-cycle (LGA1954 coming)

3.4 How to check compatibility: socket + chipset + BIOS

Checking compatibility requires checking *three* levels, not just the socket. The first level is, of course, the **socket**: CPU and motherboard must have the same socket. The second level is the **chipset**, which is the "controller" on the motherboard that manages peripheral connections (ports, chipset PCIe lanes, overclocking functions): even with the same socket, not all chipset/CPU combinations are valid or convenient. On AMD, for example, high-end chipsets (X series, like X670E or X870E) enable more features and overclocking, while mid-range ones (B series)

are cheaper but limited; on Intel, there is a similar distinction, with "Z" chipsets (e.g., Z890) unlocking overclocking and cheaper "B" chipsets that do not. The third level, often forgotten, is the **BIOS/UEFI version** (the motherboard firmware): a motherboard released before a certain CPU might not recognize it until its firmware is updated. It's a very common mistake to buy a motherboard that is "compatible on paper" and find it unresponsive because it has a BIOS that is too old for the newly purchased CPU.

The correct way to proceed is to consult the **CPU support list** (the official list of supported CPUs) for that specific model on the motherboard manufacturer's website, which indicates the minimum BIOS version required for each CPU. It's a two-minute check that saves days of frustration.

3.5 Backward compatibility and BIOS updates without CPU (flashback)

The "BIOS too old" problem has an elegant solution offered by many mid-range and high-end motherboards: **BIOS Flashback** (also called *Q-Flash Plus* or similar depending on the brand). This function allows you to **update the BIOS without having a CPU, RAM, or graphics card installed**: just download the BIOS file to a USB stick, insert it into a specific port, press a dedicated button on the board, and the board updates itself using only power. It is invaluable in the classic case: I bought a new CPU and an older motherboard that doesn't recognize it; without flashback, I would be stuck, because to update the BIOS I would need a CPU that is already supported to install temporarily. With flashback, I solve everything in ten minutes. Anyone buying a new platform or planning upgrades should consider the presence of this feature an almost indispensable requirement when choosing a motherboard.

4. PCI Express: the complete treatment

This is the densest section of the chapter, because **PCIe** is the "nervous system" that connects the CPU to the graphics card, fast SSDs, and other expansions, and it is a source of enormous misunderstandings.

4.1 What is PCIe: a point-to-point serial bus

PCIe stands for *Peripheral Component Interconnect Express*: it is the high-speed interconnection standard that allows CPUs and peripherals to exchange data. Its fundamental architectural characteristic is that it is a **serial** and **point-to-point** bus. "Serial" means that data travels in a line over narrow but very fast paths,

instead of in parallel over many wires as in older standards: paradoxically, at these frequencies, sending data in series over a few very fast channels is more efficient and reliable than spreading it over many parallel wires that risk "desynchronizing." "Point-to-point" means that each device has its own dedicated connection with the host, without having to share it and without contending for bandwidth with others, as was the case with shared buses of the past.

4.2 Lanes: what x1, x4, x8, x16 mean

The basic unit of PCIe is the **lane**. A lane is a single bidirectional serial connection, physically composed of two pairs of wires: one pair for transmitting and one for receiving, so that data can flow in both directions simultaneously (*full duplex*).

Lanes can be aggregated to increase bandwidth: the designations **x1**, **x4**, **x8**, **x16** indicate how many lanes are grouped for a given connection. An "x16" slot uses 16 lanes in parallel and therefore offers sixteen times the bandwidth of a single lane; an "x4" slot uses four, and so on.

This has a dual nature, *physical* and *electrical*, which causes much confusion. A slot can be **physically** as long as an x16 (i.e., have the mechanical space for 16 lanes) but be **electrically** connected only to x4 or x8, meaning it actually has a lower number of wired lanes. This is a very common situation on inexpensive motherboards: the second long slot, which looks like an x16, actually operates at x4. The device fits and works anyway, but at the reduced bandwidth of the lanes actually present. For this reason, when reading the technical specifications of a motherboard, one must always distinguish the physical length of the slot from the actual number of electrical lanes (often indicated with notations such as "x16 (x8 electrical)").

4.3 Generations: PCIe 3.0, 4.0, 5.0 and a brief mention of 6.0

Each new generation of PCIe **doubles** the bandwidth per lane compared to the previous one, for the same number of lanes. This generational doubling is the key to understanding everything else. The following table summarizes the approximate unidirectional bandwidth per configuration (the actual "useful" values are slightly lower due to encoding overhead, but the order of magnitude and ratios are correct):

Generation	Bandwidth per lane (~)	x4 (typical NVMe)	x16 (typical GPU)	Diffusion (Jul. 2026)
PCIe 3.0	~1 GB/s	~4 GB/s	~16 GB/s	Legacy, still common on

				chipsets
PCIe 4.0	~2 GB/s	~8 GB/s	~32 GB/s	Consolidated standard, excellent q/p ratio
PCIe 5.0	~4 GB/s	~16 GB/s	~64 GB/s	Current consumer top (high-end GPU and SSD)
PCIe 6.0	~8 GB/s	~32 GB/s	~128 GB/s	Emerging, currently in server/datacenter environments

PCIe 6.0, which doubles again, is technically defined and is arriving in the enterprise world, but on consumer desktops in mid-2026 it is not yet relevant: consumer CPUs and motherboards reach PCIe 5.0. We will discuss this in more detail in the chapter on workstations and servers, where extreme bandwidth truly matters.

4.4 Backward compatibility between generations and slots

One of PCIe's great virtues is **full backward and forward compatibility**. A newer generation device inserted into an older generation slot (for example, a PCIe 5.0 GPU in a PCIe 4.0 slot) works without issues, simply negotiating the speed of the lower of the two generations. Conversely, an older device in a newer slot works at its own speed. Furthermore, a device with fewer lanes fits into a longer slot: an x4 card works in an x16 slot (using only the lanes it needs). However, the mechanical inverse does not work: an x16 card does not physically fit into a shorter x1 slot, unless the slot is "open" at the back. The practical message is reassuring: PCIe "makes do" almost always, adapting to the lowest common denominator, and there is no risk of damage by mixing generations; at most, you lose a bit of bandwidth.

4.5 Who provides the lanes: CPU versus chipset, and why it matters

This is the most subtle and important point for making informed choices. PCIe lanes in a system come from **two different sources**, with very different characteristics. Some lanes are provided **directly by the CPU**: these are the fastest and lowest-latency, connected directly to the processor without intermediaries,

and are reserved for the most bandwidth-hungry devices, typically the main graphics card slot (x16) and one or two primary NVMe slots. Other lanes are provided by the motherboard's **chipset**, which is connected to the CPU via a limited-bandwidth link (a "pipe" called *DMI* on Intel or a dedicated link on AMD).

The crucial point is that **all chipset lanes share that single pipe to the CPU**. If many devices connected to the chipset are loaded simultaneously (secondary SSDs, network cards, fast USB), they can saturate the link and contend for bandwidth, which does not happen on the direct CPU lanes. This is why a golden rule of assembly is: **the graphics card and the main NVMe SSD go into the slots directly connected to the CPU**, not into the chipset slots. The motherboard manual always indicates which slots are "CPU-attached" and which are "chipset-attached"; ignoring this distinction can lead, for example, to installing a fast SSD in a slot that halves its performance.

4.6 How many lanes do consumer CPUs have versus HEDT and server

Modern consumer CPUs provide a **limited** number of direct PCIe lanes: typically **20 to 28 lanes** (for example, 16 for the GPU, 4–8 for NVMe SSDs, and some for the chipset connection). This might seem few, and indeed it is: it's a positioning choice, as AMD and Intel reserve an abundance of lanes for higher-end, more expensive platforms. **HEDT** (*High-End Desktop*) and **server** platforms — such as AMD Threadripper and EPYC, or Intel Xeon — offer an enormously greater number of lanes, on the order of **64, 128, or more**. This abundance is what allows a workstation to simultaneously mount multiple GPUs, dozens of SSDs, capture cards, 100-gigabit network cards, and more, all at full bandwidth and without contention. This is a topic we will explore in more detail in the chapter dedicated to workstations; for now, suffice it to say that *the number of lanes is one of the invisible walls separating the consumer world from the professional one*, and it is often the real reason why certain "extreme" users need to upgrade their platform.

4.7 Bifurcation: what it is and when it's needed

Bifurcation is the ability to **split a single x16 slot into multiple smaller independent connections**, for example, into two x8s, or four x4s. The CPU has a certain number of physical lanes; bifurcation allows them to be reallocated differently from the default grouping. The typical consumer use case is to install an expansion card that hosts multiple NVMe SSDs (a "quad M.2 card") into a single x16 slot, asking the motherboard to divide those 16 lanes into four x4 groups, one for each SSD. It is also useful for those who want two graphics cards at x8/x8

instead of just one at x16. Bifurcation is not mandatory, and not all motherboards support it (it needs to be checked in the BIOS and specifications): it's a feature to check *before* purchasing if you have one of these scenarios in mind. For the average user, who installs a GPU and one or two SSDs in their respective slots, it is not a relevant factor.

4.8 Practical impact: GPU on x8 vs x16, NVMe Gen4 vs Gen5

Let's conclude with the most frequent practical questions. The first: **does a graphics card on x8 lose performance compared to x16?** The answer, surprising to many, is: almost always no, or very little. Modern graphics cards, even high-end ones, rarely saturate the bandwidth of a recent-generation x16 slot; if the slot is of a modern generation (PCIe 4.0 or 5.0), running it at x8 of that generation still leaves enormous bandwidth, and the actual frame rate loss is typically in the order of a few percentage points, often imperceptible. It only becomes more significant on very old generations or with cards at the limit of their bandwidth. This reassures those who, to install more devices, find their GPU running at x8.

The second: **is an NVMe PCIe 5.0 (Gen5) SSD worth it compared to a Gen4?** On paper, Gen5 doubles sequential bandwidth, and in benchmarks involving huge transfers, the difference is noticeable. In real-world daily use, however, the advantage is much less perceptible than the numbers suggest, because most common operations (system boot, program opening, game loading) depend more on random small-file performance than on peak sequential bandwidth. In addition, Gen5 SSDs tend to cost more, run hotter, and require more serious heatsinks. For the typical user, a good Gen4 currently offers the best value for money; Gen5 makes sense for specific professional video editing workloads with gigantic files or for those who simply want the absolute best regardless.

5. Commercial Nomenclature: How to Read a CPU Name

Knowing how to "decode" a CPU's name is a practical skill that prevents many errors. Let's look at the two manufacturers.

5.1 Intel: From Core i to Core Ultra

For almost fifteen years, Intel used the **Core i3 / i5 / i7 / i9** scheme, where the number indicates the increasing performance tier: i3 entry, i5 mid-range, i7 high-end, i9 enthusiast. The model was read as follows: after the designation came the

generation (for example, "14" in "i7-14700K" indicated the fourteenth generation), followed by the model identification number and a letter **suffix**.

From 2023–2024, Intel changed its branding: the **Core Ultra** nomenclature was introduced, with **Core Ultra 5 / 7 / 9** tiers (and "Core" without Ultra for lower tiers), abandoning the old "i". On desktops, this new family is the **Core Ultra 200S "Arrow Lake"** series (for example, "Core Ultra 9 285K"), with the 2026 refresh identified by the **"Plus"** suffix (for example, "Core Ultra 7 270K Plus"). The reading logic remains similar: tier (5/7/9), then the model number, then the suffixes.

Intel suffixes are the part that most often causes confusion. Here's their meaning:

Intel Suffix	Meaning
K	Unlocked multiplier: overclockable . Higher frequencies, requires "Z" chipset and good cooling.
KF	Like K (unlocked) but without iGPU : requires a dedicated graphics card.
F	Without functional iGPU : requires a dedicated graphics card. Costs a bit less.
T	Low power version (reduced TDP), for compact/silent systems; lower performance.
(no suffix)	Standard model, with iGPU, not overclockable but efficient.
S	Historically indicated "special edition" versions; today "S" primarily identifies the <i>desktop series</i> (200S).

A "Core Ultra 5 245KF" is therefore read as: mid-range (5), model 245, **K**=overclockable, **F**=without integrated graphics (requires a dedicated GPU).

5.2 AMD: The Ryzen Logic

AMD uses the **Ryzen 3 / 5 / 7 / 9** scheme, with the same principle of increasing tiers: Ryzen 3 entry, Ryzen 5 mid-range, Ryzen 7 high-end, Ryzen 9 enthusiast. After the tier comes a **four-digit** number where the first digit indicates the **series/generation**: 5000, 7000, 9000. Be aware of a trap: the 8000 series for desktops (8700G APU, etc.) is, in terms of architecture, somewhat out of sequence

compared to the 7000/9000, and is primarily dedicated to APUs with powerful graphics. In AMD's case, the final suffixes are also crucial:

AMD Suffix	Meaning
X	Higher frequencies and better boost than the base model; all desktop Ryzen CPUs are overclockable anyway.
X3D	Equipped with 3D V-Cache : huge L3 cache, top for gaming (e.g., 9800X3D, 9850X3D).
G	APU : powerful integrated Radeon graphics, plays at 1080p without a dedicated graphics card (e.g., 8700G).
GE	Like G but low power (efficient version of APUs).
F	No integrated graphics (rare in AMD desktop; e.g., 8700F), requires a dedicated GPU.
(no suffix)	Standard "non-X" model, slightly lower frequencies but often excellent price/performance ratio.

A "Ryzen 7 9800X3D" therefore reads as: high-end (7), 9000 series, **X3D**=with 3D V-Cache designed for gaming. Important and sometimes counter-intuitive note: in AMD desktops, *all* CPUs have an unlocked multiplier and are in principle overclockable (unlike Intel, where a "K" is needed); the "X" only indicates better binning and boost out of the box, not "unlocking."

5.3 Threadripper and Xeon: just a mention

Beyond consumer lines, there are professional lines: **Ryzen Threadripper** (AMD) and **Xeon** (Intel) are HEDT and server processors with many cores and many PCIe lanes, dedicated sockets, support for large amounts of RAM (often with ECC error correction), and prices of a different magnitude. They are not designed for gaming or home use, but for professional workstations, computing, and datacenters. We will cover them extensively in the chapter dedicated to workstations; here it suffices to know that they exist and represent a world separate from what is described in this chapter.

6. Brand Comparison: Intel vs. AMD

It should be noted that "who is better" constantly changes with each generation, and the following considerations reflect the situation in mid-2026: they should be re-verified at the time of purchase.

6.1 Current Strengths

In **pure gaming**, AMD currently has a rather clear advantage thanks to **X3D** models with 3D V-Cache, which are considered the fastest gaming CPUs on the market (the Ryzen 7 9800X3D and the more recent 9850X3D from 2026 dominate frame rate rankings). Intel, after a disappointing Arrow Lake debut in gaming, has recovered ground with the "200S Plus" refresh in 2026 and with software optimizations (such as the game code optimization tool introduced with the refresh), but in top-tier gaming, AMD remains the benchmark.

In **multi-thread productivity** (rendering, compilation, computation), the two brands are comparable depending on the model and price: high-core Ryzen 9s and Intel CPUs with an abundance of E-cores are both very valid, and it's advisable to look at specific benchmarks for your software. On the **energy efficiency** front, Intel has made significant progress with Arrow Lake, which consumes and heats up much less than the disastrous previous generations (13th/14th gen Core CPUs were known for high power consumption and degradation issues); AMD remains very competitive, and its X3D models in particular offer very high gaming performance at contained power consumption.

On the **price/performance ratio**, Intel's 2026 refresh brought very aggressive models back into play: the Core Ultra 5 250K Plus at around \$199 and the 270K Plus at around \$299 were welcomed as excellent deals for productivity. AMD responds with its consolidated lineup and, above all, with the long-term platform value, which we'll discuss immediately.

6.2 Platform: Total Cost and Longevity

The comparison should not only be made on the CPU, but on the **total platform cost**, which includes CPU + motherboard + RAM. Here, the issue of **socket longevity** plays a decisive role. Buying AMD **AM5** today means purchasing a platform with guaranteed support until at least 2029: an AM5 motherboard will also be able to host future CPUs (Zen 6/Ryzen 10000, expected in 2027), allowing for a future CPU-only upgrade without changing the motherboard and RAM. Buying Intel **LGA1851** today, on the other hand, means purchasing a socket that is already end-of-life, which will be replaced by LGA1954 in the second half of 2026:

future CPU upgrades will be limited to the current family, and to move up a generation, a motherboard change will be required. This is, for many, the single strongest argument in favor of AMD for those thinking long-term.

However, there is a huge contingent factor to consider in 2026: the aforementioned **DDR5 memory price crisis**. With RAM prices even multiples of normal, the "platform cost" is currently dominated by memory more than by the CPU or motherboard. In some cases, a good kit of fast RAM costs more than the CPU itself. This advises, at this particular historical moment, not to oversize RAM beyond what is necessary and to give even greater weight to choosing a long-lasting platform, in order to dilute an investment that is particularly expensive today over time.

6.3 Use Cases

For **pure gaming**, especially at high frame rates and at 1080p/1440p, the most sensible choice today is an AMD **X3D** CPU (even a previous generation model often represents a good deal). For **streaming and gaming together**, where extra cores are needed for encoding as well as gaming, both Ryzen 7/9 and Intel CPUs with many E-cores are suitable, provided there is a good thread count. For **editing and rendering**, cores and threads matter: one looks for the best multi-thread score per euro, and here both brands have strong offerings depending on the budget. For **virtualization**, cores, threads, and the amount of supported RAM matter (and, if possible, ECC support, which is easier to find on consumer AMD): multi-core CPUs are preferable. For a **budget build**, today there are excellent options from both sides, including AMD "G" APU's that allow starting without a graphics card, and low-end Intel with a good q/p ratio; here the choice depends heavily on current local prices.

7. How to Evaluate and Choose a CPU

7.1 Benchmarks: What to Look For and What Not to Trust

Benchmarks are standardized tests that measure performance. They should be interpreted carefully, choosing those relevant to *your* specific use. **Cinebench** measures rendering performance and is a good indicator for productive multi-threaded workloads, providing both single-core and multi-core scores.

Geekbench is a more general synthetic benchmark, useful for a quick overview but less representative of specific real-world workloads. For gaming, the most

significant test is **1080p gaming benchmarks** with a very powerful graphics card: at low resolution, the GPU is not the bottleneck, so the measured differences truly reflect the CPU's power. It might seem counterintuitive to test gaming CPUs at 1080p instead of 4K, but this is precisely how the CPU's contribution is isolated: at 4K, many different CPUs yield the same frame rate because the GPU is the limiting factor.

What not to trust? Do not trust **manufacturer-provided benchmarks** without independent verification, as they are chosen to highlight their strengths. Do not trust a **single synthetic number** as a universal metric: a CPU might shine in one test and disappoint in the software you actually use. Do not trust **GHz** as already extensively explained. The right approach is to consult independent reviews that test the applications and games you are interested in, looking at averages across multiple titles and *frame time* values (consistency, not just the average), which better reflect real fluidity than average frame rate.

7.2 CPU/GPU Balancing: Avoiding Bottlenecks

A **bottleneck** occurs when one component is too slow compared to another and limits its performance: the chain is only as strong as its weakest link. In a gaming PC, a **CPU bottleneck** means the graphics card could produce more frames, but the CPU isn't feeding it data fast enough (typical at high frame rates and low resolution, or with weak CPUs and powerful GPUs); a **GPU bottleneck** is the opposite, with the CPU waiting for the graphics card (typical at high resolutions). A certain degree of GPU bottleneck is actually desirable in high-resolution gaming, as it means you are fully utilizing the graphics card.

Avoiding a bottleneck means **balancing** the CPU and GPU according to the intended use. A common mistake is to spend almost the entire budget on the GPU, pairing it with a cheap CPU (or vice versa): the expensive component will be throttled by the other. The practical rule is to size the two components to the same "tier" level, considering the resolution at which you will be gaming: at 1080p with high refresh rates, a strong CPU is needed; at 4K, a slightly more modest CPU can be afforded, and the budget concentrated on the GPU.

7.3 Overclocking: What's Needed and If It's Still Worth It

Overclocking (OC) is the practice of running the CPU at frequencies higher than factory settings to extract more performance. To do this on Intel, you need a CPU with a **K** suffix (unlocked multiplier) and a motherboard with a **Z** chipset; on AMD, all desktop CPUs are unlocked, but an adequate chipset is still needed

(generally B or X series) and, in both cases, **robust cooling** and a good power supply, because OC increases power consumption and heat.

Is it still worth it? Much less than in the past. Modern CPUs already perform intelligent and aggressive "auto-overclocking" through boost algorithms, pushing themselves to the limit of available thermal headroom: the remaining manual gain is often modest and comes at the cost of significantly higher power consumption and heat, as well as stability risks. Today, it makes more sense to invest in a good cooler (which allows the CPU to *sustain* its automatic boosts for longer) and, on the memory front, activate the RAM's **XMP/EXPO** profiles (which are a "certified overclock" of the memory, simple and safe) rather than dedicating oneself to manual CPU overclocking. For the vast majority of users, extreme overclocking has become a hobby for enthusiasts rather than a necessity.

7.4 Budget-tier recommendations

The following prices are indicative orders of magnitude for the CPU alone (European market, July 2026) and **should be verified**, especially in the current context of high volatility. Remember also to always add the motherboard and RAM for the real cost of the platform.

Tier	CPU Budget (indicative)	Use Case	Recommended Approach
Entry	~100–150 €	Office, study, light gaming, secondary PC	Recent 6-core CPU; consider AMD "G" APU to start without a dedicated GPU; inexpensive (B) chipset, stock cooler often sufficient.
Mid	~200–350 €	Solid gaming, daily productivity, light streaming	6–8 performing cores; for gaming, an AMD X3D is often the best investment; good quality B motherboard, decent air cooler.
High-end	400 €+	Top-tier gaming, heavy	Top models (X3D for gaming, many

		editing/rendering, light workstation	cores for productivity); high-end motherboard if expansions are needed; liquid or high-level air cooling.
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A cross-cutting tip: for *gaming*, often the best use of money is to get a mid-to-high-end X3D CPU and focus the rest of the budget on the graphics card, because the GPU plays the lion's share in frame rates at high resolutions. For *productivity*, conversely, investing in CPU cores yields more.

8. Common buyer mistakes

We conclude with the mistakes that, in practice, ruin the most builds and waste the most money.

The first, almost a classic, is **buying a CPU with an "F" (or "KF") suffix without having a dedicated graphics card**. Attracted by the small saving, one ends up with a PC that gives no video signal, because the CPU lacks integrated graphics. If a dedicated GPU is not planned, this suffix should be absolutely avoided.

The second is **socket or BIOS compatibility error**: buying a CPU and a motherboard with different sockets, or a motherboard that is "compatible on paper" but with a BIOS too old to recognize the CPU, without the flashback function to update it cold. This is avoided by checking the socket, chipset, and the motherboard's official CPU support list, with the minimum required BIOS version.

The third is **pairing a powerful CPU with an inadequate cooler** (or relying on the included "stock" cooler when it's not up to the task). The result is thermal throttling: the CPU overheats, cuts frequencies, and never maintains the boost you paid for. High-end CPUs, especially when overclocked or under full load, require serious cooling; saving money here means throwing away performance paid for elsewhere.

The fourth is **neglecting RAM in relation to the IMC**. The **IMC** (*Integrated Memory Controller*) is the part of the CPU that communicates with the RAM, and

system performance heavily depends on the CPU-memory pairing. Installing slow RAM, in single channel (a single module instead of two, losing *dual channel* which doubles bandwidth), or forgetting to activate the **XMP/EXPO** profile in the BIOS (thus leaving the RAM at very low base frequencies) are errors that bottleneck an otherwise capable CPU. AMD architectures, in particular, are historically sensitive to RAM speed and latency. In 2026, however, the flip side must be added: with DDR5 prices skyrocketing, overdoing it with ultra-fast and ultra-high-capacity RAM can cost more than the CPU itself; the right balance is to aim for a good dual-channel kit at a sensible frequency, activate its XMP/EXPO profile, and not unnecessarily oversize.

The fifth error is the **CPU/GPU imbalance** already discussed: spending everything on one component and bottlenecking it with the other. The sixth, more subtle, is **paying for PCIe lanes or features that won't be used** (very expensive motherboards bought for simple gaming) or, conversely, **installing the fast SSD or GPU in the wrong slots** (chipset slots instead of those directly connected to the CPU), losing bandwidth without realizing it. Finally, an error in perspective: **chasing the latest generation at all costs**, when often the previous generation model, now discounted, offers 90% of the performance at a fraction of the price — a particularly valid consideration today, with new AMD generations postponed to 2027 and the current Intel socket already at end-of-life.

Operational Summary — CPU Decision Checklist

Before purchasing, answer these questions in order:

1. **Define the primary use.** Pure gaming, productivity/rendering, mixed use, or office/budget? This question guides everything else (for gaming, look at AMD X3D; for productivity, count the cores).
2. **Set the budget for the entire platform, not just the CPU.** Sum CPU + motherboard + RAM. In 2026, carefully check the price of DDR5, which can be the most significant item.
3. **Choose the platform with longevity in mind.** AM5 (AMD) is supported at least until 2029 and will host future CPUs; LGA1851 (Intel) is at the end of its cycle. If you plan future CPU-only upgrades, this weighs heavily.

4. Verify compatibility on three levels: identical socket, chipset adequate for the use (Z/X for overclocking, B for saving money), and supported BIOS version (check the official motherboard CPU support list). Prefer motherboards with BIOS Flashback.

5. Check the CPU suffix. Do you avoid "F/KF" if you don't have a dedicated GPU? Do you need a "G" APU to start without a graphics card? Do you want an "X3D" for gaming?

6. Size your cooling solution for actual power consumption (PL2/PPT), not TDP. Pair your chosen CPU with an adequate cooler, or you will lose performance due to throttling.

7. Balance CPU and GPU based on gaming resolution to avoid bottlenecks. Don't concentrate all your budget on a single component.

8. Plan RAM and PCIe correctly: RAM in dual channel with XMP/EXPO profile active, without over-sizing in 2026; main NVMe SSD and GPU in slots directly connected to the CPU, not to the chipset.

9. Trust independent benchmarks for the loads you will actually use (1080p gaming to isolate the CPU, Cinebench for productivity), not GHz or manufacturer slides.

10. Consider discounted previous generations. They often offer much of the performance for much less, especially in a market phase like the current one.

Final note: references to models, sockets, and prices reflect the market status in July 2026 (Arrow Lake Refresh "200S Plus", Ryzen 9850X3D, AM5 until 2029, Intel LGA1954/Nova Lake socket coming soon, Zen 6/Ryzen 10000 expected in 2027, DDR5 price crisis). Always check availability and updated price lists before purchasing.

Chapter 2 - The Motherboard

***Note on data updates.** The architectural information in this chapter (chipset function, socket hierarchy, VRM principles, form factor, PCIe and USB standards) is stable and changes slowly. **Prices**, however, are extremely volatile and vary by country, currency, availability, and promotions: any figure indicated here should be understood as an **indicative order of magnitude** and should be verified at the time of purchase. The list of the **latest generations** (in particular AMD's 800 series chipsets and the Intel LGA1851 platform) is also what was available at the time of writing: when you encounter a point where data may have already changed, you will find it explicitly marked with the phrase **[VERIFY]**.*

1. What the motherboard does

1.1 Definition and role as a hub

The **motherboard** (often abbreviated as *mobo*) is the large printed circuit board that forms the backbone of every computer. If we were to use a metaphor, it is not the "brain" (that's the CPU) nor the "muscles" (the GPU): it is rather the **nervous and circulatory system combined**, i.e., the physical and electrical infrastructure on which all other components are mounted and through which they communicate with each other and receive power.

Before going into detail, it is useful to clarify an acronym that will appear frequently: **PCB** (*Printed Circuit Board*) is the rigid sheet, usually green or black, composed of multiple layers of fiberglass alternated with very thin copper traces. These traces are the microscopic "wires" that carry signals and current from one end of the board to the other. A quality motherboard uses more copper layers (often 4 to 8, sometimes more in premium models) and a greater copper thickness, which improves the integrity of high-frequency signals and the ability to carry current without overheating.

The role of the motherboard is that of a **central hub**: it connects the CPU, RAM, graphics card (GPU), other expansion cards, storage devices (SSDs and hard drives), power supply unit (PSU), and all external peripherals (keyboard, mouse, monitor, network, audio). None of these components "speak" directly to each other: everything passes through the motherboard, which acts as a switchboard and a road network. The quality and characteristics of the motherboard therefore

determine not so much the raw power of the system (which is provided by the CPU and GPU), but rather **how much and what you can connect, at what speed, with what stability, and with what possibilities for future expansion.**

1.2 Main Components on the PCB

Looking at a motherboard from above, several distinct areas can be recognized. We will describe them here as an introduction, and then delve deeper into them in dedicated sections.

The **CPU socket** is the receptacle, generally in the upper-center of the board, where the processor is installed. It is a rectangle full of tiny contacts; its shape, number of pins, and fastening mechanism determine which CPUs are compatible. The socket is the absolute first compatibility constraint: a CPU designed for one socket will not physically fit into a different socket.

The **RAM slots** (also called *DIMM slots*, where **DIMM** stands for *Dual In-line Memory Module*, the format of memory modules for desktops) are the long vertical slots next to the socket, typically two or four. Memory modules are inserted here.

The **PCIe slots** (*Peripheral Component Interconnect Express*) are the horizontal slots in the lower part of the board. This is where the graphics card and other expansion cards (sound cards, network cards, capture cards, additional controllers) are installed. They come in different lengths depending on the number of available electrical "lanes".

The **chipset** is a small microchip, often hidden under a metal heatsink in the lower half of the board. It is the "co-processor" that manages communications with peripherals and provides additional connectivity. It is so important that we will dedicate the longest section of the chapter to it.

The **VRM** (*Voltage Regulator Module*) is the row of electronic components, often covered by a robust heatsink, that surrounds the CPU socket on one or two sides. Its task is to transform the power supplied by the PSU into a clean and stable voltage suitable for the CPU.

The **M.2 slots** are small horizontal slots, usually covered by metal plates, where modern "gum stick" format SSDs are installed. The **SATA ports** (*Serial ATA*) are the "L"-shaped connectors for connecting traditional 2.5 or 3.5-inch hard drives and SSDs.

Finally, there are the **power connectors** (the large 24-pin connector for the motherboard and one or two 8-pin connectors, called *EPS*, for the CPU), the **rear I/O panel** (the set of ports that protrude from the back of the case), and a multitude of **internal headers**, which are small pin headers where fans, RGB lighting, front USB ports of the case, and power buttons are connected. All these elements will be revisited in detail later.

2. The Chipset

2.1 What the Chipset Is and What It Manages

The **chipset** is, historically, the set of logic circuits that govern the flow of data between the CPU and the rest of the system. In computers from twenty to thirty years ago, the chipset was divided into two distinct physical parts: a **Northbridge**, close to the CPU, which managed fast connections (memory and graphics card), and a **Southbridge**, further away, which managed slow connections (disks, USB, audio, network). With technological evolution, the Northbridge has been progressively **integrated into the CPU itself**: today, the memory controller (**IMC**, *Integrated Memory Controller*, the circuit that drives RAM) and the controller for the main PCIe lanes for the graphics card live directly in the processor. What we call "chipset" in modern motherboards is, therefore, essentially the heir to the old Southbridge: a single chip dedicated to secondary connectivity.

What does the chipset of a current motherboard actually manage?

Fundamentally, it **provides additional connectivity lanes** compared to those the CPU offers alone. The CPU, for cost and physical space reasons, exposes a limited number of PCIe lanes (typically 16 lanes reserved for the graphics card plus 4 or 8 lanes for one or two NVMe SSDs) and few other resources. Everything else — the numerous USB ports, SATA ports for disks, the network card, PCIe lanes for secondary slots, often the audio chip and Wi-Fi — is managed or enabled by the chipset. In practice, the chipset is a **connectivity multiplier and distributor**: it takes a single fast connection from the CPU and "expands" it into dozens of ports and slots.

This means that the choice of chipset directly determines **how many peripherals you can connect and with what advanced capabilities** (overclocking, USB4, PCIe 5.0), while it **does not** determine the system's computing power. Two boards with different chipsets but the same CPU and RAM offer, with the same configuration,

almost identical performance in normal workloads: the difference lies in features and expandability, not in "frames per second."

2.2 Communication between CPU and chipset

The chipset must constantly communicate with the CPU, and this communication occurs over a dedicated high-speed link. It is important to understand this because that link is a **potential bottleneck**: all traffic from peripherals managed by the chipset (disks, USB, secondary slots) must pass through it to reach the CPU.

In **Intel**, this link is called **DMI** (*Direct Media Interface*). DMI is, technically, a bus derived from PCIe technology. In recent platforms, DMI 4.0 is used with a variable number of lanes depending on the chipset's tier: entry-level chipsets use a narrower DMI (for example, the equivalent of 4 PCIe 4.0 lanes), while high-end chipsets use a wider DMI (the equivalent of 8 PCIe 4.0 lanes), which doubles the available bandwidth between the chipset and the CPU. This explains why a high-end chipset not only offers more ports but also allows them to work together without saturation.

In **AMD** systems, on the AM5 platform, the connection between the CPU and the chipset is simply a **4-lane PCIe 4.0 link**. An important architectural detail concerns AMD's "dual" chipsets: in X670/X670E and X870E motherboards, AMD uses **two identical chips** (codename Promontory 21) connected **in cascade** (daisy chain), meaning one attached to the other. The first chip communicates with the CPU, the second communicates with the first. This doubles the total number of available ports and lanes but introduces a small increase in latency for devices connected to the second chip, and still maintains that 4-lane link as the overall bottleneck to the CPU. This is the compromise AMD chose to offer extensive connectivity by reusing a single, high-volume manufactured chip.

2.3 The AMD chipset hierarchy (AM5 socket)

The **AM5** socket is AMD's current desktop platform, introduced with the Ryzen 7000 series processors and also used by subsequent series (8000 with integrated graphics and 9000). One of AMD's most appreciated features is **socket longevity**: AM5 has been declared supported for several years, which generally allows for future CPU upgrades while keeping the same motherboard (after a BIOS update, which we will discuss). All AM5 chipsets use **DDR5** memory (there is no DDR4 variant on AM5) and provide PCIe 5.0 from the CPU side.

Before reading the table, three fundamental interpretative keys. The letter "E" at the end of the name (as in B650E or X670E) stands for *Extreme* and indicates that **PCIe 5.0 lanes are guaranteed for both the graphics card and at least one SSD**; in versions without "E", PCIe 5.0 for the graphics card is often optional or absent, while for storage it may be present. The **800 series** (B850, X870, X870E), newer than the 600 series, is distinguished primarily by the wider availability of **USB4** (a very high-speed port, up to 40 Gbps, compatible with Thunderbolt): it is **mandatory on X870/X870E**, while on B850 it remains optional but common on most boards. Finally, on AM5, **CPU overclocking is enabled almost everywhere except on the entry-level A620 chipset**, and this is one of the most significant historical differences compared to Intel.

AMD Chipset (AM5)	Tier	CPU Overclock	PCIe 5.0 GPU	PCIe 5.0 storage	USB4	Notes
A620	Entry	No	No (GPU to PCIe 4.0)	No/limited	No	Fewer lanes and ports; ideal for budget builds. On some boards, limits on top CPU TDP.
B650	Mainstream	Yes	Optional (often PCIe 4.0)	Yes (at least 1 M.2)	Optional/rare	The "sensible" chipset for most users.
B650E	Mainstream+	Yes	Yes (guaranteed)	Yes	Optional	Like B650 but with PCIe 5.0 also for the GPU.
X670	High (dual chip)	Yes	Optional	Yes	Optional	Many more USB/SATA ports

						and PCIe lanes.
X670E	High (dual chip)	Yes	Yes	Yes	Optional	Maximum connectivity for the 600 series, full PCIe 5.0.
B850	Mainstream+ (800 series)	Yes	Optional	Yes	Optional but frequent	Update of B650E; USB4 more widespread. [VERIFY] availability/price
X870	High (800 series)	Yes	Frequent / guaranteed on many boards	Yes	Mandatory	Single chip but with USB4 as standard. [VERIFY]
X870E	Top (800 series, dual chip)	Yes	Yes	Yes	Mandatory	The ultimate in AM5 expandability. [VERIFY]

[VERIFY] The exact PCIe 5.0 guarantees on the video card for X870 (non-E) vary from model to model: some boards implement it, others do not. If there is a strict requirement (for example, a PCIe 5.0 GPU or SSD to be fully utilized), always check the technical specifications of the **individual model**, not just the chipset name.

A recurring practical clarification: the difference between a B650 and an X670E **does not concern how fast your game will be**. It concerns how many NVMe SSDs you can install, how many fast USB ports you will have, whether you can use two

video cards or expansion cards simultaneously at full bandwidth, and how much margin you will have for extreme overclocking. For a normal single-video-card gaming PC, a good B650/B650E is more than sufficient.

2.4 The Intel Chipset Hierarchy (LGA1700 and LGA1851)

Intel has recently undergone a socket change, so it's necessary to distinguish between two platforms.

The **LGA1700** socket hosts 12th, 13th, and 14th generation processors (code names Alder Lake, Raptor Lake, and Raptor Lake Refresh). It is a "mature" platform, widely diffused and often available at convenient prices. A historical peculiarity of LGA1700 is that, depending on the board, it can use either **DDR4 or DDR5** memory (never both on the same board): this makes it interesting for those who want to reuse existing DDR4 RAM and save money, with a small performance penalty compared to DDR5.

The **LGA1851** socket is the most recent platform, introduced with **Core Ultra 200S series** processors (code name Arrow Lake). It supports **only DDR5** and introduces a different organization of lanes (the CPU offers PCIe 5.0 for the video card and for storage, with improved native support for Thunderbolt/USB4). Pay attention to a clear constraint: **LGA1851 is not backward compatible with LGA1700 CPUs and vice versa**; the number (1700 vs 1851) precisely indicates the number of contacts on the socket.

Intel's golden rule on overclocking is diametrically opposed to AMD's: **only "Z" series chipsets (Z690, Z790, Z890) allow CPU overclocking**. All other Intel chipsets allow at most memory overclocking (via Intel's **XMP**, *Extreme Memory Profile*, profiles — the equivalent of AMD's EXPO), and on this point there has been an evolution: while historical entry-level chipsets like H610 do not even enable XMP, more recent mid-range chipsets (B760, B860) do.

Intel Chipset	Platform	Tier	CPU OC	Memory OC (XMP)	DMI to CPU	Notes
H610	LGA1700	Entry	No	No	x4	Often 2 RAM slots; few USB/SAT A; no PCIe 5.0

						from chipset.
B660	LGA1700	Mainstream	No	Yes	x4	Good balance point 12th/13th gen.
B760	LGA1700	Mainstream	No	Yes	x4	B660 refresh, better connectivity out of the box.
H670	LGA1700	Mid-range	No	Yes	x8	More lanes than B660, no CPU OC.
H770	LGA1700	Mid-range	No	Yes	x8	H670 refresh, more fast USB.
Z690	LGA1700	High-end	Yes	Yes	x8	Full OC, maximum lanes (12th/13th gen).
Z790	LGA1700	High-end	Yes	Yes	x8	Z690 refresh, more USB 3.2 Gen2x2.
B860	LGA1851	Mainstream	No	Yes	x8 [VERIFY]	Mainstream Arrow Lake; USB4/Thunderbolt more common. [VERIFY]

Z890	LGA1851	Top	Yes	Yes	x8	Flagship Arrow Lake: full OC, many PCIe 4.0 lanes from chipset, native Thunderbolt 4/USB4.
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[VERIFY] The precise details of Intel's 800 series (B860, and the entry H810 not listed but existing) and the exact width of their DMI may differ from "traditional" values; check the specific model's datasheet, especially for the number of M.2 slots and USB4/Thunderbolt support.

2.5 Decision table: which chipset for which user

This table pragmatically summarizes the pairing between user type and recommended chipset. It is intentionally simplified: it serves as a guide, not a replacement for checking individual models.

User Profile	Recommended AMD (AM5)	Recommended Intel
Office/multimedia, minimal budget, no OC	A620	H610 / B660
Entry-mid gaming, single GPU, 1–2 SSDs	B650	B760
Mid-high gaming/creator, overclocking, 2+ fast SSDs	B650E / B850	Z790 / Z890
Workstation, maximum connectivity, many drives, USB4	X670E / X870E	Z890
Those who want to reuse DDR4 RAM and spend little	(not available: AM5 is DDR5 only)	LGA1700 with DDR4 board

2.6 Overclocking: the net summary

Summarizing the logic of overclocking, because it is one of the first choices that influences the chipset: on **AMD AM5**, **CPU overclocking is allowed on all chipsets except the A620**, so even an inexpensive B650 allows pushing an unlocked processor. On **Intel**, **CPU overclocking strictly requires a "Z" chipset** (Z690, Z790, Z890); with any other chipset, the CPU runs at its factory frequencies (including automatic turbo modes, which are not considered manual overclocking). **Memory** overclocking, however, is more permissive: almost all modern chipsets allow it via EXPO (AMD) or XMP (Intel), with the exception of the cheapest Intel chipsets like H610. Since on many platforms the biggest performance gain, for the same cost, comes precisely from activating the high-speed memory profile, checking that the chipset allows it is often more important than being able to overclock the CPU.

3. The VRM (power delivery section)

3.1 What is the VRM and why does it exist

The **VRM** (*Voltage Regulator Module*) is the circuit that converts the **12 volts** supplied by the power supply into the much lower and more precise voltage required by the CPU, typically between about **1.0 and 1.4 volts**. This may seem like a trivial task, but it is not at all: a modern high-performance CPU can draw **over 200-250 watts** at full load, and since power is the product of voltage and current, at low voltage this means enormous currents, on the order of **150-250 amperes**. The VRM must supply this current stably, cleanly, and reactively: if the CPU goes from idle to maximum load in a few microseconds, the VRM must follow that demand without causing the voltage to collapse or overload, otherwise the system becomes unstable (crashes, reboots) or the CPU reduces its frequency to protect itself (*throttling*).

3.2 Power Phases

To handle such high currents, the VRM is divided into multiple **phases**. A power phase is a small, complete circuit essentially composed of several elements. There are **MOSFETs** (*Metal-Oxide-Semiconductor Field-Effect Transistor*), electronic switches that turn on and off thousands of times per second; modern VRMs often use integrated components called **DrMOS** or *power stages*, which enclose MOSFETs and their driving logic in a single package, improving efficiency and temperature. Then there's the **choke** (or inductor), that small square block that

stores energy and "smoothes" the current. There are **capacitors**, which stabilize the voltage by filtering out disturbances. And finally, there's a **PWM controller** (*Pulse Width Modulation*), the "orchestra conductor" that coordinates the staggered switching of the various phases.

The advantage of having multiple phases is twofold. First, the total current is **distributed**: if ten phases share 200 amperes, each handles only 20 amperes, remaining cool and reliable. Second, by having the phases work in a **staggered** manner over time, the resulting voltage is more continuous and cleaner. However, be aware of a common myth: **the number of phases is not in itself a measure of quality**. A motherboard advertised with "16 phases" but built with cheap components and inadequate dissipation can perform worse than one with "8 phases" but high-current components and a good heatsink. It should also be known that many motherboards use **doublers**, circuits that make one phase appear as two: the board claims "12+2 phases" but the controller actually drives half that number. This is not necessarily a defect, but it is why simply counting phases is an unreliable marketing metric.

3.3 How to recognize an adequate VRM

Since the nameplate data is ambiguous, the best way to evaluate a VRM is to **cross-reference three signals**. The first is the **presence and mass of the heatsinks**: a serious VRM is covered by robust aluminum blocks (sometimes connected by a *heatpipe*, a small tube that transfers heat) with a large surface area and fins; a "naked" VRM, with exposed chips, is almost always an indicator of an inexpensive board intended for low-power CPUs. The second signal is the **amperage rating of the individual phases**, a data point that technical reviewers often report (for example, "50A or 70A power stage"): phases of 60-90A each are an excellent sign. The third, and in practice the most reliable for non-engineers, is to **read specialized technical reviews**, which directly measure VRM temperatures under prolonged load with a demanding CPU. If a review shows that the VRM of a certain board stays below 70-80 °C with the CPU you intend to use at maximum load, that board is adequate; if it exceeds 100-110 °C or throttles, it is not.

The practical rule is simple: **the more powerful the CPU and the more you intend to overclock it, the more the VRM matters**. For an entry-level CPU in an office PC, almost any VRM will do. For a high-end, multi-core CPU pushed to its

limits, a weak VRM becomes the limiting factor for the entire system and can even reduce its lifespan due to excessive heat.

4. The Form Factor

4.1 What the form factor is and why it matters

The **form factor** is the dimensional standard of the motherboard: it defines the physical measurements, the position of the mounting holes, and the general layout of the components. It is a crucial parameter because it must be **consistent with the case**: a case declares which form factors it accepts, and it is not possible to mount a board larger than what the case allows. The form factor also influences expandability (number of available slots), ease of cooling, and, in part, the price.

4.2 The four main form factors

ATX (*Advanced Technology eXtended*) is the reference standard in the desktop world, with dimensions of **305 × 244 mm** (approximately 30.5 × 24.4 cm). It is the "full" format: it typically offers four RAM slots, multiple PCIe slots, ample space for the VRM and heatsinks, and maximum connectivity. It is the most balanced choice for most fixed builds because it does not impose compromises on expandability.

Micro-ATX (often abbreviated as *mATX* or *μATX*) measures **244 × 244 mm**, meaning it is a square that is shorter downwards compared to ATX. It usually retains four RAM slots but reduces the number of PCIe slots (typically one or two usable). It is the format with the **best price/performance ratio**: mATX boards often cost less than equivalent ATX boards while offering everything needed for a normal PC with a single graphics card. It is an excellent choice for those who want to save money without sacrificing four RAM slots.

The **Mini-ITX** measures just **170 × 170 mm**: it is the compact format par excellence, designed for small and portable PCs. However, it imposes significant compromises: it has **only one PCIe slot** (meaning only one video card, no additional cards) and, above all, **only two RAM slots**. For the same features, Mini-ITX boards also tend to cost **more** than ATX boards, because concentrating everything into such a small space is engineeringly expensive. It is chosen for its form factor, not for saving money.

The **E-ATX** (*Extended ATX*) is the largest format, with a width of up to **305 × 330 mm** (exact measurements vary slightly between manufacturers, and this is a point to check carefully against the case specifications). It is reserved for workstations and high-end systems that require enormous VRMs, many slots, and extreme connectivity. It requires large and specifically compatible cases, and comes at a high cost.

Form factor	Dimensions (mm)	Typical PCIe slots	Typical RAM slots	Pros	Cons
E-ATX	up to 305 × 330	Many	4 (sometimes 8 on HEDT)	Maximum expandability and VRM	Expensive, requires large and compatible cases
ATX	305 × 244	3–4	4	Ideal balance, widespread standard	Larger footprint than mATX
Micro-ATX	244 × 244	1–2	4	Excellent price, compact but spacious	Fewer PCIe slots
Mini-ITX	170 × 170	1	2	Extreme compactness, portability	Limited expandability, high price, limited RAM

4.3 Practical implications

The choice of form factor is intertwined with the choice of case and the goals of the build. For a "normal" gaming or work PC, **ATX or Micro-ATX** cover the vast majority of needs; Micro-ATX in particular is often the smartest choice in terms of value for money. **Mini-ITX** makes sense when compactness is a priority (a living room PC, one to take to events), accepting its limitations and premium price. **E-ATX** is only appropriate for professional configurations with very high expandability, where many lanes, many drives, and an oversized VRM are truly needed. A common mistake is to buy an elegant but small case and then discover

that the desired ATX board doesn't fit, or that there's no space for the CPU cooler: **the form factor must be decided together with the case, not separately.**

5. Slots and ports

5.1 PCIe slots: number, generation, and lane allocation

As anticipated, **PCIe** slots house the video card and other expansion cards. To understand them, two concepts must be introduced: **generation** and **lanes**.

Generation (PCIe 3.0, 4.0, 5.0) determines the bandwidth per lane: each generation doubles the previous one. A PCIe 3.0 lane offers about 1 GB/s, a 4.0 about 2 GB/s, a 5.0 about 4 GB/s. **Lanes** are the parallel electrical pathways: an "x16" slot has sixteen lanes, an "x4" has four, and so on. Total bandwidth is the product of lanes and bandwidth per lane: an **x16 slot in PCIe 4.0** therefore offers about 32 GB/s, an **x16 in PCIe 5.0** about 64 GB/s.

A point that confuses many beginners is the difference between the **physical length** and **electrical lanes** of a slot. A slot can be physically as long as an x16 (to accommodate a graphics card, which requires the long connector) but only be electrically wired as x4 or x1. On technical specifications, this is written, for example, as "PCIe x16 (in x4 mode)": this means that the graphics card physically fits, but only receives a quarter of the bandwidth. This is an important detail when you want to use multiple slots simultaneously.

The most important distinction, however, is between **slots connected directly to the CPU** and **slots connected to the chipset**. The **first x16 slot**, intended for the main graphics card, is almost always connected **directly to the CPU**: it is the fastest, with full bandwidth and minimal latency, and it is the one you should use for the GPU. Secondary slots, on the other hand, are generally connected **to the chipset**, which means their bandwidth shares the DMI/PCIe link to the CPU we discussed: if you use them intensively along with other chipset peripherals, they may contend for bandwidth. For a normal single-graphics-card PC, this is completely irrelevant; it only becomes relevant in configurations with multiple high-bandwidth expansion cards.

5.2 M.2 slots and lane sharing

M.2 slots house modern SSDs. An NVMe SSD (*Non-Volatile Memory Express*, the fast protocol used by high-performance M.2 SSDs) typically uses **four PCIe lanes**

to achieve its high speeds. Motherboards offer one to four (or more, on top models) M.2 slots, of different generations: the first M.2 slot is often connected directly to the CPU and can be PCIe 5.0 or 4.0, while the others are connected to the chipset and may be of a lower generation.

Here, a concept as important as it is underestimated comes into play: **lane sharing**. Since the total PCIe lanes are a finite resource, manufacturers often **share** the same lanes among multiple connectors, with the rule that **they cannot all be used simultaneously**. The classic case: a certain M.2 slot **shares lanes with some SATA ports**, so installing an SSD in that slot **automatically disables two or four SATA ports**. Another case: a second M.2 slot shares lanes with a secondary PCIe slot, so populating one reduces or disables the other. Furthermore: on some boards, populating all M.2 slots causes the graphics card slot to drop from x16 to x8.

This is one of the **most costly mistakes** made by inexperienced buyers: they buy a board with "four M.2 slots and six SATA ports" convinced they can use them all, and then discover that filling the M.2s turns off half the SATA ports, or that adding a drive removes lanes elsewhere. The solution is to **always read the board's manual (or the "detailed specifications" section) before purchasing**, identifying the lane sharing table, which lists exactly which connectors are mutually exclusive. On platforms with few lanes (e.g., entry-level chipsets), lane sharing is more stringent; on high-end chipsets, with many more lanes from the chipset, it is less frequent but still possible.

5.3 RAM slots: 2 vs 4

The number of RAM slots has two implications. The first concerns **dual channel**: modern desktop platforms achieve full memory bandwidth by using **two channels**, which means installing modules **in pairs**. With four slots, you can start with two modules and add two more later; with two slots (typical of Mini-ITX), you are already "full" if you want dual channel, and to increase capacity you will have to **replace** the modules rather than add them. The second implication concerns **upgrading**: four slots offer more future flexibility, while two slots force you to discard existing modules to increase RAM.

A technical detail often overlooked: **filling all four slots makes it more difficult to achieve very high memory frequencies**, because the integrated memory controller (IMC) in the CPU has to drive more modules and the signal is more complex to manage. For this reason, those aiming for maximum RAM speeds often

prefer **two high-capacity modules** rather than four lower-capacity ones, even if they have four slots.

5.4 SATA Ports

SATA ports are used to connect mechanical hard drives and "traditional" 2.5-inch SSDs. They are still useful for large, inexpensive storage (large hard drives use SATA). The number of SATA ports varies from four to eight depending on the motherboard; as already seen, some of these ports can be shared with M.2 slots via lane sharing. Those who plan to connect many mechanical drives (for example, for a small home server or multimedia storage) must carefully count the SATA ports **actually usable simultaneously with the M.2 SSDs** they intend to install.

5.5 The Rear I/O Panel

The **I/O panel** (Input/Output) is the row of ports that protrudes from the back of the case. It is one of the most tangible differences between price ranges, and it is good to decipher its acronyms.

USB ports follow a confusing nomenclature due to continuous renamings. **USB 2.0** are slow (480 Mbps) and only suitable for keyboards, mice, and undemanding peripherals. **USB 3.2 Gen 1** (formerly called USB 3.0 or USB 3.1 Gen 1) offer 5 Gbps; **USB 3.2 Gen 2** offer 10 Gbps; **USB 3.2 Gen 2x2** reach 20 Gbps using two lanes. At the top are **USB4** and **Thunderbolt**, which reach **40 Gbps** and support advanced functions such as connecting monitors and high-speed external enclosures; USB4 and Thunderbolt 4 are largely interoperable. The **Type-C connector** is the now standard reversible physical form for faster ports, while the classic rectangular **Type-A** remains common for intermediate speeds.

On the panel you will also find **video** outputs (HDMI, DisplayPort): these are only needed **if you use the integrated graphics (iGPU)** of the CPU, i.e., without a dedicated graphics card; if you have a graphics card, you will connect the monitor to it and the motherboard's outputs will remain unused. Then there is **networking**: modern motherboards offer at least one **2.5G** (2.5 Gbps) Ethernet port, while high-end models may have **10G** (10 Gbps); **Wi-Fi** with its antenna is also often present. The **audio** section includes analog jacks and, on better motherboards, an optical output and higher quality audio chips.

Two very practical functions, found mostly in mid-range motherboards and above, deserve a mention. **BIOS Flashback** (also called *Q-Flash Plus*, *Flash BIOS*

Button depending on the manufacturer) is a button that allows you to **update the BIOS without a CPU, RAM, or graphics card installed**: it is essential when you buy a "new old stock" motherboard with a BIOS too outdated to recognize your new CPU. **Clear CMOS** is a button (or a jumper) that **resets the BIOS settings** to their factory defaults: indispensable for restoring startup after a failed overclock.

5.6 Internal Headers

Headers are pin connectors on the PCB where internal cables are connected. The main ones are the **front USB headers**, which power the case's USB ports (Type-A and Type-C on the front); **ARGB and RGB headers** (*Addressable RGB*, individually addressable LED lighting), for connecting fans and light strips; **fan connectors** (usually 4-pin PWM, which allow speed control); and the header dedicated to the **AIO pump** (*All-In-One*, the pre-assembled liquid cooling system), which provides stable and full power to the pump. Counting the available headers is important for those with many fans or an elaborate lighting system: an inexpensive board might offer too few, forcing the purchase of additional *splitters* or *hubs*.

6. Nomenclature and Brands

6.1 Why Names Matter

Manufacturers organize their boards into **lines** (or *series*), which represent increasing price and feature tiers. Learning to read these lines allows you to immediately place a board in the correct tier without having to analyze every specification. Be careful, however: **the same line can contain models of different prices** depending on the chipset (a "Strix B650" costs less than a "Strix X670E"), so the line indicates *relative positioning*, not an absolute price.

6.2 ASUS

ASUS structures its range quite linearly. The **Prime** line is the entry/mainstream, essential and affordable. **TUF Gaming** is the mid-range, with an emphasis on robustness and reliable components, offering an excellent quality/price compromise. **ROG Strix** (*Republic of Gamers*) is the mid-to-high range, with more robust VRMs, more connectivity, and refined aesthetics. At the top are **ROG Maximus** (for Intel platforms) and **ROG Crosshair** (for AMD platforms), the flagship boards designed for extreme overclocking and maximum features. Finally, there is the **ProArt** line, aimed at content creators, with a sober aesthetic and connectivity (Thunderbolt, fast networking) designed for workstations.

6.3 MSI

MSI uses four-letter acronyms in ascending order of tier. **PRO** is the entry/business line, sober and functional. **MAG** (*MSI Arsenal Gaming*) is the mid-range gaming tier, where very popular models like the "Tomahawk" fall. **MPG** (*MSI Performance Gaming*) is the mid-to-high range. **MEG** (*MSI Enthusiast Gaming*) is the top, with the flagship boards. The mnemonic rule is that, as you go up, the second letter tells the progression: from PRO to Arsenal, Performance, Enthusiast.

6.4 Gigabyte

Gigabyte separates the entry-level from the gaming lines. **UD** (*Ultra Durable*) is the essential entry-level. Boards simply labeled **Gaming** occupy the mid-range. The premium brand is **Aorus**, structured in increasing levels: **Aorus Elite** (mid-range), **Aorus Pro** (upper mid-range), **Aorus Master** (high-end), and **Aorus Xtreme** (flagship). There is also the **Aero** line for content creators. Moving up the Aorus scale, VRMs, cooling, connectivity, and build quality all improve.

6.5 ASRock

ASRock has historically offered aggressive pricing. Entry-level boards carry essential names or the designation **Pro**. **Steel Legend** is the mid-range with a distinctive aesthetic. **Phantom Gaming** (often abbreviated as *PG*) is the mid-range gaming line, while names like **Riptide**, **Lightning**, or **Nova** identify specific variants. At the top is **Taichi**, the flagship line recognizable by its gear-themed aesthetic, with a complete feature set and robust VRMs.

6.6 Price Ranges and Brand Reputation

As a general guideline (with the caveat that **prices are volatile** and should always be checked), entry-level lines are in the most economical price bracket, mid-range lines in the intermediate, and flagship lines in the high-end, with top-tier boards potentially costing as much as a mid-range graphics card. The most important practical advice is: **don't pay for a tier you won't use**. A flagship board only makes sense if you truly utilize its features (extreme overclocking, exotic connectivity, many drives); otherwise, that money is better invested in the CPU, GPU, or RAM.

Regarding **brand reputation**, three factors matter more than aesthetics. The first is the **quality and usability of the BIOS/UEFI** (the firmware interface used to configure the board): a well-made, stable, and feature-rich BIOS makes life much easier, especially for overclocking and memory management. The second is the quality of support and the **RMA** (*Return Merchandise Authorization*) procedure,

which varies in speed and efficiency by region. The third is the **actual quality of components** (VRMs, heatsinks, network, audio) beyond marketing. Since all major brands (ASUS, MSI, Gigabyte, ASRock) produce both excellent boards and mediocre models, the golden rule is **not to trust the brand alone but to evaluate each specific model through independent technical reviews**.

7. How to Choose the Ideal Motherboard

7.1 The Step-by-Step Procedure

Choosing a motherboard never starts with the motherboard itself: it starts with the **CPU**, and proceeds in a cascading manner. This is the correct method, ordered by priority.

The first step is to **decide on the CPU**, because the CPU determines everything else. The **socket** immediately follows from the CPU (AM5 for recent Ryzens, LGA1700 or LGA1851 for Intel depending on the generation): this is a rigid and non-negotiable constraint. Once the socket is chosen, you move on to the **chipset**, evaluating which functions are truly needed: overclocking yes or no, how many lines and ports, PCIe 5.0 or USB4 necessary or superfluous. Then the **form factor** is defined in coordination with the case (ATX, Micro-ATX, Mini-ITX): this choice must be made together with the case, not afterwards. At this point, the **necessary connectivity** is checked by comparing how many USB ports, how many M.2 slots, how many SATA ports, and what network are needed for your actual use. Only then is the **VRM** checked, sizing it to the chosen CPU (the more powerful and overclocked the CPU, the more robust the VRM must be). Finally, the **budget** is applied, choosing from the models that meet the requirements the one with the best quality/price ratio, without paying for unnecessary features.

7.2 Compatibility Checks

Once a candidate is identified, its compatibility must be checked on five fronts, because an error here can render the system unusable.

Compatibility with the **CPU** is not limited to the socket: you need to check that the board supports that specific CPU **with the installed BIOS version**. Boards sold before the release of a new CPU might have a BIOS that is too old to recognize it, making an update necessary via the **BIOS Flashback** function (which, as seen, also works without a compatible CPU). Manufacturers publish a **supported CPU**

list for each board: this should always be consulted when pairing a very new CPU with a board that might have been in storage for some time.

Compatibility with **RAM** is verified on the **QVL** (*Qualified Vendor List*, the list of memory modules tested and certified by the manufacturer for that board) and on the supported memory speeds. The QVL is not an absolute constraint (modules not listed often work), but choosing one present in the list reduces the risk of stability problems, especially at high frequencies.

Compatibility with the **GPU** mainly concerns physical space (*clearance*): modern video cards are large and heavy, and you need to ensure that, once installed, the GPU does not cover M.2 slots or SATA ports that are needed, and that there is space in the case. In compact form factors, this aspect becomes critical.

Compatibility with the **case** is the form factor consistency already discussed: the case must accept the dimensions of the board. Compatibility with the **cooler** (CPU heatsink) concerns the bulk: very tall air coolers or large liquid radiators must find space without interfering with the VRM heatsinks, tall RAM modules, or the case panel.

7.3 Typical Buyer Mistakes

Some errors occur with such frequency that they warrant an explicit warning. The first is **paying for a top-tier chipset without using its features**: buying an X670E or a Z890 and then installing a mid-range CPU, a single graphics card, and a single SSD means throwing away money that would have yielded much more in CPU, GPU, or RAM. The second, already discussed, is **ignoring M.2 and SATA lane sharing**, ending up with ports that disable each other after purchase. The third is **pairing a weak VRM with a top-tier CPU**: an inexpensive board with an undersized VRM will limit a powerful CPU and can overheat under prolonged load. To these are added more trivial but frequent errors: forgetting to check BIOS support for a new CPU, choosing a form factor incompatible with the case, underestimating the size of the cooler, and being guided solely by the number of VRM phases or aesthetics instead of technical reviews.

Operational Summary — Motherboard Decision Checklist

Use this checklist in order, from top to bottom. Each point must be checked before moving to the next.

1. CPU and socket (rigid constraint). Have I already chosen the CPU? The socket derives from it: AM5 (recent Ryzens, DDR5 only), LGA1700 (Intel 12th–14th gen, DDR4 or DDR5) or LGA1851 (Intel Core Ultra 200S, DDR5 only). The board must have exactly that socket.

2. BIOS Support. Does the board support my specific CPU with the factory BIOS? If the CPU is very recent, does the board have **BIOS Flashback** to update it without a compatible CPU? Have I checked the supported CPU list?

3. Chipset and actual features. Do I need **CPU overclocking**? If yes: for Intel, a **Z** chipset is needed; for AMD, any chipset except A620 is sufficient. Do I need **RAM overclocking** (EXPO/XMP)? Almost all allow it, except Intel H610. Do I really need **PCIe 5.0** (GPU/SSD) or **USB4**, or am I paying for features I won't use?

4. Form factor and case. Have I chosen the format (ATX / Micro-ATX / Mini-ITX / E-ATX) **together with the case**? Does the case accept these dimensions? Remember that Mini-ITX only has 2 RAM slots and 1 PCIe slot, and that Micro-ATX often offers the best value for money.

5. Connectivity and lane sharing. Have I counted how many **M.2 slots**, how many **SATA ports**, and how many **USB ports** I need? Have I read the **lane sharing** table to verify that the M.2 and SATA I want to use **do not disable each other**? Do I have enough **RAM slots** (2 vs 4) for future upgrades I anticipate? Is the **network** (2.5G/10G/Wi-Fi) adequate?

6. VRM. Is the chosen CPU powerful and/or will I overclock it? If yes, does the board have a **robust VRM** (substantial heatsinks, high-current phases) confirmed by **technical reviews** (temperatures under prolonged load)? Remember that the number of phases alone is not an indicator of quality.

7. Physical compatibility. Does the **GPU** fit and not cover ports/slots I need? Does the **CPU cooler** (tall air or liquid radiator) find space without interfering with VRM, RAM, and case? Is the chosen **RAM** on the QVL or otherwise compatible with the declared speeds?

8. Brand and model. Have I placed the board in the correct tier by reading the line (Prime/TUF/Strix/Maximus-Crosshair for ASUS; PRO/MAG/MPG/MEG for MSI; UD/Gaming/Aorus for Gigabyte; Pro/Steel Legend/Phantom Gaming/Taichi for ASRock)? Have I evaluated the **individual model** with independent reviews, instead of trusting only the brand or aesthetics?

9. Final budget. Among the models that pass all the previous points, am I choosing the one with the **best value for money**, avoiding paying for a chipset or a tier whose features I will not use? Would that money be better spent elsewhere (CPU/GPU/RAM)?

Final golden rule. The motherboard does not increase "fps": it enables, connects, and powers. Buy the **cheapest board that meets all your real requirements** with a VRM adequate for the CPU — no more, no less.

*End of chapter. Architectural data is updated to the best of available knowledge at the time of writing; prices, availability, and details of the latest chipsets (AMD 800 series, Intel LGA1851 platform) marked with **[VERIFY]** must be confirmed on the official technical specifications of individual models at the time of purchase.*

Chapter 3 - RAM Memory

***Note on data updates.** This chapter is written with knowledge updated to early 2026. Technical concepts (how RAM works, what CAS latency is, why dual channel doubles buses) are stable over time and do not age. Conversely, three categories of information change rapidly and are explicitly indicated in the text with the label **[volatile data]**: (1) prices, (2) what "the latest generation" on the market is and the maximum frequencies reached, (3) specific recommended kits and motherboard compatibility lists (QVLs). Before purchasing, always verify these three points from recent sources.*

1. What is RAM

1.1 Definition: volatile working memory

RAM is an acronym for *Random Access Memory*. The term "random" has nothing to do with chance or luck: it simply means that the processor can read or write any memory cell in a nearly constant time, regardless of its physical location. This distinguishes it from sequential access memories (like old magnetic tapes), where to reach data at the end, one had to scroll through everything that preceded it. In RAM, however, every address is directly accessible, as if each square on a giant chessboard had its own postal address and the postman could teleport anywhere in one go.

The most important characteristic to understand, and the one that most often confuses those new to hardware, is that RAM is **volatile** memory. Volatile means that it retains data only as long as it is electrically powered: the instant the computer is turned off, or even just restarted, all the contents of RAM vanish. This is not a defect; it is a design choice. RAM is not for *storing* data over time; it is for *working on it* in the present. It is the workbench, not the archive.

1.2 The fundamental difference from storage

To truly understand the purpose of RAM, it's essential to contrast it with **storage**, i.e., mass storage: mechanical hard drives (HDD) and solid-state drives (SSD). Storage is **non-volatile**: it retains data even when the computer is off. This is where the operating system, installed programs, documents, photos, and games reside. When we buy "a computer with 1 TB," that terabyte is almost always storage.

The most effective, and not coincidentally most used, metaphor is that of the desk and the filing cabinet. Imagine an employee who needs to process paperwork. The **filing cabinet** (storage) contains all the office's paperwork, thousands of folders, organized and preserved forever, but inconvenient to consult: to retrieve a document, one must stand up, walk, open a drawer, and search. The **desk** (RAM), on the other hand, is the surface where the employee places only the documents they are currently working on: they are at hand, immediately accessible, but space is limited, and at the end of the day, the desk is cleared.

This metaphor explains three things at once. First, why RAM is much faster than storage: retrieving a sheet already on the desk is enormously quicker than going to find it in the filing cabinet. Numerically, RAM operates with latencies in the order of nanoseconds (billionths of a second) and bandwidths of tens of gigabytes per second, while even a fast NVMe SSD has latencies thousands of times higher. Second, why RAM is more expensive per gigabyte: it is an intrinsically more refined and faster technology, so 32 GB of RAM and 32 GB of SSD have very different prices. Third, why much less is needed: I keep five files on my desk, but I store five thousand in the filing cabinet.

When you launch a program, the operating system **copies** the necessary code and data from storage (filing cabinet) to RAM (desk). The CPU works by reading from and writing to RAM, not directly to storage, because otherwise it would be dramatically slowed down. When you save a file, the process is reversed: the data is written from RAM to storage to be made permanent. This is precisely why, if the power goes out before you've saved, unsaved work is lost: it was only on the desk, never put back in the filing cabinet.

A common conceptual error is confusing "low RAM" with "low space." If storage is full, the computer indicates that there is no more room to save files. If RAM is insufficient, however, the computer doesn't immediately crash: it resorts to a trick called **swap** (or *paging file*), which uses a portion of storage as emergency RAM. Since storage is orders of magnitude slower, the result is a noticeable slowdown, with the HDD or SSD constantly working and the interface becoming choppy. Many users who complain about a "slow" computer actually have an insufficient RAM problem that forces continuous reliance on swap.

1.3 How CPU and RAM communicate: the integrated memory controller

For the CPU to use RAM, someone must act as an intermediary: translating the processor's requests ("give me the content of address X," "write this value to

address Y") into precise electrical signals on the memory buses, adhering to very strict timings. This task falls to the **memory controller**.

Until about twenty years ago, on the PC platform, the memory controller resided in a separate chip on the motherboard called the **northbridge**, part of the so-called chipset. The CPU communicated with the northbridge via a bus (the Front Side Bus), and the northbridge communicated with the RAM. This introduced an extra intermediary, with its own latency. The breakthrough came when designers moved the memory controller **inside the CPU itself**: hence the acronym **IMC**, *Integrated Memory Controller*. AMD introduced it in Athlon 64 processors (2003), and Intel generalized it with the Nehalem architecture (2008); since then, it has been the absolute standard.

Integration has two important practical consequences. The first is reduced latency: by eliminating the passage through an external chip, the CPU accesses RAM faster. The second, less intuitive but crucial for buyers, is that **RAM compatibility and limits largely depend on the CPU**, not just the motherboard. It is the processor's IMC that determines which memory types it supports (DDR4 or DDR5), how many frequencies it can stably handle, and how many channels it manages. Two CPUs mounted on the same motherboard can have different memory limits because they have different IMCs. The motherboard defines the boundaries (number of slots, bus layout, signal quality), but the controller in the CPU does the work. We will keep this principle in mind throughout the chapter: **RAM, CPU, and motherboard form a three-part system; none of the three is chosen in isolation.**

2. Capacity

2.1 How much RAM is really needed, per use case

The question "how much RAM do I need?" does not have a single answer, because it depends on what the computer needs to do. The underlying logic is simple: you need enough RAM to keep everything you are working on simultaneously on your "desk" (RAM), without forcing the system into continuous swapping. Let's look at the ranges, keeping in mind that the indicated values are approximate and that software requirements grow over time [**volatile data**]: a "generous" amount today becomes "the bare minimum" in a few years.

With **8 GB**, a system can still function for light tasks: browsing with a few tabs open, word processing, email, video streaming. It must be said clearly, however, that 8 GB in the current landscape is the absolute floor and often already insufficient: modern browsers consume a lot of memory, the operating system reserves a slice, and just a few applications open together are enough to start using swap. For a new PC intended to last, 8 GB is not recommended unless for machines with a very limited role.

With **16 GB**, advanced office use and gaming are comfortably covered. This amount has been the sweet spot for years, and it remains an excellent base for a user who plays games, keeps many tabs open, uses productivity applications, and perhaps some light creative tools. In gaming, most current titles run well with 16 GB, although the most recent and ambitious games are starting to benefit from more memory, especially if you play while a browser, voice chat, and recording software are running in the background.

With **32 GB**, you enter the "set for years" territory for a demanding user. This is the recommended choice for serious gamers who also engage in heavy multitasking, those who work with large-file photo editing, amateur or semi-professional Full HD video editing, light virtualization, and software development with IDEs and containers. In 2026, 32 GB has become the "sweet spot" for mid-to-high-end PCs, partly because RAM prices, while fluctuating, have often made the jump from 16 to 32 GB less painful [**volatile data: check current price**].

With **64 GB and beyond**, you're catering to heavy professional work: 4K/8K video editing with many tracks and effects, 3D rendering, very high-resolution photography with many layers, running multiple virtual machines (VMs) simultaneously, local databases, and compiling enormous software projects. A rapidly growing use case is **local AI**: running large language models (LLMs) on your own machine requires abundant memory. Here, a distinction must be made: models primarily run on the **VRAM** of the graphics card (the dedicated memory of the GPU, a topic for another chapter), but system RAM is still needed to load models, manage context, perform inference on the CPU when VRAM is insufficient, and prepare data. For serious work with medium-sized models, 64 GB is a good starting point, and 128 GB is by no means excessive. Those doing fine-tuning, dataset processing, or managing complex pipelines can easily justify even larger quantities.

The following table summarizes the recommendations. It should be read as a compass, not as dogma: actual needs are measured by observing memory usage during typical use (in Windows with Task Manager, in Linux with `htop` or `free -h`), and if the system consistently approaches saturation or uses a lot of swap, it means more RAM is needed.

Capacity	Recommended Typical Use	Notes
8 GB	Basic office, browsing, very light use	Absolute minimum, often already tight on a new PC
16 GB	Advanced office, gaming, moderate multitasking	Excellent "safe" base for most users
32 GB	Gaming + streaming, photo/video editing, dev, light VMs	Current mid-to-high-end sweet spot
64 GB	4K video, 3D, many VMs, local AI (medium models)	Professional/prosumer territory
128 GB+	Heavy rendering, datasets, AI/fine-tuning, intense virtualization	Workstation/creator, often on HEDT platforms

2.2 Why the number of modules matters: 1×16 vs 2×8 vs 4×8

A very common mistake is to only consider the total ("I need 16 GB") while ignoring **how** those 16 GB are divided into physical modules. A RAM module, the small rectangular board inserted into a slot, is technically called a **DIMM** (*Dual In-line Memory Module*). The same total capacity can be achieved with different configurations, and they are not equivalent.

Let's take 16 GB. I can get them with **1x16** (a single 16 GB module) or with **2x8** (two 8 GB modules). From a capacity standpoint, they are identical, but from a **performance** standpoint, they are not, and the difference is far from negligible. The reason is called **dual channel**, and we will delve into it in the dedicated section; here, suffice it to anticipate the principle. Modern desktop platforms have a memory controller capable of communicating with RAM on **two channels** in parallel, as if they had two highway lanes instead of one. To utilize both lanes, at least **two modules** are needed. With a single module (1x16), you travel on only one lane: the RAM is said to operate in **single channel**, and the available

bandwidth is halved. With two modules (2x8), both lanes are activated, operating in **dual channel**, and the bandwidth doubles.

The practical impact is significant, especially in two scenarios. The first is gaming with integrated graphics (APUs, i.e., processors with included GPUs): there, system RAM also acts as graphics memory, and switching from single to dual channel can improve the frame rate by as much as 30-70%. The second is any workload limited by memory bandwidth, from compression to scientific computing. Even in gaming with a dedicated GPU, dual channel provides a measurable, albeit less dramatic, advantage. The practical rule derived from this is among the most important in the chapter: **for the same capacity, always prefer two modules over one**. Buying 1x16 "to leave a slot free and add RAM in the future" is almost always a false economy, because in the meantime, you forgo dual channel.

So, if two modules are better than one, aren't four better than two? Here the reasoning becomes more complex, and it's the reason why the **4x8** configuration exists and should be treated with caution. Adding a third and fourth module does **not** add channels: mainstream desktop platforms remain two-channel even when filling four slots. The four modules are distributed two per channel. The problem is electrical: populating all slots means putting more "load" on the CPU's integrated memory controller, which has to drive more chips. This makes it more difficult to maintain high frequencies stably. In practice, a kit of 2 modules often reaches high frequencies without problems, while the exact same 4 modules may force a reduction in frequency to remain stable. The phenomenon is particularly marked on **DDR5**, whose high-frequency signal management is more delicate: many motherboards, with four DDR5 modules populated, see the guaranteed maximum frequency drop significantly. We will discuss this again in common errors. The operational summary: **for a given capacity, a two-module kit is almost always preferable to a four-module kit**. If 32 GB are needed, a 2x16 kit almost always beats a 4x8; if 64 GB are needed, a 2x32 is more manageable than a 4x16.

There's an additional, often overlooked, reason to buy RAM in a **single kit**: the modules within the same kit have been tested and selected (the technical term is *binned*) to work together at a specific frequency and latency. Buying two modules today and another two of the same model a year from now doesn't guarantee that all four will operate at the advertised frequency, even if the product code is identical, because the silicon chips might come from different batches with

slightly different characteristics. If you plan to reach a certain capacity, it's best to buy it all at once in a single kit with the desired number of modules.

3. Speed and Latencies

This is the most technical and most misunderstood section. RAM speed isn't just one number: it's a combination of two quantities that pull in opposite directions, **frequency** (how often operations occur) and **latencies** (how much time passes before an operation produces its result). Understanding how they combine is what separates the informed buyer from someone who only looks at the biggest number on the box.

3.1 Frequency: MT/s vs. "MHz", what it really means

On DDR RAM packaging, there's a large number, like 3200, 6000, 6400. Historically, this number has been called "MHz" (megahertz), but it's an improper use that causes enormous confusion, and it's worth clarifying once and for all.

The correct unit is **MT/s**, which stands for *Mega Transfers per second*, millions of data transfers per second. The acronym **DDR** means *Double Data Rate*, and it describes the fundamental trick of this technology: the memory transfers data **twice** for each cycle of its clock signal, once on the rising edge of the signal and once on the falling edge. It's like a metronome that, instead of marking one beat per tick, marks two, one when the hand goes up and one when it goes down.

The consequence is that the memory's **real clock speed** is half of the advertised number. A "DDR5-6000" RAM has an effective clock speed of 3000 MHz, but since it transfers data twice per cycle, it performs 6000 million transfers per second, i.e., 6000 MT/s. Calling it "6000 MHz" is technically incorrect (the clock speed is 3000 MHz), but it has become common jargon. The correct form is **6000 MT/s**. In the rest of the chapter, we will use MT/s, but know that when you read "6000 MHz" on a forum or a box, it refers to the same thing.

Why does frequency matter? Because it determines, along with bus width and the number of channels, the **bandwidth**, which is how much data per second the RAM can pour into the CPU. More MT/s means more bandwidth. A simple way to estimate it on a 64-bit (8-byte) channel is: $\text{bandwidth} \approx \text{MT/s} \times 8 \text{ bytes}$. A DDR5-6000 on a single channel thus provides about 48 GB/s, which doubles in dual channel. Bandwidth-hungry workloads (integrated graphics, some AI processing, compression, streaming of large datasets) directly benefit from higher frequency.

3.2 Timings: CL, tRCD, tRP, tRAS and what CAS latency is

While frequency tells you how **often** RAM works, **timings** tell you how **quickly** it responds to a single request. They are a series of numbers, usually reported as a sequence like "36-38-38-96" or by highlighting the first one, "CL36". They represent delays, measured in **clock cycles**, that the memory must wait between one internal operation and the next. Since they are delays, here **lower is better**: it's the opposite of frequency.

To understand them, a bit of anatomy is needed. Internally, a RAM chip is organized as a grid of cells arranged in **rows** and **columns** (like a spreadsheet). To read data, the controller must first "activate" the correct row and then indicate the column. Each of these steps takes time, and timings quantify precisely those times.

The first and most cited is **CL**, *CAS Latency*, where CAS stands for *Column Address Strobe*. It is the number of clock cycles that elapse between the moment the controller requests a specific column (of an already active row) and the moment the first data is available at the output. In simple terms: if the row is already open, how long do I wait to get the data? It is the most frequent latency in real-world operation, which is why it is highlighted (the "CL36" on the box).

The other three main ones complete the picture. **tRCD**, *RAS to CAS Delay* (where RAS is *Row Address Strobe*), is the delay between activating a row and being able to request a column within it: in practice, how long it takes to "open" a new row before being able to read. **tRP**, *Row Precharge time*, is the time required to "close" the currently active row (precharge it) before being able to open another one: when the data I'm looking for is in a different row from the one open, I pay tRP to close plus tRCD to open the new one. **tRAS**, *Row Active time*, is the minimum time a row must remain active before it can be closed, a constraint that ensures proper cell refresh. There are dozens of other secondary and tertiary timings (tRC, tRFC, tFAW, and many others) that overclocking enthusiasts manually adjust, but for purchase decisions, the four main ones, and especially CL, are more than sufficient.

3.3 How to calculate real latency in nanoseconds

This is the point that unmasks marketing and that every buyer should master. A low CL, by itself, doesn't tell you how fast the RAM is, because CL is expressed in **clock cycles**, and the duration of a cycle depends on the frequency. A clock cycle at a high frequency lasts less than a cycle at a low frequency. So the same CL "weighs" different real times at different frequencies.

Real latency, the one that truly matters because it's measured in absolute time, is calculated as follows:

$$\text{Latency (ns)} = (\text{CL} \times 2000) \div (\text{frequency in MT/s})$$

The 2000 comes from the fact that the real clock is half of the MT/s (factor 2) and that we convert to nanoseconds (factor 1000): $2 \times 1000 = 2000$. Let's see the formula in action by comparing seemingly very different kits.

Kit	Frequency	CL	Real Latency (ns)
DDR4	3200 MT/s	16	$(16 \times 2000) / 3200 =$ 10.0 ns
DDR4	3600 MT/s	18	$(18 \times 2000) / 3600 =$ 10.0 ns
DDR5	6000 MT/s	30	$(30 \times 2000) / 6000 =$ 10.0 ns
DDR5	6000 MT/s	36	$(36 \times 2000) / 6000 =$ 12.0 ns
DDR5	6400 MT/s	32	$(32 \times 2000) / 6400 =$ 10.0 ns
DDR5	7200 MT/s	34	$(34 \times 2000) / 7200 =$ 9.4 ns

The table teaches several things at once. First: an apparently "worse" CL (CL30 versus CL16) does not mean slower RAM at all, because at double the frequency, that CL30 produces the **same** real latency of 10 ns. This is exactly the case for DDR4-3200 CL16 and DDR5-6000 CL30: identical latency, but DDR5 has double the bandwidth. Second: at the same frequency (6000 MT/s), the difference between CL30 and CL36 is real, 10 ns versus 12 ns, and for this reason, a "6000 CL30" kit is considered better than a "6000 CL36" kit. Third: latency in ns is the **great equalizer** that allows comparing kits of different generations and frequencies on a common ground. When someone asks you if RAM is "fast," the serious answer is not the frequency or CL alone, but the combination: bandwidth (frequency $\times$ channels) and real latency (ns).

3.4 Frequency versus latency: which matters more and when

Since frequency and latency pull in opposite directions (as frequency increases, CL often increases, and vice versa), the practical question is: which of the two should you invest in? The answer depends on the workload, and understanding why helps avoid wasteful spending.

Bandwidth-sensitive workloads benefit most from frequency. This includes integrated graphics, which, as mentioned, uses RAM as video memory and is bandwidth-hungry; some parallel processing; and copying large blocks of data. Here, between fast RAM with relaxed timings and slower RAM with tight timings, the former tends to win.

Latency-sensitive workloads benefit most from low timings. The emblematic case is **gaming**, especially when the CPU is the bottleneck (low resolutions, high frame rates, games with complex logic and many scattered memory accesses). Game engines make many small and unpredictable accesses: what matters is not so much how much data per second, but how quickly the single requested data arrives. For this reason, in gaming, with the same bandwidth, a kit with lower real latency performs better. A "6000 CL30" kit (10 ns) is often preferable to a "6400 CL36" kit (11.25 ns) even though the second has a higher frequency, because in gaming, latency carries more weight.

The practical rule: for general use and gaming, aim for **low real latency** at a balanced frequency; for integrated graphics or pure bandwidth loads, prioritize **frequency**. And in any case, don't be mesmerized by the large MT/s number while ignoring the CL.

3.5 XMP and EXPO: why RAM runs at base frequency without them

Here's one of the points that generates the most disappointment for first-time buyers: **you buy "6000 MT/s CL30" RAM, install it, and the computer runs it at 4800 MT/s with very high timings**. This is not a defect, it's not a scam: it's the expected behavior, and to correct it, you need to activate a profile. Let's understand why.

Official memory specifications are defined by an entity called **JEDEC** (*Joint Electron Device Engineering Council*), which standardizes "safe" frequencies and timings guaranteed to work on any compliant system. These JEDEC values are deliberately conservative. When you turn on your PC, the RAM communicates its characteristics to the motherboard via a small chip on board called **SPD** (*Serial Presence Detect*), and by default, the system sets the JEDEC values, which are the safe and low ones. This is why that "6000" starts at 4800: 4800 is the basic JEDEC specification, 6000 is the "boosted" speed for which the kit was selected.

To run the RAM at the frequency and timings advertised on the box, you need to activate a **preloaded overclocking profile**, also stored in the SPD. On the **Intel**

platform, this profile is called **XMP** (*Extreme Memory Profile*); on the **AMD** platform, for recent generation Ryzen CPUs on the AM5 socket, it is called **EXPO** (*EXtended Profiles for Overclocking*). These are two commercial names for the same concept: a set of optimized settings that, with a single click in the BIOS, bring the RAM to the performance you paid for. Many high-end kits include **both** profiles to cover both platforms; some include only one, and it's good to check before purchasing, especially for an AMD build (in the absence of EXPO, an XMP profile can almost always still be used, but timing compatibility is not 100% guaranteed).

Activating them is simple but requires entering the **BIOS/UEFI**, the motherboard's firmware (usually by pressing `Del` or `F2` at startup). There, you look for the XMP or EXPO entry (often on the main page or in an "OC"/"Tweaker"/"Ai Overclock" section), enable it by selecting the desired profile, save, and restart. The system will restart at the correct frequency. This is a step that many users forget, ending up with expensive RAM performing like cheap RAM. **If you bought high-frequency RAM and haven't activated XMP/EXPO, you're not using it as you should.** It will come back in common errors because it is, statistically, the number one error.

An honest technical clarification: XMP and EXPO are, to all intents and purposes, forms of **memory overclocking**, meaning operation beyond standard JEDEC specifications. These profiles are validated by the kit manufacturer and in the vast majority of cases are extremely stable, but precisely because they "push" beyond the standard, it is possible, in rare cases, to encounter instability related to the fact that the memory controller of *your* specific CPU does not like that frequency. In those cases, you step down a frequency notch or slightly relax the timings. This is also why compatibility lists (QVLs) exist, which we will discuss shortly.

4. Dual channel (and beyond)

4.1 How dual channel works and the real bandwidth gain

We have already anticipated the concept when discussing the number of modules; now we will delve deeper. **Dual channel** is the ability of the memory controller to communicate with RAM over **two independent parallel buses**, effectively doubling the overall bandwidth of the interface to memory. If a single DDR channel transfers data 64 bits at a time, two channels transfer 128. Reusing the

highway metaphor: two lanes instead of one, with double the traffic that can flow at the same instant.

The theoretical bandwidth gain is, on paper, 100%: doubling the channels doubles the theoretical bandwidth. In practice, the performance benefit is not always 100% because not all workloads are limited by memory bandwidth, but in all scenarios where they are, the increase is real and often noticeable. The most striking case, already mentioned, is **integrated graphics**: an APU upgraded from single to dual channel can gain tens of percentage points in frame rate, because its GPU was literally suffocating due to lack of bandwidth. Even with a dedicated GPU, and in productivity tasks, dual channel provides a measurable advantage. The operational conclusion is the same: **dual channel is not a luxury; it is the normal configuration and should always be ensured.**

4.2 Which slots to populate: the A2/B2 case and installation errors

A surprisingly frequent physical error lurks here. Mainstream motherboards typically have **four memory slots**, but to activate dual channel with **two** modules, it's not enough to just insert them into any two slots: they must be placed in the correct slots. The four slots are divided between the two channels in an alternating manner, and the manufacturer specifies a precise configuration for dual channel with two modules.

In the vast majority of motherboards, the slots, numbered starting from the CPU, correspond to two channels (A and B) with two slots each, arranged as **A1, A2, B1, B2** or indicated as DIMM_A1, DIMM_A2, DIMM_B1, DIMM_B2. The recommended configuration for two modules is almost always the **second slot of each channel**, i.e., **A2 and B2**, which are physically the **second and fourth slots starting from the CPU** (thus leaving a gap between the two modules). This arrangement, which seems strange at first glance because it leaves an empty slot in the middle, is the one that offers the best high-frequency signal quality, an aspect that has become critical with DDR5. Some manufacturers use the opposite scheme (A1/B1), but the principle is always the same: **two modules should be placed in two slots of the same "color" or according to the manual's scheme, never randomly.**

The typical error is to insert the two modules into the first two adjacent slots (slots 1 and 2), which often belong to the **same** channel: the result is that the RAM operates in **single channel** despite having two modules, halving the bandwidth without any error message appearing. The computer turns on, works, and the user notices nothing, except for a performance drop that they will hardly connect

to the cause. The golden rule: **always consult the motherboard manual for the correct arrangement**, and when in doubt, the A2/B2 configuration (second and fourth slots) is the correct one in the vast majority of cases. After assembly, verify in Windows with Task Manager (Performance tab → Memory, where the wording about channel usage appears with some tools) or, even better, with utilities like CPU-Z, which explicitly reports "Channel #: Dual" in the Memory tab.

4.3 Beyond dual channel: quad and octa channel

Consumer desktop platforms stop at dual channel, but superior platforms exist. Workstations and servers based on **HEDT** (*High-End Desktop*) platforms and dedicated sockets (such as AMD Threadripper and Threadripper PRO lines, or Intel Xeon counterparts) offer **quad channel** (four channels), **hexa/octa channel** (six or eight channels) and, in more recent server generations, even more than eight. Each additional channel adds its bandwidth: an octa channel system has memory bandwidth that crushes any consumer desktop, and it's what's needed to feed CPUs with dozens of data-hungry cores, for massive datasets, for massive virtualization, and for AI/HPC workloads.

On these platforms, installation rules also change (channels must be populated in precise groups to activate them all) and often the type of RAM (memory **registered** with **ECC**, which we will briefly mention). Since the topic is vast and concerns a specific niche, we will cover it in depth in the **chapter dedicated to workstations and servers**. Here it is enough to know that if one day you build a machine for heavy computing, the number of memory channels will be one of the key parameters, and that the leap from the two channels of the desktop to the four/eight of professional platforms is one of the main technical reasons why those platforms exist.

4.4 Single rank versus dual rank

A final concept, subtle but influential on performance, is the module's **rank**. A *rank* is a set of memory chips on the module that are accessed together as a single 64-bit block (or 72 with ECC). A module can be **single rank** (only one block) or **dual rank** (two independent blocks on the same module), and more rarely quad rank.

The practical difference is that the controller, with more ranks available, can "interleave" accesses: while one rank is completing an internal operation, the controller can start another on the other rank, hiding part of the latencies. This phenomenon, called **rank interleaving**, means that the same capacity and

frequency perform slightly better in a dual rank configuration. In many gaming and productivity tests, a system with dual rank RAM (for example, two dual rank modules, thus four total ranks) shows a gain of a few percentage points compared to single rank, all else being equal.

However, there's a downside: more ranks mean more electrical load on the memory controller, so dual rank RAM is typically harder to run at extreme frequencies than single rank. It's another example of the constant compromise in this field between performance and signal stability. For the average buyer, the issue is secondary and largely outside their direct control (often larger capacity modules are dual rank by design), but it's useful to know to interpret reviews and benchmarks that mention it. A practical indication, however, emerges clearly: **two modules are often preferable to one for this reason too**, because two single rank modules offer the controller two ranks to interleave with, while a single rank module offers only one.

5. DDR4 versus DDR5

5.1 What changes under the hood

DDR4 and DDR5 are two successive **generations** of the DDR standard. DDR5 is the more recent of the two in the mainstream world (the next standard is already being defined, but it is not yet in consumer PCs at the time of writing [**volatile data: check current generation**]). The generational leap is not just a simple increase in number: frequencies, voltages, internal architecture, and power management all change. Let's look at the key points.

Frequencies make a clear jump. DDR4 on desktop primarily operates in the 2133–3600 MT/s range, with overclocked peaks beyond; DDR5 starts at 4800 MT/s and easily scales to 6000, 6400, and beyond, with extreme kits being much faster. The available bandwidth with DDR5 is therefore much greater from the outset.

The standard operating **voltage** drops from 1.2 V for DDR4 to 1.1 V for DDR5, an efficiency advantage for the same performance. But the most interesting change concerns **where** voltage regulation occurs. On DDR4, RAM power management is on the motherboard; on DDR5, however, each module integrates its own on-module **PMIC** (*Power Management Integrated Circuit*). Moving voltage regulation onto the module improves the quality and stability of local power delivery, a

requirement that has become necessary at DDR5's high frequencies. It is also one of the reasons why DDR5 modules cost more for the same capacity.

DDR5 also introduces **ECC on-die**. ECC (*Error Correcting Code*) is a mechanism that detects and corrects errors in memory bits. Be careful not to get confused: DDR5's **on-die** ECC corrects errors that occur *inside the chip* and is designed to ensure the reliability of increasingly dense chips; it is **not** the same as the "full" ECC of server memory, which also protects data as it travels between RAM and CPU and requires specific CPU and motherboard support. In other words: a desktop DDR5 module has on-die ECC but is not "ECC memory" in the enterprise sense of the term. This is a distinction that confuses many and is worth keeping in mind.

Finally, each DDR5 module is internally divided into **two independent 32-bit sub-channels** (plus ECC bits), instead of DDR4's single 64-bit channel. A single DDR5 module therefore offers the controller two parallel sub-channels, improving access efficiency and the ability to handle multiple requests simultaneously. This is one of the architectural reasons why DDR5 performs better even with the same theoretical bandwidth.

The table summarizes the generational comparison. Typical frequency values are indicative, and the "high" range of DDR5 will increase over time **[volatile data]**.

Feature	DDR4	DDR5
Typical desktop frequencies	2133–3600 MT/s	4800–6400+ MT/s
Standard voltage	1.2 V	1.1 V
Power management	On the motherboard	PMIC on-module
ECC on-die	No	Yes (internal to chip, not "server" ECC)
Module structure	1x 64-bit channel	2x 32-bit sub-channels
Electrical key (notch)	DDR4 position	DDR5 position (different)

5.2 Compatibility: DDR4 and DDR5 are not interchangeable

This is the point where you cannot make a mistake, or risk purchasing unusable components. **DDR4 and DDR5 are physically and electrically incompatible.** They have the same number of pins but arranged differently, and most importantly, the **notch** (the indentation on the gold edge of the contacts that prevents incorrect insertion) is in a different position: a DDR5 module will not fit

into a DDR4 slot and vice versa. This is a deliberate physical protection, precisely to prevent incorrect insertions.

But the issue goes beyond the slot: it depends on the **CPU + motherboard combination**. Each motherboard supports **either** DDR4 **or** DDR5, never both on the same slot (there have been very rare hybrid cases with separate slots, but these are negligible exceptions). And since the memory controller is integrated into the CPU, the processor must also support the correct standard. In practice: when you choose the platform, you simultaneously decide on the CPU, motherboard, and RAM type as a single block. You cannot buy DDR5 RAM and then install it on a DDR4 motherboard, nor "upgrade" from DDR4 to DDR5 without changing the motherboard (and often the CPU). This is exactly the principle of the "system of three" stated at the beginning: **RAM, CPU, and motherboard are chosen together**. The error of buying the wrong type of RAM for one's platform is among the most costly and frustrating, and unfortunately among the most common among those who assemble without checking.

5.3 When DDR4 still makes sense and what is the DDR5 sweet spot

With DDR5 now mature and with much more reasonable prices compared to its launch, **for a new build, the default choice is DDR5 [volatile data: check current price ratio]**. The latest Intel and AMD platforms were born DDR5, offering more bandwidth and a better prospect for longevity.

However, there are still scenarios where **DDR4 still makes sense**. The first is **upgrading an existing system** already on a DDR4 platform: if you have a good DDR4 motherboard and CPU, adding or replacing DDR4 RAM is much cheaper than overhauling the entire platform to switch to DDR5. The second is a **very tight budget entry-level build**, where a previous-generation DDR4 platform (CPU + motherboard + RAM) can cost significantly less than an equivalent DDR5, with a performance difference that is not decisive for light uses. The third is **component recovery**: if you already have quality DDR4 RAM, reusing it makes sense. Outside of these cases, however, looking to the future means DDR5.

On the DDR5 front, there's an established **sweet spot**, which is the point of best performance-to-price-to-stability ratio. This deserves a technical explanation because it's one of the most instructive examples of how RAM and CPU are linked. For **AMD AM5** platforms (Ryzen 7000 series and later), the reference kit is **DDR5-6000 CL30**. The reason is not arbitrary: it depends on the relationship with the **Infinity Fabric**, the internal bus that connects the various blocks of the processor

(core clusters, memory controller, cache) in Ryzen CPUs. On these CPUs, the best performance is achieved when the memory controller (called **UCLK**), the Infinity Fabric (**FCLK**), and the memory clock (**MCLK**) run in a synchronous ratio, specifically a **1:1** ratio between the memory clock and the controller clock. At 6000 MT/s, the RAM operates with an effective clock of 3000 MHz, and the memory controller can maintain the 1:1 ratio stably on most processors. Going higher (e.g., 6400 or 6800 MT/s), many Ryzen CPUs can no longer maintain the 1:1 ratio, and the controller is forced to switch to a **2:1** ratio, introducing a divider that **increases latency** and often negates, or even reverses, the advantage of the higher frequency. This is why "6000 CL30" is considered the golden spot on the AM5 platform: it maximizes frequency while maintaining synchronicity, with a real latency of 10 ns. Increasing the frequency paradoxically results in *worse* performance in most cases, unless you're lucky with the silicon and know how to tweak advanced settings.

On recent **Intel** platforms, the memory controller is more tolerant and scales better with higher frequencies, so it makes sense to aim higher there (e.g., 6400, 7200 MT/s and beyond) if the budget allows, always considering real latency. But the general concept remains: **the "best" frequency is not the absolute highest, it's the highest that your specific CPU can handle while maintaining synchronicity and stability**. Always check updated recommendations for the exact platform you are buying, as these sweet spots evolve with new generations **[volatile data]**.

6. Brands and How to Choose

6.1 Who Makes the Chips and Who Assembles the Modules: Two Different Jobs

One of the most ingrained misunderstandings is thinking that brands like Corsair or G.Skill "produce" RAM. In reality, in the world of memory, there are **two distinct levels**, and understanding them clarifies a lot about why some kits cost and perform more than others.

At the first level are the actual **memory chip manufacturers**, the silicon where data is physically stored. There are very few in the world, because building semiconductor factories (fabs) costs billions: essentially **Samsung**, **SK Hynix**, and **Micron**. These three giants produce the DRAM chips that end up, in one way or

another, on the vast majority of modules on the market, regardless of the brand printed on them.

At the second level are the **module makers**: companies that buy chips from the three manufacturers, mount them on a printed circuit board (the module's PCB) along with the PMIC, SPD, and any heatsink, test them, select them, define the XMP/EXPO profiles, and then market them under their own warranty and brand. The list here is longer: **Corsair, G.Skill, Kingston, Crucial, TeamGroup, Patriot, ADATA**, and others. A clarification: **Crucial** is **Micron's** consumer brand, so it is both a chip manufacturer and an assembler, a special case that guarantees an integrated supply chain. **Samsung** and **SK Hynix** also sell modules under their own brand, but they are less present in consumer retail than specialized assemblers.

The practical consequence is that **the brand on the box doesn't tell the whole story**: two kits from different brands might use the exact same Hynix chips, and two kits from the same brand might use chips from different manufacturers depending on the batch. What truly distinguishes kits is the **selection (binning)** of the chips, the quality of the PCB, the effectiveness of the XMP/EXPO profiles, and the support/warranty.

6.2 Why G.Skill, Corsair, and Kingston dominate

If these three brands (with Crucial and TeamGroup close behind) are the most recommended, there are concrete reasons that go beyond mere popularity.

The first is **binning**, which is the selection of the best chips. Not all chips coming out of a fab are equal: some can handle higher frequencies with lower timings, others less so. The more serious assemblers buy large volumes, test them, and allocate the best chips to high-end kits (which they sell for more) and standard chips to budget kits. A "premium" G.Skill or Corsair kit with high frequency and low latency is such because it uses selected chips capable of those performances with margin. This is exactly what you pay the premium for.

The second is **warranty** and support. These brands typically offer long warranties, often lifetime within the specified limits, and reliable replacement service: a non-trivial aspect for a component that is expected to last as long as the PC.

The third, crucial during purchase, is **QVL compatibility**. QVL stands for *Qualified Vendor List*: it is the list, published by motherboard manufacturers, of specific

RAM kits that have been **tested and validated** on that board at a certain frequency. The dominant brands appear extensively in QVLs precisely because they are the most widespread and most tested, which drastically reduces the risk of incompatibility or instability with XMP/EXPO. Buying a kit present in your motherboard's QVL is the easiest way to rest easy.

6.3 Hynix A-die and M-die: why enthusiasts seek them out

Among overclocking enthusiasts, "**Hynix A-die**" or "**M-die**" chips are spoken of with reverence. What are they and why do they matter? As mentioned, chips come from a few manufacturers, and each one produces different "revisions" of silicon over time, identified by codes. In the DDR5 world, **SK Hynix** chips have earned a reputation for being the best for overclocking, and among their revisions, the **A-die** is considered the elite: it handles very high frequencies and very tight timings with great margin, while the **M-die** is a step below but still excellent. In practice, a kit equipped with Hynix A-die chips has the potential to be pushed far beyond its stated specifications, which appeals to those who perform manual tuning.

For the average buyer, the hunt for a specific "die" is a nerdy detail: if you simply enable XMP/EXPO without intending to manually overclock, the chip revision matters little, because the kit will perform as advertised regardless. The topic only becomes relevant if you intend to **push the RAM beyond specifications** by manually adjusting timings: in that case, seeking out kits known to use Hynix A-die (information available in databases and overclocking communities, as manufacturers do not always declare it) maximizes the chances of achieving record results. It's an example of how the same product has a different "reading level" depending on how deep one wants to go.

6.4 RGB and heatsinks: aesthetics versus substance, and the height problem

Most mid-to-high-end kits today feature **RGB** lighting (colored and programmable LEDs) and conspicuous metallic **heatsinks** (*heatspreaders*). It's important to distinguish what is functional from what is purely aesthetic.

RGB is, from a performance standpoint, completely irrelevant: it's pure aesthetics. There's nothing wrong with wanting a build that lights up in a coordinated way, but it should be purchased with the awareness that you're paying a premium for appearance, not for speed. In fact, RGB kits tend to be taller (to accommodate the LED bar) and more expensive for the same specifications compared to versions

without. Those seeking the best performance/price ratio can save money by choosing plain models, which often have identical chips.

Heatsinks have a real function but are often overrated on consumer kits: they dissipate heat from the chips, which is useful at high frequencies and in aggressive overclocking, but for RAM at standard specifications (even with XMP/EXPO at 6000 MT/s), a modest heatsink is more than sufficient, and the large heatspreaders on some kits are largely for show. RAM heat is modest compared to CPUs and GPUs.

However, there's an aspect of module size that is far from aesthetic and often leads to a common physical error: **clearance from the air CPU cooler**. Large air CPU coolers, with their big fans, often protrude over the RAM slots. If the modules are very tall (due to elaborate heat spreaders or RGB bars), they can **physically collide** with the CPU cooler, preventing its installation or forcing the fan to be moved upwards (worsening cooling) or even making it impossible to close the case. This is a very real problem that ruins many last-minute assemblies.

Countermeasures: always check the **module height** (in millimeters, stated in the specifications) and the **RAM compatibility of the CPU cooler** (many cooler manufacturers publish the maximum RAM height supported under the fan), or opt for **low-profile** kits when using a bulky air cooler. With AIO liquid cooling, this problem is almost nonexistent, because above the CPU there is only the pump/waterblock, which does not invade the RAM slots.

6.5 The step-by-step selection method

Putting the whole chapter together, here is the logical order for choosing RAM, a process that avoids all typical errors if followed correctly.

Always start with the **platform**: which CPU and which motherboard? This determines whether the RAM will be **DDR4 or DDR5** (this is not negotiable, it depends on the socket and chipset) and which frequencies make sense to aim for (remembering the 6000 CL30 sweet spot on AM5 and Intel's greater scalability). Once the platform is defined, decide on the **capacity** based on the use case (16/32/64 GB according to section 2). Then choose a **kit of two modules** to ensure dual channel, avoiding single modules and, when possible, avoiding four modules in favor of two larger ones. At this point, identify the right **frequency/latency sweet spot** for the platform, thinking in terms of **real latency (ns)** and not just the large number in MT/s. After that, check for the presence of the kit (or an equivalent one) in the motherboard's **QVL**, or at least that the correct profile (XMP

for Intel, EXPO for AMD) is supported. Only at the end does the **budget** come into play, choosing from the kits that have passed the previous filters the one with the best quality/price ratio, taking into account the height of the modules if an air cooler is used and deciding whether to pay the eventual aesthetic premium for RGB. This order, **platform → capacity → 2-module kit → frequency/latency sweet spot → QVL → budget/aesthetics**, is the operational core of the entire chapter.

6.6 Budget tiers: entry, mid, high-end

Translating the method into recommendations by tier, with the caveat that specific kits and prices are constantly changing [**volatile data: check current models and prices before buying**].

For the **entry-level** (minimum budget) tier, the goal is to have dual channel and sufficient capacity without unnecessary expenses. For an economical DDR5 build, this means a **2x8 GB (16 GB)** kit at JEDEC specifications or slightly higher, from a reliable brand like Crucial, Kingston, or TeamGroup, without RGB. For a build that reuses or leverages affordable used parts, a **DDR4** platform with a good 2x8 3200 CL16 kit remains perfectly valid for light gaming and work. The priority here is: two modules, reputable brand, XMP/EXPO enabled.

For the **mid-range** (average user) tier, the default choice is a **DDR5 2x16 GB (32 GB)** kit at the platform's sweet spot: for AM5, the famous **6000 CL30 EXPO**; for Intel, a 6000 CL30 or a step higher if the budget allows. Brands like G.Skill, Corsair, Kingston, with or without RGB as preferred, and ideally present in the QVL. This configuration offers the best balance of cost, performance, and longevity for the majority of new PCs today.

For the **high-end** (uncompromising performance, creators, and professionals) tier, capacity and/or quality increases. For top-tier gaming and serious productivity, **2x16 GB or 2x32 GB (64 GB)** of a selected low-latency kit; for workstations, 4K editing, many VMs, or local AI, **64 or 128 GB**, considering, on platforms that allow it, even 4-module kits or HEDT solutions with more channels (refer to the workstation chapter). Here, it makes sense to invest in Hynix A-die chips if overclocking is a goal, in high-frequency kits on Intel platforms that benefit from them, and to pay attention to aesthetics if it's part of the objectives. The principle, however, remains unchanged: even in high-end, **real latency and synchronization with the CPU matter more than the big number on the box**.

7. Common Buyer Mistakes

Here, we collect and explain the most common mistakes, many of which have already been encountered throughout the chapter. Knowing them in advance is the best way to avoid making them.

The first, and statistically most common, is **not enabling XMP or EXPO**. You spend money on high-frequency RAM and leave it running at the base JEDEC speed because you haven't entered the BIOS to enable the profile. The result is having paid for performance you're not using. Remedy: enter the BIOS/UEFI, enable XMP (Intel) or EXPO (AMD), save, and verify the frequency with CPU-Z. It's a click that's worth tens of euros in RAM.

The second is **buying the wrong type of RAM for the platform**, i.e., DDR4 when DDR5 is needed or vice versa. This stems from not understanding that the memory type depends on the CPU and motherboard and is not interchangeable. Remedy: *always* verify, before ordering, which standard your motherboard (and CPU) supports, because the wrong module simply won't fit into the slot.

The third is **mixing different kits**. Buying two modules today and another two a year later, perhaps of the same model but from a different batch, or worse, mixing different brands, capacities, and frequencies. Since the modules were not selected together, coexistence at the declared frequency is not guaranteed: there's a risk of instability, failure to activate XMP/EXPO, or operation at the slowest module's frequency. Remedy: buy the target capacity in a **single kit** of the desired number of modules right from the start.

The fourth is **installing modules in the wrong slots**, typically the first two adjacent ones, resulting in single channel even with two modules. No error message warns you: the PC works, but with half the bandwidth. Remedy: follow the motherboard manual; if in doubt, use **A2 and B2** (second and fourth slots from the CPU) and check for "Dual" in CPU-Z.

The fifth is **installing four modules on platforms that handle them poorly**, especially on DDR5, where filling all four slots often forces the controller to lower its frequency to remain stable. You end up with more capacity but lower effective frequency, sometimes even struggling to enable XMP/EXPO. Remedy: prefer **two modules of larger capacity** (2x16 instead of 4x8, 2x32 instead of 4x16) and, if four modules are absolutely necessary, account for a possible frequency reduction and check for support in the QVL.

The sixth is **choosing RAM that is too tall under a bulky air CPU cooler**, resulting in physical collision that prevents installation or compromises cooling. Remedy: check the height of the modules in mm and the clearance declared by the cooler manufacturer, or choose **low-profile** modules.

The seventh, more conceptual, is **being hypnotized by frequency while ignoring real latency**, buying a kit with very high MT/s and terrible timings, perhaps on a platform (like AM5) where that frequency also forces the controller out of the 1:1 ratio, resulting in worse performance than a "slower" but balanced kit. Remedy: think in terms of **real latency (ns)** and respect your platform's sweet spot.

Finally, the eighth is **oversizing or undersizing capacity** without considering your actual use: buying 64 GB for browsing (waste) or 8 GB for video editing (continuous frustration from swapping). Remedy: measure memory usage in your typical use and size accordingly, with a reasonable margin for the future.

Operational Summary — RAM Selection Checklist

Use this checklist in the order indicated: each step depends on the previous ones.

1. **Determine the platform.** Identify the CPU and motherboard. Check if they support **DDR4 or DDR5** (not interchangeable). This is the starting constraint, before any other choice.
2. **Define capacity based on usage.** Light office: 16 GB. Gaming/multitasking: 16–32 GB. Editing, dev, light VMs, serious gaming: 32 GB. 4K video/3D/many VMs/local AI: 64 GB or more. When in doubt and for future-proofing, 32 GB is the safe baseline today.
3. **Choose a kit of TWO modules.** Never a single module (you would lose dual channel). Avoid four modules when you can achieve the same capacity with two of larger size, especially on DDR5.
4. **Identify the platform's frequency/latency sweet spot.** On **AMD AM5: DDR5-6000 CL30 EXPO** (1:1 ratio with the Infinity Fabric). On **Intel: 6000 CL30** as a baseline, with the possibility of higher frequencies where the platform scales well. Always think in terms of **real latency (ns) = CL × 2000 ÷ MT/s**, not just the MT/s number.

5. **Check the QVL and profiles.** Verify that the kit (or an equivalent one) is in the motherboard's **Qualified Vendor List**, or at least that it offers the correct profile: **XMP** for Intel, **EXPO** for AMD.

6. **Check physical constraints.** If you're using a bulky **air** CPU cooler, check the **module height** (clearance) or choose **low-profile** kits. With liquid AIOs, this is almost never an issue.

7. **Choose brand and budget.** Prefer reliable assemblers (**G.Skill, Corsair, Kingston, Crucial, TeamGroup**). Decide if **RGB** is worth the aesthetic surcharge (it doesn't provide performance). For the same specifications, aim for the best price-to-performance ratio.

8. **After installation: activate XMP/EXPO.** Enter the BIOS/UEFI, enable the profile, save, restart. **Verify** frequency and channels with **CPU-Z** (Memory tab: correct frequency and "Dual Channel"). Without this step, the fast RAM you paid for won't perform as it should.

9. **Do not mix kits.** If you plan to reach a certain capacity, buy everything **at once in a single kit**. Adding modules from different batches in the future is a common source of instability.

***Volatile data reminder.** Before purchasing, always update: (1) **prices** and the cost-effectiveness ratio of DDR4/DDR5; (2) what the **current generation** is and the maximum available frequencies; (3) the **specific kits** recommended and the updated **QVL** for your motherboard. The concepts in this chapter remain valid; products and price lists change quickly.*

Chapter 4 - The Graphics Card (Consumer GPU)

***Note on data and reference period.** This chapter is updated as of mid-2026. This is an anomalous time for the graphics card market: since the last quarter of 2025, there has been a severe **memory shortage** (DRAM and GDDR), caused by data center demand for artificial intelligence, which has pushed list prices far above **MSRPs** (Manufacturer's Suggested Retail Price). For this reason, whenever I quote a price in this chapter, I refer to it as "list price/MSRP" and warn that the actual retail price, especially in Italy and Europe (where VAT and logistics are added), can be significantly higher and change week by week. Data on generations, models, and prices should therefore be verified at the time of purchase: they are the part of this chapter that ages most quickly. The technical principles, however, remain valid over time.*

1. What the GPU does

Description and function

The **GPU** (*Graphics Processing Unit*) is the computer component specialized in **massive parallel computing**. To understand what this means, it's helpful to contrast it with the **CPU** (*Central Processing Unit*). The CPU is designed to execute a few very complex instructions in rapid succession: it has a relatively low number of "cores" (processing units), each very powerful and flexible, suitable for tasks that require continuous and unpredictable logical decisions, such as running the operating system or program logic. The GPU follows the opposite philosophy: it contains thousands of small, simple cores, each individually less powerful, but capable of working all together at the same instant on the same type of operation. This is the ideal architecture when the same calculation needs to be applied to millions of distinct data points simultaneously.

The historical use case that gave birth to the GPU is **3D rendering**, which is the transformation of a mathematically described three-dimensional scene (vertices, triangles, textures, lights) into a two-dimensional pixel image to be displayed on a screen. A 1920×1080 pixel monitor contains just over two million pixels; at 3840×2160 (so-called 4K), it exceeds eight million. If each frame needs to be recalculated sixty, one hundred twenty, or more times per second, the number of operations per second becomes astronomical. No CPU could handle it: an

architecture that calculates millions of pixels in parallel is needed, and that is exactly what the GPU does.

Over time, however, it was realized that this same parallel computing engine is also useful outside of gaming. Today, the GPU is used for **video encoding and decoding** (compressing and decompressing video streams, fundamental for streaming and editing), for **scientific computing**, and especially for **local artificial intelligence**: training and especially executing (performing *inference*) neural networks, from image generation models like Stable Diffusion to large language models (**LLM**) run on one's own PC. The reason is the same as for rendering: neural networks boil down to enormous matrix multiplications, which is the same calculation repeated millions of times on different data. It is this convergence between graphics and AI that has made the GPU the most strategic — and most contested — component in the entire hardware sector.

The Physical Components of a Graphics Card

When we speak of "graphics card" in the strict sense, we mean the entire product: a printed circuit board with various components on it, which is inserted into the computer. The actual graphics chip is only one part. Let's look at them one by one.

The **GPU die** is the actual silicon chip: the graphics processor. This is where the computing cores (which NVIDIA calls *CUDA cores* and *Tensor cores*, AMD *Stream Processors* and *AI Accelerators*), the units dedicated to ray tracing, and the memory controllers reside. It is the heart of the product and determines its raw computing power.

VRAM (*Video RAM*) is the dedicated memory of the card, physically separate from the system RAM. It contains textures, 3D models, rendering buffers, and — in the case of AI — the weights of the neural model. It is such a central element that we dedicate the entire next section to it.

The **VRM** (*Voltage Regulator Module*) is the power delivery system of the card. The GPU receives a 12-volt supply from the power supply, but the chip operates at much lower voltages (approximately 1 volt) and very high amperages. The VRM converts and stabilizes this energy, distributing it cleanly to the die. A good quality VRM, with many power "phases," allows for higher clocks, lower temperatures, and longer lifespan; a cheap VRM is often the real limit of low-end cards and poorer versions.

The **PCB** (*Printed Circuit Board*) is the green (or black) board on which everything is mounted. Its quality, the number of copper layers, and the layout of the traces influence electrical stability and overclocking capability.

The **dissipation system** dissipates the heat produced by the die and the VRMs. It is typically composed of a *vapor chamber* or *heat pipes* that transport heat to a stack of aluminum fins, cooled by one or more fans. In more powerful cards, the dissipation is the most voluminous and heaviest part of the product, and this explains why two cards with the same chip can have very different dimensions, noise levels, and prices.

Typical errors at this level

The most common beginner's mistake is to confuse the **GPU die** (the chip) with the **video card** (the complete product). This leads to a second misunderstanding: thinking that two cards with the same name (for example, two "RTX 5070s") are identical. They are not: they share the chip, but the third-party manufacturer that assembles them chooses the cooler, the VRM, the factory clocks, and the power supply. This is the difference between *reference* versions and *custom* versions, which we will discuss extensively in section 6.

2. VRAM and memory bus

What VRAM is and why it's different from system RAM

VRAM is the very high-speed memory soldered directly onto the video card, around the GPU die. It serves to store all the data that the graphics chip needs to work on at any given moment: textures (the "skins" applied to 3D models), scene geometry, various intermediate rendering buffers, and, in AI applications, the neural network weights. The difference compared to system RAM (DDR4 or DDR5 modules inserted into the motherboard and used by the CPU) is twofold. Firstly, VRAM is enormously faster in terms of bandwidth: a mid-range video card moves hundreds of gigabytes per second, while a system RAM kit stops at a few tens. Secondly, VRAM is *dedicated* and local: it is a few millimeters from the chip that uses it, avoiding the bottleneck of the **PCIe** bus (*Peripheral Component Interconnect Express*, the channel that connects the video card to the rest of the computer). If the GPU had to read textures from system RAM by passing through PCIe every time, performance would plummet.

The generations of VRAM encountered today are mainly three. **GDDR6** (*Graphics Double Data Rate 6*) is the mature and economical standard, still used today by AMD across the entire RX 9000 series and by NVIDIA on the lower tiers of past generations. **GDDR6X** is a faster variant developed by NVIDIA with Micron, appearing on high-end RTX 30 and 40 series. **GDDR7** is the most recent generation, adopted by NVIDIA across the entire RTX 50 series: it offers greater bandwidth for the same bus width and is more efficient. As of this writing, AMD on the RX 9000 series (RDNA 4) has chosen to stick with GDDR6, relying on wider buses and internal cache to compensate — a choice that also has the advantage of lower costs, not insignificant during the memory crisis.

The memory bus and the bandwidth formula

The **memory bus** is the width of the channel connecting the GPU die to the VRAM, measured in bits. Typical values are 128, 192, 256, and 384 bits (with extreme cases like the 512 bits of the RTX 5090). It should be imagined as the width of a highway: more lanes (more bits) allow more data to pass through simultaneously. By itself, however, the bus says nothing: the speed at which each lane travels, i.e., the effective frequency of the memory chips, also matters.

The overarching metric is **bandwidth** (memory bandwidth), which is how many gigabytes per second the GPU can exchange with its VRAM. The conceptual formula is:

$$\text{Bandwidth (GB/s)} = (\text{effective memory speed in Gbps} \times \text{bus width in bits}) \div 8$$

We divide by 8 because one byte consists of 8 bits. Let's take a concrete example: a card with GDDR6 memory at 20 Gbps on a 256-bit bus has a bandwidth of $(20 \times 256) \div 8 = \mathbf{640 \text{ GB/s}}$. A card with the same type of memory but a 128-bit bus stops at $(20 \times 128) \div 8 = \mathbf{320 \text{ GB/s}}$, which is half. This is why the bus is as important a factor as the amount of VRAM.

The following table shows how, for the same memory speed, the bus width determines the final bandwidth. The values are indicative and serve to establish orders of magnitude.

Bus Width	18 Gbps Memory (GDDR6)	20 Gbps Memory (GDDR6)	28 Gbps Memory (GDDR7)
128 bit	288 GB/s	320 GB/s	448 GB/s
192 bit	432 GB/s	480 GB/s	672 GB/s

256 bit	576 GB/s	640 GB/s	896 GB/s
384 bit	864 GB/s	960 GB/s	1344 GB/s

It's immediately clear why GDDR7 is interesting: it allows for high bandwidths even with narrower buses, which enables manufacturers to save on the number of chips and PCB complexity. And it's also why you can't compare the bandwidth of two cards by only looking at the bus bits without considering the memory type.

How much VRAM do you really need

The amount of VRAM required depends on three intertwined factors: the **resolution** at which you game or work, the **quality of textures** and effects, and the presence or absence of demanding features like **ray tracing** (realistic physical calculation of lights, shadows, and reflections). A general rule applies: the higher the resolution, the more high-definition textures must reside in memory simultaneously, thus more VRAM is needed. Added to this is the fact that ray tracing requires additional data structures in memory, and some *frame generation* technologies (AI-driven generation of intermediate frames) consume additional VRAM.

As a practical reference, valid in mid-2026 and destined to grow over time as games become more demanding:

Usage Scenario	Minimum Recommended VRAM	Comfortable VRAM
Gaming 1080p, medium-high details	8 GB	12 GB
Gaming 1440p, high details + ray tracing	12 GB	16 GB
Gaming 4K, maximum details + ray tracing	16 GB	20–24 GB
Stable Diffusion / image generation	8–12 GB	16 GB+
Quantized local LLMs (7–8 billion parameters)	8–12 GB	16 GB
Medium-sized local LLMs (13–34 billion)	24 GB	24–48 GB

In the realm of **local AI**, VRAM is even more critical than in gaming, and for a clear reason: a neural model must fit *entirely* into the graphics card's memory to

run at full speed. If the model doesn't fit, it either won't start at all, or it will be "split" between VRAM and system RAM, leading to a performance collapse. For this reason, for those working with AI, the amount of VRAM often becomes the number one selection criterion, even more so than the raw power of the chip. A 7–8 billion parameter language model quantized to 4-bit occupies approximately 5–6 GB and fits comfortably on a 12 GB card; a 70 billion model, however, requires tens of gigabytes and remains the preserve of professional cards or multi-GPU configurations.

The case of "crippled" GPUs on the bus

There is a recurring commercial trap that this chapter must make explicit: the card with **lots of VRAM but a narrow bus**. A manufacturer, to make a low-end model appealing, might equip it with 16 GB of memory — a number that looks great on the box — but connect it with only a 128-bit bus. The result is that the card has space to accommodate abundant textures, but the insufficient bandwidth does not allow it to *move them* fast enough. It's like having a huge warehouse with only one narrow door: the goods are there, but they can't enter and exit at the required speed.

This scenario is particularly disappointing in two cases: at high resolutions, where bandwidth is the limiting factor, and in games with continuous texture streaming. The inexperienced buyer purchases "16 GB" convinced they've made a future-proof deal, only to discover that a card with "only" 12 GB but a wider bus outperforms it in real-world performance. The lesson is clear: **VRAM should never be evaluated in isolation, but always in conjunction with the bus and memory type**, meaning by looking at the overall bandwidth. A good habit is to always check three numbers together — capacity (GB), bus width (bit), and memory type (GDDR6/6X/7) — before being swayed by capacity alone.

3. NVIDIA Nomenclature: How to Read the Name

The Name Structure

NVIDIA sells its consumer cards under the **GeForce RTX** brand. The acronym **RTX** (*Ray Tracing eXtension*) indicates, from the 20 series onwards, the presence of dedicated hardware units for ray tracing and AI calculations; the old acronym **GTX** identified cards without such units. The actual name is a four-digit number that follows a very regular pattern, such as **RTX [series][tier]**.

Let's take the example of the **RTX 4070**. The first two digits, "40", indicate the **generation** (in this case, the 40 series, Ada Lovelace architecture). The next two digits, "70", indicate the **tier** within that generation. So "RTX 4070" reads as "tier 70 card of the fourth RTX generation". By the same criterion, an RTX 5080 is the tier 80 of the fifth generation, an RTX 3060 is the tier 60 of the third generation, and so on. Once this pattern is understood, the name of any NVIDIA card becomes immediately readable.

The Series (Generations)

Each generation corresponds to a different **architecture**, meaning an internal chip design. The three most relevant for today's buyers are:

The **30 series (Ampere)**, released in 2020, is now old but still very popular in the used market. It introduced the second generation of RT cores and the first truly viable DLSS.

The **40 series (Ada Lovelace)**, released in late 2022, brought a great leap in efficiency and DLSS 3 technologies with frame generation. It is no longer in production, so it is mainly found used or as residual stock, but it remains interesting in terms of price/performance ratio precisely because it was built with less expensive memory than current ones.

The **50 series (Blackwell)**, the current consumer generation at the time of writing, announced at CES in January 2025 and launched in the following months. It introduces the fourth generation of RT cores, the fifth generation of Tensor cores, GDDR7 memory across the entire range, and DLSS 4 technologies with **Multi-Frame Generation** (generating multiple intermediate frames instead of just one).

***Warning — evolving data.** In mid-2026, there is persistent talk of a mid-generation refresh called **RTX 50 SUPER** (with more VRAM: rumors suggest 18 GB on the 5070, 24 GB on the 5070 Ti and 5080, and a 12 GB 5060). At the time of writing, however, these models **have not yet been launched** and their debut has been postponed several times precisely due to memory shortages, with some sources moving it to 2027 or considering it uncertain. They should therefore be treated as unconfirmed rumors, to be verified at the time of purchase.*

The Tiers

The last two digits encode the positioning and, with it, what can be expected in terms of performance and price. The scheme, valid as a general rule across generations, is as follows:

The **50 tier** is the absolute entry-level, designed for light 1080p gaming and multimedia use. The **60 tier** is the large volume of the market: the "popular" card for high-quality 1080p and for 1440p with some compromises. The **70 tier** is the heart of the mid-range segment, dedicated to high-refresh 1440p and 4K with the help of upscaling. The **80 tier** is the high-end, designed for 4K gaming with high details. The **90 tier** is the enthusiast, the absolute top of the consumer range, with maximum VRAM and power, oriented towards demanding 4K, content creation, and AI — and with prices that cross into professional territory.

The Suffixes: Ti and SUPER

In addition to the base number, NVIDIA uses two suffixes to create intermediate variants. The suffix **Ti** (historically *Titanium*) indicates an enhanced version of the tier it is attached to: an RTX 4070 Ti is positioned between the 4070 and the 4080, and is faster than the plain 4070. The suffix **SUPER** plays a similar role, typical of mid-generation updates: it indicates an improved revision of an existing model, often with more cores, more VRAM, or more bandwidth. In some cases, both coexist (for example, "RTX 4070 Ti SUPER"). The practical rule to keep in mind is that, within the same generation, the ascending order of performance tends to be: base model → SUPER → Ti → Ti SUPER, then moving up in numerical tier.

Summary Table of Recent NVIDIA Generations

The table shows the main models, with VRAM, bus, and MSRP at launch. **Please note that, due to memory shortages, actual prices in mid-2026 are often much higher than the indicated MSRPs, especially in the high-end tiers.**

Model	Gen. / Architecture	VRAM	Mem. Type	Bus	MSRP at Launch (USD)
RTX 3060	30 / Ampere	12 GB	GDDR6	192 bit	~329
RTX 3070	30 / Ampere	8 GB	GDDR6	256 bit	~499
RTX 3080	30 / Ampere	10 GB	GDDR6X	320 bit	~699
RTX 4060	40 / Ada	8 GB	GDDR6	128 bit	~299

RTX 4070	40 / Ada	12 GB	GDDR6X	192 bit	~549
RTX 4070 Ti SUPER	40 / Ada	16 GB	GDDR6X	256 bit	~799
RTX 4080 SUPER	40 / Ada	16 GB	GDDR6X	256 bit	~999
RTX 4090	40 / Ada	24 GB	GDDR6X	384 bit	~1599
RTX 5060	50 / Blackwell	8 GB	GDDR7	128 bit	~299
RTX 5060 Ti	50 / Blackwell	8 / 16 GB	GDDR7	128 bit	~379–429
RTX 5070	50 / Blackwell	12 GB	GDDR7	192 bit	~549
RTX 5070 Ti	50 / Blackwell	16 GB	GDDR7	256 bit	~749
RTX 5080	50 / Blackwell	16 GB	GDDR7	256 bit	~999
RTX 5090	50 / Blackwell	32 GB	GDDR7	512 bit	~1999

Typical NVIDIA Naming Errors

The most common mistake is comparing two cards solely by tier, ignoring the generation: a newer generation RTX 4070 can outperform or match an older, higher-tier RTX 3080, thanks to more efficient architecture and new upscaling technologies. The second mistake concerns the RTX 5060 Ti, sold in two versions, 8 GB and 16 GB, under the same name: the 8 GB version is penalized in modern games and at higher resolutions, and for a few more euros, the 16 GB version is almost always the better choice. The third mistake is being misled by the 90-tier into thinking it's "double" the 80: 90-tier cards offer superior performance, but with a price/performance ratio that significantly worsens as you go up, because at that tier, you also pay for exclusivity and abundant VRAM, which is more useful for professionals than gamers.

4. AMD Naming Convention: How to Read the Name

The Name Structure

AMD sells its consumer graphics cards under the **Radeon RX** brand. The naming scheme resembles NVIDIA's: **RX [series][tier]**. Let's take the **RX 7800 XT**. The first two digits, "78", should be read as "7000 series, 800 tier": the 7000 series is the

generation, and 800 is its positioning within that generation. The "XT" suffix indicates, as we will see, the more powerful variant of that model. As with NVIDIA, the tier increases with the number: an RX 7600 is entry-level, an RX 7900 is high-end.

The Series (Generations)

There are three relevant AMD generations today, all based on the **RDNA** (*Radeon DNA*) architecture:

The **RX 6000 series (RDNA 2)**, from 2020, is now dated but still available on the used market, offering a good price/performance ratio in pure rasterization, but with weaker ray tracing compared to contemporary NVIDIA counterparts.

The **RX 7000 series (RDNA 3)**, from late 2022, introduced a *chiplet* design (the chip divided into multiple separate dies) and FSR 3 technology with frame generation. It still includes the high-end models RX 7900 XT and XTX, as the subsequent generation abandoned the *enthusiast* segment.

The **RX 9000 series (RDNA 4)**, the current generation at the time of writing, launched in March 2025. The jump in numbering (from 7000 to 9000, skipping 8000) aligns the GPU branding with that of Ryzen processors. RDNA 4 significantly improved ray tracing (third-generation accelerators) and introduced **FSR 4**, the first version of AMD's upscaling based on machine learning. It's important to note that this generation **does not have an enthusiast-tier model**: AMD chose to focus on the *mid-range* segment, with the RX 9070 XT at the top of the lineup.

The suffixes: XT, XTX, GRE

AMD uses suffixes to distinguish power variants with the same number. The model **without a suffix** is the base version. The **XT** suffix indicates a more powerful version, with more compute units or higher clocks: the RX 9070 XT is faster than the plain RX 9070. The **XTX** suffix, which appeared on the 7000 series, indicates the maximum variant, a step above the XT (as in the RX 7900 XTX compared to the 7900 XT). The **GRE** suffix (*Golden Rabbit Edition*, a name initially created for the Chinese market for the Year of the Rabbit) identifies a slightly depowered version with a narrower bus, positioned between two models of the same series to cover an intermediate price range; the RX 9070 GRE, for example, has 12 GB on a 192-bit bus compared to the 16 GB on a 256-bit bus of the plain 9070, and in 2026 it was made globally available after an initial China-exclusive debut.

Approximate AMD ↔ NVIDIA equivalence table

This table matches models by **price range and indicative performance**, not by exact technical equivalence. Relative performance varies from game to game and, more importantly, changes depending on whether pure rasterization (where AMD is competitive) or ray tracing (where NVIDIA maintains an advantage) is considered. It should therefore be read as a compass, not as a law.

Tier / Indicative Price	AMD (RDNA 4)	NVIDIA (Blackwell)
Entry (~\$300)	RX 9060 XT 8 GB	RTX 5060
Entry-mid (~\$350)	RX 9060 XT 16 GB	RTX 5060 Ti 16 GB
Mid (~\$550)	RX 9070	RTX 5070
Mid-high (~\$600)	RX 9070 XT	(between 5070 and 5070 Ti)
High (~\$750)	— (no model)	RTX 5070 Ti
High / enthusiast (\$900+)	— (no model)	RTX 5080
Absolute enthusiast (\$2000+)	— (no model)	RTX 5090

The most eloquent data in the table is the row of dashes: in the current generation, AMD **does not compete at all** in the high-end and enthusiast segments, having left them uncovered. Anyone looking for a card beyond the power of the RX 9070 XT today effectively turns only to NVIDIA.

Typical errors in AMD nomenclature

A recurring error is confusing XT and XTX, or assuming that the GRE suffix indicates something better when it actually indicates a reduced variant. A second error, mirroring the one seen for NVIDIA, concerns the RX 9060 XT sold in 8 GB and 16 GB versions: the 8 GB version is less advisable for modern games, and the price difference with the 16 GB is often small. A third error is assuming that AMD is always cheaper: this is true in many segments, but market conditions in 2026 have made prices volatile even for AMD, although Radeon models have held their MSRP better than their NVIDIA counterparts, thanks to memory supplies secured in advance.

5. NVIDIA vs AMD: Full Comparison

This is the ultimate decision-making section. I compare the two manufacturers on the planes that truly matter, so that the reader can understand *when* one is convenient and *when* the other, without bias.

Pure Rasterization: Price/Performance

Rasterization is the traditional technique for generating images, without the realistic physical calculation of light. It still determines most of the performance in games. In this area, **AMD is historically very competitive**, often offering more frames per second per euro spent than NVIDIA in the same range. In the current generation, the RX 9070 XT offers rasterization performance close to that of the more expensive RTX 5070 Ti, at a lower list price: it is the classic example of AMD's advantage in price/performance ratio in pure gaming.

Ray Tracing: NVIDIA Advantage, RDNA 4 Recovery

Ray tracing simulates the physical behavior of light to achieve realistic reflections, shadows, and lighting. It is much heavier to compute than rasterization and requires dedicated hardware units (NVIDIA's *RT cores* and AMD's *Ray Accelerators*). Historically, **NVIDIA has had a marked advantage** in this field: its cards lose less performance when ray tracing is activated. With RDNA 4, however, **AMD has recovered a lot of ground**, significantly closing the gap thanks to third-generation accelerators. The NVIDIA advantage remains, especially in games with *path tracing* (the most extreme and realistic form of ray tracing), but the gap is no longer as abysmal as in previous generations.

Upscaling: DLSS vs FSR vs XeSS

Upscaling is a technology that allows a game to be rendered at a lower resolution (e.g., 1080p) and then enlarged to a higher resolution (e.g., 4K) using intelligent algorithms, gaining frames with minimal quality loss. It has become a central competitive factor, as much as raw power. The three main technologies are:

NVIDIA's **DLSS** (*Deep Learning Super Sampling*) is considered the benchmark for quality and diffusion. It is based on neural networks executed on Tensor cores and is exclusive to NVIDIA cards. DLSS 4, exclusive to the 50 series, introduces **Multi-Frame Generation**, capable of generating more intermediate frames via AI to multiply the frame rate; however, it should be remembered that frame generation adds latency and does not replace "true" performance.

AMD's **FSR** (*FidelityFX Super Resolution*) is the answer to DLSS. Up to version 3, it was *vendor-agnostic*, meaning it also worked on NVIDIA and Intel cards, but with lower quality. With **FSR 4**, AMD has moved to a machine learning-based approach that has greatly improved quality, bringing it closer to DLSS, but at the cost of limiting it to RDNA 4 cards equipped with the necessary AI accelerators.

Intel's **XeSS** (*Xe Super Sampling*) is the third option, created for Intel Arc cards but also usable in a reduced form on other brands. It offers good quality and is the "third wheel" in the sector.

Encoder: NVENC vs AMF

The **encoder** is the hardware unit that compresses video in real-time, essential for **streaming** (Twitch, YouTube) and for exporting in **video editing**. NVIDIA has **NVENC** (*NVIDIA Encoder*), universally considered the best for quality and stability, so much so that it is a recurring reason why streamers and videomakers choose NVIDIA. AMD uses **AMF** (*Advanced Media Framework*): historically inferior, it has improved in recent generations, but NVENC maintains a recognized advantage, especially in high-efficiency encoding (HEVC and AV1) at low bitrates.

CUDA vs ROCm: why NVIDIA dominates for AI and productivity

This is perhaps the most important point for a reader who, in addition to gaming, is looking at **artificial intelligence and productivity**, and it is the basis on which the entire section dedicated to workstations will rest. **CUDA** (*Compute Unified Device Architecture*) is NVIDIA's proprietary software platform for general-purpose computing on GPUs. It is not just an API: it is an entire ecosystem, mature for over fifteen years, on which almost all AI software is built — from PyTorch to TensorFlow, from image generation frameworks to tools for local LLMs. The AMD counterpart is called **ROCm** (*Radeon Open Compute*): it is open and continuously improving, but it remains less mature, less widespread, and with more fragmented software support. The practical result is clear: **for AI, accelerated creative productivity, and scientific computing, NVIDIA dominates** not so much because of the hardware but because of the software. Those who buy a card thinking of using it to train or run models, edit videos with AI effects, or work with professional applications, will in most cases find far fewer frictions with NVIDIA. This is a consideration that, for certain usage profiles, completely overturns AMD's price/performance advantage in pure gaming.

Drivers, power consumption, prices

On the **driver** front, both brands today offer stable software, but NVIDIA has historically enjoyed a reputation for greater reliability at the launch of new games, while AMD has sometimes suffered from problems in the first few weeks after a card's release, which were then resolved with updates. In terms of **power consumption**, the two brands are equivalent in their respective segments, with model-by-model fluctuations. Regarding **prices**, as already mentioned, AMD tends to offer more for the same money in rasterization, while NVIDIA charges for the ecosystem (DLSS, NVENC, CUDA); in the context of 2026, moreover, real prices are governed more by memory availability than by official price lists.

The third player: Intel Arc

Intel Arc, Intel's line of graphics cards, should be mentioned as a third player. The **Battlemage** series (cards like the Arc B580 12 GB) has gained an excellent reputation in the entry-level segment, offering more VRAM than the competition at aggressive prices and quality AV1 encoders. Arc's historical limitations are its drivers, which are less mature, especially in older games, and an AI software ecosystem that lags behind CUDA. Furthermore, in mid-2026, the future of Intel's gaming line appears uncertain: some rumors suggest a downsizing of high-end models, diverted towards the professional market. For the budget-conscious buyer, however, an entry-level Arc card remains one of the best VRAM/price options today, provided they accept some compromises on drivers.

6. AIB partners and custom versions

Founders Edition / reference vs custom

When NVIDIA and AMD design a card, they also produce an "official" version: NVIDIA calls it the **Founders Edition** (FE), AMD calls it the **reference** version. These are the cards with the chip manufacturer's reference design. Alongside these, there is a large number of **AIB partners** (*Add-In Board partners*, third-party manufacturers who buy the chips and assemble their own cards): ASUS, MSI, Gigabyte, Sapphire, PowerColor, and many others. Each produces its own **custom** versions, with different coolers, PCBs, VRMs, clocks, and aesthetics. The same RTX 5070, therefore, exists in dozens of physically different variants, all with the same chip but with build quality, temperatures, noise, and price that can vary considerably.

NVIDIA partners and their ranges

Each AIB partner organizes its cards into product lines ordered by increasing quality. Knowing these commercial names is essential to avoid paying too much or too little. The main NVIDIA partner brands, with their respective ranges from low to high, are:

ASUS offers the **Dual** line (entry, simple dual-fan cooler), the **TUF Gaming** (mid-range robust, excellent quality/price balance), and the **ROG Strix** (high-end, top-notch cooling and VRMs, high prices). **MSI** offers the **Ventus** (entry), the **Gaming / Gaming Trio** (mid-range, with a triple-fan cooler), and the **Suprim** (top-of-the-range, premium build). **Gigabyte** has the **Windforce / Gaming** line (entry and mid-range) and the **Aorus** (high-end). The picture is completed by brands like **Zotac** (often aggressive on price, with very compact models), **Palit** and **Gainward** (good quality/price ratio, generally cheaper), **PNY** (widespread and reliable, no frills), and **INNO3D**.

AMD partners and "reference brands"

The AMD landscape has some peculiarities. **Sapphire** is considered the quintessential reference AMD partner — the equivalent of what the best premium partners are for NVIDIA — with the **Pulse** line (mid-range, excellent balance) and the **Nitro+** (top-of-the-line, highly regarded). **PowerColor** is the other big historical name associated with AMD, with the **Fighter** (entry), **Hellhound** (mid-range, quiet and well-balanced), and **Red Devil** (high-end) lines. Completing the picture are **XFX**, a historical exclusive AMD partner, and **ASRock**, which more recently entered the graphics card market with good results. Sapphire, PowerColor, and XFX are often referred to as AMD's "reference brands" because they work almost exclusively with Radeon, unlike ASUS, MSI, and Gigabyte, which produce for both brands.

How much is it really worth paying more?

The practical question is: is it worth spending more for a premium version? The difference between a base card and a top-of-the-line card focuses on four aspects. The **cooler**, larger and with more heat pipes, keeps temperatures lower. The **factory clocks**, slightly higher on premium versions ("OC" acronyms indicate a factory *overclock*), typically lead to modest performance gains, in the order of a few percentage points. The **noise**, often lower on premium versions at the same temperature, thanks to better fans and more refined curves. The **aesthetics**, with RGB lighting and more refined finishes. The reasonable practical rule is this: **mid-**

range versions (TUF, Pulse, Hellhound, Gaming) offer the best compromise and are the ones to recommend in the vast majority of cases; top-of-the-line versions (ROG Strix, Suprim, Nitro+, Red Devil, Aorus) only make sense for those seeking absolute silence, extreme overclocking, or aesthetics, and are willing to pay a premium that rarely translates into significantly superior performance. Beware, however, of the cheaper and more spartan versions of each brand on high-end cards: an undersized cooler on a powerful chip leads to high temperatures, throttling (automatic performance reduction to protect against heat), and noise.

Typical errors with custom versions

The most expensive mistake is paying the premium for a top version thinking you'll get a performance leap that isn't actually there: between the base version and the premium version of the *same* card, the frame rate difference is almost always negligible. The opposite mistake is buying the cheapest version of a powerful card to save money, ending up with inadequate dissipation and noisy fans. A third mistake is being guided only by RGB and aesthetics, ignoring the parameters that really matter: temperatures, noise, and VRM quality.

7. Physical size and compatibility

Length, thickness, and case clearance

A video card is a bulky item, and one of the most frustrating beginner mistakes is buying one without checking if it fits into your **case**. Two main dimensions need to be checked. The **length**, expressed in millimeters, must be less than the maximum *clearance* (available space) declared by the case manufacturer: high-end cards can exceed 320–350 mm, and not all cases accommodate them. The **thickness**, measured in "slots," indicates how many expansion slots the card occupies on the back of the case: cards are divided into **2-slot**, **2.5-slot**, **3-slot**, and even **4-slot** models. The larger the cooler — meaning the more powerful or premium the card — the more slots it occupies, and the greater the risk that it will cover other motherboard connectors or not fit into a compact case. The operating rule is to note the desired card's length and thickness, compare them with the case's specifications, and leave a few millimeters of margin for cables.

Power Connectors: 8-pin PCIe vs 12VHPWR / 12V-2x6

The video card receives power from the **power supply unit** (PSU) through dedicated connectors. Historically, **8-pin PCIe** connectors have been used, of which a card may require one, two, or three depending on its power consumption. With the RTX 40 series, NVIDIA introduced a new high-density single connector called **12VHPWR** (*12-pin High Power*), later revised into an improved version called **12V-2x6**, capable of delivering up to 600 watts through a single cable. However, this connector has a well-known **problematic history**: on very high-end cards (especially RTX 4090 and 5090), there have been cases of connector overheating and melting, often linked to imperfect insertion or uneven current distribution among the pins. Practical countermeasures include firmly inserting the connector all the way, avoiding tight bends of the cable right behind the plug, and preferring native PSU cables over the included multi-8-pin **adapters**, which are bulkier and considered less reliable. Anyone purchasing a high-end card with this connector should ideally equip themselves with a modern **ATX 3.0 / ATX 3.1** power supply with a native 12V-2x6 cable.

PSU Requirements by GPU Tier

Each card has a typical power consumption, often indicated as **TDP** (*Thermal Design Power*, the thermal design power, used as an approximation of electrical consumption under load) or as TBP (*Total Board Power*, the consumption of the entire card). The power supply must have sufficient power not only for the card but for the entire system, with a safety margin. The following table provides conservative recommendations for the recommended PSU wattage based on the GPU tier, considering a complete system with a comparable CPU.

GPU Tier (examples)	Typical GPU Consumption	Recommended PSU (complete system)
Entry (RTX 5060 / RX 9060 XT)	130–180 W	550–650 W
Mid (RTX 5070 / RX 9070)	200–250 W	650–750 W
Mid-high (RTX 5070 Ti / RX 9070 XT)	260–320 W	750–850 W
High (RTX 5080)	~360 W	850 W
Enthusiast (RTX 5090)	~575 W	1000–1200 W

The practical advice is not to skimp on the power supply and not to get one that is just barely sufficient: a 30-40% margin above the estimated peak consumption

improves efficiency, reduces PSU fan noise, and extends the component's life. It is also preferable to choose units from reliable brands with **80 Plus** certification (Gold or higher), because a poor power supply is one of the most insidious causes of instability and, in the worst cases, hardware damage.

GPU Sag and Supports

Modern graphics cards are heavy, and over time they tend to bend downwards under their own weight: this phenomenon is called **GPU sag**. Besides being unsightly, significant sag puts strain on the PCIe slot and the connector. The solution is an **anti-sag support** (a small prop, often included with premium cards or available separately) that supports the free end of the card. It is an inexpensive accessory and recommended for all mid-range and high-end cards, especially in configurations with vertical cases or with the card mounted horizontally.

8. Which Slot to Install It In

The First PCIe x16 Slot Connected to the CPU

The motherboard has multiple **PCIe** expansion slots, but they are not all the same, and the graphics card must be installed in the correct one. The correct slot is the **first PCIe x16 slot directly connected to the CPU**. The "x16" designation indicates the number of physical and electrical *lanes* (data paths) available: sixteen lanes are the maximum for a consumer graphics card and ensure full bandwidth. The fact that it is connected *directly to the CPU*, and not to the chipset, is crucial: it means that data travels between the graphics card and the processor via the shortest and fastest route, without intermediate steps that add latency. This slot is almost always the first from the top, closest to the CPU socket, and is typically metal-reinforced precisely to support the weight of the graphics card.

What Happens in Secondary Slots

Secondary slots, lower down on the motherboard, have inferior characteristics. Even when they are physically as long as an x16 slot, electrically they often offer only 4 lanes (**x4**) and — a decisive detail — are connected to the **chipset** rather than directly to the CPU. This means two penalties combined: less bandwidth (a quarter of the lanes) and a longer, shared path to the processor, passing through the connection between the chipset and the CPU, where traffic from other devices also travels. Installing a powerful graphics card in a secondary slot can therefore limit its performance, negligibly for gaming at normal resolutions but noticeably

in AI and intense data transfer scenarios. Secondary slots are designed for other expansion cards (network cards, video capture cards, additional controllers), not for the main graphics card. The mistake to avoid, typical of those assembling for the first time or not checking the motherboard's instructions, is to insert the graphics card into the first physical x16 slot they find without verifying that it is the one connected to the CPU: the correct slot is always clearly indicated in the motherboard manual.

9. How to Choose

Start with the Monitor: Resolution and Refresh

The correct starting point for choosing a video card **is not the budget or the brand, but the monitor**. The display's resolution and refresh rate (measured in hertz) define how much power is truly needed. Buying an enthusiast card for a 1080p 60 Hz monitor is a waste: the card would produce many more frames than the monitor can display. Conversely, installing an entry-level card on a high-refresh 4K monitor leads to disappointing frame rates. The logic is: first, establish the goal (for example, "1440p at 144 Hz with high details"), then choose the card that achieves it, with a small margin for future games. As a general guide: for 1080p at high refresh, a 60-series card is sufficient; for 1440p at high refresh, a 70-series card is ideal; for 4K with high details, at least a 70 Ti / 80-series card or equivalent top-tier AMD cards are needed.

The Budget: The GPU as the Component to Allocate the Largest Share To

In a PC primarily designed for **gaming**, the video card is the component that most influences perceived performance, and it is therefore correct to allocate the largest share of the budget to it — typically between 35% and 45% of the total configuration cost. This does not mean buying the most expensive card possible by sacrificing everything else: it means balancing. A sound rule is that the video card and the processor must be in a coherent range with each other, because excessive imbalance creates a **bottleneck**, a situation where a component that is too weak slows down the other, preventing it from performing optimally.

Common Buyer Mistakes

Let's conclude with the most frequent pitfalls, which negatively summarize everything we've seen. The first is a **top GPU with an entry-level CPU**: pairing a very powerful card with an inexpensive processor that cannot "feed" it data fast

enough, wasting a good part of the card's potential, especially at lower resolutions. The second is **insufficient VRAM for the intended use**: buying a card with too little memory for the resolution or AI workload you intend to tackle, resulting in performance drops and stuttering as soon as games or models exceed the available capacity. The third is a **case that is too small**: falling in love with a card without checking its length and thickness against the internal space of the cabinet, and discovering at the last minute that it doesn't fit. The fourth is an **undersized PSU**: pairing a barely sufficient or low-quality power supply with a demanding card, causing sudden shutdowns under load or hard-to-diagnose instability. The fifth, more subtle, is **chasing VRAM quantity alone** while ignoring the bus and memory type, as already discussed in section 2. Each of these errors stems from the same fundamental flaw: evaluating a component in isolation, forgetting that a video card lives within a system and must be chosen in harmony with everything else — monitor, CPU, case, and power supply.

Operational Summary — Video Card Decision Checklist

This checklist condenses the entire chapter into a sequence of decisions to be made in the correct order, from the first to the last step.

1. **Define your goal based on the monitor.** Establish the resolution and refresh rate of your display (or the display you intend to purchase): 1080p, 1440p, or 4K, and at how many hertz. This, not the budget, is the true starting point.
2. **Identify your usage profile.** Gaming only? Also streaming and video editing? Artificial intelligence, image generation, or local LLMs? If you fall into the latter two categories, the software ecosystem (NVIDIA's CUDA) and the amount of VRAM weigh more than pure price/performance.
3. **Choose the tier consistent with your goal.** 60/9060 series for 1080p, 70/9070 series for 1440p, 70 Ti-80 series for 4K. Do not overspec for your monitor or underspec for your target.
4. **Check VRAM along with bus and memory type.** Always check three numbers together: capacity in GB, bus width in bits, memory type (GDDR6/6X/7). Be wary of cards with a lot of VRAM but a narrow bus. For AI, aim for as much VRAM as possible within your price range.

5. Decide between NVIDIA and AMD based on your profile. Gaming only in rasterization and tight budget → AMD is often better value. Heavy ray tracing, quality streaming, AI, and productivity → NVIDIA is the more solid choice. Absolute minimum budget → also consider Intel Arc.

6. Choose the right custom version. Prefer mid-range lines (TUF, Pulse, Hellhound, Gaming, Windforce). Only upgrade to top-tier (ROG Strix, Nitro+, Suprim, Red Devil) if you need absolute silence, overclocking, or aesthetics, knowing that the performance gain is minimal.

7. Check physical compatibility. Measure the length and thickness (number of slots) of the card and compare them with your case's clearance, leaving room for cables.

8. Size the power supply with margin. Choose a reliable brand PSU, 80 Plus Gold certified or higher, with a 30–40% margin over the estimated peak system power consumption. For cards with a 12V-2x6 connector, prefer an ATX 3.0/3.1 power supply with a native cable.

9. Verify balance with the CPU. Ensure that the processor is in a consistent tier with the graphics card to avoid bottlenecks. Better two balanced components than one excellent and one mediocre.

10. Install in the first PCIe x16 slot connected to the CPU. This is indicated in the motherboard manual, usually the first from the top and reinforced. Add an anti-sag bracket if the card is heavy.

11. Check prices at the time of purchase. Given the ongoing memory shortage in 2026, ignore theoretical MSRPs and check actual retail prices, comparing multiple retailers. If your budget is limited, also consider a well-maintained previous-generation card, which during a memory crisis can offer a better price/performance ratio than new.

Final note on data currency: models, VRAM, prices, and availability in this chapter reflect the situation in mid-2026, a period marked by a memory shortage that makes prices particularly volatile. The conceptual sections (GPU operation, bandwidth formula, selection criteria, compatibility) remain valid over time; model and price tables should be re-verified at the time of purchase.

Chapter 5 - Storage

1. Overview: What is storage and why is it different from memory

Every time you turn on a computer, something we take for granted happens: the machine "remembers" itself. It finds the operating system, the installed programs, the documents written the day before, the photos from ten years ago. This ability to remember beyond shutdown is the task of **storage**, i.e., mass memory, and to truly understand what it does, one must first understand what RAM *doesn't* do.

RAM (Random Access Memory) is **volatile**: it retains data only as long as it is powered. The exact moment we cut power, the contents of the DRAM cells vanish, because each bit is stored as a tiny electrical charge in a capacitor that discharges in milliseconds if not continuously "refreshed." RAM is extremely fast and completely amnesic. Storage is the exact opposite: it is **non-volatile**, meaning it retains information even without power, for months or years, but it pays for this persistence with latencies that, compared to RAM, are astronomical. To give a mental scale: an access to RAM costs about 80–100 nanoseconds; an access to a good NVMe SSD costs 50,000–80,000 nanoseconds (i.e., 50–80 microseconds); an access to a mechanical hard disk costs 8,000,000–15,000,000 nanoseconds (8–15 milliseconds). Between RAM and hard disk, there are five orders of magnitude: it's the difference between picking up a book from your desk and ordering it by mail.

The computer is therefore built as a **memory hierarchy**, where each level decreases in speed and increases in capacity and persistence: CPU registers → L1/L2/L3 cache → RAM → storage → (possibly) remote archiving or offline backup. Storage is the step where the pyramid becomes permanent. Everything we want to find again tomorrow must, sooner or later, land there.

The role of storage in the system is twofold. On one hand, it is the **archive**: it contains the operating system, executables, and user data. On the other hand, it is the **bottleneck of perceived responsiveness**. This second point is what users most underestimate. When a PC is turned on, when a heavy program is opened, when a video game loads a level, or when a development project compiles thousands of source files, what determines the seconds of waiting is almost never the raw power of the CPU: it is the speed with which data travels from storage to RAM. A high-end processor paired with a mechanical disk gives a feeling of slowness that no amount of cores can compensate for, because the CPU spends most of its time waiting (in jargon, in *I/O wait*, input/output wait). This is why, in

the priority scale of an assembly, the transition from HDD to SSD is historically the upgrade with the best cost-to-perceived-improvement ratio ever in the history of personal computers.

1.1 The evolution: HDD → SATA SSD → NVMe SSD

The history of consumer storage over the last twenty years is the story of two successive revolutions, and it is useful to retrace it because it explains why today's market has the shape it does, with its remnants and its pitfalls.

Phase One: The Reign of the HDD (up to ~2010). For decades, the only economically sensible mass storage was the magnetic hard disk drive, an electromechanical device with moving parts. All PC software architecture — the way operating systems order read requests, the very concept of "defragmentation," caching strategies — was designed around the physical limitations of a mechanical arm moving over a spinning platter.

Phase Two: The Arrival of SATA SSDs (2008–2015). SSDs (Solid State Drives) replace moving parts with flash memory. But the first consumer SSDs emerged in a world where the standard interface was **SATA** (Serial ATA, Serial Advanced Technology Attachment), an interface designed *for hard drives*. SSDs thus entered the PC disguised as hard drives: same cable, same connector, same 2.5-inch form factor, same software protocol (AHCI, which we will discuss). The result was still a perceived revolution — boot times plummeted from a minute to ten seconds — but it was a half-revolution, because the SSD was speaking a language designed for a mechanical device. The SATA III bandwidth limit, about 550 MB/s real-world, was saturated almost immediately by the first generations of decent SSDs, and from then on, all SATA SSDs in the world — from the cheapest to the most expensive — reported more or less the same sequential numbers. The bottleneck had shifted from the drive to the interface.

Phase Three: NVMe and PCIe (2015–present). The solution is to remove the legacy interface and connect the flash memory directly to the fastest bus available in the system, **PCIe** (Peripheral Component Interconnect Express), the same bus to which the graphics card is connected. A new protocol, **NVMe** (Non-Volatile Memory Express), is defined over PCIe, designed from scratch for flash memory: massive parallelism, multiple queues, reduced latencies, no mechanical heritage. The result, within a few years, is a bandwidth leap from 550 MB/s to 3,500 MB/s (PCIe 3.0), then 7,400 MB/s (PCIe 4.0), then 14,000 MB/s (PCIe 5.0), with latencies decreasing accordingly.

Today, in 2026, the consumer landscape is this: NVMe SSDs are the de facto standard for system drives, SATA SSDs survive as inexpensive secondary storage and in older machines, and HDDs have retreated into a precise but still very important niche — high-capacity, low-cost-per-terabyte archiving. None of these three technologies are "dead." They are simply specialized.

***[volatile data]** Since the end of 2025, the storage market has been experiencing an anomalous phase of sharp price increases. The demand for memory generated by artificial intelligence infrastructures has siphoned off NAND and DRAM production capacity towards the enterprise segment, resulting in double-digit percentage price increases for consumer SSDs and reduced availability for some product lines. During the same period, Micron announced its exit from the consumer market with the Crucial brand, to focus on datacenters and OEM customers: the Crucial models mentioned in this chapter remain valid as a technical reference and may still be available as residual stock, but should no longer be considered an evolving product line. All prices indicated below should therefore be taken as orders of magnitude to be verified at the time of purchase, not as reliable values.*

2. HDD vs. SSD: Two Different Physical Architectures

2.1 How a Hard Disk Works

An **HDD** (Hard Disk Drive) is, literally, a precision turntable sealed in an aluminum box. Inside, there are one or more **platters**, rigid aluminum or glass disks coated with a very thin layer of ferromagnetic material. The platters are mounted on a single spindle connected to a motor that rotates them at a constant speed: typically 5,400 Revolutions Per Minute (RPM) in archival and laptop models, 7,200 RPM in desktop and NAS models, up to 10,000 or 15,000 RPM in older, now obsolete enterprise drives.

Information is stored by orienting the magnetization of tiny regions of the coating: one direction is "0", the other is "1". **Heads**, mounted at the end of mechanical arms that move radially over the surface of the platters, like the arm of a record player, read and write these orientations. Each surface of each platter has its own head, and all arms move together, in unison. The heads never touch the platter: they "fly" a few tens of nanometers from the surface, supported by the air cushion generated by the rotation. If a head touches the moving platter — due

to a bump, for example — a notorious *head crash* occurs, which scrapes away the magnetic coating and destroys data in that area.

Data is organized into concentric circles called **tracks**, divided into **sectors** (historically 512 bytes, now 4096 physical bytes in the *Advanced Format* standard). The set of tracks that are at the same distance from the center on all platters forms a **cylinder**.

The slowness of the HDD directly stems from this architecture. To read data, the disk must perform two physical movements in sequence:

1. **Seek time**: the arm must move the head to the correct track. This typically costs 8–12 milliseconds on a desktop drive. 2. **Rotational latency**: once on the correct track, one must wait for the desired sector to pass under the head due to the platter's rotation. On average, one waits half a rotation. At 7,200 RPM, one rotation takes approximately 8.3 ms, so the average latency is about 4.2 ms; at 5,400 RPM, it rises to about 5.6 ms.

Summing up, a "cold" access to a random disk location costs on the order of **10 milliseconds**. And here's the crucial point: that time is **physical**, dictated by mechanics, and cannot be reduced by any electronic improvement. A modern 24 TB HDD reads sequential data at 280 MB/s, a speed that is by no means negligible—it's half that of a SATA SSD. But if we ask it to read four thousand small files scattered across the disk, as an operating system does at startup, it has to pay 10 ms for each: the throughput drops to values in the order of **1–2 MB/s**, and this is why a PC with a hard disk seems bogged down in molasses. It's not slow to read: it's slow to *seek*.

Random access performance is measured in **IOPS** (Input/Output Operations Per Second). A 7,200 RPM HDD performs 75–120 IOPS. A mid-range NVMe SSD performs **one million**. The ratio is approximately 1 to 10,000, and no number in this chapter is more important than this.

2.2 How an SSD works

An **SSD** (Solid State Drive) has no moving parts. It is a specialized small computer, composed of three fundamental elements.

The first is **NAND flash memory**, the chips that physically contain the data. NAND stores information by trapping electrons in an isolated structure called a *floating gate* (or, in more modern implementations, a *charge trap*) within a transistor cell.

The presence or absence of charge—or, more precisely, the *level* of charge—determines the stored value. Since the insulator retains electrons even without power, the memory is non-volatile. This is a remarkable but not eternal property: an SSD left unplugged for years slowly loses charge and, eventually, data. For normal use, this is irrelevant; for long-term storage in a drawer, it is not (we will return to this).

Cells are organized into **pages** (typically 4–16 KB), and pages into **blocks** (hundreds of pages, for a few megabytes). Here lies the fundamental asymmetry of flash, which explains almost everything strange an SSD does: **data is read and written by pages, but erased only by entire blocks**. You cannot overwrite an already written page: you must first erase the entire block containing it. Consequently, when the operating system "modifies" a file, the SSD actually writes the new version to a free page elsewhere, marks the old one as obsolete, and sooner or later will have to clean up by consolidating valid data and erasing blocks now full of garbage. This cleanup is called **garbage collection**, and together with the **TRIM** command (with which the operating system communicates to the SSD which blocks belong to deleted files and therefore no longer need to be preserved) it is what allows the drive to maintain decent performance over time.

The second element is the **controller**, the SSD's processor. It manages everything: the translation between the logical addresses seen by the operating system and the actual physical locations in the NAND (a table called a *mapping table*, managed by the **FTL** (Flash Translation Layer) firmware), garbage collection, error correction via **ECC** (Error Correcting Code), and **wear leveling**, which is the uniform distribution of writes across all cells so that none wear out before the others. The controller is, no less than the NAND, what determines the quality of an SSD: two units with the same memory chips but different controllers can behave radically differently under load.

The third element — present only in mid-to-high-end models — is the **DRAM cache**: a DRAM chip (often 1 GB per 1 TB of capacity) that the controller uses to keep the mapping table in fast memory. We will discuss this in detail in section 5, because it is one of the most important discriminators between a good SSD and a mediocre one.

It follows that accessing data on an SSD does not require any physical movement: the controller consults the table, identifies the chip and page, and queries them

electrically. Latency is in the order of tens of microseconds and is **substantially identical** whether the data is "at the beginning" or "at the end" of the disk, and whether a large file or a thousand small files are being read. It is the disappearance of seek cost, not higher bandwidth, that is the real reason why SSDs have changed the computer user experience. From this also derives the fact that **defragmenting an SSD is useless** and even harmful, because fragmentation is only a problem for those who have to move a read/write head, and unnecessary writes consume cells.

2.3 When HDD still makes sense

Having said all this, it would be a mistake to conclude that the hard disk is obsolete. The HDD still has one robust advantage and one secondary advantage.

The robust advantage is the **cost per terabyte**. A 20 TB hard disk today costs, as an order of magnitude, less than a third of what the same capacity in SSD would cost (*variable data, and made more favorable to HDD by the ongoing NAND price increase*). For a library of 40 TB of videos, backups, and archives, the difference is not a detail: it is the difference between a feasible project and an unfeasible one.

The secondary advantage is **long-term data retention without power**. A properly stored hard disk retains data for years without being powered on; an SSD, which relies on electrical charges trapped in an insulator, tends to degrade more quickly if left disconnected for very long periods, especially if the cells are already worn and if storage temperatures are high. For a "cold" backup to put in a drawer and forget, an HDD is still the most sensible choice — provided, of course, that it is connected and checked periodically, and that you never rely on a single copy.

The scenarios where HDD is still the right answer are therefore:

- ****Cold storage****: media collections, photo archives, rarely accessed and sequentially consulted material.
- ****Backup****: destination drive for periodic backups, where capacity matters more than latency.
- ****Home NAS and small servers**** (Network Attached Storage): here, HDDs reign, both for cost and because the bottleneck is often the network itself. A gigabit network transfers a maximum of about 110 MB/s: an HDD saturates it effortlessly, and putting expensive SSDs behind such a connection brings no bandwidth benefit (it does, however, bring a benefit in latency and IOPS,

which matters if the NAS hosts virtual machines or databases, not if it hosts movies).

- ****Video surveillance****: continuous, sequential recording of 24/7 video streams, for which specific drives exist.

What the HDD should **not** do anymore, in 2026, is host the operating system. There is no debate or budget margin on this: even the cheapest PC must boot from an SSD.

2.4 HDD / SATA SSD / NVMe SSD Summary Table

Characteristic	HDD 7,200 RPM	SATA SSD	NVMe PCIe 4.0 SSD
Sequential Read	150–280 MB/s	~550 MB/s	5,000–7,400 MB/s
Sequential Write	150–280 MB/s	~520 MB/s	4,000–7,000 MB/s
Random Read 4K IOPS	75–120	80,000–100,000	600,000–1,500,000
Typical Latency	8–15 ms	~100 µs	40–80 µs
Moving Parts	Yes	No	No
Shock Resistance	Low	High	High
Noise Level	Audible	Silent	Silent
Typical Power Consumption	5–9 W	2–4 W	4–9 W (peaks)
Max Consumer Capacity	24–30 TB	8 TB	8 TB (rare), 4 TB common
Cost per TB	Lowest	Intermediate	Highest
Ideal Use	Archive, backup, NAS	Secondary storage, old PC upgrade	OS, games, work

3. SATA vs. NVMe: Interface and Protocol

Here, a distinction that many buyers confuse is essential. A storage unit is characterized by **two different things**:

- the ****physical and electrical interface****: the wires on which bits travel (SATA, PCIe);
- the ****logical protocol****: the language with which the operating system and the unit communicate over those wires (AHCI, NVMe).

Historically, the two things went hand in hand — SATA used AHCI — and so the habit of treating them as synonyms spread. But, as we will see with M.2, today they are not at all, and that is precisely where purchasing errors lie.

3.1 SATA III and AHCI

SATA (Serial ATA) is the interface introduced in the early 2000s to replace the old parallel ATA with its ribbon cables. In its third revision, **SATA III** (or SATA 6 Gb/s), it offers a raw bandwidth of 6 gigabits per second over a single serial channel. Six gigabits seem like a lot, but the line coding used (8b/10b: for every 8 bits of data, 10 are transmitted, for synchronization reasons) takes up 20% of the bandwidth, and the protocol overhead eats up the rest. The practical result is a ceiling of about **550–560 MB/s** in sequential read. This number is **the wall**: any SATA SSD, today as ten years ago, stops there. There is no "faster" SATA SSD in sequential, because the limit is not in the drive but in the pipe.

Physically, SATA uses **two separate cables**: a thin, seven-contact data cable that runs from the drive to a SATA port on the motherboard, and a fifteen-contact power connector that comes from the power supply. This is a trivial detail but it's the primary cause of panic for novice builders who connect only the data cable and don't understand why the drive isn't detected.

Logically, SATA uses **AHCI** (Advanced Host Controller Interface), a protocol designed in the era of hard drives. AHCI has a characteristic that today seems grotesque: it manages **only one command queue**, with a maximum depth of **32 commands** (a feature known as NCQ, Native Command Queuing). The historical reason is obvious: what was the point of sending a thousand simultaneous requests to a device that has only one read/write head and can serve only one at a time? The single queue served to reorder requests to minimize arm movements, not to exploit parallelism that didn't exist. But flash memory is intrinsically parallel: an SSD has dozens of NAND chips connected to the controller on multiple independent channels, and could serve many requests simultaneously. Forcing it into AHCI means bottlenecking it. Furthermore, AHCI requires the CPU to perform management tasks (register accesses, interrupt handling) that introduce non-negligible software latency when operations per second reach hundreds of thousands.

3.2 NVMe

NVMe (Non-Volatile Memory Express) is the protocol born to solve exactly these problems. It runs **directly on the PCIe bus**, completely bypassing the

motherboard's SATA controller, and at a logical level, it is built around three ideas.

The first is **massive parallelism**: NVMe supports up to **65,535 queues**, each up to **65,535 commands** deep. These are hyperbolic numbers that no consumer use even remotely approaches, but the point is not the number: it's that the protocol is no longer a bottleneck. An SSD can receive thousands of in-flight requests and reorder them internally to distribute them across its NAND channels.

The second is **core affinity**: each CPU core can have its own dedicated queue, without having to contend for locks with other cores. This eliminates a serialization point that, at high I/O frequencies, had become dominant.

The third is **reduced software path**: NVMe requires far fewer register accesses per command compared to AHCI, supports MSI-X interrupts, and drastically reduces overhead per operation.

The combined result is that an NVMe SSD is not just "faster in bandwidth": it is **faster in latency** and, above all, handles loads with deep queues much better, i.e., when many requests arrive together.

3.3 Real performance comparison: where it matters and where it doesn't

Now we come to the part that deserves the utmost honesty, because it's where marketing lies by omission.

The numbers printed on the box are the **sequential ones**: "up to 7,400 MB/s read." These are obtained by reading a huge continuous stream, with deep queues, under ideal conditions, on an empty and cold drive. They correspond to real-world use in very few cases: copying a gigantic file from one fast NVMe to another fast NVMe, working with uncompressed ultra-high-resolution video streams, loading enormous datasets into memory.

The workload that **dominates daily use**, however, is completely different: **small random reads and writes** (the famous "random 4K"), often with **few requests in flight at a time** (low queue depth, QD1–QD4). Opening an application, starting the operating system, loading game textures, compiling a project, opening a project in an IDE: these are all sequences of thousands of small, scattered reads.

And here's the uncomfortable truth: in **low queue depth random 4K**, the difference between a good SATA SSD and an excellent NVMe PCIe 5.0 is not tenfold, as sequential numbers might suggest. It's typically **two or three times** –

significant, but not miraculous. And the difference between a mid-range NVMe PCIe 4.0 and a top-tier PCIe 5.0, under the same load, is often **just a few percentage points**, completely invisible to the naked eye.

To summarize brutally, but usefully:

Transition	Perceived Improvement
HDD → SATA SSD	Huge. The PC feels like a different computer.
SATA SSD → NVMe PCIe 3.0/4.0	Modest but real. Slightly faster boot and load times, much faster file copies, much more headroom under heavy load.
NVMe Gen4 → NVMe Gen5	Almost negligible for the average user. Measurable in benchmarks, invisible in use, except for specific professional workloads.
DRAM-less QLC SSD → DRAM TLC SSD	Noticeable when it counts (prolonged writes, full disk, heavy multitasking).

This last point deserves emphasis because it is the core of the chapter: **the class of an SSD matters more than its PCIe generation**. A good Gen3 with DRAM and TLC NAND will outperform, in real-world use, a cheap DRAM-less Gen4 with QLC NAND almost every time the workload gets serious, despite the latter having double the sequential numbers on the box.

DirectStorage technology on Windows and its console equivalents deserve a separate mention. This technology allows games to load compressed assets from the SSD directly into GPU memory, reducing the need to pass through the CPU. On paper, it's the technology that would finally justify faster NVMe's in gaming. In practice, adoption has been much slower than expected, and the measured benefits on the few titles that implement it remain modest on systems with decent CPUs. It's a reason not to buy an HDD or a SATA for games; it is **not** a reason to pay for a Gen5.

4. Form factor, M.2 slot, and physical compatibility

4.1 Physical formats

Consumer storage drives today come in three main physical forms.

The **3.5-inch** format is for desktop hard drives: an aluminum brick approximately $147 \times 102 \times 26$ mm, which screws into a case cage, powered via SATA power and connected via SATA data. It is the only format where enormous capacities (12, 16, 20, 24 TB) are found.

The **2.5-inch** format is for SATA SSDs and laptop hard drives: approximately 100×70 mm, with a thickness of 7 mm (sometimes 9.5 mm on HDDs). Here too, two SATA cables. It is the format for classic upgrades on machines a few years old.

The **M.2** format is a flat card that plugs directly into a slot on the motherboard and is secured with a screw (or, on recent boards, with a tool-less snap fastener). It has no cables: power and data pass through the slot. It is the dominant format for modern SSDs and is also the main source of confusion, for the reason that follows.

4.2 The Number One Trap: M.2 SATA vs. M.2 NVMe

The M.2 slot is a physical form factor, not a protocol. There are M.2 SSDs that, despite having an identical shape, speak **SATA** and not **NVMe**: internally, they have a SATA controller, are limited to the usual 550 MB/s, and use the AHCI protocol. They appeared during the transition years and can still be found in marketplaces and in "too good to be true" offers.

The result is that an inexperienced buyer might purchase an "M.2 SSD," install it, and find themselves with the performance of a ten-year-old SATA drive, without understanding why. Worse: many motherboards have M.2 slots that support **only NVMe** and not **SATA**. In that case, the M.2 SATA drive is simply not detected, and the user spends an afternoon looking for non-existent faults.

The practical rule: **when buying an M.2 SSD, you must explicitly read "NVMe" and "PCIe" in the description.** If it's not written, assume it isn't. And when choosing a motherboard, you should read the manual to see what each M.2 slot supports, because on many boards, the primary slot is NVMe only, while a secondary one accepts both.

4.3 M.2 Keys

The M.2 connector has **notches** (*keys*) that prevent it from being inserted into incompatible slots. There are three that are relevant to storage.

The **B key** has the notch in a position that, electrically, corresponds to a maximum of **two PCIe lanes** plus SATA signals. By itself, it is now rare in modern SSDs.

The **M key** has the notch in another position and corresponds to **four PCIe lanes**. This is what fast NVMe SSDs use: if a drive has only one notch, and it's on the right when looking at the connector, it's an M key.

The **B+M key** has **two notches** and is therefore mechanically compatible with both B and M slots. B+M units are mostly M.2 SATA or NVMe limited to two PCIe lanes, because the double notch implies the lowest common electrical denominator. **An SSD with two notches is never a high-end NVMe x4**. This is a very useful visual clue: if you see two notches in the product photo, you are looking at a slow or otherwise limited unit.

It should be added, for completeness, that the M.2 form factor does not only host SSDs: Wi-Fi cards (usually A or E key, shorter), 5G modules, and more. These slots are physically similar but electrically different, and the keys serve precisely to prevent inserting the wrong thing in the wrong place.

4.4 Dimensions: 2280 and Others

The numerical designation of an M.2 encodes width and length in millimeters: **2280** means 22 mm wide and 80 mm long. It is by far the most common format and, except in special cases, it is the one you should buy.

Designation	Dimensions	Typical Use
2230	22 × 30 mm	Handheld consoles (Steam Deck, ROG Ally), ultrabooks
2242	22 × 42 mm	Compact laptops, some mini-PCs
2260	22 × 60 mm	Rare, some laptops
2280	22 × 80 mm	Desktop and laptop standard: 95% of the market
22110	22 × 110 mm	Enterprise/workstation, often with PLP

The **22110** format (110 mm in length) is almost exclusively found in enterprise environments, where the extra space is used to house the capacitors of the **PLP** (Power Loss Protection): a bank of capacitors that, in the event of a sudden power outage, provides the necessary energy to write data that was still in the volatile cache to the NAND, preventing corruption. This is a valuable feature in servers

and practically absent in consumer products, where at most a "partial" PLP is found that only protects metadata.

Please note: desktop motherboard slots usually have mounting holes for 2280 and sometimes for shorter lengths, but almost never for 22110. If you want to repurpose a 110 mm enterprise SSD, you need to verify that the motherboard can physically accommodate it.

4.5 PCIe Generations in SSDs

Each **PCIe** generation doubles the bandwidth per lane. A consumer NVMe SSD almost always uses **four lanes** (x4).

Generation	Bandwidth per Lane	x4 Bandwidth (Theoretical)	Typical Real Sequential Speed
PCIe 3.0	~985 MB/s	~3.9 GB/s	3,000–3,500 MB/s
PCIe 4.0	~1.97 GB/s	~7.9 GB/s	5,000–7,400 MB/s
PCIe 5.0	~3.94 GB/s	~15.8 GB/s	10,000–14,900 MB/s
PCIe 6.0	~7.88 GB/s	~31.5 GB/s	Not yet in consumer <i>[volatile data]</i>

PCIe is **backward compatible in both directions**: a Gen4 SSD inserted into a Gen3 slot will work, limited to Gen3 speed; a Gen3 SSD in a Gen5 slot will work, at Gen3 speed. Nothing breaks, nothing is lost except bandwidth that would not have been used anyway. This is important to know because it allows you to buy a Gen4 drive today even on a Gen3 platform, with a view to a future motherboard change.

Who really needs Gen5? The honest answer is: very few consumer users. Concrete benefits are seen in scenarios of transferring very large files between two Gen5 drives, in very high-resolution video editing with lightly compressed codecs, in professional data analysis workloads where tens of gigabytes are read in streaming, and in some AI workloads (loading model weights, datasets). For gaming, browsing, office productivity, programming, the difference compared to a good Gen4 is within the margin of error.

Gen5 also has two concrete drawbacks. The first is the **price**, significantly higher for the same capacity. The second is **heat**: first and second-generation Gen5 controllers consume a lot (peaks of 10–12 W have been observed compared to 6–8

W for a Gen4), get very hot, and **absolutely** require a serious heatsink, sometimes bulky enough to interfere with the graphics card or the CPU cooler. Some Gen5 models with more efficient controllers have improved the situation, but the issue remains.

4.6 Heatsinks and Thermal Throttling

Thermal throttling is the automatic reduction of performance that a device applies to itself when it exceeds a certain temperature, to prevent damage. In NVMe SSDs, this primarily concerns the **controller**, which is the component that truly heats up.

There's a counter-intuitive subtlety: while the controller wants to stay cool, **NAND performs better when hot** (cells write more easily at high temperatures) but suffers when *retaining* data when hot and powered off. For this reason, sensors and thresholds are primarily calibrated for the controller.

When is a heatsink needed?

- A **Gen3** SSD practically never needs one in a case with minimal airflow.
- A **Gen4** SSD benefits from one under prolonged loads; many mid-to-high-end motherboards include an integrated one, which is more than sufficient. Buying an aftermarket one is rarely necessary.
- A **Gen5** SSD **always** needs one. Without adequate dissipation, it will throttle within a few seconds of sustained writing and end up being slower than a well-cooled Gen4: one of the saddest spectacles in modern computing.

Two practical warnings. First: if the motherboard includes its own M.2 heatsink, there's **no need** to buy an SSD with a pre-installed heatsink — in fact, the two will conflict, and one will need to be removed. Second: if the SSD is intended for a **PlayStation 5**, a heatsink is mandatory but must respect the height limits imposed by the slot (approximately 11.25 mm total), so you need to buy models explicitly stated as compatible, not just any desktop heatsink.

Finally, a detail on installation: do aftermarket heatsinks require removing the SSD's label? **No**. The label of many SSDs contains a copper or aluminum layer that aids in dissipation, and removing it can, in some cases, void the warranty. The thermal pad should be applied over the label.

4.7 Lane sharing: when an SSD disables SATA ports

This is where the storage chapter intertwines with the motherboard chapter, and it should be read carefully because it's the source of the most insidious assembly errors.

PCIe lanes are a finite resource. The CPU provides a certain number (typically 20–28 usable on modern consumer platforms: 16 for the graphics card, 4 or 8 for one or two direct M.2 slots, plus the link to the chipset). The **chipset** (the "southbridge," the support chip on the motherboard) provides others, but all of these must pass through the connection between the chipset and the CPU, which is itself limited and shared with USB, network, audio, and the rest.

Since lanes are never enough, motherboard designers **multiplex** them: certain resources are alternatives to each other. Typical configurations found in manuals are:

- Populating the second or third M.2 slot ****disables two or four SATA ports**** (because those lanes were shared).
- Populating a certain M.2 slot ****reduces the PCIe slot of the graphics card from x16 to x8**** (impact on GPU performance: in most cases negligible on Gen4/Gen5, but not on Gen3).
- Populating a second PCIe slot ****disables**** an M.2.

The practical consequence: you install the new NVMe in the wrong slot, and suddenly the old SATA SSD or archive hard drive **disappears from the BIOS**. The PC works, but half of the storage is gone. Those unfamiliar with the phenomenon spend hours blaming cables, the power supply, or the drive itself.

The rule: **before buying, open the motherboard manual PDF and look for the "M.2 and SATA configuration" table (or similar)**. It's always there, usually on a page nobody reads, and it clarifies exactly which slot shares what. A second practical criterion: the M.2 slot **directly connected to the CPU** (usually the first one, immediately below the video card slot) is the fastest and typically **does not** steal SATA ports; that's where the system drive goes.

5. NAND technology and key components

5.1 SLC, MLC, TLC, QLC: how many bits per cell

We've said that a NAND cell stores information as a trapped electrical charge level. The crucial question is: **how many distinct levels can we distinguish in a cell?** Because the more levels we distinguish, the more bits we can store in the same physical cell, and thus the less each gigabyte costs.

- **SLC** (Single-Level Cell): 2 levels, **1 bit** per cell.
- **MLC** (Multi-Level Cell): 4 levels, **2 bits** per cell.
- **TLC** (Triple-Level Cell): 8 levels, **3 bits** per cell.
- **QLC** (Quad-Level Cell): 16 levels, **4 bits** per cell.
- **PLC** (Penta-Level Cell): 32 levels, 5 bits — announced for years, not yet a significant consumer product.

The density gain is obvious. The cost, less so. Imagine having to distinguish the water level in a glass: if you only need to say "empty or full" (SLC), you can be quite wrong and still be right. If you need to distinguish sixteen different levels (QLC) in the same glass, every extra drop or less counts, the measurement takes more time, and the slightest evaporation (i.e., charge loss due to wear or time) makes you slip into the wrong level.

There are three important consequences:

1. **Durability.** Each write/erase cycle slightly damages the cell's insulator. The more levels that need to be distinguished, the less degradation can be tolerated before the levels become ambiguous. Hence the order of magnitude of tolerable cycles: SLC ~100,000, MLC ~10,000, TLC ~1,000–3,000, QLC ~300–1,000. 2. **Write speed.** Writing requires bringing the charge to the exact level, with subsequent verifications. More levels mean slower cell programming. "Native" QLC writing (i.e., not cache-assisted) is dramatically slow: values in the order of **80–150 MB/s** are seen, which is worse than a hard drive. 3. **Price.** In the opposite direction: QLC is the cheapest per gigabyte, which is why it exists.

Type	Bits/cell	Indicative P/E cycles	Native write speed	Typical use today
SLC	1	~100,000	Very high	Enterprise only, cache
MLC	2	~10,000	High	Practically disappeared

				from consumer
TLC	3	~1,000–3,000	Medium	The consumer quality standard
QLC	4	~300–1,000	Low	High-capacity budget SSDs

The operational conclusion is clear: **for the system drive, buy TLC**. QLC has its place — a large 4 TB secondary drive intended to host a game library that is read a lot and written to little is a perfectly legitimate use of QLC, and in that scenario, the savings are real. But a QLC drive as the main unit of a PC that writes continuously (updates, browser caches, temporary files, compilations, video editing) is a choice that comes at a cost in both performance and longevity.

A technical note should be added for completeness: today, all consumer NAND is **3D NAND** (or V-NAND, in Samsung's terminology), meaning the cells are no longer arranged on a single plane but stacked in vertical towers of tens or hundreds of layers (it has gone from 32 layers to over 200–300 layers in recent generations). This has allowed for increased density without further shrinking the cells, which paradoxically has *improved* reliability compared to older planar NAND with tiny nodes. This is why modern TLC is more reliable than MLC from ten years ago.

5.2 DRAM cache versus DRAM-less and HMB

An SSD controller must maintain the **mapping table**: the table that associates each logical address (what the operating system sees) with its actual physical location in the NAND. This table is large: as a rule of thumb, about **1 MB of table for every GB of capacity**, so a 1 TB SSD has a mapping table of about 1 GB.

Where is it stored?

An SSD **with DRAM cache** mounts a dedicated DRAM chip (typically 1 GB for 1 TB) and stores the entire table there. Every time the controller needs to translate an address, it consults the DRAM: latency in the order of nanoseconds. Immediate.

A **DRAM-less** SSD does not have that chip. The table lives in the NAND itself and is consulted from there. Consulting the NAND to *find* where the data is, before reading the data, means making two accesses to the NAND instead of one: latency doubles, or worse.

To mitigate the problem, **HMB** (Host Memory Buffer) was introduced: an NVMe protocol feature that allows the SSD to **borrow a small portion of system RAM** (typically 32–64 MB, in some cases a few hundred) to store part of the mapping table. It works, and that's why the best modern DRAM-less SSDs (WD SN770, for example) are surprisingly good. But it has three limitations: (a) 64 MB is not 1 GB, so only the "hot" part of the table can be cached, and on a disk used in a disorganized way, misses increase; (b) access to system RAM still goes through the PCIe bus, so it's slower than onboard DRAM; (c) HMB requires operating system support — it's present in modern Windows and Linux, but in particular environments or old installations, it might not be, and in that case, the SSD falls back to very poor performance.

When does the difference become noticeable? Not in sequential benchmarks, where a DRAM-less drive can shine. It's noticeable in **workloads with many random requests on a full and fragmented disk**, which is exactly the condition of a system disk after two years of use. It's noticeable in latency under mixed load. It's noticeable in performance consistency.

Practical rule: for the system disk, an SSD with DRAM cache is preferred. If the budget doesn't allow it, at least choose a reputable DRAM-less SSD with HMB (SN770, not some nameless newcomer), and in any case with TLC NAND. The combination to avoid as a primary disk is **DRAM-less + QLC**: it's the bottom of the barrel, and it's exactly what many "bargain" 1 TB drives offer at rock-bottom prices.

5.3 Dynamic SLC cache and speed collapse

There's a phenomenon that baffles many users: they copy a 200 GB file to a new SSD, see 5,000 MB/s for the first thirty seconds, then suddenly the speed **drops** to 900 MB/s, or 200, or 90. Is the SSD faulty? No. It's working exactly as designed.

The mechanism is **dynamic SLC cache** (*pseudo-SLC cache*, or SLC caching). The controller takes a portion of the TLC or QLC NAND and makes it operate **in SLC mode**, meaning it writes only one bit per cell instead of three or four. In SLC mode, writing is very fast because there's no need to precisely position the charge among eight or sixteen levels, but only to distinguish between "charged" and "not charged." All incoming writes are therefore dumped into this fast area. Then, during idle moments, the controller **reprocesses** the data in the background, rewriting it in native TLC/QLC mode and freeing up the cache.

The problem is that the SLC cache is **dynamic**: its size depends on the free space. An empty 2 TB SSD can have an SLC cache of 200–600 GB; the same SSD 85% full might only have 20–30 GB, or almost nothing. And when the cache runs out, it writes **directly** to the native NAND, at its real speed. For a decent TLC, this means 900–1,500 MB/s (still good), but for a QLC, it means **80–150 MB/s**: worse than a hard drive, for hours, with the system seemingly frozen.

From this, three operational lessons follow:

1. **Sustained write tests matter more than peak numbers.** Serious reviews show a graph of write speed over a transfer of hundreds of gigabytes: you immediately see where the floor gives way and at what height. 2. **Don't fill the SSD.** We'll discuss this in the next section, but this is where the most dramatic effect is seen. 3. **QLC should be avoided for heavy and continuous write loads.** A user who downloads, edits videos, moves large archives, with an almost full QLC will have a terrible experience.

5.4 TBW and MTBF: how to interpret declared endurance

Two acronyms appear in specifications and are regularly misunderstood.

TBW (Terabytes Written) is the total amount of data that the manufacturer guarantees can be written to the drive within the warranty period. A mid-range 1 TB SSD typically declares 600 TBW; a high-end one 1,200 TBW; a cheap QLC might stop at 200–400 TBW. TBW increases proportionally with capacity, because the more cells there are, the more writes can be distributed.

The question everyone asks is: **is it a lot or a little?** Let's do the math. A normal home user writes between 10 and 30 GB per day to the system drive (updates, cache, documents, downloads). Let's take the worst case, 30 GB per day: that's about 11 TB per year. With 600 TBW declared, the drive would last **over fifty years**. Even a developer who compiles continuously, or a videographer who writes 200 GB per day, remains in the order of **eight to ten years**.

The conclusion, which must be stated clearly because it contrasts with widespread anxiety, is that **for the consumer user, TBW is almost never the limiting factor**. A modern SSD, in the vast majority of cases, becomes obsolete or is replaced long before its cells wear out. The real exceptions are enterprise workloads (databases with continuous writes, server caches, storage nodes) and some extreme professional workloads. If you fall into those categories, TBW is the first number to look at — and you probably need an enterprise unit, not a consumer one.

It should be remembered that TBW also has a **contractual** role: the warranty expires upon reaching the TBW *or* the declared number of years, whichever comes first. It is a legal limit even before it is a physical one: once the TBW is exceeded, the unit usually continues to function normally.

A related indicator is **DWPD** (Drive Writes Per Day), more commonly used in enterprise settings: it states how many times per day the entire capacity of the disk can be rewritten for the entire duration of the warranty. A consumer SSD is around 0.3 DWPD; a "write intensive" enterprise one can reach 3 or 10 DWPD.

MTBF (Mean Time Between Failures) is a number that should be interpreted with great caution, as it misleads almost everyone. An SSD typically declares 1,500,000 hours of MTBF. Divided by 8,760 hours in a year, that's **171 years**. No one seriously believes that an SSD will last 171 years, and indeed, that's not what the number means.

MTBF is a **statistical population measure**, not one of individual longevity. Roughly speaking, it means: if I put 1,500,000 units into operation for one hour each, I expect one failure. Or: if I put 1,000 into operation for one year, I expect about 5–6 failures ($8,760,000 \text{ unit-hours} / 1,500,000 \approx 5.8$). This data is useful for those who manage fleets of thousands of disks to dimension spare parts. It **says absolutely nothing** about how long *your* disk will last, and it does not account for aging wear, because it is measured during the product's "useful" life phase, excluding infant mortality and final wear.

The advice: look at the TBW and the **warranty years** (5 years is the standard for good quality; 3 years indicates an entry-level product). Ignore MTBF as a purchasing criterion.

One last thing, which is the most important of this entire section: **no durability number is a substitute for backup**. SSDs, unlike hard drives, tend to fail **suddenly and totally**, often due to a controller or firmware issue, without the noises and progressive bad sectors that gave warning on HDDs. A dying SSD often becomes an unreadable brick, and professional data recovery from NAND is enormously more difficult and expensive than from a magnetic platter. The **3-2-1 rule** (three copies of data, on two different media, one of which is off-site) is not a recommendation for paranoids: it is the bare minimum for anyone who cares about their data.

6. Capacity and configuration strategy

6.1 How much capacity

The question "how much space do I need?" has an answer that has changed rapidly in recent years, primarily for one reason: **game sizes have exploded**. Modern AAA titles regularly occupy 100–150 GB, and some exceed 200 GB with high-resolution texture packs. Just one of these games consumes a tenth of a 1 TB drive.

The absolute minimum acceptable for a system drive today is **500 GB**, and it must be said that this is already tight: Windows 11 with its paging files, hibernation, and updates occupies 60–80 GB alone, and common programs consume another hundred. With 500 GB, one can manage, if disciplined and having a second drive for data.

The **sensible reference capacity in 2026 is 1 TB** for a general-purpose PC, and **2 TB** for a gaming or creative work PC. Below 1 TB, continuous compromises are made; above 2 TB, the cost increases, and it's worth evaluating whether it's better to add a second drive (even slower and cheaper) rather than doubling the capacity of the primary one.

There's also a little-known technical reason why **larger capacities are often faster**: a 2 TB SSD has more NAND chips than a 500 GB one, and the controller can therefore distribute operations across more channels in parallel. It's typical for the 250 GB or 500 GB model in a line to have significantly lower declared write speeds than the 1 or 2 TB model in the exact same line. Buying the smallest size in a series often means buying the worst version in terms of performance, not just space.

6.2 The Fill Percentage

This is one of the most useful and least known facts in the entire chapter: **a full SSD is a slow SSD**, and not by a small margin.

There are three competing reasons.

The first is the **dynamic SLC cache** mentioned above: less free space means less fast cache, and the drive falls back to the native NAND speed sooner.

The second is **garbage collection**. The controller needs free blocks to consolidate data and delete obsolete ones. If there is very little free space, it has to perform

acrobatics: read a nearly full block, copy valid pages elsewhere, erase, rewrite. The number of physical writes for each logical write — a parameter called **write amplification** — skyrockets. This means not only slowness but also **accelerated wear**: a drive kept constantly full wears out faster.

The third is **over-provisioning**. Every SSD reserves a portion of NAND from the factory that is not visible to the user (this is why a "1 TB" SSD shows 931 GB: partly it's the difference between decimal TB and binary TiB, partly it's reserved space). This margin is precisely for garbage collection and replacing defective blocks. Leaving free space is equivalent to increasing over-provisioning, and it improves everything.

The practical rule: do not exceed 80% fill, and try to stay below 75% on the system drive. A 1 TB SSD should practically be considered a 750 GB usable SSD. This is, incidentally, an excellent argument for buying the next size up: 2 TB used at 80% offers a comfortable 1.6 TB, with intact performance.

6.3 Sensible Configurations

Let's look at storage architectures that make sense, by use profile.

Minimum configuration (office PC, browsing, study). A single 1 TB NVMe, TLC, with or without DRAM. No second drive. Simple, quiet, efficient. If the budget is truly minimal, even a 1 TB SATA is better than a 256 GB NVMe: **space matters more than marginal speed.**

Standard configuration (gaming PC, general use). A 2 TB NVMe, TLC with DRAM, as the sole drive. This is the most elegant solution: nothing to manage, nothing to move, excellent performance everywhere. It's what I would recommend by default to most people.

Two-tier configuration (gaming with large library, creator). A 1 TB TLC NVMe with DRAM for the operating system and work applications, plus a second 2–4 TB NVMe (here QLC becomes acceptable, and sometimes it's the rational choice) for the game library and data. The advantage is that heavy and continuous system writes are on the good drive, while the large drive serves predominantly read loads, where QLC performs well.

Full configuration (professional, local archive). 1–2 TB NVMe for OS and applications, 2–4 TB NVMe or SATA for active projects, 8–20 TB HDD for archive and local backups. This is the classic three-tier configuration — hot, warm, cold —

and it still makes perfect sense. It must be accompanied by an **off-machine backup**: an external drive or a NAS, because an internal HDD does not protect against theft, fire, power surge, or ransomware.

A note on **RAID** (Redundant Array of Independent Disks) in a home environment: RAID 0 (striping, two disks working in parallel) is almost always a bad idea on modern SSDs — it doubles the risk of total data loss for sequential bandwidth that no one uses — and RAID 1 (mirroring) is a high-availability mechanism, **not a backup**: if you accidentally delete a file, or if ransomware encrypts it, it is deleted or encrypted on both disks instantly. RAID protects against hardware failure of a disk, and nothing else.

7. Reference Brands and Models

7.1 Who produces NAND and who assembles

This distinction is key to navigating an otherwise chaotic market, and it's the same logic already encountered in the RAM chapter.

There are **very few NAND memory manufacturers** in the world: Samsung, SK Hynix (which absorbed Intel's NAND division to create **Solidigm**), Kioxia (Toshiba's former memory division), Western Digital/SanDisk (historically in a production joint venture with Kioxia), Micron (consumer brand Crucial), and YMTC in China. All other brands you see on shelves — Kingston, Corsair, Sabrent, Teamgroup, Adata, Silicon Power, Lexar, Patriot, PNY, and dozens of others — **do not produce NAND**. They buy memory chips and controllers on the market, assemble them on a PCB, and put firmware and a label on them.

This is not a defect in itself. Some assemblers make excellent products (the Kingston KC3000 and Sabrent Rockets have been excellent SSDs, built on high-end Phison controllers and good NAND). But it has a precise and dangerous consequence: **the assembler can change the internal components without changing the product name**.

It's the phenomenon known as **component lottery** or *silent revision*: an SSD is launched with TLC NAND and controller X, gets excellent reviews, and six months later the exact same model number is produced with QLC NAND and a different controller, with much worse sustained performance. The buyer reads the enthusiastic launch reviews and receives a different product at home. The most

cited case in the consumer sphere is that of the **Kingston NV1/NV2**, openly declared as subject to component variation, with units found on the market in radically different configurations; but the phenomenon has affected, to varying degrees, many brands, including the famous case of the silent revision of the Samsung 970 EVO Plus and the variations on the WD Blue SN550.

The defense is twofold. First: **prefer, at the same price, brands that produce their own NAND** (Samsung, WD/SanDisk, Solidigm, Micron/Crucial as long as available, Kioxia), because the supply chain is vertical and silent revisions are rarer and more documented. Second: **be wary of models whose specifications do not state the type of NAND**. If a datasheet doesn't say whether it's TLC or QLC, whether it has DRAM or not, it's almost always because you wouldn't like the answer, or because the manufacturer wants to keep its options open.

7.2 Overview of SSD Brands

Samsung. It is the historical reference, for a simple reason: it produces NAND, controllers, and DRAM in-house. The **970 EVO/Pro** (PCIe 3.0) series were the gold standard for years; the **980** series introduced an annoying ambiguity (the plain "980" is a Gen3 **DRAM-less**, while the "980 Pro" is a Gen4 with DRAM: almost identical names, products in different categories — a textbook case of misleading nomenclature). The **990 PRO** series is today a reference Gen4, fast, efficient, with excellent consistency under load; the **990 EVO Plus** variant is cheaper and uses a hybrid configuration of PCIe lanes. Samsung tends to cost more than average: you pay for the brand, but also for real consistency. Historical note: the first 990 PRO and 980 PRO units had health degradation issues resolved via firmware update — a reminder that **updating SSD firmware** (with official tools: Samsung Magician, WD Dashboard, Crucial Storage Executive) is a good habit, not an operation for power users.

Western Digital / SanDisk. The other big vertical name. The **WD Black SN850X** is probably the most recommended Gen4 SSD with DRAM: excellent performance, great thermal management, competitive price, and in many configurations it is the rational choice for a system drive. The **WD Black SN770** is the champion of the DRAM-less category: it uses HMB intelligently, has TLC NAND, and is surprisingly close to SSDs with DRAM in real-world use, at a lower price. It is the exception that proves that "DRAM-less" does not automatically mean "poor quality," as long as the rest is done well. The **SN850X** also comes in a heatsink version and a PS5-compatible version. The **WD Blue** line (SN580 and similar) is

the honest entry-level. The group recently separated its flash activities under the **SanDisk** brand, so one can expect to see the same products under transitioning names *[variable data]*.

Crucial (Micron). Historically the champion of price-to-performance. The **P3** and **P3 Plus** are inexpensive QLC drives (good as secondary drives, not recommended as primary); the **P5 Plus** was an excellent Gen4 with DRAM; the **T500** is an excellent and very competitive Gen4 with DRAM; the **T700** and **T705** were among the first top-tier Gen5 drives, very fast and very hot, requiring substantial heatsinks. **[volatile data]** As mentioned at the beginning, Micron has announced its exit from the consumer market with the Crucial brand: these products remain technically valid but should no longer be considered a line with a future, and availability must be checked on a case-by-case basis. Anyone buying residual Crucial stock today gets a technical bargain but must account for less guaranteed support and replacement over time.

Kingston. An assembler, not a NAND manufacturer. The **KC3000** was an excellent Gen4 (Phison E18 controller, TLC NAND, DRAM) and the **Fury Renegade** is its gaming counterpart. At the opposite extreme, the **NV2** series (and previously the **NV1**) is the textbook example of the component lottery: same name, variable components, unpredictable performance. It's not an "bad" product in absolute terms, but it's a product where **you cannot know what you are buying**, which in 2026 is a sufficient reason to stay away if alternatives at the same price exist.

Solidigm. This is Intel's former NAND division, now under SK Hynix. It produces its own NAND. It has become particularly interesting in the niche of **very high-capacity, low-cost SSDs per TB**: the **P44 Pro** series is an excellent Gen4 with DRAM (technically a close relative of the SK Hynix Platinum P41), while the high-capacity QLC series (P41 Plus, and lines from 4 TB upwards) are among the most sensible when a lot of space is needed at a low price for predominantly read workloads.

Seagate. Known primarily for hard drives, it has a good line of **FireCuda** SSDs (530, 540), based on Phison controllers and TLC NAND, with very high declared TBW and a long warranty. These are valid products, often positioned at premium prices.

Sabrent. An American assembler that has built a solid reputation with its **Rocket** line (Rocket 4 Plus, Rocket 5), based on top-tier Phison controllers. It is often the

first to market with new generations. Good products, but the rule always applies: check the revision and actual components.

SK Hynix. Less common in Europe but technically excellent: the **Platinum P41** and the subsequent **P51** are drives with proprietary NAND and controllers, among the most efficient on the market (low power consumption, low heat, excellent performance). If found at a good price, it is a very solid choice.

7.3 SSD Orientation Table (indicative)

Tier	Reference Models	Features	Recommended Use
Entry / Second Drive	Crucial P3 Plus, WD Blue SN580, Solidigm P41 Plus	Gen3/Gen4, DRAM-less, often QLC	Secondary storage, game libraries
Quality Entry	WD Black SN770 , Kingston NV3 (with reservations)	Gen4, DRAM-less HMB, TLC	System drive on a tight budget
Mid-range (the sweet spot)	WD Black SN850X , Samsung 990 EVO Plus, Crucial T500, SK Hynix P41/Solidigm P44 Pro	Gen4, DRAM, TLC	System drive: the recommended choice
High-end Gen4	Samsung 990 PRO, Seagate FireCuda 530/540	Max Gen4, DRAM, TLC	Workstations, for those who want the best without Gen5
Gen5	Crucial T705, Samsung 9100 PRO, Sabrent Rocket 5, Corsair MP700 Pro	Gen5, DRAM, TLC, mandatory heatsink	Only for massive professional transfer workloads

Specific models and their availability change rapidly; this table should be used to understand categories, not as a definitive shopping list. [volatile data]

7.4 Hard Drives: Seagate, WD, Toshiba, and the SMR Trap

In the hard drive market, there are **only three manufacturers** left in the world: **Seagate**, **Western Digital** (with WD and HGST brands), and **Toshiba**. Everything you buy, under any label, comes from them.

WD historically uses a color code for its lines, which is useful to know:

WD Line	Color	Purpose
WD Blue	Blue	General desktop, light use
WD Black	Black	Performance desktop, 7,200 RPM, large cache
WD Red / Red Plus / Red Pro	Red	NAS (caution: the plain "Red" is SMR!)
WD Purple	Purple	Video surveillance, continuous writing
WD Gold	Gold	Enterprise, datacenter

Seagate has a similar structure: **BarraCuda** (desktop), **IronWolf** and **IronWolf Pro** (NAS), **SkyHawk** (surveillance), **Exos** (enterprise). Toshiba uses alphanumeric designations (N300 series for NAS, X300 for performance desktop, MG for enterprise).

And now, the most important trap in this section.

CMR versus SMR. These are two ways of writing magnetic tracks on the platter.

CMR (Conventional Magnetic Recording — sometimes called PMR) writes tracks side-by-side and not overlapping. Each track can be rewritten independently of the others. This is the "normal" behavior expected from a drive.

SMR (Shingled Magnetic Recording) exploits a physical asymmetry: the **write** head is wider than the **read** head. If tracks are written partially overlapping, like roof shingles, more can be packed onto the same platter, gaining 10–25% capacity with the same hardware. Reading works perfectly, because the narrower read head can read the uncovered portion of the track.

The problem is **rewriting**. Since tracks overlap, a single track cannot be rewritten without destroying adjacent ones: the entire "band" of shingled tracks must be read, modified in memory, and then rewritten entirely. The drive masks this phenomenon with a CMR cache zone, but when the cache runs out — i.e., under

prolonged random writing — performance **collapses**, and not by a little: it drops to **10–20 MB/s**, with latencies of seconds. The drive appears broken.

Where does it become a disaster? In a **NAS with RAID**. When a failed drive is replaced, the array must **rebuild** the data onto the new drive: this is a heavy, continuous write load, lasting hours or days. An SMR drive, under these conditions, slows down so much that the RAID controller can consider it unresponsive and **eject it from the array**, ruining the rebuild. In an already degraded array, with only one redundancy drive remaining, this can mean **total data loss**.

What makes the situation even more serious is that, in 2020, WD, Seagate, and Toshiba were discovered selling SMR drives **without declaring it**, including them in lines explicitly intended for NAS (particularly some 2–6 TB WD Red drives). This led to class actions, public apologies, and a clarification of product lines: since then, WD has created "Red **Plus**" to indicate CMR, leaving the plain "Red" for SMR. But the lesson remains: **labels are not enough; the specific model must be verified**.

Operational rule: for a NAS, for a RAID, or for any use involving significant writes, only CMR should be purchased. Before buying a hard drive, search for the exact model number (e.g., WD60EFPX, ST8000VN004) along with the word "CMR" or "SMR": the three manufacturers now publish official lists, and enthusiast communities maintain updated ones. SMR only makes sense for write-once, read-many archival storage, typically in external USB drives intended for backing up large archives that are rarely rewritten.

One last note: some ultra-high capacity enterprise hard drives are **helium-filled** instead of air-filled. Helium, being less dense, reduces turbulence around the platters, allowing more platters to be stacked (9, 10, 11) and reducing power consumption and temperatures. These are excellent, hermetically sealed drives, but expensive and often noisy (7,200 RPM enterprise models are not designed to sit under a living room desk).

7.5 The selection process, in order

Summarizing the method for choosing storage in a logical sequence, proceed as follows:

1. **Usage.** What should this drive do? Operating system? Game library? Archive? Backup? Everything else follows from this.
2. **Interface.** Operating system and

applications → NVMe, always. Secondary data storage → NVMe if there are free slots, SATA if not. Mass archive → HDD. 3. **Capacity.** Estimate actual occupancy, add margin for the future, and **divide by 0.8** to adhere to the fill rule. Need 1 TB of stuff? Buy 1.5–2 TB. 4. **NAND and DRAM Type.** System drive → **TLC**, preferably **with DRAM**. Secondary drive predominantly for reading → QLC acceptable. 5. **TBW and Warranty.** Verify that the TBW is consistent with the expected load (for consumer use, it almost always is) and prefer 5-year warranties. 6. **Brand.** Given identical specifications, prefer manufacturers who produce their own NAND and do not make silent revisions. 7. **Price.** Only at this point look at the price, and compare the **cost per gigabyte** among the remaining candidates. Price is the last filter, not the first, because a cheap drive that fails on points 4 and 6 is not a bargain: it's a postponed problem.

8. Common buyer mistakes

Here, we explicitly list the most common errors seen — many already anticipated, but worth listing because each costs money, time, or data.

Buying a QLC DRAM-less SSD as a primary drive. This is the number one mistake, and it's marketing-driven: the box says "PCIe 4.0, up to 5,000 MB/s," the price is great, it seems like a bargain. Then you discover that those 5,000 MB/s only last as long as the SLC cache holds out, and that when the drive fills up — which is after a year of normal use — write speeds drop below 200 MB/s and the system becomes sluggish under load. **The system drive is the last place to save 20 euros.**

Mounting the NVMe in the wrong slot and disabling SATA ports. Already thoroughly discussed: the motherboard manual has the lane sharing table, and it should be read **before** buying, not after the drives have disappeared from the BIOS.

Paying for a Gen5 SSD without needing it. A Gen5 costs significantly more than a Gen4 of the same capacity, runs much hotter, requires a bulky heatsink, and in consumer use — including gaming — offers a perceived improvement ranging from "none" to "imperceptible." That price difference, spent instead on **capacity** (2 TB Gen4 instead of 1 TB Gen5) yields a real and daily benefit. Gen5 makes sense for those who habitually move tens or hundreds of gigabytes between fast drives, and for few others.

Buying an M.2 SATA believing you're buying an NVMe. Same slot, different protocol, one-tenth the performance. Check that "NVMe" and "PCIe" appear in the technical specifications, and be wary of drives with two notches.

Getting an SMR HDD for a NAS. Concrete risk of total data loss during a RAID rebuild. Verify the exact model against official CMR/SMR lists.

Buying the smallest size in a series. The 250 or 500 GB model of a line is often slower in writing than the 1 or 2 TB model of the same line, because it has fewer chips to parallelize across. You pay little and get little, in two senses.

Filling the drive to 95% and then complaining that the PC is slow. The remedy is free: free up space, or buy more capacity.

Defragmenting an SSD. Useless and harmful. Modern Windows knows this and, when it "optimizes" an SSD, it actually sends TRIM commands — which is a completely different and legitimate thing. Do not install third-party defragmentation utilities on an SSD.

Believing that RAID 1 is a backup. It is not. It protects against a single drive failure. It does not protect against accidental deletions, ransomware, power surges, theft, or fire.

Ignoring firmware updates. Several serious reliability issues on branded SSDs have been resolved by firmware updates. It's worth installing the manufacturer's official utility and checking once in a while.

Never checking S.M.A.R.T. status. S.M.A.R.T. (Self-Monitoring, Analysis and Reporting Technology) is the self-diagnosis system integrated into all drives: it exposes counters such as power-on hours, terabytes written, reallocated sectors, and remaining life percentage. Free tools like CrystalDiskInfo (Windows) or `smartctl` (Linux) allow you to read them in seconds. A quick look now and then, and especially **before buying used**, avoids unpleasant surprises: a used SSD with 80% of its life consumed is not a bargain at any price.

Trusting a single storage medium. It's worth repeating one last time: SSDs die without warning, and recovery is expensive, if not impossible. Any storage configuration without a backup strategy is, simply put, incomplete.

9. Operational Summary — Storage Decision Checklist

Step 1 — Define the role of each drive.

- ☐ Operating system and applications → **NVMe SSD**, no exceptions.
- ☐ Games and active projects → NVMe (or SATA if no slots available).
- ☐ Archive and local backup → **High-capacity CMR HDD**.
- ☐ Remote/external backup → planned? (3-2-1 rule)

Step 2 — Check motherboard compatibility.

- ☐ How many M.2 slots are there and what PCIe generation do they support?
- ☐ Open the manual, read the **lane sharing** table: which M.2 slot disables which SATA ports or reduces the GPU slot?
- ☐ Is the chosen slot for the system drive **directly connected to the CPU**?
- ☐ Does the motherboard already include an **M.2 heatsink**? (If yes, do not buy an SSD with an integrated heatsink.)
- ☐ Is the format **2280**? (Check mounting holes if using different formats.)

Step 3 — Choose SSD specifications.

- ☐ Is it **NVMe** and not M.2 SATA? (Look for "NVMe" and "PCIe" in the technical specifications; be wary of drives with two notches.)
- ☐ Is the NAND **TLC**? (QLC only allowed for secondary drives with predominantly read operations.)
- ☐ Does it have **DRAM cache**? (If DRAM-less, is it a reputable model with HMB like the SN770?)
- ☐ Is the **TBW** consistent with the expected workload? (For consumer use, almost always yes.)
- ☐ Is the **warranty** 5 years?
- ☐ Are there **sustained write** tests in reviews? Where does the speed drop?
- ☐ Is the PCIe generation adequate: **Gen4 is the sweet spot**; Gen5 only if the workload truly justifies it.

Step 4 — Size the capacity.

- ☐ Estimate actual expected usage.
- ☐ Add margin and divide by **0.8** (do not exceed 80% full).
- ☐ Minimum 1 TB for general use, **2 TB recommended** for gaming or creative work.
- ☐ Verify that the chosen capacity is not a depowered version of the series.

Step 5 — Choose the brand.

- ☐ Given similar specifications, prefer a vertical NAND manufacturer (Samsung, WD/SanDisk, SK Hynix/Solidigm, Kioxia, Micron).
- ☐ Verify that the model is not known for ****silent revisions**** of components.
- ☐ Check that the technical specifications explicitly state NAND type and presence of DRAM.

Step 6 — If buying an HDD.

- ☐ Verify it is ****CMR**** (mandatory for NAS and RAID), checking the exact model number.
- ☐ Choose the right line for the use (NAS / surveillance / desktop / enterprise).
- ☐ Consider noise and vibrations if the PC is in a quiet environment.

Step 7 — After installation.

- ☐ Are all drives visible in the BIOS/UEFI? (If not → suspect lane sharing.)
- ☐ Is the NVMe drive recognized at the correct PCIe generation? (Verifiable with CrystalDiskInfo or manufacturer utilities.)
- ☐ Is ****TRIM**** active? (On modern Windows and Linux, it is by default.)
- ☐ Firmware updated with the official utility?
- ☐ Temperatures under load checked? (Above 70 °C sustained on the controller, better dissipation is needed.)
- ☐ ****Backup configured.**** No storage configuration is complete without this point.

Quick tips by budget tier (*indicative and currently very unstable prices — [volatile data]*)

Tier	Recommended configuration	Logic
Entry	1× 1TB Gen4 TLC NVMe, even quality DRAM-less (e.g., WD Black SN770)	A single, good drive. Better a decent 1TB than a "premium" 500GB or a poor 2TB.
Mid	1× 2TB Gen4 TLC NVMe with DRAM (e.g., WD Black SN850X, Samsung 990 EVO Plus, Crucial T500)	The right choice for most people. Simple, fast, capacious.

High-end	1× 2TB Gen4/Gen5 NVMe with DRAM for OS + 1× 4TB NVMe (even QLC) for libraries + 1× 8–20TB CMR HDD for archive	Three tiers: hot, warm, cold. Gen5 only if the workload justifies it.
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Final note. If I had to summarize the entire chapter in one sentence, it would be this: *the quality of an SSD is measured by TLC, DRAM, and sustained write performance, not by the big number printed on the box.* Those who internalize this principle will buy well even without knowing anything else; those who ignore it will continue to pay for sequential bandwidth they will never use, and wonder why their new computer seems bogged down after a year.

Chapter 6 - The Power Supply Unit (PSU)

Foreword: the invisible component

Of all the parts that make up a personal computer, the power supply is the one that receives the least attention and, statistically, causes the most damage. It's an object that doesn't appear in benchmarks, doesn't produce frames per second, doesn't have a Cinebench score, isn't displayed in any RGB-lit showcase (or rather: some manufacturers have tried, with questionable results). It sits at the bottom of the case, often in a separate chamber, covered by a metal shroud, and the average user forgets about it the same day they close the side panel.

Yet every single electron that powers the CPU, GPU, RAM, SSDs, and fans passes through it. The power supply — in English, *Power Supply Unit*, hence the acronym **PSU** which we will use from now on — is the only component that, failing badly, can drag the rest of the system with it. A graphics card that breaks, breaks alone. A PSU that breaks badly can take out the motherboard, CPU, memory, and drives in a fraction of a second, because it's the only thing in the case physically connected to a wall delivering 230 volts of alternating current.

This chapter treats the PSU as it deserves: starting with what it physically does, moving on to how to size it, how to read a label, how to distinguish a good product from a bad product disguised as good, and ending with a repeatable purchasing procedure. As with previous chapters, data that may become outdated (prices, models on the market, generations of graphics cards) are explicitly marked with the **[volatile data]** tag: hardware moves fast, physical principles much less so.

1. What the PSU does

1.1 AC→DC conversion

The electrical outlet at home, in Italy and throughout Europe, supplies alternating current (**AC**) at a nominal voltage of 230 volts and a frequency of 50 hertz.

"Alternating" means that the voltage is not constant: it oscillates sinusoidally, passing fifty times per second from a positive peak value to a negative peak value. This is excellent for transporting energy over long distances — which is why the entire global electrical grid works this way — but it is totally unusable by a digital circuit.

Computer components want direct current (**DC**): a stable, steady voltage that does not oscillate. A transistor inside a CPU has no idea what to do with a 50 Hz sine wave. It wants a fixed 12 volts, or 5, or 3.3, and it wants them stable within a few percentage points, always, without interruption.

The PSU is the machine that performs this translation. The process, greatly simplifying a chain that in reality has dozens of stages, works like this. The alternating current enters and first encounters an **EMI filter** (*ElectroMagnetic Interference*): a group of capacitors and inductors whose task is twofold: to prevent disturbances from the electrical grid from entering the power supply and — equally important and often forgotten — to prevent disturbances generated by the power supply itself (which is a circuit that switches tens of thousands of times per second) from returning to the grid and disturbing other appliances. In inexpensive power supplies, this stage is the first to be sacrificed: components are removed, empty pads are left on the printed circuit board, and the result is a device that, by law, should not even be allowed to bear the CE marking.

After passing the filter, the sine wave encounters a **rectifier bridge**, a group of four diodes that "flips" the negative half-waves, making them positive. At this point, we have a voltage that is always positive but pulses violently: it is not yet direct current; it is a kind of camel hump repeated a hundred times per second. Then the **PFC** (*Power Factor Correction*) stage intervenes, which in modern power supplies is almost always **active** and not passive. The PFC's task is to ensure that the current drawn from the grid is in phase with the voltage and has a waveform as sinusoidal as possible: without it, a power supply would draw current in narrow, violent pulses, polluting the grid and forcing the energy provider to size everything for an apparent power much greater than that actually consumed. A well-designed active PFC brings the power factor above 0.95; it is also the stage that charges the large primary electrolytic capacitor, the big, cylindrical one that is immediately visible when opening a power supply and typically operates around 400 volts DC.

From those 400 volts DC, we need to step down to the 12 volts required by the computer, and this must be done with complete galvanic isolation between the mains and the PC — otherwise touching the case would mean touching the electrical grid. This task falls to the **primary switching stage**, which chops the continuous 400 V into a high-frequency square wave (tens or hundreds of kilohertz), sends it through a transformer, and collects a low-voltage version on

the other side. Working at high frequency is the trick that allows modern power supplies to be small: a transformer working at 50 Hz to deliver 850 watts would be a brick weighing several kilograms, while one working at 100 kHz fits in the palm of a hand. The topologies used have names that the reader will encounter in technical reviews — half-bridge, full-bridge, and especially **resonant LLC**, today the de facto standard in the mid-range and high-end because it allows transistors to switch when the current through them is close to zero, drastically reducing losses.

Finally, on the secondary side, the low-voltage but still alternating current is rectified. In quality power supplies, this is done with **synchronous rectification**: instead of diodes (which dissipate a threshold voltage, about 0.4-0.7 V, multiplied by tens of amperes, i.e., tens of watts wasted as heat), actively driven MOSFETs are used, which behave like almost ideal switches. This is one of the main reasons why a Gold or Platinum power supply is more efficient than a Bronze one.

1.2 The rails: 12V, 5V, 3.3V (and the ghosts of the past)

The term **rail** indicates a power line at a specific voltage. A modern ATX power supply provides the following voltages to the computer:

Rail	Voltage	Current Use	Percentage of Total
+12V	12 volt	CPU (via VRM), GPU, fans, hard drive motors, practically everything	90-99%
+5V	5 volt	SATA disk control electronics, USB ports, some motherboard circuits	Few watts
+3.3V	3.3 volt	DIMM slots (via regulators), M.2 and PCIe slots (partially), various logic	Few watts
+5VSB	5 volt standby	Keeps the system "alive" when the PC is off: Wake-on-LAN, power	1-3 A

		button, charging USB	
-12V	-12 volt	Historical remnant of RS-232 serial ports. Practically unused today	0.3 A or less

The natural question is: why does 12V dominate so overwhelmingly? The answer lies in Ohm's law and power. Power is voltage multiplied by current ($P = V \times I$); for the same power, the higher the voltage, the lower the current. And current is what costs: losses in a conductor are proportional to the *square* of the current ($P_{\text{lost}} = R \times I^2$). Transporting 600 watts at 12 volts requires 50 amperes; transporting them at 5 volts would require 120, with almost six times greater losses and much thicker cables. 12 volts is the historical compromise between "high enough to limit current" and "low enough to remain in extra-low safety voltage."

There is, however, a second, more modern and more important reason. No PC component actually uses 12 volts as such. The CPU today operates at around 1.0-1.3 volts; the GPU is in the same range; DDR5 memory operates at 1.1 volts. Those 12 volts are further converted *on-site* by the **VRMs** (*Voltage Regulator Modules*, the voltage regulation modules we discussed extensively in the motherboard chapter) present on the motherboard and graphics card. The VRM is a multi-phase DC-DC converter that takes the 12 volts input and steps it down to the voltage required by the chip, with a precision and response speed that no half-meter long cable could guarantee. In other words: the PSU does not power the CPU. The PSU powers the VRM, which powers the CPU. The PSU delivers "raw" 12-volt power as close as possible to the point of use, and fine conversion happens there.

This architecture has a practical consequence on how power supplies are internally constructed. In older power supplies, designed in the 2000s, the three main voltages were all generated by the transformer with separate windings and "group regulated" (**group regulation**): the control circuit looked at a weighted average of 12V and 5V and tried to keep it within limits. The flaw is evident: if the load is heavily unbalanced — and in 2026 it always is, because 12V takes 95% of the power while 5V and 3.3V are almost unloaded — the voltages diverge. The 12V tends to drop under load, the 3.3V tends to rise, and in pathological cases, it goes outside the ATX $\pm 5\%$ tolerance. The modern solution, adopted by any decent

power supply, is **DC-DC**: the transformer generates *only* 12 volts, and the 5V and 3.3V are derived from them via small buck converters placed on dedicated boards. The result is voltage regulation that remains within 1-2% under any load condition. If you read in a technical review that a power supply "uses a group-regulated platform," you are reading — in 2026 — that it is an obsolete design, almost always belonging to the lowest price range.

One last concept: **single rail** versus **multi rail**. Some power supplies virtually divide their 12V into several separate "rails," each with its own overcurrent protection. These are almost never physically distinct rails (there is only one transformer): it is a subdivision made downstream, with separate current sensors on the various cable groups. The theoretical advantage is safety: if a cable shorts, the protection for that single branch trips at a low threshold, instead of waiting for the entire power supply to reach its global threshold. The disadvantage is that with very power-hungry graphics cards, there is a risk of tripping the protection of a single rail while still remaining within the total power. In practice: both architectures are valid if well implemented, a well-protected single rail is perfectly fine for a home PC, and this is a discussion that in the 2010s consumed much more energy in forums than in actual cases.

1.3 Why it's the component you DON'T skimp on

We come to the point that makes the entire chapter worthwhile. A shoddy power supply is not "a slightly worse power supply." It is a device that can fail in qualitatively different ways from any other component.

The first risk is excessive **ripple**. Ripple is the residual AC voltage that remains on the DC output: instead of being a flat 12-volt line, the voltage continuously oscillates above and below that value. The ATX standard dictates that ripple should not exceed 120 millivolts peak-to-peak on the 12V rail and 50 mV on the 5V and 3.3V rails. A good power supply stays well below this — 20-40 mV on the 12V is typical for a high-end product. A poor power supply can reach 150-200 mV or more, especially under heavy load. The consequences are neither immediate nor spectacular: the PC works, but the VRMs on the motherboard and graphics card receive dirty voltage, have to work harder to clean it, downstream capacitors heat up and age more quickly, and in extreme cases, seemingly inexplicable instabilities occur — crashes under load, memory errors, random reboots — which the user spends months attributing to RAM, video drivers, or Windows.

Ripple is a slow and silent killer, and it does not appear in any advertised specifications.

The second risk is **out-of-tolerance regulation**. If under transient load the 12V drops to 11.2 volts, the graphics card's VRM can go into protection mode or the system might simply shut down. The classic symptom — "the PC only reboots when I play games" — is, in the vast majority of cases, an inadequate power supply, not a faulty GPU.

The third risk is catastrophic: the **lack of functional protections**. A well-designed power supply has a series of monitoring circuits (which we will examine in detail in section 6) that, in case of an anomaly, shut everything down in milliseconds. An inexpensive power supply often has these protections only on paper: the supervisory integrated circuit might be a model that supports them, but the passive components that enable them were not installed to save a few cents. In that case, an internal fault can send incorrect voltages to components — 12 volts on the 5V line, for example — leading to instant and irreversible destruction of everything connected. This is not a theoretical risk: independent laboratories that test inexpensive power supplies by deliberately destroying them regularly produce this type of result, and in the real world, there are endless threads of users who have lost their motherboard, CPU, and drives due to a thirty-euro PSU bought on a marketplace.

The fourth risk is the most trivial and most concrete: **fire**. A power supply operates with 230 volts input, with capacitors charged to 400 volts, and dissipates tens of watts as heat. The components separating the primary from the secondary, the filters, fuses, and isolation distances on the printed circuit board are safety elements. In power supplies lacking real certifications — and a CE mark unilaterally affixed by an Asian manufacturer is not a certification, it's a self-declaration — these elements are the first to fail.

Finally, there's an economic consideration that should convince even those who aren't swayed by technical arguments. The power supply is the component with the longest lifespan in the entire system. A CPU

A number illustrates the point. A card with a TBP of 350 watts can exhibit peaks of 600-700 watts over a measurement window of one-tenth of a millisecond. No household instrument sees these: a plug-in meter updates its reading once or twice per second, and on that scale, the peak is invisible, completely lost in the

average. But the power supply absolutely sees it, and — crucially — its overcurrent protection circuit sees it too. If the **OCP** (*Over Current Protection*) is set too tightly and its response is fast, the power supply interprets the peak as a fault and shuts down. This led to the phenomenon, which became famous with NVIDIA RTX 30 series cards in 2020-2021 **[volatile data]**, of perfectly honest 750-watt power supplies shutting down PCs with cards whose average consumption was well below that threshold.

This is precisely the problem that the ATX 3.0 standard (section 5) was created to solve. And it's why the correct practical rule is: **take the realistic peak consumption calculated previously and add a 30-40% margin**. Not because the system actually consumes 40% more, but because that margin absorbs transients without triggering protections, and because it places the power supply's typical operating point in the zone where it is most efficient and quiet.

2.3 Indicative Tables and Calculators

Online calculators — the serious ones, like be quiet!'s calculator, Seasonic's, or Cooler Master's classic **[volatile data: tools change URLs and are discontinued]** — are useful for cross-referencing, not as an oracle. They should be used with the understanding that they tend to be conservative (the manufacturer has an interest in selling you a larger power supply) and that they often rely on nominal TDPs rather than actual consumption. The OuterVision calculator is historically the most detailed because it allows specifying overclocking and custom loads.

The following table offers a starting point for typical configurations in 2025-2026. The TBP values for the cards are those declared at product launch and should be verified **[volatile data: the range of video cards changes every 18-24 months]**.

Configuration Type	GPU (Indicative TBP)	CPU (Peak)	Estimated System Peak	Recommended PSU
Office / HTPC, integrated graphics	—	65-90 W	~150 W	400-450 W
Entry gaming (e.g., RTX 5060 / RX 9060 XT)	~150-180 W	~120 W	~350 W	550-600 W
Mid gaming (e.g., RTX 5070 / RX 9070)	~250 W	~150 W	~450 W	650-750 W

Mid-high (e.g., RTX 5070 Ti / RX 9070 XT)	~300 W	~180 W	~530 W	750-850 W
High-end (e.g., RTX 5080)	~360 W	~200 W	~600 W	850-1000 W
Enthusiast (e.g., RTX 5090)	~575 W	~250 W	~900 W	1000-1200 W
Workstation / dual GPU	2× 300-575 W	250-350 W	1200-1600 W	1600 W+ (and system verification)

Regarding the last row, there's a specific note for those living in Italy: the standard residential contract provides for 3 kW of committed power, with a tolerance up to approximately 3.3 kW before the limiter trips. A system with two high-end video cards, under full load, can dangerously approach half of that budget on its own, and must be considered alongside everything else in the house. This is not an issue for 99% of configurations, but it is a real constraint for those building inference or rendering machines.

2.4 Oversizing: pros and cons

The main **pro** is efficiency. A power supply's efficiency curve is not flat: it's low at very light loads (below 10-15%), rises quickly, peaks around 40-60% of the nominal load, and drops slightly as it approaches 100%. Sizing the power supply so that the system's typical consumption — not the peak, but the *typical* consumption, i.e., what you get when gaming or working — falls within that central window means operating at the point of maximum efficiency.

The second pro is noise, and it's what is most noticeable in daily practice. Practically all decent power supplies today have a **zero-RPM** or semi-passive mode: below a certain load threshold (typically 30-40%), the fan remains off, and the power supply cools by convection. An 850 W power supply in a system that typically draws 300 W will be absolutely silent most of the time. The same system with a 550 W power supply will have the fan constantly running.

The third pro is longevity: a power supply that constantly operates at 90% of its nominal load works with hotter capacitors, and the life of an electrolytic capacitor approximately halves for every 10 °C increase in temperature. The fourth is the

possibility of upgrading the graphics card in three years without having to change the power supply as well.

There are two **cons**. The first is obvious: cost. The second is the drop in efficiency at very low loads. A system that consumes 60 watts at idle, powered by a 1200 W unit, is operating at 5% load, well below the point where 80 PLUS certification measures anything. Modern high-quality units still perform well there — this is precisely what Cybenetics certification measures, as we will see — but inexpensive units falter. The practical advice is not to overdo it: installing a 1000 W unit on a system that consumes 300 W peak does not bring benefits, costs more, and improves nothing. The sweet spot is as described above: realistic peak + 30-40%.

3. The 80 PLUS Certification

3.1 What it actually measures (and what it doesn't)

80 PLUS is a voluntary certification program, born in 2004 in the United States, which measures **only one thing**: the efficiency of AC to DC conversion, expressed as the ratio between the power delivered at the output and the power drawn from the wall socket. A power supply that delivers 500 watts to the PC while drawing 555 watts from the socket has an efficiency of 90%: the 55 watts difference have been converted into heat inside the case.

It is crucial to understand what 80 PLUS **does not** measure, because confusion on this point is the source of most incorrect purchases:

It does not measure ripple. It does not measure voltage regulation. It does not measure transient response. It does not measure the presence or calibration of protections. It does not measure the quality of capacitors. It does not measure fan noise. It does not measure *hold-up time* (the time, a minimum of 17 milliseconds according to the ATX standard, for which the power supply must continue to deliver valid voltages after a power outage, to give the system time to manage an orderly shutdown). It does not measure lifespan.

An 80 PLUS Gold label certifies that the power supply is efficient. It says absolutely nothing about whether it is *good*. These are two orthogonal axes, and there are poorly built Gold power supplies and very well-built Bronze power supplies on the market. We'll come back to this in a moment.

A historical note should be added to help calibrate trust: for years, the program was managed by an organization (Clearesult/Ecova) that tested only one sample provided by the manufacturer, without random checks on series production. Various independent tests have found products on the market that did not achieve the efficiency declared on their labels. It is not a fraudulent program, but it is a program with limited control capabilities.

3.2 Levels and Percentages

Efficiency thresholds are defined at three (or four) load points. A very important point, almost always ignored: **the thresholds vary depending on the mains voltage**. The "American" 115-volt table, the one that circulates in forums, is not the one that applies in Italy. We operate at 230 volts, and for the same power output, the input current is half, so losses are lower and requirements are stricter. The correct table for Europe is the "230V EU internal" one.

Minimum Required Efficiency — 230V EU (the table that matters in Italy)

Level	10% Load	20% Load	50% Load	100% Load
80 PLUS (White)	—	82%	85%	82%
80 PLUS Bronze	—	85%	88%	85%
80 PLUS Silver	—	87%	90%	87%
80 PLUS Gold	—	90%	92%	89%
80 PLUS Platinum	—	92%	94%	90%
80 PLUS Titanium	90%	94%	96%	91%

Minimum Required Efficiency — 115V (for comparison, not applicable to the Italian grid)

Level	10%	20%	50%	100%
80 PLUS (White)	—	80%	80%	80%
Bronze	—	82%	85%	82%
Silver	—	85%	88%	85%
Gold	—	87%	90%	87%
Platinum	—	90%	92%	89%

Titanium	90%	92%	94%	90%
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Note that Titanium is the only level that imposes a requirement at 10% load. This is a significant choice: it recognizes that a modern PC spends most of its life in idle or near-idle, and that low-load efficiency has a real impact on electricity bills and heat.

How much is it worth, in monetary terms, to go up a level? Let's do an honest calculation. Suppose a PC that consumes an average of 300 watts output, used four hours a day for 365 days. With a Bronze power supply at 88% efficiency, it draws 341 W from the socket; with a Gold at 92%, it draws 326 W. The difference is 15 watts, for 1460 hours a year, which is about 22 kWh. At an energy price of 0.25 €/kWh **[volatile data]**, we are talking about approximately 5.50 euros per year. The surcharge of a Gold over a Bronze pays for itself in three or four years, not in six months. The correct conclusion is not "Gold is not worth it": it is that **you don't buy a Gold for efficiency itself**, you buy it because the Gold and higher tiers concentrate the best designs, with better components and longer warranties. Efficiency is an indicator correlated with quality, not the reason for purchase.

3.3 Cybenetics: The Certification that Measures What Matters

Cybenetics is an independent certification body, founded by Aris Mpitiopoulos (one of the most rigorous technical reviewers in the industry), created precisely to fill the gaps of 80 PLUS. It issues two distinct certifications.

The first is **ETA**, which measures efficiency. The methodological difference compared to 80 PLUS is substantial: instead of measuring at three fixed points, Cybenetics calculates a **weighted average across the entire load range**, from 10% to 100%, explicitly including light loads; it also measures the efficiency of the 5-volt standby rail and "vampire power" consumption (i.e., with the PC off but the power supply connected to the mains). Levels range from ETA-A+++ (the highest) down through A++, A+, A, B, C, D, E. A Cybenetics ETA-A unit roughly corresponds to an 80 PLUS Gold/Platinum, but the data is much more representative of real-world behavior.

The second, and for many users the most valuable, is **LAMBDA**: it measures **noise** in an anechoic chamber across the entire load range, with levels ranging from LAMBDA-A++ (average below 15 dB(A), meaning a virtually inaudible power supply) down to LAMBDA-E. It is the only certification in the world that quantifies

power supply noise with a repeatable method, and for those building a silent PC, it is worth more than any other data.

Cybenetics also publishes **complete test reports** on its website: ripple, voltage regulation, transient response, hold-up time, efficiency at every load, and protection verification. These are free PDF documents, twenty or thirty pages long, that tell you everything there is to know about a power supply. Consulting the Cybenetics database before a purchase is, today, the single most useful action a buyer can take.

3.4 Why a bad Gold can be worse than an excellent Bronze

Now we can close the loop with a concrete and illustrative example. Let's imagine two 650-watt power supplies.

The first is a Gold from a lesser-known brand, built by a third-tier OEM. It achieves 90% efficiency — it did so using low-cost components but with an efficient topology, which is perfectly possible — but it has 85 °C Chinese capacitors on the secondary, a 90 mV ripple on the 12V (within ATX specifications, but at the limit), poorly calibrated OCP protection that trips on GPU transients, a noisy sleeve bearing fan without a zero-RPM mode, and a three-year warranty.

The second is a Bronze from Seasonic or be quiet!, built on a mature platform. It achieves 88% efficiency. It has 105 °C Japanese capacitors, a 30 mV ripple, correctly calibrated and verified protections, an FDB fan with zero-RPM, and a five or seven-year warranty.

The second power supply is, by every metric that matters — stability, quietness, longevity, safety — a better product. It consumes 15 watts more at 650. The lesson, repeated because it is the most important in the chapter: **80 PLUS is a coarse filter, not a selection criterion**. It serves to exclude junk (never buy below Bronze, and practically never buy below Gold in 2026 because there's no reason to), not to choose between two candidates.

4. Modularity

4.1 Three architectures

Non-modular: All cables exit the power supply, permanently soldered or crimped. They are all there, always, including the four SATA connectors you'll never use and the two Molex connectors no one has used since 2012. What isn't

used must be hidden somewhere in the case — typically in an empty hard drive cage or behind the back panel, where it obstructs airflow and complicates assembly. The advantage is cost: for the same platform, a non-modular PSU costs 10-20 euros less than a full modular one, because it eliminates the connectors, the connector PCB, and the associated contact resistances. It's a reasonable choice only in the absolute budget range or in builds where cable management is not important.

Semi-modular: The cables that are *always* needed, in any configuration — the 24-pin motherboard cable and the 8-pin CPU EPS cable — are fixed; everything else (PCIe, SATA, Molex) is detachable. This is a smart compromise and is the dominant choice in the mid-range. The savings compared to full modular are modest (5-15 euros), and the practical disadvantage is almost nil, except in very small cases where even routing two fixed cables of the wrong length is an annoyance.

Full modular: Every single cable, including the 24-pin, is detachable. There are three advantages. The first is cable management: only the necessary cables are connected. The second, less obvious and much appreciated by those who have built PCs in compact cases, is that you can **install the bare power supply in the case and then connect the cables**, instead of having to shove a metal block with a meter and a half of dangling spaghetti into the compartment. The third is the possibility of using custom or "sleeved" cables, for aesthetic reasons or to have custom lengths.

4.2 The most important warning in this chapter: cables are not interchangeable

This should be written in bold and read twice, because it is the error that destroys the most hardware related to power supply.

Modular cables are NOT interchangeable between different brands, and often not even between different models from the same manufacturer.

The

The operational rules that follow are three, and they do not allow exceptions. First: **use only the cables provided in the box of the power supply unit you are using**. Second: when replacing a power supply, **also replace all cables**, even those that "look the same" and even if the new power supply is of the same brand as the old one. Third: if you purchase third-party cables (CableMod, Corsair Premium Kit, and similar), **verify that they are declared compatible with that**

specific model, and not with that brand generically. Serious custom cable manufacturers have configurators that ask for the exact model of the power supply precisely for this reason.

A note on an adjacent area: in 2023, CableMod had to recall its angled adapters for the 12VHPWR connector from the market, after a significant number of overheating and melting cases [**volatile data: historical episode, but the principle remains**]. The moral is not that third-party cables are always dangerous, but that you don't improvise on the high-current path, and that an extra adapter means two extra electrical contacts where parasitic resistances can form.

5. Standards and Connectors

5.1 ATX 2.x versus ATX 3.0/3.1

The **ATX** (*Advanced Technology eXtended*) standard, introduced by Intel in 1995, defines, among other things, the "Power Supply Design Guide": the document that establishes what voltages a power supply must provide, with what tolerances, with what connectors, and with what behavior under abnormal conditions. The **ATX 2.x** revision, in its various sub-versions, governed the market for almost twenty years.

The **ATX 3.0** revision, published by Intel in 2022, stems from a very specific problem: the transients of modern video cards that we discussed in section 2.2. The central novelty is that ATX 3.0 imposes explicit requirements on the ability to sustain **power excursions** well beyond the nominal power for very short intervals, without the protections intervening and without the voltages going out of tolerance. In practical terms, an ATX 3.0 power supply must withstand excursions up to approximately **200% of its nominal power** for durations in the order of 100 microseconds, with scalar requirements for longer durations, and must tolerate that a single video card connected to the high-power 12-volt connector requests up to **three times** its nominal power for an instant. It is a severe requirement, which in fact forces the designer to oversize the power components and, above all, to **calibrate the protections with a timed response** instead of an instantaneous one: the protection must distinguish between a legitimate 80-microsecond peak and a true short circuit.

The second major contribution of ATX 3.0 is the introduction of the 16-pin **12VHPWR** (*12 Volt High Power*) connector, designed to carry up to 600 watts on a single cable.

The **ATX 3.1** revision, from 2024, does not revolutionize anything but corrects the problem that the 12VHPWR had revealed in the field. It replaces that connector with the **12V-2x6** variant (also known as H++), makes it mandatory for the power supply to be able to deliver the declared power through it, and tightens some requirements on transients. In 2026, a new mid-range or high-end power supply should be ATX 3.1 **[volatile data: the standard evolves]**.

Is

The **12VHPWR / 12V-2x6** is the 16-pin connector: twelve power pins (six 12V and six ground) and four side signal pins, the **sense pins**. The sense pins are the conceptually interesting part: two of them (Sense0 and Sense1) form a two-bit code that the power supply uses to communicate to the graphics card **how much power it is authorized to draw** through that cable — 150, 300, 450, or 600 watts. A card that detects a 150 W code will limit its consumption accordingly. The connector, in its first 12VHPWR incarnation, has had a troubled history: numerous documented cases of housing melting, first with RTX 4090s in 2022-2023 and subsequently with RTX 5090s in 2025 **[volatile data]**. The identified causes were essentially two: incomplete insertions (the connector was not pushed all the way in, contacts only partially touched, contact resistance increased, heat melted the plastic) and current imbalances among the six 12-volt conductors, with a single wire forced to carry current well beyond its capacity. The **12V-2x6** revision addresses the first cause elegantly: **it shortens the sense pins and lengthens the power contacts**, so that if the connector is not fully inserted, the sense pins do not make contact, and the card — not receiving authorization — limits itself to a few watts instead of trying to draw 600 through partial contacts. It does not solve the second cause, which is a design problem on the graphics card side (current balancing), but it is a substantial safety improvement.

The operating rule for the 12V-2x6 is clear: **use the native power supply cable, insert it until it clicks, check that there are no visible gaps between the connector and the card body, do not bend the cable sharply in the first 3-4 centimeters from the connector, and do not use the four-PCIe connector adapter included in the graphics card box if you can avoid it.** Those adapters

work, but they introduce additional contacts and are the configuration where most incidents have occurred.

SATA power connectors (15-pin, flat, with the unmistakable L-shape) power mechanical drives and 2.5" SSDs. They carry 12V, 5V, and 3.3V. 4-pin **Molex** connectors are an artifact from the eighties, surviving to power old accessories, fan hubs, and pumps: they carry 12V and 5V, are inconvenient to insert and extract, and by 2026 their use is residual.

5.3 "PCIe Gen5 ready": what it means (and what it doesn't mean)

This is a marketing term, not a formal standard, and needs to be decoded. On power supply boxes, it usually means two things together: that the unit complies with ATX 3.0 or 3.1 (thus handling transients as per specification) and that it has at least one **native 12V-2x6 connector** wired directly to the power supply, without adapters.

What it *doesn't* mean is equally important: it has nothing to do with the version of the motherboard's PCI Express slot. A power supply doesn't "know" if the slot is Gen4 or Gen5 and doesn't care: it provides 12 volts, period. The term refers to the power connector (sometimes improperly called "PCIe 5.0 power connector") and its ability to sustain power spikes. A power supply not "Gen5 ready" will perfectly power a PCIe 5.0 graphics card that uses traditional 8-pin connectors.

6. Internal Quality

6.1 The OEMs: Who really builds your power supply

This is the section that separates the informed buyer from everyone else, and the concept to grasp is that **the vast majority of brands that sell power supplies do not manufacture them**. Corsair does not have power supply factories. Neither does be quiet!, nor NZXT, nor Cooler Master, nor MSI, nor — with some nuances — EVGA. What they do is design or commission a product, define its specifications, test it, and put their brand on it, while the detailed electronic design and physical production are entrusted to an **OEM** (*Original Equipment Manufacturer*).

The significant OEMs in the consumer market are relatively few. **Seasonic** is the only one that is simultaneously a top-tier OEM and a brand that sells under its own name; it also produces for others (historically for NZXT, Antec, and others).

Super Flower is Taiwanese, has an excellent reputation, and is the platform behind many high-end EVGA units and the Leadex brand. **CWT** (*Channel Well Technology*) is huge and produces for Corsair and many others; its quality depends entirely on the platform's tier, ranging from very good to mediocre. **FSP** (*Fortron Source Power*) is another giant, with a catalog ranging from economical to excellent, and also produces under its own brand. **HEC, Andyson, Sirfa/High Power, Solytech, Great Wall, Gspower, Enhance, Delta** complete the landscape with very different quality levels. In the lower end of the market, manufacturers like **Sama** or various minor Chinese OEMs provide the platforms for no-name products.

The direct and counterintuitive consequence of this structure is the following: **the same brand can simultaneously sell excellent power supplies and terrible power supplies**, because they are produced by different OEMs on different platforms. A brand's high-end series might be built by Seasonic, and the same brand's budget series by a third-tier OEM, with the same label and logo. **Buying "a Corsair" or "a Cooler Master" means nothing. You need to buy a specific model.**

This, incidentally, is exactly why community tier lists (section 7.2) exist and are so useful: they list *models*, not brands.

6.2 Protections

A well-designed power supply includes an integrated supervisor circuit that continuously monitors the outputs and intervenes by shutting everything down in case of an anomaly. The protections that must be present — and whose presence is declared in the technical specifications, but whose *correct calibration* is only verified in independent tests — are as follows.

OVP (*Over Voltage Protection*) intervenes if an output voltage rises above a preset threshold (typically around 13.5-14 volts on the 12V rail). This protection literally saves the computer: without it, a fault in the regulation circuit can send destructive voltages to components.

UVP (*Under Voltage Protection*) does the opposite: it shuts down the power supply if a voltage drops below a threshold (around 10-10.5 volts on the 12V). This prevents components from operating in brownout conditions, which can cause data corruption on drives.

OCP (*Over Current Protection*) monitors the current supplied on each rail (or on each virtual rail, in multi-rail power supplies) and intervenes if it exceeds the threshold. It is the most delicate protection to calibrate, for the reasons seen regarding transients: too tight and the power supply shuts down the PC every time the video card has a peak, too loose and it doesn't protect.

OPP (*Over Power Protection*) looks at the *total* power delivered, not the current of a single rail, and intervenes if the system demands more than the power supply can provide. It is the global safety net.

SCP (*Short Circuit Protection*) is the most fundamental: it detects a dead short circuit between a rail and ground and shuts down immediately. It must always be present, without exception.

OTP (*Over Temperature Protection*) uses an internal sensor, usually on the secondary heatsink, and shuts down the power supply if the internal temperature exceeds a critical threshold. It is the only protection that is often *missing* even in otherwise decent units, and it is the one that matters most if the power supply is mounted with the fan facing down in a case with little space, or if a fan fails.

Finally, there are "ancillary" but non-trivial protections: **SIP** (*Surge and Inrush Protection*) protects against current spikes at startup and mains transients, and **NLO** (*No Load Operation*) allows the power supply to operate safely without a connected load, which is useful during testing.

The practical point: almost all datasheets declare all these acronyms. The declaration costs nothing. What matters is that they are *implemented* and *calibrated*, and this can only be discovered by reading an independent test (Cybenetics, HWBusters, TechPowerUp, Tom's Hardware) where the reviewer deliberately tries to trigger them.

6.3 Capacitors, fans, materials

Capacitors, along with the fan, are the element that ages. The primary electrolytic capacitor (the large one, around 400 V) and the secondary capacitors work in a hot environment and must maintain their capacity for years. The classic discriminant is the origin: Japanese capacitors — the brands to look for in reviews are **Nippon Chemi-Con**, **Rubycon**, **Nichicon**, **Panasonic** — have better tolerances, lower equivalent series resistance, and, above all, a much longer declared lifespan compared to low-end Taiwanese or Chinese capacitors (Teapo, CapXon, Elite, Su'scon). The other parameter is the **nominal temperature**: **105 °C** capacitors are

the quality standard, while 85 °C ones are a sign of cost-cutting. In the secondary, modern quality power supplies widely use **solid polymer capacitors**, which do not have liquid electrolyte, do not dry out, and tolerate heat better.

The **fan** deserves attention because it's the component you hear. The bearing type determines noise and lifespan: a **sleeve bearing** is the cheapest, gets noisier with age, and has a reduced lifespan if mounted horizontally; a **needle bearing** is an improvement over the sleeve bearing, and is better; a **ball bearing** lasts a long time but is noisier; an **FDB** (Fluid Dynamic Bearing) is now the quality standard: quiet, with a declared lifespan of tens of thousands of hours, and insensitive to mounting orientation. **Magnetic levitation** (maglev) fans are a further evolution, found in the premium segment.

Zero-RPM mode (or semi-fanless, or "hybrid mode") deserves a mention. This is the function where the fan remains completely still below a certain load and temperature threshold. The advantage is obvious: a power supply that isn't spinning is a power supply that makes no noise and doesn't draw in dust. The potential disadvantage is that, by remaining still for a long time, the power supply accumulates heat, and when the fan turns on, an annoying on/off cycle can be perceived if the hysteresis is poorly designed. Quality units manage this with wide hysteresis. Many power supplies have a physical switch on the back to disable zero-RPM mode: if your power supply operates in a hot case or near its limit, disabling it is a prudent choice.

6.4 Warranty as an indicator

Here's a simple, quantitative criterion that doesn't require reading a single technical review: **the length of the warranty is the best single indicator of the manufacturer's confidence in their product**. A manufacturer offering ten or twelve years is stating, with a contractual commitment that costs them real money, that they expect that unit to function for a decade. No one gives away twelve years of warranty on a product built with substandard capacitors.

Warranty	What it suggests	Typical range
2-3 years	Economical product, basic components	Entry / to avoid for serious builds
5 years	Decent product, mid-range	Mid, acceptable
7 years	Good product, quality components	Mid-high
10 years	High-quality product,	High-end

	Japanese capacitors	
12 years	Top of the consumer market	Flagship

Pay attention to a non-trivial detail: in Italy and Europe, there is a two-year legal guarantee of conformity towards the seller, which is different from the **manufacturer's commercial warranty**. The latter is what matters here, and it is offered directly by the manufacturer, often requiring product registration. Keep your invoice and, if the manufacturer requires it, register the power supply upon purchase.

7. Brands and reference lines

This section should be read with a caveat: **manufacturer ranges change every 18-24 months**, and an excellent model can be replaced by a successor built by a different OEM and of lower quality, while retaining the same commercial name. What follows is therefore a map of *historical trends* and *line reputation*, not a price list. **[volatile data: always verify the exact model on an updated tier list before buying]**

7.1 Main brands

Seasonic is the absolute benchmark for reputation, as it is both an OEM and a brand. The **Prime** line (in Gold, Platinum, Titanium variants, and more recently the PX/TX versions with 12V-2x6 connector) represents the pinnacle of the consumer market, with a twelve-year warranty. The **Focus** line (GX, PX) is the upper-mid range, offering excellent quality/price ratio, with a ten-year warranty. Seasonic's more economical lines (Core, S12) are more ordinary and do not enjoy the same reputation: even with Seasonic, the name on the box is not enough.

Corsair has a broad and stratified range, and is the perfect example of why you need to look at the model. The **CX** line is the economic range, historically built by CWT on basic platforms: acceptable but not exciting, with a shorter warranty. The **RM** line, and especially **RMx**, is the core of their offering: Gold units, quiet, well-built, with a ten-year warranty; they are among the most frequently recommended for a mid-to-high-end build. The **HX** and **HXi** lines are a step up (Platinum, digital monitoring in the "i" variant), and **AX/AXi** represents the pinnacle (Titanium). The **SF** line covers the SFX format for compact cases and is considered excellent.

be quiet! is a German brand that, as the name suggests, focuses on silence. The **Pure Power** line is the entry-mid: honest, functional, without fireworks. The **Straight Power** line is the upper-mid range and is highly regarded. The **Dark Power** and **Dark Power Pro** lines are the top tier: excellent performance, almost absolute silence, high price. The brand's transversal strength lies in its fans, which are self-produced and among the best in the industry.

EVGA has a peculiar history: its high-end units, built on Super Flower platforms (the **G**, **P**, **T** series, and particularly the SuperNOVA G2/G3/G5/G6/G7 and P2/T2), were for years among the most highly recommended. Its economical units (**N1**, **W1**, **BQ**, **BR** series) were built by lower-tier OEMs and range from mediocre to definitely to be avoided. It should be noted that EVGA announced its exit from the graphics card market in 2022 and has progressively reduced its presence in the market [**volatile data: check current product availability and after-sales support**].

Cooler Master is a brand with a very heterogeneous range. The **V** line (V Gold, V Platinum, V SFX) is of excellent quality. The **MWE** lines are entry-level, and their quality has varied significantly between revisions: some are acceptable, others are not. This is a textbook case where the series name is not enough, and the exact revision needs to be known.

Thermaltake also has an equally heterogeneous range. The mid-to-high-end **Toughpower GF/PF/GT** line is generally good; the economical lines (Smart, Litepower) are to be avoided for any system with a serious graphics card.

NZXT has adopted a simple approach over the years: the **C** series (C750, C850, C1000 Gold and Platinum) is built on Seasonic platforms and enjoys a good reputation. It is an example of a brand that, not having a fallback economic range, has a consistently homogeneous quality.

MSI entered the power supply market relatively late. The **MPG A-GF** line (and subsequent ones) is of good quality. Some budget **MAG** lines have received very critical reviews in the past, with cases of units failing safety tests [**volatile data**]. It's another brand that needs to be evaluated model by model.

Then there are brands that have recently gained ground in the high-end market, such as **ASUS** with the ROG Thor and Loki lines, **Super Flower** which also sells under its own brand (Leadex), **XPG/ADATA**, **Lian Li**, **Phanteks**, **FSP** under its own brand, and **Antec** (whose Signature line is very well-regarded). And on the

other side, there's the entire universe of brands that shouldn't even be considered: the 30-40 euro units with fanciful wattages printed on the box, sold on general marketplaces, often with names that change every six months. The criterion is brutal but effective: **if you can't find a single independent technical review of that model, don't buy it.**

7.2 Community Tier Lists

Since no buyer can read a technical review for each of the three hundred models on the market, the community has produced an exceptionally useful tool: **PSU tier lists**. Historically, the most well-known is the one maintained by the **Cultists Network** (the successor to the old LinusTechTips tier list, which was retired) **[volatile data: tier lists change maintainers and locations over time; search for the current version]**.

The operation is simple: power supplies are cataloged by exact model and wattage, and assigned to a tier (Tier A, B, C, D... or similar schemes) based on available independent tests, warranty return data, and the OEM platform. The practical rule of thumb is: **for a system with a dedicated graphics card, do not go below mid-to-high tiers; for a system with a high-end card, stay in the top tiers**. The lists also explicitly indicate dangerous models, those not to be bought at any price.

Other resources to be aware of are the **Cybenetics database** (which offers the complete reports mentioned), reviews from **HWBusters**, **TechPowerUp**, **Tom's Hardware**, and the destructive tests by **GamersNexus**. All are free, and collectively cover almost all models worth considering.

8. How to choose, step by step

8.1 The Procedure

First step: calculate the wattage. Add the GPU's TBP, the CPU's peak power consumption (PL2 or PPT), and 80-100 watts for everything else. Add a 30-40% margin. Round up to the next commercial increment (typical increments are 550, 650, 750, 850, 1000, 1200, 1600 W).

Second step: decide on the standard. If the graphics card uses a 16-pin connector, demand ATX 3.0 or 3.1 with **native 12V-2x6**. If it uses 8-pin PCIe connectors, ATX 3.x is still preferable but not an absolute requirement.

Third step: set the efficiency threshold. In 2026, with current availability and prices [volatile data], there's no reason to go below **80 PLUS Gold** for a system with a dedicated graphics card. Bronze is only acceptable in a low-power office build. Platinum and Titanium make sense for those who keep their PC on many hours a day or who seek maximum silence.

Fourth step — the one nobody does and that makes all the difference: verify the specific model. Look up the exact model (not the series, the *model with its wattage*) on an updated tier list and, if possible, read the Cybenetics report or a technical review. Five minutes of research at this point are worth more than all the previous steps combined.

Fifth step: choose modularity. Full modular if the budget allows and the case is compact or has a window. Semi-modular as a reasonable compromise. Non-modular only for budget builds in large cases.

Sixth step: check the warranty. Below five years, in a serious build, it's a red flag. Ten years is what's expected from a mid-to-high-end product.

Seventh step: verify physical details. The length of the PSU (1000W and above units can be longer than 160-180mm and might not fit in all cases), the form factor (standard ATX versus **SFX** or **SFX-L** for Mini-ITX cases), the number of actually available PCIe and SATA connectors, and the cable length (in cases with a bottom-mounted PSU and top-mounted EPS connector, a short CPU cable is a serious problem: at least 60-65cm are needed, and in many full tower cases, an extension cable is advisable).

8.2 Budget-based recommendations

[volatile data: prices indicated are orders of magnitude for the Italian market and change frequently]

In the **entry-level tier** (approximately 60-90 euros, for 450-650W systems), the goal is to buy a decent product without chasing the absolute best. This tier includes lines like Corsair RM (smaller wattages), be quiet! Pure Power, Seasonic Focus GX in lower wattages, MSI MPG A-GF. The advice: aim for a semi-modular 650W Gold unit from a known brand, with at least five years of warranty. This is already enough for a mid-range graphics card, and you can upgrade the GPU later without changing everything.

In the **mid-range tier** (approximately 100-150 euros, for 750-850W systems), the market is crowded, and the choice is wide. Corsair RMx, be quiet! Straight Power, Seasonic Focus GX/PX, NZXT C series, Cooler Master V Gold, ASUS TUF Gaming: these are all competent products. The advice: **850W Gold, full modular, ATX 3.1 with native 12V-2x6, ten years warranty**. This configuration covers 90% of serious gaming builds by 2026, with room for a future graphics card upgrade.

In the **high-end tier** (approximately 180-350 euros, for 1000-1200W and above systems), you're buying something that needs to handle a top-tier graphics card with brutal transients, and you're buying silence and longevity. Seasonic Prime PX/TX, be quiet! Dark Power Pro, Corsair HX/AX, ASUS ROG Thor. The advice: **1000-1200W Platinum or Titanium, full modular, ATX 3.1, twelve years warranty if possible**. If the machine is a workstation that runs hours a day under load, Titanium truly pays off.

A note for those building in **SFF** (*Small Form Factor*, compact cases): the **SFX** and **SFX-L** formats have a smaller market and higher prices for the same power output. Corsair SF, Cooler Master V SFX, Lian Li SP, FSP Dagger are the usual references. Never use an SFX→ATX adapter as an excuse to cram a cheap unit into a small case: quality SFX units have higher power density and thermal sizing designed for that volume, and cheap ones in that format are particularly dangerous.

8.3 Common mistakes, explained

The 30-euro no-name power supply. This is the cardinal error. A 30-euro power supply advertised as "700W" does not deliver 700 watts: it typically delivers 300-400 real watts before going into protection or failing, has out-of-spec ripple, absent or non-functional protections, low-quality capacitors, and in destructive tests, it's not uncommon for it to fail by catching fire. The buyer's logic — "I spent 1500 euros on components, I'll save on the power supply" — is exactly the opposite of economic rationality. Rule: the power supply should account for between 7% and 12% of the total build budget.

Cutting it too close with wattage. Buying a 650W PSU for a system that consumes 620W at peak is technically possible but practically wrong: the power supply will operate at 95% load, with the fan always at maximum, capacitors at their thermal limit, and zero margin for GPU transients. The first sudden shutdown under load will occur within a few weeks.

Trusting only the 80 PLUS sticker. We've discussed this at length. Efficiency is not quality.

Reusing modular cables from the old power supply. The most destructive error. It's worth re-reading section 4.2. When changing power supplies, you change *all* the cables.

Daisy-chaining on powerful graphics cards. Many power supplies provide PCIe cables with **two connectors on the same cable** (one 6+2 in the middle and another 6+2 at the end). The reason is economic: only one PSU-side connector is needed to power two GPU-side connectors. The problem is that those two connectors share the same three 12-volt conductors of the cable, and an 18 AWG conductor is reasonably sized for 7-8 amps. Two 8-pin PCIe connectors should carry up to 300 watts combined, which is 25 amps at 12 volts, distributed over only three wires instead of six: this leads to over 8 amps per wire, with conductor heating and voltage drop. For a card that draws little power (150-180 W), nothing happens. For a card that draws 300 W or more, daisy-chaining is a bad idea, and manufacturers' recommendations are unanimous: **a physically separate PCIe cable for each connector on the graphics card**, drawn from a different power supply outlet. If the power supply doesn't have enough PCIe outlets to do this, it's a sign that it's undersized for that card.

Using the adapter included with the GPU instead of the native cable. Adapters from 3× or 4× PCIe to 12VHPWR work, but they add a point of contact and a point of mechanical failure on a path that carries up to 600 watts. If the power supply has a native 12V-2x6 cable, use it. If it doesn't, this is a good argument for changing the power supply.

Ignore the label's technical specifications. On a serious power supply, the side label shows the power distribution across the rails. The number to look for is the **combined power on the 12V rail**: on an honest 850W unit, the 12V alone should be able to deliver 830-840W (i.e., about 70 amps). If a "700W" power supply only declares 400-500W on the 12V, with the rest of the power attributed to 5V and 3.3V (which will never use more than 30-50W), that power supply is, functionally, a 450-550W unit sold as 700W. This is a marketing trick typical of budget units and older generations, and reading the label exposes it in ten seconds.

Buying a used power supply. Capacitors age, the fan wears out, the warranty is not transferable, and you don't know the conditions it operated under. It's the only PC component for which the used market is categorically inadvisable.

Forgetting upstream protection. A power supply, no matter how good, is not a surge protector for the mains. In areas with an unstable electrical grid — and in many parts of Italy, it is — a **surge protector power strip** costs twenty euros, and a **UPS** (Uninterruptible Power Supply) with AVR costs about a hundred euros and protects against voltage drops and outages. For a two-thousand-euro build, it's reasonable insurance. Pay attention to a technical detail: power supplies with active PFC can have problems with inexpensive UPS units that output an approximated square wave during a blackout; for a modern PC, a UPS with a **pure sine wave** output is needed.

Operational Summary — PSU Decision Checklist

A. Sizing 1. Retrieve the **GPU's TBP** from the manufacturer's website (not the retailer's box). 2. Retrieve the **CPU's peak power consumption** (PL2 for Intel, PPT for AMD), not the nominal TDP. 3. Add **80-100W** for the motherboard, RAM, drives, fans, pump. 4. Sum and add a **30-40% margin** for transients and to operate at peak efficiency. 5. Round up to the next commercial size. Verify with an online calculator as a cross-check, not as an oracle.

B. Standards and Connectors 6. Graphics card with a **16-pin connector**? → Demand **ATX 3.1 with native 12V-2x6**. No adapters. 7. Count the **actually available PCIe/12V-2x6 connectors**: one for each graphics card connector, on separate cables. 8. Verify there are enough **SATA connectors** for planned drives and an **EPS 8+8** if the motherboard requires it.

C. Quality 9. Minimum certification: **80 PLUS Gold** for any system with a dedicated GPU (230V EU table, not 115V). 10. Search for the **exact model** on an updated tier list (Cultists Network or equivalent). If the model doesn't appear anywhere, don't buy it. 11. If available, read the **Cybenetics report**: check ripple, regulation, transient response, protections, and the **LAMBDA** level if silence matters. 12. Verify the presence (and in tests, the functionality) of **OVP, UVP, OCP, OPP, SCP, OTP**. 13. Verify that the label declares a **power on the 12V rail equal to at least 95% of the total nominal power**.

D. Ergonomics and Physics 14. Choose **modularity**: full modular if budget and case justify it. 15. Check **unit length** against available space in the case, and **form factor** (ATX / SFX / SFX-L). 16. Check **cable length**, especially the CPU EPS in large cases. 17. Prefer **FDB** fan with **zero-RPM mode**.

E. Warranty and Protection 18. **Minimum 5 years**; 10 years is the expected standard in the mid-to-high range. Register the product if required. 19. Provide **upstream protection**: surge protector strip or, better, a **pure sine wave UPS**.

F. Things Not to Do, Ever 20. Never reuse **modular cables** from another power supply, not even from the same brand. 21. Never use **daisy-chaining** of PCIe connectors on graphics cards over ~200 W. 22. Never buy a **no-name** power supply or one without independent technical reviews. 23. Never buy a **used** power supply. 24. Never insert the 16-pin connector **partially**: it must click, with no visible gaps, and the cable should not be sharply bent in the first few centimeters.

Final note on data: the models, prices, manufacturer ranges, and generations of graphics cards cited in this chapter reflect the market status at the time of writing and are inherently subject to rapid obsolescence. The physical and design principles — AC/DC conversion, the dominance of the 12V rail, the role of transients, the distinction between efficiency and quality, the non-interchangeability of modular cables — do not change.

Chapter 7 - The Case and Ventilation

Foreword: The Component Everyone Underestimates

There's an implicit hierarchy in the mind of anyone building their first computer. At the top are the CPU (Central Processing Unit) and the GPU (Graphics Processing Unit), because these are the parts discussed in reviews and featured in benchmarks. Then comes RAM (Random Access Memory, volatile working memory) and storage, because at least they have a number that grows and can be compared. Then the motherboard, chosen for its socket and price. Then the power supply, often chosen poorly. And finally, as an afterthought, almost an aesthetic accessory, the case: the "cabinet," the "chassis," that box of sheet metal and glass that holds everything together.

This hierarchy is wrong, and for a very concrete reason: the case is the only component that affects **all others simultaneously**. It doesn't calculate anything, doesn't produce frames, doesn't store data, but it determines the temperature at which the CPU and GPU operate, the noise the system produces in the room, the amount of dust that accumulates on the heatsinks, the useful life of the components, and — something only discovered once assembly begins — how painful or pleasant the assembly and maintenance experience will be.

A wrong case won't make your computer explode. It does something more insidious: it silently slows it down. A GPU that operates at 68 °C in a well-ventilated chassis, maintaining its maximum boost, can reach 84 °C inside an "aquarium" enclosed by three glass panels. At that point, the boost algorithm—which is dynamic and temperature-sensitive—lowers frequencies to stay within thermal limits. The result is a 5-10% performance loss that the user doesn't see because they never saw the "before." They simply bought an 800 euro graphics card and are only using 730 euros worth of its performance.

This chapter covers the case and everything surrounding it—formats, compatibility, airflow, fans, filters, brands—with the same level of detail reserved for "noble" components. By the end, you will have the tools to open a chassis's technical specifications on an e-commerce site and understand, in ninety seconds, whether it's suitable for your configuration or if it's a trap.

Note on volatile data. Models, prices, and availability of cases and fans change rapidly and vary greatly across markets. Every euro figure reported in

*this chapter is indicative of the Italian market and should be verified at the time of purchase. Data marked **[volatile data]** are most subject to aging.*

1. The role of the case

1.1 Mechanical and electrical protection

The oldest and most obvious function of the case is that of a structural container. PC components are fragile in a non-intuitive way: the motherboard is a fiberglass and copper sandwich with hundreds of solder joints that crack if the PCB (Printed Circuit Board) is flexed; a large air cooler can weigh 1.5 kg and, if the system is transported vertically in a car, exerts enormous leverage on the four socket mounting points; a modern triple-slot graphics card weighs up to 2 kg and is suspended from a PCIe (Peripheral Component Interconnect Express, the high-speed expansion bus) connector and two screws on the rear bracket.

The case provides the rigid frame that absorbs these stresses. It does so through the **motherboard tray**, the vertical metal sheet to which the motherboard is screwed via **standoffs**: small threaded cylinders that raise the PCB a few millimeters from the sheet, preventing electrical contact between the copper traces on the back of the board and the metal of the chassis. It's good to remember this immediately, because it's one of the classic beginner's mistakes: **mounting the motherboard without standoffs, or leaving a standoff in a position where the board doesn't have a hole, causes a short circuit** which, in the best case, prevents power-on and, in the worst case, permanently damages the board.

Then there's a less visible electromagnetic function. A PC is a source of EMI (Electromagnetic Interference): inside the case, there are signals switching billions of times per second, and a closed, grounded metal enclosure acts as an imperfect but effective **Faraday cage**, reducing both emissions to the outside and susceptibility to external disturbances. This is why cases have expansion slots closed by metal brackets and why there are regulations (FCC in the United States, CE marking in Europe) that chassis must comply with. In practice, for the home user, this translates into a simple rule: **the brackets for unused PCIe slots must be reinstalled**, not because fluid physics dictates it, but because they serve to close the shell and — secondarily — to prevent air from entering or exiting through unintended paths.

1.2 Thermal Management: The True Function

This is the core of the chapter. From a thermodynamic perspective, a computer is an **electric energy to heat converter**. Practically 100% of the power drawn from the outlet (minus a negligible amount that exits as LED light and Wi-Fi radio waves) eventually ends up as heat in the room's air. A system that draws 500 W from the outlet is, from an environmental heating perspective, indistinguishable from a 500 W electric heater.

The problem is not the heat itself: it's **power density**. The CPU dissipates its TDP (Thermal Design Power, the thermal power designed, i.e., the heat the cooling system must be able to dissipate under sustained conditions) from a surface of a few square centimeters; the GPU does the same from a die the size of a postage stamp. The CPU cooler and the graphics card cooler are tasked with taking that concentrated heat and transferring it to the air passing through them. But the air passing through them **is the air inside the case**.

This is the central idea, and it bears repeating because it's the source of 90% of errors: the CPU cooler and the GPU cooler do not cool with room air. They cool with case air. The case's job is to make the case air as similar as possible to room air, meaning it should be continuously replaced before it heats up. If the case doesn't exchange air, a condition of **recirculation** occurs: the GPU expels air at 55 °C, that air finds no way out, is re-aspirated by the GPU's own fans and the CPU cooler, and the system stabilizes at a much higher thermal equilibrium. The coolers still work perfectly — they are simply working with incoming air at 45 °C instead of 24 °C, and those 21 degrees of difference are almost entirely transferred to the chip temperatures.

The parameter describing this phenomenon is called **ambient-to-case delta T**: the difference between the temperature of the air inside the case (measurable with sensors, or estimable from motherboard temperature sensors) and the room temperature. In a well-ventilated case under load, this delta is 3-6 °C. In a closed and poorly configured case, it can exceed 15-20 °C. **Every degree of delta T is an extra degree on the CPU and GPU**, added to everything else.

1.3 Acoustics

The second axis on which the case works is noise. Here there is a fundamental compromise that needs to be understood immediately, because no marketing will ever explicitly state it: **airflow and silence are in tension**. PC noise has three origins:

1. **Aerodynamic noise** from fans, generated by air turbulence passing through blades, grilles, and heatsink fins; 2. **Mechanical noise** from fan bearings, liquid cooling pumps, and mechanical hard drives; 3. **Structural vibrations** transmitted by the frame, which acts as a sound box if the metal sheets are thin.

A traditional "silent" case — think of historical models from be quiet! or the Fractal Design Define — addresses the problem with **insulation**: side and front panels internally lined with sound-absorbing material (bituminous or polyurethane foam), a closed front with a solid door, and reduced openings. This works on the noise that escapes, but it worsens airflow, so the fans have to spin faster to compensate, and faster fans make more noise. The net gain exists, but it is much smaller than commonly believed.

The modern school — which has dominated the market from 2019-2020 onwards — has inverted the reasoning: **a very open (mesh) case with large, slow fans is preferred**. The reasoning is that aerodynamic noise grows in a strongly non-linear way with rotational speed (as a first approximation, acoustic power scales as the fifth power of the peripheral speed of the blades: doubling the RPM means a noise increase of about 15 dB, which is "much more than double" to human perception). If the case offers little resistance, the fans can spin at 700-900 RPM instead of 1400-1600 to move the same amount of air, and the net result is **cooler AND quieter**, even if the residual noise is less "shielded". This is currently the winning strategy for the vast majority of users, and independent tests consistently confirm it.

There are still edge cases where insulation makes sense: very low-power systems (an office PC with a 65W CPU and no dedicated graphics card) placed in very quiet rooms, where the residual noise of low-speed fans is still perceptible and the acoustic barrier makes a difference. But a gaming system with a 300W GPU cannot be made silent by enclosing it: it can only be made hot.

1.4 Aesthetics

It must be said honestly: for a huge segment of the market, the case is **the only component that is visible**, and it is therefore legitimate for aesthetics to weigh in on the choice. The point is not to deny this need, but to order it correctly relative to the others. The practical rule I propose, and which I will return to in the selection procedure, is: **aesthetics is a selection criterion within the set of cases that work, not a criterion that defines the set**. First, filter by compatibility and airflow, then choose the most aesthetically pleasing among

those remaining. The set is quite broad: in 2025-2026, there are dozens of beautiful *and* well-ventilated cases, so it is no longer necessary to compromise. Those who choose the "aquarium" display case before looking at temperatures are simply reversing the order of factors, and they will pay for it in degrees.

1.5 Assembly and Maintenance Ergonomics

The last role, one that is only appreciated after the second or third assembly: a good case allows for easy assembly. The features that make a difference are concrete and verifiable in video reviews:

- ****space behind the motherboard tray**** for cable management (from the scarce 20 mm of economic cases to the 30-35 mm of well-designed models);
- ****tool-less side panels****, snap-on or with magnetic closure;
- ****rolled edges**** on the sheet metal, which prevent cutting fingers — a detail that immediately distinguishes a well-crafted chassis from one built on a budget;
- ****reusable expansion slot covers****, not tear-off: some economic cases have PCIe slot covers fixed with a metal bridge to be broken, and once broken, they cannot be reattached;
- ****removable dust filters without disassembling the PC****, ideally from the front or top, not from underneath requiring the case to be lifted;
- ****removable drive trays**** and access to the rear without having to lay the machine down.

None of this appears in a specifications table, and all of it determines whether working on the computer again in two years will be an afternoon task or a nightmare.

2. Form Factors: Chassis and Motherboard

2.1 Motherboard Form Factors, Reviewed from a Case Perspective

The motherboard form factor — covered in the dedicated chapter — is the primary constraint that determines the case, because the chassis must have the standoffs in the correct positions and the physical space to contain it. There are four form factors relevant to the consumer market, defined by the **ATX** standard (Advanced Technology eXtended, the standard introduced by Intel in 1995 that governs dimensions, mounting holes, and connector placement):

Form Factor	Dimensions (mm)	Expansion Slots	Typical Use
E-ATX ("Extended ATX")	305 × 330 (nominal, but variable)	7	Workstations, HEDT, high-end boards
ATX	305 × 244	7	Reference standard, full desktop
Micro-ATX (mATX)	244 × 244	4	Compact budget, offices
Mini-ITX (ITX)	170 × 170	1	SFF (Small Form Factor) systems

An important clarification and source of real trouble: **"E-ATX" is not a rigorous standard**. Formally, it indicates 305 × 330 mm, but manufacturers also use the label for boards measuring 305 × 277 mm or 305 × 267 mm (the so-called "EE-ATX" or simply enlarged ATX formats). A case declared "E-ATX compatible up to 280 mm" will accept a 277 mm board but not a 330 mm one. **Always check the actual width of the board in millimeters**, not the acronym. This is one of the most costly mistakes because the problem is only discovered when everything is already in hand.

Compatibility is **descending**: a case that accepts ATX also accepts mATX and ITX, because the standoff holes are a subset. The reverse is not true. Mounting an ITX in an ATX case is perfectly legitimate (there will be a lot of empty space, which is not a thermal defect, if anything the opposite), but it is a waste of volume and money.

2.2 Case Form Factors

The commercial names of chassis are less standardized than one might think. There is no official definition of "mid tower." A convention is used, and the most honest parameter for comparing chassis is **volume in liters**, which many manufacturers and specialized websites now declare.

Category	Typical Height	Typical Volume	Max Mobo	Typical Fans	Notes
Full tower	500-650 mm	60-100+ L	E-ATX / SSI-EEB	8-12	Custom loop, many drives, workstatio

					n
Mid tower	420-500 mm	40-60 L	ATX (often "narrow" E-ATX)	5-9	The reference format
Mini tower / mATX	350-420 mm	25-40 L	mATX	3-5	Space/cost compromise
SFF (Small Form Factor)	variable	8-25 L	Mini-ITX	1-4	Extreme compactness, many compromises
Cube / "dual chamber"	variable	30-60 L	ITX → E-ATX	variable	Dual-chamber layout, aesthetics

Full tower. Makes sense in three cases: custom liquid cooling (with multiple 360/420 mm radiators mounted simultaneously), a large number of mechanical drives (home NAS, video archives), or true E-ATX workstation motherboards. Outside of these scenarios, it's wasted space and money: a full tower is not automatically cooler than a good mid tower, because volume itself doesn't cool — **airflow** cools. A full tower with three fans is worse than a mid tower with six.

Mid tower. This format solves 90% of use cases. It accepts ATX boards, 360 mm radiators, 340-400 mm long graphics cards, and 160-175 mm tall air coolers. This is where manufacturers' research and development is concentrated, where there is the most competition, and thus the best quality/price ratio. **If you don't have an explicit and articulable reason to leave this category, stay in it.**

Mini tower / mATX. Makes sense when the constraint is desk space or budget. An mATX motherboard costs less than an equivalent ATX, and the four expansion slots are more than sufficient for anyone without particular needs (in 2026, expansion slots are almost exclusively for the GPU: audio, network, and USB are integrated). The compromise is thermal: less space for fans, often only 2-3 positions, and tighter clearances.

SFF (Small Form Factor). This is a discipline in itself, almost a sport. Below 15 liters, every millimeter is contested: the graphics card must be short or "2.5 slot," the CPU cooler is often a low-profile 47-67 mm tall, the power supply is an **SFX** or

SFX-L (compact format, 100 × 63.5 mm versus 150 × 86 mm for standard ATX) which costs 40-60% more than an ATX of equal power, cables must be shortened or replaced with custom sets. The result can be splendid — a 12-liter system with full desktop performance — but the journey is expensive, requires in-depth research, and does not forgive planning errors. **It is the wrong format for a first build.** Industry reference chassis include the Fractal Design Terra and Ridge, the Cooler Master NR200P (the most accessible and "forgiving"), the Lian Li A4-H2O and Q58, and the boutique scene (SSUPD, Formd, Dan Cases) **[volatile data]**.

2.3 Which format for which need

- ****Office PC / HTPC / browsing****: mATX or ITX. Low heat, low space, low cost.
- ****Entry/mid-range gaming PC (single GPU, air cooler or AIO 240)****: mid-tower mesh. There is no rational alternative.
- ****High-end gaming PC (top-tier GPU, AIO 360, many fans)****: quality mid-tower or dual-chamber like Lian Li O11. Note: dual-chamber cases often ****lack front intake**** and must be configured with air intake from the bottom and/or side, as we will see.
- ****AI/ML or rendering Workstation (multiple GPUs, many PCIe lanes, lots of RAM)****: full tower, and here the constraint becomes ****the width to accommodate two 3-slot graphics cards with air space in between****, in addition to the PSU capacity. It must be clearly stated that two flagship GPUs side-by-side in a consumer case are a thermally problematic configuration: the upper card draws hot air from the lower card. Solutions include blower models (rare in consumer), liquid cooling, or chassis with increased slot spacing.
- ****NAS / home server****: specific chassis with cages for 6-12 3.5" drives, medium-flow fans in front of the cages, or rack formats if you have a cabinet.

3. Compatibility and internal space: the checklist that saves your purchase

This is the most important operational section of the chapter. All the numbers listed here are published in the case specifications; the problem is that almost no one reads them **before** ordering. Take a sheet (or a file) and fill in these fields for your configuration.

3.1 Maximum GPU length

This is the measurement, in millimeters, from the rear bracket to the front end of the graphics card. The case declares a "**GPU clearance**" or "max VGA length".

Operational rule: take the actual length of the specific model you buy — not the chip, the model. An ASUS RTX 5070 can be 305 mm long, and the same RTX 5070 from another manufacturer can be 240 mm. The length is a characteristic of the cooler, not the GPU.

Beware of three pitfalls:

1. The value declared by the case is often "with front fans not installed".

Many chassis declare two numbers: for example, "400 mm, 360 mm with front radiator". If you mount a radiator or thick fans in front, you lose 25-55 mm.

2. Power cables. The **12V-2×6** connector (the evolution of 12VHPWR, the 16-pin connector of modern high-power cards) exits from the top side of the card and requires a minimum bending radius: the standard requires not to bend the cable within 35 mm of the connector. If the side panel is 15 mm from the card, the cable will be crushed — and this is exactly the mechanism that caused connector melting incidents. Some cards have the connector rotated or positioned differently precisely for this reason. **Check the width clearance, not just the length.**

3. Slot thickness. Modern cards occupy 2, 2.5, 3, or even 4 slots in height. A case with 7 PCIe slots physically accommodates any card, but if you have an expansion card (a sound card, a 10 Gbit adapter, a capture card), you might find that the GPU covers its slot.

3.2 Maximum CPU cooler height

Measurement in millimeters from the motherboard surface to the top of the cooler. Typical values:

Cooler Category	Height	Required Case
Low-profile (Noctua NH-L9, Thermalright AXP)	37-67 mm	SFF, HTPC
Compact single tower	120-140 mm	mATX, mid tower
Standard single tower (e.g., Peerless Assassin, NH-U12A)	155-160 mm	mid tower (almost all)
Dual tower (Noctua NH-D15, Thermalright FA/PA)	160-168 mm	wide mid tower
"Large" dual tower	170-175 mm	explicitly verify

The rule: **leave at least 5 mm of clearance** compared to the value declared by the case. Manufacturing tolerances, gasket thickness, stand-off height, and the bulging of the glass side panel can eat up those few millimeters. A cooler that touches the side panel not only won't close: it transmits vibrations to the glass and turns the entire side of the case into a speaker.

3.3 Radiators: dimensions, positions, thicknesses

The **radiator** is the heat exchanger of a liquid cooling system (**AIO**, All-In-One, closed and pre-filled systems, or custom loops). It is measured by the number and size of the fans it accommodates:

Abbreviation	Fans	Approximate Length	Approximate Cooling Capacity
120	1 × 120 mm	~155 mm	poor, not recommended
240	2 × 120 mm	~275 mm	CPUs up to ~180 W
280	2 × 140 mm	~315 mm	~15-20% better than a 240
360	3 × 120 mm	~395 mm	high-end CPUs (up to 250-300 W)
420	3 × 140 mm	~455 mm	maximum consumer, large cases

Possible **mounting positions** and their implications:

- ****Top (upper), exhaust.**** This is the canonical position for an AIO. Hot air rises, the radiator expels it upwards, the pump remains lower than the highest point of the circuit (important: ****the pump should never be the highest point****, otherwise the air that inevitably accumulates in the circuit collects in the pump and produces the classic gurgling sound and premature wear). The compromise: the radiator slightly heats the air that... exits, so no problem; however, it takes up space from the VRMs (Voltage Regulator Module, the CPU power delivery stage on the motherboard) and can interfere with the VRM heatsinks themselves or with tall RAM modules. ****Check the "CPU cooler to top" clearance of the case and the radiator + fan thickness**** (typically 27 + 25 = 52 mm, but 38 and 45 mm thick radiators exist).

- ****Front (front), intake.**** Maximum performance for the CPU, because the radiator receives fresh air from outside. The cost is that all the air entering the case has been preheated by the radiator (by 3-8 °C under load), and this penalizes the GPU. It is a legitimate choice if the priority is the CPU. It reduces the available GPU length.
- ****Side / bottom.**** Typical of dual-chamber cases. To be evaluated on a case-by-case basis.

One last honest note: **for the vast majority of consumer CPUs, a good air cooler for 40-60 euros (Thermalright Peerless Assassin, Phantom Spirit, DeepCool AK620) is within 2-4 °C of a 130 euro AIO 360**, it doesn't have a pump that can fail, it doesn't have liquid that can evaporate over time, and it doesn't have a single point of failure. An AIO makes sense for aesthetics, to free up space around the socket, for truly extreme CPUs, or when the case height does not allow for a tower. Don't buy it thinking it's automatically superior.

3.4 Drive Bays

We distinguish three families:

- ****3.5" (mechanical HDDs).**** These require a drive cage with trays, ideally cushioned (anti-vibration grommets). The trend in recent years is to reduce their number: many modern cases offer 2, some "airflow-first" only 1, some SFF zero. If you have an archive of mechanical drives, ****count them first****.
- ****2.5" (SATA SSDs and old laptop HDDs).**** These take up little space and mount on trays behind the motherboard tray or on the PSU shroud. Typically 2-4 positions.
- ****M.2 (NVMe SSDs).**** These do not concern the case: they mount directly on the motherboard. This is why modern cases have fewer and fewer bays — most new systems have no physical drives other than NVMe.

It's worth noting that front 3.5" drive cages, when present, **are an obstacle to airflow**: they are placed exactly behind the intake fans. Modern cases make them removable precisely for this reason. If you don't need them, remove them.

3.5 Power Supply: position, length, shroud

Virtually all modern cases mount the power supply unit (PSU) **at the bottom**, with its fan facing downwards and a dust filter underneath. This thermally isolates the PSU from the rest of the system: it draws cool air from beneath the case and exhausts directly out the back, without exchanging heat with the internal

air. This is a huge improvement over 2000s cases, where the top-mounted PSU drew hot air from the CPU.

Two practical implications:

1. **The case must have sufficiently tall feet** (at least 20-25 mm) for the PSU to breathe. On a long-pile carpet, even 25 mm might not be enough: the carpet obstructs it and the PSU heats up. **Do not place the PC on a carpet.** 2. **The maximum PSU length** is stated by the case (typically 160-200 mm). Standard ATX PSUs are 140-160 mm deep; high-power ones (1200 W+) can reach 180-200 mm. If the case states 180 mm "without front drive cage" and you want the cage, the actual number decreases.

The **PSU shroud** (or "basement cover") is the horizontal sheet metal cover that hides the power supply and cables. It's aesthetic but also functional: it creates a separate compartment for the tangle of cables. Some cases have a **perforated** shroud, which allows fans to be mounted above it; others have a solid one, which can be a problem in a specific case: if you have a low front intake fan and the shroud separates it from the rest, that air won't reach the GPU.

3.6 Cable management

The space behind the motherboard tray is where all the cables run: the 24-pin ATX, the 8-pin EPS CPU cables that need to go up to the top left, the PCIe/12V-2×6 GPU cables, SATA, fan cables, and front panel cables.

- ****Space behind the tray****: under 20 mm is tight, 25 mm is acceptable, 30+ mm is comfortable. This is where budget cases cut corners.
- ****Rubber grommets****: rubber rings in the tray holes that hide edges and hold cables in place. Their absence is not a functional defect, but an aesthetic and practical one.
- ****Channels and Velcro straps****: well-designed modern cases (Lian Li, Fractal, Phanteks) integrate channels with snap-on covers and pre-mounted Velcro straps. They make a real difference in assembly time.
- Tidy cable management ****does not dramatically improve temperatures**** in a modern case (most cables are behind the tray, out of the airflow), but a bundle of cables left hanging in front of the intake fans certainly worsens them. And most importantly: the PC will need to be reopened, and finding it organized is a blessing.

3.7 Vertical GPU mounting and PCIe risers

Vertical mounting — the graphics card rotated 90° and facing the glass panel — is purely aesthetic and requires two things: a vertical bracket (sometimes included, often bought separately) and a **PCIe riser cable**, which is a flexible extension that carries the signal from the motherboard slot to the rotated graphics card.

There are two serious caveats, and they should be taken seriously:

- 1. The PCIe generation of the riser must match.** Risers are sold as "PCIe 3.0", "PCIe 4.0", or "PCIe 5.0". The PCIe bus doubles its signaling frequency with each generation, and signal integrity over a flexible cable becomes progressively more difficult. **A PCIe 3.0 riser used with a PCIe 4.0/5.0 card causes transmission errors**, which manifest as crashes, artifacts, black screens, or — more insidiously — as an automatic downgrade of the link to a lower speed that costs you performance without telling you. Always check in BIOS/UEFI or with GPU-Z that the link is negotiated at the expected generation and width (x16). A quality PCIe 5.0 riser costs 60-100 euros **[volatile data]**; if the budget doesn't allow it, do not mount vertically.
- 2. Vertical mounting suffocates the card.** Rotated against the side panel, the GPU is 15-30 mm from the glass, and its fans draw air from there. In many cases, this means +8/+15 °C compared to horizontal mounting. Cases designed for vertical mounting (some Lian Li, Hyte Y60/Y70) leave more space or use the side panel as an air intake. **If your case is not designed for this, vertical mounting is a thermal downgrade paid at a high price.**

4. Airflow: the serious discussion

4.1 Mesh vs. glass: how much does it really matter

The most frequent question and the one with the clearest answer. The **front panel** is the point from which, in most layouts, fresh air enters. If the front is a tempered glass or solid acrylic panel, air does not pass **through** it: it must squeeze into the side (typically 10-20 mm on each side) or top gaps. Intake fans, which are designed to work with low static pressure, encounter an obstacle and see their actual flow rate plummet: anechoic chamber tests show flow reductions in the order of 40-70% compared to the same fan in free air.

The effect on components, in comparative tests between the mesh variant and the glass variant **of the exact same case** (an experiment performed multiple times by specialized press, for example on the old NZXT H510 vs H510 Flow, or on the Corsair 4000D vs 4000D Airflow), is typically:

- ****GPU: 8 to 15 °C higher**** with a closed front panel, resulting in a drop in sustained boost;
- ****CPU: 3 to 8 °C higher****, less dramatic because the CPU cooler is higher up and closer to the rear exhaust fan;
- ****Noise: equal or higher**** with a closed front panel, because the fans have to spin faster and because air being choked in the gaps generates turbulence.

The operational conclusion is brutal and admits no nuances: **the front panel must be mesh, or at least have a significant and direct open surface in front of the fans**. Glass on the front is a design flaw disguised as an aesthetic feature. The **side** glass, however, is almost always harmless, because in the vast majority of layouts, the side is not a primary airflow path.

A small technical note to avoid naivety: not all "mesh" is created equal. A grid with small, dense round holes and a lot of material between holes can have an **open area** of 40%, while a well-designed honeycomb mesh can reach 65-75%. And perforated panels with decorative "tight-knit" patterns for aesthetics are often worse than they seem. If a review mentions "restrictive mesh," it's a warning.

4.2 Positive, Negative, Neutral Pressure

Let's define precisely. In a steady state, the mass of air entering the case must equal the mass exiting — this is conservation of mass, there's no choice. What changes is **where** it exits.

- ****Positive pressure****: The airflow of intake fans is greater than that of exhaust fans. The excess air exits through gaps, bracket holes, and crevices. The case is "inflated."
- ****Negative pressure****: Exhaust airflow exceeds intake. The missing air is sucked in through every gap, every unfiltered hole, the back of the case, the PCIe bracket bay.
- ****Neutral pressure****: Essentially in equilibrium.

Why do we care? **For dust**. In positive pressure, all incoming air passes through the intake fans, and these are the only point where **dust filters** exist. Filtered air enters, and the air exiting through the gaps pushes out dust particles. The case

remains reasonably clean, and dust accumulates on the filters, where it can be removed in thirty seconds.

In negative pressure, air enters everywhere, and "everywhere" is unfiltered. Dust settles on heatsinks, fins, cards — where cleaning is much more laborious.

Recommendation: slight positive pressure. In practice, this means having an intake airflow 10-30% higher than the exhaust. Be careful not to confuse the number of fans with airflow, however: three intake fans behind a dense filter can move less air than two exhaust fans in open air. The filter is resistance, and resistance reduces airflow.

It must also be said honestly, because it's treated like a religion online: **the thermal difference between the three configurations is 1-3 °C**, which is almost irrelevant. It's maintenance, not temperature, that's the serious argument for positive pressure. A well-ventilated negative pressure case is cooler than a poorly-ventilated positive pressure one. The priority remains the **volume of air exchanged**, not the sign of the differential.

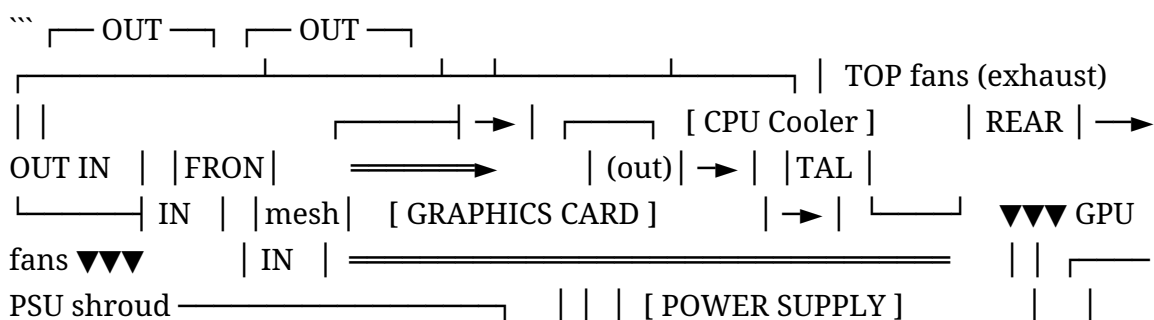
4.3 Airflow Path

Two principles guide the layout:

Principle 1: Hot air rises. Due to natural convection, heated air expands, decreases in density, and rises. This effect, by itself, is weak compared to forced ventilation, but it gives the correct direction: **intake low and front, exhaust high and rear** means working with physics rather than against it.

Principle 2: The path must be short and direct. Fresh air must reach the heatsinks (GPU at the bottom, CPU in the middle-top) without mixing with already used hot air.

The canonical layout, valid for 90% of mid towers:



┌───┴───┐ | IN (air from below, filtered) ``

- **Front intake (2-3 fans)**: fresh air directly hitting the graphics card, which is the component that dissipates the most heat.
- **Bottom intake (optional, 1-2 fans)**: if the case allows, fans on the bottom in front of the GPU are extremely effective, as they blow fresh air directly into the graphics card fans. It's one of the most underrated and most effective configurations overall. Requires high feet and a filter.
- **Rear exhaust (1 fan, 120 or 140 mm)**: immediately behind the CPU cooler, expelling the air that has just passed through it before it recirculates.
- **Top exhaust (1-3 fans)**: collects rising hot air. If you mount an AIO radiator, this is its natural location.

Some counterintuitive but verified notes:

- **Do not put intake fans at the top.** They fight against convection and recycle hot air from the AIO.
- **The glass side panel is not a thermal problem** as long as the front is open. Those who want a glass side can have it without guilt.
- **In a dual-chamber case like the Lian Li O11**, the front is often closed, and the air intakes are **the bottom and the side**. These cases work very well, but **only if you configure them as designed**: intake from bottom + side, exhaust from top + rear. Those who mount all fans at the top for exhaust and no air intake create a vacuum and get terrible temperatures.

4.4 Typical configurations by number of fans

Fans	Configuration	Notes
2	1 front IN + 1 rear OUT	The bare minimum. Only acceptable for low-power systems (< 250 W total).
3	2 front IN + 1 rear OUT	Sensible basic configuration. Slight positive pressure. Most common in entry-level cases.
4	3 front IN + 1 rear OUT	Excellent compromise. Still positive pressure. Standard for mid-range

		"airflow" cases.
5	3 front IN + 1 rear OUT + 1 top OUT	Balanced. Good for high-end air-cooled systems.
6	3 front IN + 1 rear OUT + 2 top OUT	Balanced/neutral. Typical configuration with a 240/280 AIO at the top.
7+	3 front IN + 2/3 bottom IN + 1 rear OUT + 3 top OUT	Enthusiast configuration. Marked positive pressure, excellent GPU temperatures.

Marginal returns decrease rapidly. The jump from 2 to 4 fans is worth many degrees. The jump from 6 to 9 is worth 1-2 °C and adds noise. The fourth fan is an excellent investment; the ninth is a decoration.

4.5 Dust filters

A filter is a fine mesh (nylon or steel) placed in front of air intakes. It does two things: it traps dust and **resists airflow**. Resistance costs flow, typically 10-20% with a clean filter and much more with a dirty filter.

Where they are really needed:

- ****Under the PSU****: mandatory, and must be ****removable from the back or side****, not from underneath (otherwise you have to lift the PC every time).
- ****Front (in front of intake fans)****: yes, and it's the main defense.
- ****Bottom (if there are intake fans)****: yes.
- ****Top (if fans are exhausting)****: ****no, or only magnetic and removable****. A filter on an exhaust surface filters nothing — air exits — and only adds resistance. It only serves to prevent dust from settling inside ****when the PC is off****. If your case has a magnetic filter on top and the PC is almost always on, you can remove it and gain a few percentage points of exhaust.

Maintenance: filters should be cleaned every 1-3 months depending on the environment (case on the floor + pets + smoke = every month). Compressed air or a simple rinse, with the precaution of **drying them completely** before reassembling.

5. Case fans

5.1 Sizes: 120, 140, 200 mm

A fan is, physically, a propeller that moves a volume of air. For the same **airflow** (volume of air per second), a larger fan can rotate more slowly, because each rotation moves more air. And since aerodynamic noise depends strongly on the **peripheral speed of the blades**, rotating more slowly means making less noise.

This is why, **for the same airflow, a 140 mm fan is quieter than a 120 mm fan, and a 200 mm fan is quieter than both**. It's a simple and reliable principle.

However, there are practical counter-arguments:

- ****140mm fans have fewer mounting positions.**** Many cases accept 3 x 120mm but only 2 x 140mm on the front, and in that case, ****3 x 120mm move more air than 2 x 140mm****, negating the advantage.
- ****200mm fans have a very limited model selection****, rotate slowly (600-800 RPM), have ****very low static pressure****, and are therefore unusable on radiators or behind dense filters. They are excellent for free intake in cases designed for them, but they are a niche.
- ****140mm radiators**** (280, 420) are slightly more efficient than their 120mm counterparts (240, 360) for the same length, but are supported by fewer cases.

Rule of thumb: use 140mm fans when the case accepts the same number as 120mm fans; otherwise, use 120mm fans and be happy.

5.2 Specifications: reading them without being misled

Every fan has a spec sheet with four key numbers. They should be read together, never in isolation.

CFM (Cubic Feet per Minute). This is the **airflow**, meaning how much air the fan moves *under zero resistance conditions*, in free air. Typical values: 50-80 CFM for a 120mm fan, 60-100 for a 140mm fan. In Europe, it can also be expressed in m³/h (1 CFM ≈ 1.7 m³/h). **The stated number is the theoretical maximum and you will never see it in reality**, because there is always resistance inside the case.

Static pressure (mmH₂O, millimeters of water column). This is the fan's ability to **push air against resistance**. Typical values: 1.0-1.5 mmH₂O for an "airflow" fan, 2.0-3.5 mmH₂O for a high-speed "static pressure" fan. This is the parameter that matters when the fan needs to push air through:

- the dense fins of a **radiator**;
- the fins of an **air cooler**;
- a dense **dust filter**;
- a restrictive front grille.

When to prioritize one or the other: a fan in **free intake** or **free exhaust** (in front of a hole, essentially) benefits from airflow. A fan on a **radiator or cooler** benefits from static pressure. A fan in front of a filter is somewhere in between. In the reality of 2026, the best "hybrid" fans (Arctic P12/P14, Noctua NF-A12x25, Phanteks T30) are good in both regimes, and the distinction is less dramatic than it used to be — however, it is still true that a fan explicitly declared "high airflow" with thin, wide blades (like Noctua NF-S12, or the old "airflow" Corsair AF) performs poorly on a radiator.

RPM (Revolutions Per Minute). The rotational speed. Together with the diameter, it determines the peripheral speed and thus the noise. A fan with a 200-2000 RPM range is much more versatile than one with an 800-1200 range: the former can be made almost inaudible at low loads and pushed to maximum under stress.

dBa (decibel A-weighted). Noise, measured with a weighting that reflects the sensitivity of the human ear. **This is the least reliable number on the spec sheet**, because measurement conditions (distance, anechoic chamber, load) are not standardized among manufacturers, and because dBA does not capture the **quality** of the noise. A 30 dBA fan with a low-frequency hum is more tolerable than a 28 dBA fan with a high-pitched whine or bearing tick. **Stated dBA values are only useful for comparing fans from the same manufacturer.** For cross-brand comparisons, trust independent measurements that normalize noise (e.g., "at 35 dBA, how many CFM does it move?", which is the correct way to compare).

One last pitfall: **PQ** (pressure-flow) curves. A serious manufacturer publishes a graph showing how much airflow the fan delivers at each level of backpressure. This is the only complete data. Noctua, Arctic, and Phanteks do this; many others do not.

5.3 PWM (4 pin) vs. DC (3 pin)

The connector for system fans is **standardized**:

Pin	Function
1	GND (Ground)

2	+12 V
3	Sense / Tachometer (reads RPM)
4	PWM (control signal)

A **3-pin fan** is regulated by **DC (Direct Current)**: the motherboard lowers the supply voltage, from 12 V downwards, and the fan slows down. This is simple, but has a limitation: below a certain voltage (typically 5-6 V), the motor does not have enough torque to start or maintain rotation, so the minimum speed is relatively high (often not dropping below 40-50% of nominal speed).

A **4-pin fan** uses **PWM** (Pulse Width Modulation): the voltage always remains at 12 V, but a square wave signal (25 kHz for the Intel standard) rapidly switches the motor's drive circuit on and off. By varying the **duty cycle** (the percentage of time the signal is high), the speed is controlled. The advantage is enormous: **starting torque remains full**, so it can go down to very low speeds (10-20%, i.e., 200-300 RPM) with reliable startup, and control is much more precise and linear.

Recommendation: buy PWM fans. The extra cost is only a few euros and is money well spent. If your motherboard has a 4-pin header (all modern ones do), you can still set it to DC mode to drive 3-pin fans — most BIOS/UEFI allow this, and some automatically detect the type.

Control from BIOS/UEFI. Every modern motherboard has a section (Q-Fan on ASUS, Smart Fan on Gigabyte/MSI, Fan Xpert, etc.) where you define a **fan curve**: a function that links fan speed to a reference temperature. Three non-trivial tips:

- 1. Choose the right temperature source.** If you tie your case fans to the **CPU package** temperature, you'll have fans that constantly speed up and slow down, because the temperature of a modern CPU fluctuates by 20 °C in half a second with every micro-load. The result is the infamous **fan surging**, the PC's "panting," and it's much more annoying than a constant noise. Instead, tie them to the **motherboard / VRM / chipset** temperature (a large thermal mass that changes slowly) or, if the BIOS allows, to the GPU.
- 2. Use "smoothing" / "step up-down time":** almost all BIOSes allow you to set a delay (in seconds) before the fan reacts to a change. Set it to 3-6 seconds. This eliminates surging almost completely.
- 3. Don't aim for zero-RPM on the case.** Some fans stop completely below a certain threshold. This is fine for the GPU; for case fans, a permanent minimum flow at 400-500 RPM is inaudible and keeps the air moving.

Hubs and daisy-chains. Motherboards typically have 4-7 fan headers, and a well-populated case can have 8-9. Solutions:

- **Passive hub / splitter**: a splitter that connects multiple fans to a single header. Caution: **each header has a current limit**, typically 1 A (12 W). A PWM fan consumes 0.1-0.3 A, so 3-4 fans per header is the prudent limit. Also, only **one** fan can report the tachometer signal; the others are ignored (so the BIOS will show only one RPM value).
- **Active hub**: powered directly from the PSU (with a SATA or Molex connector) and only receives the PWM signal from the motherboard. It can handle 6-10 fans without problems. This is the correct solution for a system with many fans, and many mid/high-end cases include one.
- **Daisy-chain**: some fan series (Lian Li Uni Fan, Corsair iCUE Link, Phanteks D30) physically clip to each other and pass power and data through contacts, eliminating almost all cables. This is a very elegant solution for cable management and is one of the reasons these products are expensive. The downside: **they tie you to a proprietary ecosystem**, with a dedicated controller and dedicated software.

5.4 Bearings

The bearing is what supports the rotating shaft. It determines **mechanical noise, lifespan, and allowed mounting orientation**.

Type	Description	Typical Lifespan (MTBF)	Noise	Notes
Sleeve	Oil-lubricated cylindrical bushing	20,000-30,000 h	Quiet when new	Lubricant leaks if mounted horizontally ; wears out and becomes noisy. Inexpensive.
Ball bearing	Caged balls	50,000-75,000 h	Louder (humming/ticking)	Durable, tolerates heat and any orientation. Used in server environments.

FDB (Fluid Dynamic Bearing)	Sleeve with hydrodynamic pressure oil film and sealing	60,000-150,000 h	Very quiet	The best compromise. De facto standard in quality fans. Commercial names: SSO2 (Noctua), Rifle, Hydro, HDB.
Maglev / Magnetic levitation	Shaft supported by magnetic field, almost no friction	100,000+ h	Extremely quiet	Expensive, heavy marketing, real but marginal benefits compared to a good FDB.

Purchase rule: look for FDB (or commercial equivalents) and ignore everything else. Sleeve bearings are only acceptable in pre-installed fans in budget cases and only if mounted vertically. Maglev is a luxury.

One last honest observation: stated MTBF (Mean Time Between Failures) durations—150,000 hours is 17 years—are statistical extrapolations from accelerated tests, not promises. They are for comparison, not for planning.

5.5 Fan Brands

Brand	Reference Models	Indicative Price (each)	Positioning
Arctic	P12 / P14 (PWM PST), P12 Max	6-13 €	The absolute king of price/performance. Performance 3-5% from the top, one-fifth the price. Spartan aesthetics, slightly stiff cable. If budget matters, this is the answer.

Noctua	NF-A12x25 (the reference), NF-A14, NF-P12 redux	20-35 €	The technical gold standard. Minimal tolerances (0.5 mm blade clearance), excellent static pressure and airflow, exemplary quietness, 6-year warranty. Divisive brown-beige color (black "chromax" line exists, which costs more).
Phanteks	T30-120	25-35 €	30 mm thick fan, three speed modes, off-the-charts performance on radiators. Non-standard footprint.
be quiet!	Silent Wings 4 / Pro 4	20-30 €	Focused on silence. Excellent, a step below Noctua in static pressure.
Lian Li	Uni Fan SL/TL/SL-Infinity	20-30 € (kit of 3: 60-90 €)	Daisy-chain, spectacular RGB, good performance. Closed ecosystem.
Corsair	RS/QX, iCUE Link LX/RX	20-35 €	Good, but the value is in the iCUE software and the Link ecosystem. Expensive.
Thermalright	TL-C12 / TL-B12	5-9 €	Very cheap Chinese alternative,

			surprisingly good. Variable build quality.
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[volatile data: prices change frequently and models are refreshed every 18-24 months.]

When it makes sense to spend. The jump from a generic pre-installed fan to an Arctic P12 is worth a lot: better temperatures, lower noise, 8 euros. The jump from an Arctic P12 to a Noctua NF-A12x25 is worth an extra 25 euros for **2-4 °C and a couple of dB**. It's worth it if: (a) the PC is half a meter from your head in a quiet room; (b) the fan is on a radiator or cooler, where Noctua's superiority in static pressure is expressed; (c) you have already optimized everything else. **It's not worth it** if you still have a gasping GPU in a closed case: fix the case first.

6. Case Brands

There is no "absolute best": there are companies with different philosophies and levels of refinement. Here's the overview, with the caveat that **specific models age quickly** and need to be verified **[volatile data]**.

Lian Li. Historically a manufacturer of premium aluminum, today the strongest name in the enthusiast segment thanks to the **O11** family (O11 Dynamic, O11 Dynamic EVO, EVO XL, and derivatives). These are dual-chamber cases: the main chamber houses the motherboard and GPU, while the side chamber conceals the PSU and cables. They offer enormous radiator support (up to three 360mm radiators). Excellent construction, superb ergonomics. **Note: the front panel is closed.** They should be configured with intake from the bottom and side. The O11 Vision and the "aquarium" versions with three glass panels require even more care. The **Lancool** line (203/216/207) is also excellent; it's their airflow-first offering at much more contained prices and often provides the best value for money in the entire market.

Fractal Design. The Swedish company with the best reputation for design and finish. Three lines to know: **Meshify** (airflow-first, angular mesh front, the archetype of a well-ventilated case), **North** (Scandinavian aesthetic with a walnut wood front — and, notably, with a real mesh behind the wood, making it beautiful *and* functional), **Torrent** (the extreme airflow case, with two enormous 180mm

front fans, reference thermal performance, and a very particular layout). Also, the **Define** line for those who still want sound dampening. Construction, tolerances, and accessories are always top-notch. Mid-to-high prices.

be quiet! German company. Historically focused on silence (the **Silent Base**, **Dark Base** lines), but with a decisive shift towards airflow with the **Pure Base 500DX / 500FX** and **Shadow Base** lines. Included fans are of excellent quality (their fans are a real strong point). Solid construction, sober and "adult" aesthetic.

Corsair. Wide range of products, from the **4000D Airflow** (for years the default choice for an affordable and well-ventilated mid-tower) to the **iCUE** series with its integrated RGB ecosystem and sensors. Powerful but heavy iCUE software. Solid quality, medium prices. The "Airflow" version of each model is the one to buy; the plain version with a closed front panel is not.

NZXT. Very recognizable minimalist aesthetic (the **H** line). Historically, they got airflow wrong (the H510 with its closed front panel is a textbook example of this error) and then corrected it with the **Flow** versions. Good construction, prices a bit high for what they offer, excellent internal ergonomics. If you buy NZXT, buy a Flow model.

Phanteks. Highly regarded by enthusiasts. **Eclipse** lines (P400/P500 and derivatives) and **NV** (dual-chamber glass cases). Excellent internal engineering, many intelligent solutions for cable management, often generous fan configurations.

Cooler Master. Very wide range, variable quality. The **NR200 / NR200P** is the benchmark in accessible SFF. The older **MasterBox** series was the budget choice for years. Check model by model: qualitative consistency is not their strong suit.

Montech, DeepCool, Antec, Thermaltake. The aggressive budget segment. **Montech** in particular (AIR, King, XR series) has earned a notable reputation by offering cases with 3-6 ARGB fans included at prices ranging from 60-90 euros, with honest construction. **DeepCool** (CH and Morpheus series) is similar. They are the rational choice for those on a tight budget: **you accept thinner sheet metal and less refined finishing in exchange for excellent airflow and included fans.** This is a much smarter compromise than buying a "premium" closed-off case.

6.1 What distinguishes a well-designed case

Beyond the brand, here are the concrete signs of quality, which you can verify in a video review:

1. **Sheet metal thickness** (0.8-1.0 mm in well-made cases, 0.55-0.6 mm in budget ones). Thin sheet metal flexes, vibrates, and resonates. 2. **Rolled edges**, no sharp cuts. 3. **Tolerances**: panels fit together, screws go in without forcing, glass doesn't "rub." 4. **Assembly ergonomics**: cable channels, pre-installed Velcro straps, tie-down points, included fan hub, removable PSU bracket, removable filters without disassembling anything. 5. **Fan provision**: how many are included and of what quality. A 90-euro case with 4 decent PWM fans costs less than an 80-euro case with zero fans (to which 40 euros for Arctic P12s must be added). 6. **Front I/O**: number and type of USB ports. In 2026, demand at least **one USB-C** (ideally USB 3.2 Gen 2 at 10 Gbps or higher) and verify that the motherboard has the corresponding internal header — it's a classic mistake to buy a case with front USB-C and a cheap motherboard that doesn't have the header to connect it. 7. **Documentation and spare parts**: serious manufacturers sell spare parts (screws, filters, panels, feet). Fractal, Lian Li, be quiet! do.

7. How to choose: the procedure

Follow this order. **The order is the point**: reversing it is the cause of almost all bad purchases.

Step 1 — Motherboard format. You have already chosen the motherboard (or at least its format). This defines the minimum: ITX → everything; mATX → mid/mini tower; ATX → mid tower or larger; E-ATX → check the **width in millimeters**, not the acronym.

Step 2 — Clearance. Write down the three numbers of your configuration and compare them with the case's technical specifications, with a margin:

- **GPU** length (specific model) + 15 mm margin for cables;
- **CPU cooler** height + 5 mm;
- **radiator** size and thickness (if AIO) and its position;
- **PSU** length;
- number of **3.5" drives**.

Step 3 — Airflow. The front **must** be mesh or otherwise have a direct airflow path. If it isn't, the case is excluded — unless it's a dual-chamber designed to draw air from the bottom and side, in which case it's fine but you need to configure it correctly. Check how many fan positions there are and where.

Step 4 — Quality and features. How many fans are included? What type (PWM or 3-pin)? Is there a hub? What is the thickness of the sheet metal? How much space is behind the tray? **Calculate the real total cost:** case price + necessary additional fans.

Step 5 — Aesthetics. Now, and only now, choose the one you like best from those that survived steps 1-4.

Step 6 — Budget. If your preferred choice exceeds the budget, go back to step 5 and pick the second one. Do not go back to step 3.

7.1 Typical buyer mistakes

1. The "aquarium" case. Three glass panels, RGB lighting, zero air vents. It looks beautiful in photos and bakes components. If you still want one, know that you're paying 10-15 °C on the GPU, and at least configure it with robust exhaust and intake fans from the bottom, if possible.

2. Buying the case before the GPU. The graphics card is the bulkiest and most variable component. Choose it first, or at least decide on the tier and get a case with generous clearance (350+ mm).

3. Trusting the chip's designation instead of the model. "Will an RTX 5080 fit?" is not a question with a single answer. The right question is: "Will this specific RTX 5080, which is 336 mm long and 3.2 slots thick, fit?"

4. Cases without included fans. Some cases, especially "premium minimalist" ones, are sold with zero or only one fan. The price seems competitive until you add €40-70 for fans. **Count the total cost.**

5. Ignoring panel noise. A thin side panel that vibrates at a certain frequency can make an otherwise silent system unbearable. This defect doesn't appear in any spec sheet: you only discover it in reviews.

6. Putting all fans on exhaust. "The more air goes out, the cooler it is" is false reasoning. The case becomes depressurized, air enters from every crevice (unfiltered), and the actual airflow drops because the fans are working against

enormous resistance. You need a path: **air enters from somewhere, exits from somewhere else.**

7. Mounting fans backward. Every fan has a direction. Airflow goes **from the side with the motor support grille towards the side with the blades** — or, more simply: look at the **arrows printed on the fan frame** (one indicates airflow, the other rotation direction). In the absence of arrows: air exits from the side where the four/seven struts supporting the hub are located.

8. The PC on the floor on carpet. Obstructs the PSU's air intake and turns the bottom into a vacuum cleaner. At least a hard surface, preferably raised.

9. Choosing SFF as a first build. This leads to an expensive dead end.

10. Neglecting the USB-C header. Verify that the motherboard has the internal "USB 3.2 Gen 2 Type-C" connector (20-1 pin) if the case has a front USB-C. Otherwise, that port will remain a decorative hole.

7.2 Budget-tier recommendations

[Volatile data: prices and models. Verify before purchase.]

Entry (€50-90). The goal is to maximize airflow and included fans, accepting thin sheet metal and simple finishes. Look at **Montech** (AIR / XR series), **DeepCool** (CH series), **Cooler Master** (MasterBox), **Lian Li Lancool 207/216** if the price drops into this range. Look for: mesh front, 3-4 included fans (ideally PWM), GPU support ≥ 330 mm, cooler ≥ 160 mm. Do not look for: glass on three sides, elaborate RGB.

Mid (€90-160). This is the tier with the best value-for-money in the entire market. **Fractal Design North / Pop Air / Meshify, Lian Li Lancool 216/207, Corsair 4000D Airflow, be quiet! Pure Base 500DX, Phanteks Eclipse, NZXT H7 Flow.** Expect: decent sheet metal, excellent ergonomics, 3-4 quality fans included, 360 AIO support, neat cable management, front USB-C. **If you have no special needs, your case is here.**

High-end (160-300+ €). You pay for build quality, aluminum, modularity, fan ecosystems, multi-radiator support. **Lian Li O11 Dynamic EVO / EVO XL, Fractal Design Torrent** (the temperature champion) or **Define/Meshify 2 XL, be quiet! Dark Base, Hyte Y70.** Buy for a specific reason: custom loop, dual 360mm radiator, well-designed vertical GPU, strong aesthetic requirement. **Do not buy it**

expecting drastically better temperatures than a good 120 euro case: the difference exists but is small.

Note on SFF. In this format, there is no sensible entry-level. The serious entry ticket starts from the Cooler Master NR200P (~100 €) and rises quickly, but it must be added to the premium for an SFX power supply (+40-70 €) and often a low-profile cooler and a short GPU. **The real cost of an SFF is 150-300 € higher than an equivalent mid-tower.**

8. Operational Summary — Case Selection Checklist

Print it, or copy it to a file, and fill it out before ordering.

A. Dimensional Constraints (mandatory: if one fails, the case is excluded)

- ☐ **Motherboard format**: my mobo is ____ (ITX / mATX / ATX / E-ATX of ____ mm width). Does the case support it? **Yes / No**
- ☐ **GPU length**: my specific model is ____ mm long. The case states ____ mm **in the configuration I will use** (with front radiator/fans installed). Margin ≥ 15 mm? **Yes / No**
- ☐ **GPU thickness**: ____ slots. Does the case have enough free slots below? **Yes / No**
- ☐ **Width clearance for the 12V-2×6 cable** (if the GPU uses it): is the side panel ≥ 30 mm from the connector? **Yes / No**
- ☐ **CPU cooler height**: mine is ____ mm high. The case states ____ mm. Margin ≥ 5 mm? **Yes / No**
- ☐ **Radiator** (if AIO): size ____ (240/280/360/420), radiator + fan thickness ____ mm. Does the case support it in the ____ position (top/front)? Does it interfere with RAM or VRM heatsinks? **Yes / No**
- ☐ **PSU length**: mine is ____ mm deep. The case states ____ mm (with the cages I intend to keep). **OK / KO**
- ☐ **3.5" drives**: I need _____. The case accepts _____. **OK / KO**

B. Airflow (mandatory)

- ☐ Is the **front mesh** or does it have a direct airflow path in front of the fans? If it's closed, is it a dual-chamber designed to draw air from the bottom/side? **Yes / No** — if No, discard.
- ☐ How many **fan positions** are there? Front ____ / Top ____ / Rear ____ / Bottom ____ / Side ____

- ☐ Can I achieve at least the minimum configuration **3 IN (front) + 1 OUT (rear)**? **Yes / No**
- ☐ Are there **dust filters** on all air intakes, and are they **removable** without disassembling the PC? **Yes / No**
- ☐ Are the **feet** ≥ 20 mm high for PSU air intake? **Yes / No**

C. Quality and Features

- ☐ **How many fans are included?** _____ — what type? (PWM 4-pin / DC 3-pin) — what quality?
- ☐ **Real total cost** = case price (_____ €) + additional fans (_____ €) = _____ €
- ☐ Is a **fan hub** included? **Yes / No** — Does my mobo have enough headers? _____
- ☐ Is the **space behind the motherboard tray** ≥ 25 mm? **Yes / No**
- ☐ Are there **rubber grommets, cable channels, Velcro straps**? **Yes / No**
- ☐ Do reviews report **thin sheet metal, vibrations, poor tolerances, sharp edges**? **Yes / No**
- ☐ Does the **front I/O** have a **USB-C**? And does my motherboard have the corresponding **internal header**? **Yes / No**

D. Configuration, after purchase

- ☐ **Front/bottom fans as intake**, **rear/top fans as exhaust**. Direction arrows checked on each fan.
- ☐ **Slight positive pressure**: intake airflow $\geq$ exhaust airflow.
- ☐ Fan curves in BIOS/UEFI **tied to a slow source** (mobo/VRM/GPU), **not to the CPU package**, with 3-6 second smoothing.
- ☐ **3.5" cages removed** if not needed, to free up front airflow.
- ☐ **Unused PCIe slot brackets reinstalled** in their place.
- ☐ **Top filter** removed if the PC is almost always on and the top is all exhaust (optional).
- ☐ **Standoffs** checked: one for each motherboard hole, **no extras**.
- ☐ Validation test: after 20 minutes of combined CPU+GPU stress test, GPU temperature is **below 80 °C** and the CPU is not thermally throttling. If not, the problem is the case or fan configuration, not the cooler.
- ☐ **Maintenance reminder**: clean filters every **1-3 months**.

E. The three golden rules, if you forget everything else

1. The front must breathe. Mesh, not glass. **2. Measure the GPU before buying the case.** The specific model, in millimeters. **3. Four good fans in a 100 euro case**

beat two fans in a 250 euro case. The volume of air exchanged is what cools; everything else is secondary.

End of chapter. The next chapter — the display peripheral (the monitor) — concludes the consumer part before moving on to professional and workstation hardware.

Chapter 8 - Cooling

8.1 Why cooling is needed: physics before marketing

Every active electronic component in a computer is, from a thermodynamic point of view, an electrical energy converter into thermal energy. This statement sounds brutal but is literally true: a processor does not "consume" current to produce computation and then dissipate some heat as a side effect. A processor absorbs electrical power and returns it entirely to the environment in the form of heat. Computation — information, the result of the operation — does not take away energy. If a CPU (Central Processing Unit) draws 200 watts from the motherboard's 12-volt line, those 200 watts become 200 thermal watts to dissipate. Not 180, not 150: two hundred. The cooling system is therefore not an accessory that "helps" the processor: it is the only way out for the energy that enters it, and its effectiveness directly determines the operating frequency, stability, and lifespan of the component.

The reason this heat is a problem lies in the physics of silicon. A MOSFET (Metal-Oxide-Semiconductor Field-Effect Transistor, the elementary building block of every modern digital circuit) switches between an on and an off state; with each switch, it dissipates energy to charge and discharge the parasitic capacitances of the circuit, and it also continuously loses energy due to leakage currents that flow through the device even when it should be off. Leakage current increases in a strongly non-linear way with temperature. This is where the thermal vicious cycle begins: a hot chip dissipates more power for the same work, and by dissipating more power, it gets even hotter. Simultaneously, the mobility of charge carriers in silicon decreases as temperature increases, meaning transistors switch more slowly, and to maintain the same frequency, more voltage would be needed — which in turn produces more heat. Hence the observable fact, verifiable on any test bench, that a well-cooled CPU consumes *fewer* watts than the same poorly cooled unit, even while performing the exact same work.

The second reason heat must be controlled is long-term degradation. Phenomena such as electromigration (the physical migration of metal atoms in interconnect traces, driven by electron flow) and NBTI (Negative Bias Temperature Instability, a progressive shift in the threshold voltage of transistors) have an exponential dependence on temperature, typically described by an Arrhenius equation. In practical terms: a chip operating at 90 °C ages significantly faster than one operating at 70 °C. This does not mean that a CPU at 90 °C will break tomorrow —

manufacturers dimension limits precisely to guarantee the expected useful life even at the limit — but it does mean that the thermal margin is a margin of longevity and quietness, not just performance.

8.1.1 Throttling: The Emergency Brake

Since the chip cannot physically prevent itself from heating up, manufacturers have integrated distributed temperature sensors (DTS, Digital Thermal Sensor) and protection logic into the silicon. When the die temperature reaches a threshold called **T_{jmax}** (maximum junction temperature, i.e., the maximum allowed temperature at the transistor junction), the processor automatically reduces frequency and voltage to stay within limits. This phenomenon is called **thermal throttling**. If, despite throttling, the temperature continues to rise to a second critical threshold, **THERMTRIP** intervenes, a hardware signal that immediately shuts down the machine to prevent permanent damage.

Throttling is not a malfunction: it is the system working as intended. But it is also an unequivocal sign that the cooling system is undersized for the load. Users perceive this as sudden frame rate drops in games, rendering times that worsen as work progresses (the first few minutes are fast, then it slows down), and benchmarks that return lower scores on the second run compared to the first.

It's important to understand that modern processors no longer have a "nameplate" frequency guaranteed under all conditions. Since around 2015, both Intel and AMD have adopted opportunistic boost algorithms—Intel Turbo Boost / Thermal Velocity Boost, AMD Precision Boost 2 and PBO (Precision Boost Overdrive)—that increase frequency until *any one* of the limits is reached: current limit, power limit, temperature limit. The first limit hit is the one that governs. Consequently, the cooler doesn't just "prevent throttling": the cooler *determines the operating frequency*. A Ryzen 9 or Core i9 with a mediocre cooler won't break; it will simply run slower and be noisier. The cooler, in this architecture, is effectively a performance component.

8.1.2 TDP, PL1/PL2, PPT: decoding power numbers

Here we need to be surgical, because marketing has made these acronyms almost useless if taken literally.

TDP stands for *Thermal Design Power*. Historically, it was the thermal power, expressed in watts, that the cooling system had to be able to dissipate for the

processor to operate at its nominal specifications. The problem is that neither of the two major manufacturers today uses TDP as a synonym for "maximum consumption."

For **Intel**, modern nomenclature distinguishes:

- **PBP** (Processor Base Power), which replaced the old TDP designation: this is the power dissipated when the processor operates at its *base frequency* with all cores active. It is a conservative value and, in real-world use, almost never representative.
- **MTP** (Maximum Turbo Power): the maximum power under boost conditions. This is the figure that matters for sizing the cooler.
- **PL1** and **PL2** (Power Limit 1 and Power Limit 2): PL1 is the long-term sustained power limit, PL2 is the peak limit allowed for a time interval called **Tau**. In high-end desktop configurations, many motherboard manufacturers set PL1 = PL2 and infinite Tau by default, meaning the processor *always* operates at its peak limit. This is precisely why a processor declared "125 W" can draw 250 W in a rendering, without anything being broken.

For **AMD**, on AM4 and AM5 platforms, the key acronym is **PPT** (Package Power Tracking), the overall power limit absorbed by the package. The approximate relationship historically used is $PPT \approx TDP \times 1.35$: a processor declared 105 W has a PPT around 142 W, one declared 170 W reaches about 230 W. AMD complements PPT with two other limits, **TDC** (Thermal Design Current, sustained current deliverable by the VRM under stabilized thermal conditions) and **EDC** (Electrical Design Current, instantaneous peak current); the VRM (Voltage Regulator Module, the motherboard's voltage regulation module, discussed in the chapter on motherboards) is the circuit that physically delivers this current.

The practical consequence is only one, and it should be etched in stone: **you don't choose a cooler by looking at the CPU's TDP; you choose it by looking at the real power consumption under heavy load** (MTP for Intel, PPT for AMD, or even better, independent measurements in reviews). A user who buys a cooler "suitable up to 150 W" for a processor "with a 125 W TDP" that draws 250 W in Cinebench has made a sizing error of 60%.

8.1.3 The concept that truly matters: thermal resistance

If there's one quantity worth learning, it's this one. A cooling system can be described by a **thermal resistance**, measured in degrees Celsius per watt (°C/W). It links the dissipated power to the temperature difference between the chip and the ambient air:

$$\Delta T = P \times R_{th}$$

$$\text{or: } T_{CPU} = T_{ambient} + (\text{Power in watts} \times \text{Thermal resistance in } ^\circ\text{C/W})$$

A good dual-fan tower cooler has an overall thermal resistance (from die to air) in the order of 0.10–0.15 °C/W at medium fan speed. Let's do the math: with 200 W dissipated and an R_{th} of 0.13 °C/W, the thermal difference is 26 °C. If the air inside the case is 35 °C (not 22, because it's warmer inside the case than in the room), the CPU will settle around 61 °C. With the same cooler and a 280 W processor, the difference becomes 36 °C, reaching 71 °C. With an inexpensive cooler at 0.25 °C/W and 280 W, the difference would be 70 °C, leading to 105 °C, which means permanent throttling.

This model, simplified as it is, explains almost everything observed in reality: it explains why doubling the cooler's surface area doesn't halve temperatures (because the total thermal resistance is the sum of multiple resistances in series, and that of the thermal paste and IHS doesn't change), it explains why ambient temperature counts one-to-one (every extra degree in the room is an extra degree on the CPU), and it explains why comparing coolers should always be done with the same dissipated power and ambient temperature.

The chain of thermal resistances, from the hot spot to the air, is as follows: die → internal TIM (the material joining the die to the IHS, solder or paste) → **IHS** (Integrated Heat Spreader, the metal lid soldered over the die that we see externally) → external thermal paste → cooler base → heatpipes → fins → air. The designer can only improve the latter links; the former are decided by Intel and AMD. This is why beyond a certain point, adding mass and fans yields diminishing returns: you are optimizing one link in the chain while the bottleneck is elsewhere.

8.1.4 Target Temperatures: What's Normal and What's Not

One of the topics with the most misinformation. Let's establish some benchmarks, noting that these are typical values at the time of writing and that exact limits

should always be checked in the technical specifications of your specific model, as they change from generation to generation.

Component	Idle (desktop, no load)	Gaming Load	Prolonged Heavy Load (rendering, compilation)	Intervention Limit (T_{jmax})
Intel Core desktop CPU (12th–14th gen, Core Ultra)	30–45 °C	55–75 °C	75–95 °C	~100 °C
AMD Ryzen 5000 CPU (AM4)	35–45 °C	60–75 °C	75–90 °C	~90 °C
AMD Ryzen 7000/9000 CPU (AM5)	35–50 °C	60–80 °C	85–95 °C	~95 °C (7000- series X3D lower, ~89 °C; 9800X3D back to ~95 °C)
GPU (die/core, recent GeForce and Radeon)	30–45 °C	65–80 °C	70–85 °C	~90 °C core
GPU — GDDR6/GDDR 6X memory (T_{mem})	—	70–95 °C	up to 100–105 °C	~105 °C (throttling)
GPU — hotspot / junction	—	75–95 °C	85–100 °C	~105–110 °C

Note on volatile data: T_{jmax} values and throttling thresholds change with generations and sometimes with microcode updates. The figures in the table are indicative and valid to the best of available knowledge; the authoritative data is that provided by the manufacturer for the specific SKU.

There are two recurring misconceptions to dispel immediately.

First misconception: "my CPU is at 90 °C, it's about to break." No. AMD Ryzen 7000 and 9000 series processors, in particular, are designed to *use* all available thermal budget: the boost algorithm raises the frequency until it reaches 95 °C, and then stays there. A Ryzen 9 that consistently operates at 95 °C under full load with a good cooler is not a sick system; it's a system doing exactly what it was designed to do. The difference between a mediocre cooler and an excellent one, on these chips, is not reflected in the temperature (which is always 95 °C) but in the *sustained frequency* and the *benchmark score*. Anyone evaluating their cooler by only looking at the thermometer, on these platforms, is looking at the wrong instrument.

Second misconception: "idle temperature is what matters." Also no. Resting temperature is the least informative of all, because it depends abnormally on OS boost transients (a background service waking up is enough to make the reading spike by 15 °C for half a second) and because at very low power, the thermal resistance of the system is almost irrelevant. A cooler is evaluated under constant and prolonged load, with at least 10–15 minutes of stress testing, measuring *simultaneously* temperature, power consumption, and frequency. Without all three numbers, the comparison is meaningless.

A third element must be introduced: the **air intake temperature**. No air cooler or AIO can bring the CPU below the temperature of the air entering it; this is a thermodynamic limit, not a defect. If the air inside the case is at 40 °C because the airflow is poorly designed, the cooler starts at 40 °C and adds its ΔT . This is why the chapter on case and ventilation and this chapter are, in reality, a single chapter divided into two: **an excellent cooler in a suffocated case performs like a mediocre cooler in a well-ventilated case**.

8.2 Air Cooling

8.2.1 Description and function: how a tower really works

A modern air cooler is composed of four functional elements, and it's worth understanding them one by one because the product's quality lies entirely in the details of how they are made.

The base (cold plate). This is the plate, almost always made of copper (sometimes nickel-plated for aesthetics and oxidation resistance), that rests on the CPU's IHS. Its task is to collect heat from a small, concentrated area. Two construction

philosophies compete in this field: the **solid base** (a machined copper block, into which the heatpipes are inserted and soldered) and **direct-touch** (the heatpipes are flattened and directly touch the IHS, without an intermediate plate). Direct-touch eliminates a thermal interface but leaves aluminum or copper grooves between the tubes, resulting in less uniform heat distribution; the solid base adds an interface but distributes heat better. In practice, both well-executed solutions yield equivalent results, while both poorly executed solutions (concave base, poor soldering, rough finish) yield terrible results. A detail often overlooked is the **flatness and convexity** of the base: many manufacturers make the base slightly convex to compensate for the deformation of the IHS under the pressure of the mounting system.

The heatpipes. These are the technological core and the reason why a 1 kg tower dissipates more heat than a 3 kg solid copper block. A heatpipe is a sealed copper tube, partially vacuumed, and containing a small amount of working fluid (typically water). Inside the wall, there is a capillary structure (**wick**): sintered copper powder, metallic mesh, or longitudinal grooves. The operation is a closed cycle of phase change. At the hot end (in contact with the CPU), the fluid evaporates, absorbing the **latent heat of vaporization** — an enormous amount of energy compared to that needed to heat the same fluid by a few degrees. The vapor, which occupies a much larger volume, expands and moves very quickly towards the cold end (immersed in the fins), where it condenses, releasing the latent heat. The condensed liquid is then returned by the capillary action of the wick. Since the partial vacuum lowers the boiling point of water to a few tens of degrees, the cycle works even at normal operating temperatures.

The result is that a heatpipe has an **effective thermal conductivity** orders of magnitude higher than that of solid copper (we are talking about thousands of equivalent W/m·K, compared to about 400 W/m·K for copper). This explains why: (a) serious coolers have 6, 7, or 8 heatpipes and not just one; (b) the number of heatpipes is a crude but real indicator of transport capacity; (c) there is a limit beyond which the heatpipe "dries out": if the power is too high, the liquid evaporates faster than the wick can return it, and the conductivity suddenly collapses. This is one of the reasons why air coolers have a fairly clear power ceiling.

A little-known corollary: **orientation matters**, though less than feared. A good quality sintered wick heatpipe works in any position, including against gravity,

but in the "worst" position (evaporator at the top, condenser at the bottom, with gravity opposing liquid return) it can lose a few percentage points of efficiency. In standard cases with a vertical motherboard, the tower works horizontally, which is a neutral position; in cases with a horizontal motherboard (some HTPCs, some bench cases), it's advisable to check the manufacturer's instructions.

The fin stack. This is the heat exchange surface with the air: dozens of thin aluminum fins (rarely copper, which is more efficient but much heavier and more expensive), threaded onto the heatpipes. Here, the key parameter is **fin density**, often expressed in FPI (Fins Per Inch). Dense fins offer more surface area but create more resistance to airflow: they require high **static pressure** fans and perform poorly with slow fans. Sparse fins offer less surface area but breathe well even at low speeds: they are the choice for silence-oriented coolers. There is no single "best" setting: there is a coherent pairing between fin density and fan curve. An excellent cooler has a fin stack *designed together* with its fan.

The way the fins are attached to the heatpipes is an underestimated quality differentiator: in good products, they are **soldered** or otherwise joined with a metallurgical bond; in economic products, they are only mechanically pressed, with an air-to-metal interface that introduces thermal resistance.

The fan. Provides the airflow that carries heat away from the fins. The two fundamental parameters are **airflow** (CFM, Cubic Feet per Minute — or m³/h) and **static pressure** (mm H₂O, millimeters of water column). Airflow indicates how much air it moves in the absence of obstacles; static pressure indicates how well it can push air *through* an obstacle. A fin stack is an obstacle. We will discuss this in detail in section 8.7, as the topic is common to both air and liquid cooling.

8.2.2 Formats: How to read the sizes

Air coolers are divided into three main families, distinguished by flow geometry.

Low profile. The fin stack is horizontal, above the socket, and the fan blows downwards (**top-flow**). Typical heights range from 25 to 70 mm. They were designed for Mini-ITX cases, HTPCs, compact office systems, and slim workstations. A collateral advantage is that the downward flow also cools the motherboard VRMs and RAM modules, which a tower cooler does not. The disadvantage is that the available surface area is small, and the fan, being thin and small, must spin fast: these coolers are noisy in proportion to the power dissipated. Typical references: the **Noctua NH-L9** series (approx. 37 mm, suitable

for CPUs up to approx. 65 W), the **Noctua NH-L12S** (approx. 70 mm, much more capable), the **Thermalright AXP90** and **AXP120** family, the **ID-Cooling IS** coolers for Mini-ITX.

Single tower. A single vertical fin stack, with one fan in front (push configuration) and sometimes a second behind (push-pull). Typical heights 120–160 mm. This is the dominant format in the mid-range: a good compromise between footprint, performance, and price. References: **Thermalright Assassin X120 Refined**, **Arctic Freezer 36**, **DeepCool AK400**, **be quiet! Pure Rock 2**, ****Noctua NH-U12S / NH-U12A**

In terms of brand positioning, some clear lines can be drawn. **Thermalright** has overturned the market in recent years by offering high-end performance at low-end prices; the trade-off is a lower perceived quality of materials and mounting kits, and less structured after-sales support in Europe. **Noctua** represents the opposite extreme: high prices, but excellent mounting systems (SecuFirm2), some of the best fans on the market (NF-A and NF-P), impeccable documentation, a long warranty, and — a detail worth its weight in gold — **free or nearly free adaptation kits for new sockets**, which means a Noctua cooler can survive multiple platform generations. **be quiet!** builds on silence, with acoustically refined solutions and a very understated matte black aesthetic. **DeepCool** and **Arctic** occupy the middle ground with solid products and competitive prices; Arctic, in particular, is aggressive on price and generous with its warranty.

Commercial context note, potentially volatile: the availability of some brands may vary due to regulatory or regional distribution reasons. Before ordering, verify availability and support in your market.

8.2.4 Pros and cons of air cooling, without rhetoric

Pros. Reliability is the decisive point: an air cooler has no wear-prone parts except the fan bearing, which can be replaced in two minutes and costs 15–20 euros. There's no liquid, no gaskets, nothing that can evaporate, gel, or leak. The practical lifespan is that of the bearing, and with good bearings (rifle bearing, FDB — Fluid Dynamic Bearing, Noctua's SSO2) we're talking about 100,000–150,000 hours MTTF (Mean Time To Failure). In concrete terms: a good air cooler can accompany three or four successive builds. The cost is unbeatable: 40 euros today buys performance that required 100 euros ten years ago. Maintenance is zero, excluding periodic dust removal. And **noise**: for the same performance, a well-designed tower is almost always quieter than an AIO, because it doesn't have a

pump — which is a source of continuous noise, often at annoying frequencies (low-frequency hum, gurgling).

Cons. Bulk is the Achilles' heel. A dual-tower cooler 165 mm high and weighing 1.5 kg creates three categories of problems: (1) it might not fit in the case; (2) it might interfere with RAM modules, especially those with tall heat spreaders; (3) it might interfere with the first PCIe (Peripheral Component Interconnect Express, the expansion bus discussed in the motherboard chapter) slot or with M.2 SSD heatsinks. The weight, moreover, exerts torque on the motherboard: during transport, it's advisable to remove heavier coolers or transport the case horizontally. Finally, aesthetics: a black tower covers half the motherboard and hides the RAM; for those building a glass-sided system with lighting, it's a sacrifice. It must be honestly said that it's an *aesthetic* sacrifice, not a technical one.

8.2.5 Compatibility: the three measurements to always check

This is where most purchasing errors occur. Before buying an air cooler, three checks are mandatory.

1. The socket. The cooler kit must include the mounting system for your motherboard's socket. Relevant sockets today: **LGA1700** (Intel 12th/13th/14th generation), **LGA1851** (Intel Core Ultra 200 series desktop), **LGA1200** (10th/11th gen, legacy), **AM4** (Ryzen 1000–5000), **AM5** (Ryzen 7000/8000/9000). Two extremely valuable practical notes: AM5 has maintained the **same mounting hole pattern as AM4**, so almost all AM4 coolers are mechanically compatible with AM5. And LGA1851 has maintained the 78 × 78 mm hole pattern of LGA1700, so compatibility there is also broad — but **it is still mandatory to check the cooler manufacturer's website**, because the height of the ILM (Independent Loading Mechanism, the processor retention mechanism) and the thickness of the IHS can change, and some kits still require a different backplate.

2. The maximum cooler height allowed by the case. Every case specifies "CPU cooler clearance" or "max CPU cooler height," typically 155–175 mm in mid-towers. This should be compared with the cooler's stated height. **Do not leave 1–2 mm margins:** manufacturing tolerances, backplate thicknesses, and socket position on the motherboard introduce variations. If the case states 165 mm and the cooler measures 165 mm, it's a gamble. If the case states 170 and the cooler 165, you're safe.

3. RAM clearance. This is the real silent killer. In tower coolers, the front fan protrudes towards the memory slots. The manufacturer usually states two values: the maximum height of RAM modules with the fan in the standard position (often 32–35 mm), and the allowed height by raising the fan (at the cost of increasing the overall cooler height, falling back to point 2). "Bare" memory modules (without heatsinks, or with low-profile heatsinks) measure approximately 31–33 mm; many gaming modules with prominent heatsinks and RGB (Red-Green-Blue, addressable LED lighting) reach 42–50 mm. A Noctua NH-D15 with 48 mm tall RAM simply won't fit, or forces you to raise the fan beyond the case's limits, or remove it entirely, losing performance. **The practical rule is: if you've already chosen a large tower cooler, buy low-profile RAM.**

A fourth, less frequent but real check: some very wide tower coolers can partially cover the first PCIe x16 slot (where the graphics card goes) or the M.2 heatsinks above the first slot. This problem often occurs with Mini-ITX cases and compact mATX boards. All serious manufacturers publish compatibility lists and even downloadable 3D models; using them is not being pedantic, it's good practice.

8.2.6 Typical buyer mistakes

The first and most expensive: **sizing the cooler based on the stated TDP instead of actual power consumption.** We discussed this in 8.1.2; it's the mistake that leads to buying a 40-euro cooler for a Core i7 K that draws 250W under load.

The second: **ignoring RAM clearance**, buying "nice" memory first and then discovering the cooler doesn't fit.

The third: **believing that the number of heatpipes alone determines performance.** Six well-soldered heatpipes to a dense fin stack and well-ventilated beat eight poorly made heatpipes. The number is an indication, not proof.

The fourth: **using the stock cooler on a high-end CPU.** The coolers included in the box (AMD Wraith Stealth/Spire, Intel Laminar coolers) are designed for the bare minimum, i.e., to make the CPU function without failure. On a Ryzen 5 or a non-K Core i5, they are acceptable; on anything above that, they are a bottleneck. It should also be noted that, as of the writing date, **most high-end processors are sold without any cooler included:** it's not an option, it's a mandatory component to budget for.

The fifth: **not mounting the backplate correctly**, or tightening the screws unevenly. We will return to tightening in the section dedicated to thermal paste.

The sixth, specific to Intel: **ignoring the ILM bending issue on LGA1700**. The retention mechanism of this socket exerts pressure that tends to slightly bend the processor's IHS, creating a concavity in the center and reducing contact with the cooler base. This phenomenon typically costs 3–8 °C. There are aftermarket **contact frames** (Thermalright, Thermal Grizzly, and in some cases motherboard manufacturer kits) that replace the original ILM and distribute pressure more evenly. It's a 15–20 euro modification that recovers several degrees, but it requires removing the original mechanism — a reversible operation that must be done carefully. *Verify applicability to your socket and generation, as the problem is specific and not universal.*

8.2.7 Budget-tier recommendations (air)

Entry (CPUs up to ~100W real: Ryzen 5 non-X, Core i5 non-K, APUs). A single tower for 30–40 euros is more than sufficient. Arctic Freezer 36, Thermalright Assassin X120, DeepCool AK400. In many cases, the cooler included in the box is enough, but spending 30 euros buys a huge acoustic gain: it's probably the best value for money in the entire PC.

Mid (CPUs up to ~180–220W real: Ryzen 7, non-overclocked Ryzen 9, Core i5 K, Core i7). Economical dual tower: **Thermalright Peerless Assassin 120 SE or Phantom Spirit 120 SE**. They are, without exaggeration, the most solid advice one can give anyone in 2025–2026: performance within 2–4 °C of top-tier models at a third of the price. Higher build quality alternative: DeepCool AK620.

High-end (CPUs over 250W: Ryzen 9 in PBO, Core i9 K, workstations). Here air reaches its physical limit. Serious options are Noctua NH-D15 G2, be quiet! Dark Rock Elite, DeepCool Assassin IV. These should be chosen knowing that on a top-end Core i9 under full load, even the best air cooler will leave the CPU partially throttling: this is not a defect of the cooler, it's that the CPU dissipates more heat than air can carry away in that space. The conscious choice at this point is: accept a few percentage points less performance in exchange for reliability and silence (air), or switch to liquid cooling.

8.3 AIO Liquid Cooling

8.3.1 Description and Function

AIO stands for *All-In-One*: a pre-assembled, pre-filled, and factory-sealed liquid cooling system. The user simply mounts it; it is not opened, refilled, or maintained.

The circuit is conceptually simple. A **waterblock** (or cold plate) rests on the CPU's IHS: it's a copper block whose bottom surface is smooth (in contact with thermal paste) and whose top surface is carved into a dense grid of micro-fins, through which the liquid is forced to pass. The liquid absorbs heat, is pushed by a **pump** (in the vast majority of AIOs, the pump is integrated into the block itself, above the waterblock; in some models, it's integrated into the radiator) through a flexible **tube** to the **radiator**, which is a liquid-to-air heat exchanger: flat channels through which the liquid flows, crossed by a stack of aluminum fins. One or more **fans** blow air through the radiator, which dissipates heat into the environment. The cooled liquid returns to the waterblock through the second tube. It's a closed circuit.

The point that almost everyone misunderstands is **what the liquid actually does**. The liquid doesn't "cool" in the sense that it doesn't produce cold: it's simply a very effective *means of heat transport* because water has a very high specific heat capacity (approximately 4.18 kJ/kg·K) and because a pump can move it continuously. In an air cooler, heat must be transported from the base to the fins via heatpipes, and the fins must physically be there, above the socket, in a limited space. In an AIO, heat is *carried elsewhere*, to a point in the case where there's space for a much larger exchange surface and more fans. **The advantage of an AIO is not the liquid: it's that the liquid allows the exchange surface to be placed where there's room.**

There's a second, less obvious but valuable advantage: **thermal mass**. The liquid in the circuit (approximately 150 to 400 ml) acts as a thermal flywheel: it absorbs short power peaks without the temperature immediately rising. In burst loads (gaming, where the CPU alternates between peaks and pauses), this translates to lower peak temperatures and fans that spin up more slowly, resulting in more stable acoustic behavior. This is one of the reasons why an AIO "seems" quieter in gaming even when it isn't under constant load.

8.3.2 Radiator sizes: which for which CPU

The dissipation capacity of an AIO is determined almost entirely by the **radiator surface area** and the airflow through it. The pump and waterblock matter much less than marketing suggests: differences between good quality pumps are in the order of 1–2 °C, while between a 240 mm radiator and a 360 mm one there are 5–10 °C.

Sizes are expressed by approximate length: **120** (one 120 mm fan), **240** (two 120 mm fans), **280** (two 140 mm fans), **360** (three 120 mm fans), **420** (three 140 mm fans). 140, 320, and 480 also exist in niche markets. Note: a 280 has a *larger* surface area than a 240 (140 mm fans have approximately 36% more frontal area), and is often comparable to a 360 despite being shorter, which makes it a clever choice in cases that support it — but cases compatible with 280 mm radiators are less numerous.

A radiator **thickness** that matters: the standard is 27 mm, but some manufacturers (Arctic with the Liquid Freezer series, in particular) use 38 mm radiators, which offer more mass and surface area for the same length. The downside: greater thickness requires more static pressure from the fans and can cause clearance issues with the VRM fin stack on the motherboard or with RAM, if mounted on top of a narrow case. **Always check the maximum thickness supported by the case, summing radiator + fans.**

Radiator	Relative Surface Area	Suitable CPUs (real power consumption under load)	Notes
120 mm	1×	up to ~100 W	Often worse than a €40 tower at the same price. Not recommended unless space is constrained
240 mm	2×	up to ~180 W	Mid-range. Suitable for Ryzen 7, Core i5 K/i7
280 mm	~2.7×	up to ~230 W	Excellent compromise, if the case accepts it
360 mm	3×	up to ~280 W	The de facto standard for high-

			end
420 mm	~4×	over 300 W	For top-tier Core i9, Ryzen 9 in PBO, Threadripper. Large cases mandatory

Uncomfortable truth to say out loud: an economical 120 or 240 mm AIO is almost always a worse choice than a €40 dual-tower cooler, in every respect — performance, noise, reliability, price. It only makes sense if space physically prevents a tower (slim cases, some ITX) or if aesthetics are the priority. Liquid cooling objectively becomes superior to air **from 280 mm upwards, and on CPUs that dissipate a lot of heat.**

8.3.3 Radiator placement and the rule of air in the pump

This is the paragraph that, if ignored, leads to noisy AIOs and prematurely dead AIOs. It must be understood, not memorized.

The physical fact. An AIO, however sealed, is not hermetic indefinitely. Rubber tubes are *permeable* to water vapor: molecule by molecule, year after year, a small amount of liquid evaporates through the tube wall. Furthermore, some internal chemical reactions (and the fact that the fill is never 100% during production) generate a small amount of gas. The result is that **in every AIO, sooner or later, an air bubble forms.** It is inevitable and anticipated by designers.

The consequence. Air, being less dense than liquid, always collects at the **highest point in the circuit.** If the highest point in the circuit is the pump, the pump draws air instead of liquid. This produces three effects, in order of increasing severity: (1) an obvious and annoying noise, a gurgle or hum ("gurgling"); (2) a loss of pumping efficiency, because air does not transport heat; (3) **dry cavitation** and accelerated wear of the bearing and impeller, because the pump runs without the liquid that lubricates and cools it. A pump that runs dry for weeks will die.

The practical rule. The highest point in the circuit **must never be the pump** (i.e., the block on the CPU). From this, the correct configurations follow:

- ****Radiator on top of the case**** (at the top), with tubes exiting downwards or sideways: ideal configuration. The highest point is the top of the radiator,

which is designed to contain air (often the top channel of the radiator acts as a small air chamber). The liquid always reaches the pump full.

- ****Radiator on the front of the case, with tubes at the bottom****: correct configuration. If the radiator is at the front and the tubes exit from the ***bottom*** of the radiator, air collects in the upper part of the radiator, away from the pump. This is the configuration to choose when mounting an AIO at the front.
- ****Radiator on the front with tubes at the top****: ****wrong and to be avoided**** configuration, because this is where air from the radiator can migrate into the tubes and end up in the pump. It is the most common installation error and is what generates the famous complaint "my AIO makes a gurgling noise."
- ****Radiator at the bottom of the case****: the absolute worst. The lowest point of the circuit becomes the radiator, and the highest point becomes the pump. All the air in the circuit ends up in the pump. Never do this, even if the case mechanically allows it.

If the case geometry forces a non-ideal mounting, there is a palliative: after assembly, **tilt the case into various positions for a couple of minutes with the system on**, to make the air bubble migrate towards the radiator. This is not a permanent solution but often resolves the initial noise.

A final note on orientation: some manufacturers (Arctic, Corsair, Lian Li) offer blocks with rotatable logos or displays, so aesthetics do not force a thermally incorrect mounting. If your AIO does not allow this and you still want a precise tube orientation, it is better to choose another model than to compromise physics for a straight logo.

8.3.4 Air direction: radiator as intake or exhaust?

A question asked continuously and almost always answered incorrectly. The two options:

Radiator as intake: fans blow fresh air *from the outside* through the radiator, into the case. Advantage: the radiator receives the coldest air available (ambient temperature), so the CPU is cooler — typically 3–6 °C less. Disadvantage: the air entering the case after passing through the radiator is preheated, so the GPU, VRM, SSD, and RAM run a few degrees warmer.

Radiator as exhaust: fans draw air from the case through the radiator and expel it. Advantage: the rest of the case breathes fresh air. Disadvantage: the radiator receives air already heated by the GPU, so the CPU is warmer.

There is no universal answer; there is a criterion: **prioritize the most critical component**. If you have a Core i9 in a rendering system and a modest GPU, it makes sense to put the radiator as intake at the front. If you have a high-end 400 W GPU and an average CPU in a gaming system, it makes sense to put the radiator as exhaust on top and let the front bring fresh air to the GPU. The most common and balanced configuration in modern cases is: **radiator on top as exhaust, front fans as intake**. With a caveat: if the radiator is an intake from the front, a dust filter must be placed in front of it, because a radiator clogged with dust dramatically loses efficiency, and cleaning it is much more difficult than cleaning a fin stack of a tower cooler.

8.3.5 Brands and Models

Overview based on the best available knowledge; the AIO segment is rapidly evolving, and revisions often change, so always check reviews for the specific revision on sale.

Arctic Liquid Freezer (II and III series). The de facto benchmark for price/performance ratio, and for several revisions, also the absolute performance benchmark. Distinctive features: 38mm radiator (thicker than standard), **a 40mm VRM fan integrated into the block** that cools the motherboard's voltage regulators (something no other AIO does and that traditional AIOs, unlike low-profile coolers, remove), adjustable PWM pump, aggressive pricing. Cons: spartan aesthetics, somewhat inconvenient single cable, radiator thickness can cause clearance issues in some cases.

Corsair (iCUE Link Titan, H-series). Very refined ecosystem, extensive software, high build quality, high prices. The iCUE Link system reduces cabling with a single daisy-chain connector, which is a real practical advantage but locks you into the ecosystem.

Lian Li Galahad II. Very competitive performance, high-end aesthetics, refined lighting, medium-to-high prices. A good alternative to Corsair for those who want aesthetics without software lock-in.

NZXT Kraken (and Elite). The strong point is the large, customizable LCD display on the block; thermal performance is good but rarely top-tier, and you pay a premium for aesthetics and software (CAM).

DeepCool (LT, LE, Mystique series). Solid value for money, good performance, refined aesthetics. *Note: availability of this brand may vary by market; check local availability.*

be quiet! (Silent Loop, Pure Loop). Focused on quiet operation, with particularly discreet pumps. Performance is rarely top-tier, but acoustically excellent.

Technologically, it's worth noting that for many years, a large portion of AIOs on the market, regardless of brand, used pumps manufactured by **Asetek** under license (or by CoolIT, the other major OEM). This means that the "heart" of seemingly different AIOs was often identical, and the real differences lay in the radiator, fans, tubes, and software. This is an additional reason not to pay an excessive premium for the brand, and to focus on measured data: radiator size, performance certified by independent reviews, and noise measured at the same dissipation level.

8.3.6 Pros and Cons of AIOs

Pros. For high-end CPUs, a 360/420mm AIO outperforms air cooling: not by a small margin, in some cases by 8–12 °C under heavy load, resulting in higher sustained frequencies. It frees up space around the socket: no conflicts with RAM (any memory can be installed, even with the tallest heatsinks), no conflicts with the first PCIe slot, no weight hanging from the motherboard. It allows for compact builds with powerful CPUs, as the radiator goes wherever there's space. And, frankly, it's aesthetically cleaner: the motherboard remains visible, lighting is visible, and side-glass systems make sense.

Cons. First and foremost, the cost: a decent 360mm AIO starts at 90–110 euros and quickly goes up. The **pump is a single point of failure**: if it stops, the CPU temperature rises vertically within seconds, and the system throttles or shuts down; unlike a fan, there is no gradual degradation and no redundancy. When installing an AIO, it's good practice to connect the pump to the motherboard's **AIO_PUMP** or **CPU_FAN** connector and set the zero-speed alarm, so the BIOS blocks startup if the pump isn't running. The **lifespan** is limited: it's typically estimated at **5–7 years**, constrained by liquid evaporation through the tubes, wear on the pump bearing, and potential debris accumulation in the waterblock's

microchannels. An air cooler, over the same period, only needs a new 20-euro fan. Finally, noise: the pump adds a continuous acoustic source that doesn't exist in air cooling, and not all pumps are silent at low speeds.

8.3.7 Typical Buyer Mistakes (AIO)

Buying a **cheap 240mm AIO instead of a 40-euro tower cooler**, believing that "liquid = better." It's not: you pay more for equal or inferior performance, with an extra pump that can break.

Mounting the radiator at the bottom of the case or at the front with tubes up, for cabling or aesthetic reasons. See 8.3.3.

Not checking available space: radiator + fans can exceed 55–65 mm in thickness. On the top of many cases, this interferes with the motherboard's VRM heatsinks or RAM. The case specification usually states a maximum "clearance" for the top; it must be respected.

Connecting the pump to a generic fan connector with an aggressive PWM curve. The pump is not a fan: running it at 40% to "silence" it reduces flow and performance. Most pumps should be kept at a fixed, high speed (often 100%); some modern ones accept PWM, and the manufacturer specifies how. Read the manual.

Neglecting the break-in period. In the first few days, an AIO may produce a slight gurgling noise that subsides as air bubbles settle. If the noise persists after a week, the installation is incorrect.

8.3.8 Budget-Tier Recommendations (AIO Liquid)

Entry. None. If the budget is low, get a tower cooler. A cheap AIO is a bad deal.

Mid (CPUs up to ~200W, desire aesthetics or have tall RAM constraints). A good brand 240 or 280mm. Arctic Liquid Freezer III 240/280 is the rational choice.

High-end (CPUs over 250W). A high-end 360mm, or a 420mm if the case allows. Arctic Liquid Freezer III 360/420, Corsair, Lian Li Galahad II 360. For a Core i9 or a Threadripper, this is where liquid cooling is not an aesthetic choice but a technical necessity.

8.4 The Custom Loop: For Completeness, and to Understand Why Almost No One Should Do It

A **custom loop** is a liquid cooling circuit assembled by the user component by component. It arises from the desire to overcome the limitations of AIOs (larger radiators, multiple radiators, and especially the ability to liquid cool the GPU as well), and from a passion for the object itself.

8.4.1 The Components

Pump and Reservoir. The pump moves the liquid; the reservoir is a tank that facilitates filling, collects air, and provides a liquid margin. In modern practice, pumps and reservoirs are almost always purchased as an integrated **combo**. The two dominant pump platforms are the **D5** and the **DDC**, both derived from industrial designs (Laing/Xylem and Laing, respectively). The **D5** has high flow rate and medium head pressure (the ability to overcome circuit resistance), is larger, and very quiet and reliable. The **DDC** has higher head pressure and lower flow rate, is more compact but runs hotter and is typically noisier. In a long loop, with multiple radiators and very restrictive blocks, the DDC has an advantage; in the majority of home loops, the D5 is the sensible choice.

CPU Waterblock. A nickel-plated copper block with a microchannel base (channel width on the order of tenths of a millimeter), an acrylic or acetal top, and threaded fittings. Performance differences between good CPU waterblocks are modest (2–4 °C); mounting quality and uniform pressure are more important.

GPU Waterblock — this needs to be explained well, because this is where people get into trouble.

A stock graphics card is sold with a proprietary air cooler, screwed to the **PCB** (Printed Circuit Board, the card's circuit board) and connected to hot components via thermal paste on the GPU and **thermal pads** (sheets of soft conductive material) on memory and VRMs. To liquid cool the GPU, you must:

1. **Completely disassemble the stock cooler**, which means removing the backplate, unscrewing the plate, disconnecting the fan cables, and separating the fin stack from the PCB — an operation that requires care because the cooler adheres to the chip via hardened thermal paste.
2. **Perfectly clean the PCB** of paste and pad residue.
3. **Install a full-cover waterblock**, meaning a plate that covers not only the GPU but also the GDDR memory and VRM. Here's the crucial point: **the waterblock is specific to that PCB**, not to that GPU model. A GeForce

of a certain series produced by different manufacturers has different PCBs, with components positioned differently. A waterblock designed for the "reference" PCB (the chip manufacturer's reference design) **is not compatible** with a custom PCB card. Detailed compatibility lists exist (EK, Alphacool, and Bykski publish them) and should be consulted **before** buying the graphics card, not after. 4. **Apply thermal pads of the correct thickness** to memory and VRMs: here the margin of error is small. A pad that is too thin will not make contact (and the memory will burn out or throttle); a pad that is too thick will prevent the block from resting on the GPU die (and the GPU will throttle). Thicknesses are measured in tenths of a millimeter, and the waterblock kit provides them pre-cut and sized — use those, do not improvise.

On warranty. This is a delicate topic and needs to be stated precisely. In many non-EU countries, manufacturers declare that removing the heatsink voids the warranty, full stop. In the European Union, the situation is more nuanced: the **legal guarantee of conformity** (two years, provided for by the EU directive transposed into Italian law by the Consumer Code) covers defects of conformity present at the time of delivery, and the seller cannot exclude it entirely simply because the product has been opened. However, the seller **is not obliged to cover damage caused by the user**, and the burden of proving that the fault is not due to the modification falls, after the first few months, on the consumer. In practice: disassembling the heatsink of a 1,500 euro GPU is a risk taken by the user, and in the event of a fault, the discussion with the seller will be long and the outcome uncertain. Some manufacturers (historically few, and policies change over time — *verify case by case, the data is volatile*) explicitly state that they tolerate heatsink replacement. **The honest recommendation is: do not do it on a card you cannot afford to replace.**

Radiators. Custom loops use standard 120/140 mm module radiators, in lengths 240/280/360/420/480, and in three typical thicknesses: **30 mm** (slim), **45 mm** (standard, most common), **60 mm** (thick, for high static pressure fans). The empirical rule for sizing, cautious but reliable: **approximately 120 mm of radiator for every 150–200 W to dissipate**, with fans at moderate speed. A loop with a CPU (250 W) and GPU (400 W) therefore requires at least a 360 + a 240, and is more comfortable with two 360s.

Fittings. These are the connectors that link tubes and components. The universal standard is **G1/4"** threading. The main types are **compression fittings** (with a nut,

for soft tubing: the tube is pushed onto the barb and a nut is screwed on to compress it — secure and reversible) and fittings for **hard tubing** (with O-rings that seal against the outer wall of the tube). There are also angled fittings, drain valves (essential, and too often omitted), bulkhead pass-throughs, and various adapters.

Tubing. Two worlds: **soft tubing** (flexible tubing, in PVC or silicone, typically 10/13 mm or 10/16 mm — inner/outer diameter) is flexible, forgiving of errors, cut with scissors, and is the rational choice for beginners. **Hard tubing** (rigid tubing, in PETG, acrylic, or metal) must be measured, cut, heat-bent with special inserts, deburred, and polished: the aesthetic result is superior, the production time multiplies, and every error costs a piece of tubing. It is craftsmanship, and should be approached as such.

Liquid. Distilled water (never tap water: mineral salts deposit and conduct) with biocide and anti-corrosion additives, or a ready-made commercial **premix** (EK CryoFuel, Mayhems, Aquacomputer). Opaque colored liquid (pastel) is beautiful and problematic: it tends to sediment and clog microchannels; clear or slightly tinted liquids are far more reliable. **Golden chemical rule: never mix copper and aluminum in the same circuit.** Contact between two different metals in the presence of an electrolyte triggers **galvanic corrosion**: aluminum (less noble) corrodes rapidly, debris clogs the channels, the loop dies. Since almost all water blocks are made of copper and some inexpensive radiators are made of aluminum, this is a real and frequent mistake. Verify the materials of each individual component.

8.4.2 Brands

EK Water Blocks is historically the most popular brand, with the widest catalog of GPU water blocks and a complete ecosystem; *it should be noted that the company's stability has faced difficulties in recent years, impacting deliveries and support — a volatile data point that should be verified at the time of purchase.*

Alphacool offers excellent value for money, all-copper radiators, and a very wide range. **Watercool** (German brand, Heatkiller series) is the benchmark for absolute build quality, at high prices. **Corsair Hydro X** is the "turnkey" solution for those who stay within the Corsair ecosystem, well-made but expensive. **Bykski** and **Barrow** are Chinese brands that offer components at a fraction of the Western price, with variable but often surprisingly good quality: they are the

affordable entry point to custom loops, provided you carefully check compatibility.

8.4.3 Costs, maintenance, and for whom it's truly worthwhile

Costs. A decent CPU-only loop starts from 350–450 euros. A CPU + GPU loop realistically costs between 700 and 1,200 euros, and easily goes higher. This should be compared with a 360 mm AIO at 130 euros which, for the CPU, offers 85–90% of the benefit.

Maintenance. It's not optional. The liquid must be **replaced approximately every 12 months** (some extend to 18–24 with quality premixes), the circuit must be flushed, and filters and traps must be checked. Each intervention requires draining the loop, which means half a day of work. And there's the ever-present risk of a **leak**: a fitting not tightened properly, a damaged O-ring, a poorly cut rigid tube. A leak on a powered motherboard destroys components worth hundreds or thousands of euros. For this reason, the **leak test** — filling the loop and pressurizing it (or powering only the pump with a jumper on the 24-pin connector, keeping the rest of the system off) for at least 12–24 hours before turning on the PC — is not a recommended precaution, it is a mandatory procedure.

For whom is it worthwhile? With brutal honesty: **almost no one, from a rational point of view.** The CPU performance of a custom loop only slightly surpasses that of a good 360 AIO. The real technical advantage is GPU cooling, which can lower temperatures by 25–30 °C, eliminate fan noise from the card, and allow significant overlocks. But it's an advantage that costs 800 euros, ten hours of work, a warranty risk, and annual maintenance.

The custom loop is suitable for three categories: those seeking **absolute silence** in a very powerful system (a large loop with fans at 500 RPM is practically inaudible, and no other technology allows this); those building **extreme** systems with multiple GPUs or aggressive overlocks; and those who do it **for the pleasure of doing it**, as a hobby, exactly like restoring a vintage motorcycle. The latter is a perfectly legitimate motivation, as long as it is declared as such and not disguised as technical rationality.

8.5 Air vs. Liquid: The Final Decision

Criterion	Air (Dual	AIO 240–280	AIO 360–420	Custom Loop
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	Tower)			
Performance on CPU $\leq$ 150 W	Excellent	Equivalent	Excessive	Excessive
Performance on CPU 200–250 W	Good	Good	Very Good	Very Good
Performance on CPU > 280 W	Insufficient (throttling)	Insufficient	Very Good	Excellent
Noise at same dissipation	Excellent (no pump)	Decent	Good	Excellent (at low speed)
Cost	€30–150	€80–130	€110–220	€400–1,200+
Reliability / Failure Points	Only fan	Pump	Pump	Pump, fittings, leaks
Expected Lifespan	10+ years	5–7 years	5–7 years	Indefinite, with maintenance
Maintenance	Dusting	None	None	Annual, mandatory
Space around socket	High (RAM conflicts)	None	None	None
Aesthetics	Bulky	Clean	Clean	Maximum
Also cools GPU	No	No	No	Yes
Assembly Complexity	Low	Low/Medium	Medium	Very High

Unambiguous recommendations by CPU tier:

- ****Entry CPU (up to 100 W actual)**:** Inexpensive single tower. No reason for liquid.
- ****Mid CPU (100–200 W)**:** Inexpensive dual tower (Peerless Assassin and similar). AIO only makes sense for high RAM clearance constraints or aesthetics.
- ****High CPU (200–280 W)**:** High-end dual tower **or** AIO 280/360. Both choices are defensible; air wins on reliability and cost, liquid on temperatures and space.

- ****Flagship CPU (over 280 W: top-tier Core i9, Ryzen 9 in PBO, HEDT/Threadripper)**:** AIO 360/420 mandatory, or custom loop. Air is not enough here, and insisting means leaving performance on the table.

A final observation, which serves as a general principle for this chapter: **it is almost always smarter to spend more on the CPU and less on the cooler, than the other way around.** The cooler improves performance by a few percentage points; a superior CPU improves it by tens of percentage points. A poorly allocated budget is one where someone puts a 250 euro AIO on a 250 euro processor.

8.6 Thermal Paste: A Comprehensive Discussion

8.6.1 What it's for, and why it's not "a lubricant"

If one were to observe the top surface of the CPU's IHS and the bottom surface of the cooler's base under a microscope — both seemingly smooth and mirrored — one would see two mountainous landscapes. No mechanical process produces a perfectly flat surface: there remain micro-roughness, undulations, and sometimes a slight concavity or convexity due to manufacturing or mounting pressure.

When these two surfaces are pressed against each other, **actual** metal-to-metal contact occurs only on a minimal fraction of the apparent area: typically **less than 1–2%**. All the rest is air trapped in micro-cavities. And air is a poor thermal conductor: approximately **0.026 W/m·K**, compared to about 400 W/m·K for copper. A layer of air a few micrometers thick, extending over 98% of the contact area, constitutes a devastating thermal barrier.

The task of the **TIM** (Thermal Interface Material — this is the correct technical name for "thermal paste") is to **expel the air and fill those micro-cavities with a material that conducts heat much better than air**. Here's the point that needs to be understood once and for all: thermal paste **is not a good conductor in absolute terms** — an excellent paste has 8–13 W/m·K, which is thirty or forty times worse than copper. Paste is a **necessary compromise**: it is much worse than metal, but hundreds of times better than the air it replaces.

From this observation immediately follows the most important operational rule of the entire section: **the paste layer must be as thin as possible**. The paste should not create thickness: it should only fill the gaps. Every extra micrometer of paste

between the two metals adds thermal resistance. Anyone who spreads a millimeter of paste "to be sure" is literally building an insulator.

8.6.2 Types of TIMs

Traditional silicone-based pastes. These are a polymer matrix (silicone or similar) loaded with conductive particles: metal oxides (zinc, aluminum), ceramics, boron nitride, electrically non-conductive metal particles. They are **electrically non-conductive** (with some exceptions for high metallic loads — read the datasheet), which makes them safe: a small amount overflowing onto the socket will not cause short circuits. Stated conductivity is typically between 5 and 14 W/m·K.

A very important methodological note: **conductivity values declared by manufacturers are not comparable between different brands**, because there is no universally adopted measurement standard and each manufacturer uses the method that favors them. A paste that claims 13 W/m·K is not necessarily better than one that claims 8.5 W/m·K. **The only reliable data is empirical comparison in independent comparative tests**, where the difference between good pastes turns out to be 1–3 °C — almost irrelevant compared to the choice of cooler. Anyone who spends 60 euros on a mediocre cooler and then looks for miraculous paste is optimizing the wrong variable.

Concrete references:

- ****Arctic MX-4**** and ****MX-6****: inexpensive, very easy to apply (perfect viscosity), non-conductive, excellent performance, declared longevity of years. They are the default choice for 90% of users, and are hard to beat on price/performance ratio.
- ****Noctua NT-H1**** and ****NT-H2****: excellent, very stable over time, easy to apply and clean. A bit more expensive. The NT-H2 is the improved version.
- ****Thermal Grizzly Kryonaut****: historically among the best in pure performance, with a serious caveat: it suffers from ****pump-out**** (see below) and degrades relatively quickly with thermal cycles, especially above 80 °C. It's good for benchmarks and for those who reapply paste often; it's a questionable choice for a "mount and forget" system. Variants (Kryonaut Extreme) and more stable formulations exist in the same family.

Pump-out deserves an explanation, because it is the main degradation mechanism. Every time the system heats up and cools down, metals expand and

contract to different degrees; this produces a micro-movement between the IHS and the heatsink base which, cycle after cycle, **"pumps" the paste out of the interface**, towards the edges. Over time, voids remain in the center (where heat is highest). The result is a gradual worsening of temperatures over months or years. More fluid pastes with less stable matrices are more susceptible to the phenomenon.

Liquid metal. It is an alloy of gallium, indium, and tin (sometimes generically called galinstan), liquid at room temperature. Its conductivity is on another planet: **about 70–80 W/m·K**, almost ten times that of an excellent paste. The real gain on the IHS is typically **5–12 °C**, and on a bare die (delidded CPU, laptop, console) it can be even higher. The commercial reference is **Thermal Grizzly Conductonaut** (and Extreme).

The risks, however, are serious and must be listed without sugarcoating: 1. **It is electrically conductive.** A drop falling on an LGA socket, on an SMD component of the motherboard, or on the CPU contacts can cause a short circuit and destroy hardware. In a vertical position, the CPU is vertical: liquid metal, if in excess, can drip. 2. **It corrodes and amalgamates aluminum.** Gallium literally dissolves aluminum: it creates a fragile amalgam and destroys the metal. **Never use it on an aluminum base**, nor near aluminum components. It should only be used on nickel-plated copper or compatible surfaces. 3. **Over time it can be "absorbed" into copper** (diffusion), permanently staining the IHS and the base, and losing effectiveness. 4. It is **difficult to apply**: it must be spread in a very thin layer with a cotton swab, both on the IHS and on the base, and the surrounding area must be shielded with insulating tape or conformal coating.

In summary: liquid metal is not for the average user. It makes sense on a delidded CPU, on a laptop that throttles, on a bare die. In a normal desktop build, the risk/benefit ratio does not justify it.

PTM7950 and phase change materials. This is the innovation that has changed the game in recent years. **PTM7950** (Honeywell) is a **PCM** (Phase Change Material): at room temperature, it appears as a thin, semi-solid, non-sticky sheet that can be cut and applied. Upon initial heating (transition around 45 °C), it softens, perfectly "wets" the two surfaces by filling micro-cavities, and then partially re-solidifies upon cooling. This cycle repeats with each power-on.

The decisive advantage is not its peak conductivity (which is comparable to a good paste, around 8 W/m·K) but **the almost total absence of pump-out**: since the material re-solidifies, it is not expelled by thermal cycles. The result is exceptional long-term stability: where traditional paste degrades by 5–10 °C in three years, PTM7950 remains substantially unchanged. It has become **the de facto standard for GPUs and laptops**, precisely the applications where repasting is difficult and thermal cycles are aggressive. The small price to pay: in the first few minutes from cold, before the material has reached its transition temperature, performance is slightly worse. This is irrelevant in real-world use.

Note on a practical risk: as a highly sought-after product, PTM7950 is widely counterfeited on marketplaces. Purchase from reliable sellers.

Graphite pads. Reusable graphite sheets (the reference is **Innovation Cooling Graphite**), with high conductivity in the plane but modest through the thickness. They never dry out, do not pump out, and can be removed and reinstalled. Performance is typically 2–4 °C worse than a good paste. They are a "mount and forget" solution for those who frequently disassemble, or for those who want zero maintenance. A legitimate niche.

Thermal pads (for memory and VRMs). These are different and serve another purpose: to fill **gaps** of varying thickness (from 0.5 to 3 mm) between a component and the heatsink. They have low conductivity (1–15 W/m·K) but are compressible and maintain contact. They are used on VRAM, VRMs, and SSD chips. **They are not used under an IHS instead of paste**: there, the gap is micrometers, and a material that thins out is needed, not a pad.

8.6.3 How to apply: quantity and methods

The quantity. The traditional rule is "a grain of rice" or "a pea." These are useful approximations but should be calibrated to the size of the IHS:

- ****AM4**** (small, square IHS with chamfers): an amount equivalent to a large grain of rice or a small pea, in the center.
- ****AM5**** (IHS with characteristic openings on the sides): a pea in the center. Some prefer five dots because the usable surface is "octagonal" in shape.
- ****LGA1700 / LGA1851**** (rectangular and elongated IHS): a single dot in the center tends not to reach the short ends. A ****thin line along the major axis****, or an ****X****, or five dots is better.

The methods.

- **Central dot (pea method)**: A single drop is placed in the center and mounting pressure is allowed to spread it. This is the method recommended by most manufacturers, it works well, does not introduce bubbles, and requires no skill. **For most users, this is the right method.**
- **X / cross / five dots**: Distributes better on large or rectangular IHSs. Excellent on LGA1700.
- **Line**: Two or three lines parallel to the long axis of the IHS, for rectangular sockets.
- **Spreading**: The paste is spread with a spatula or a plastic-coated card before placing the cooler. It gives maximum control over thickness, but **risks trapping micro-air bubbles** if done incorrectly, and is the method with the most margin for error. It makes sense when using liquid metal (mandatory) or on a bare die (where the surface is small and mounting pressure is low).

Mounting pressure. This is not a minor detail: it's what expels excess paste and reduces residual thickness. **The cooler screws should be tightened in a cross pattern, a little at a time, alternating**, exactly like car wheel lug nuts: never fully tighten one screw and then move to the next, as this will tilt the base and create uneven contact. Tighten until the mechanical stop is reached (good systems, like Noctua's SecuFirm2, have a stop that prevents overtightening).

Do not lift and re-seat the cooler after mounting it: if it is removed, the paste is compromised (bubbles and voids have formed) and **must be redone from scratch**, cleaning both surfaces. It's tempting to "check" after mounting: don't do it, or if you do, be prepared to clean and reapply.

8.6.4 Common errors

Too much paste. The most common mistake. An excess does not "insulate" as much as legend would have it (because pressure expels most of it), but it spills over the sides, soils the socket, and in the case of high metallic load pastes, can cause problems. Above all, if there's too much and the pressure isn't enough to expel it, it leaves a thick and thus resistive layer.

Too little paste. A more insidious error because it's invisible: areas of the IHS remain without contact, where air acts as an insulator. This results in high and often asymmetrical temperatures (one core much hotter than the others).

Air bubbles. Arise from poorly done spreading, or from lifting and re-seating the cooler.

Not cleaning old paste. Hardened paste mixed with new paste gives poor results. Cleaning is done with **99% isopropyl alcohol** (not pink denatured alcohol, which leaves residues) and a lint-free cloth (microfiber or laboratory paper; kitchen paper leaves fibers). Clean **both** surfaces: IHS and cooler base.

Forgetting to remove the protective film from the new cooler. This deserves its own bolded line, because it is a real, frequent error that produces dramatic symptoms (immediate throttling, temperatures of 100 °C within seconds of power-on). **Almost all new coolers come with thermal paste pre-applied to the base, and with a transparent or colored plastic film for protection.** The film must be **removed** before mounting. The fact that it is transparent and well-adhered makes it easy to overlook. If a new PC throttles instantly, this is the first thing to check.

Apply paste AND use the pre-applied one together. No: either use the pre-applied paste (which is of decent quality on almost all branded coolers) *or* remove it completely with isopropyl alcohol and apply your own. Never layer them.

8.6.5 How often to replace it, and GPU repasting

On the CPU. With a quality paste (MX-4, NT-H2) and normal use, an interval of **3–5 years** is reasonable. With a paste prone to pump-out (like Kryonaut) and heavy thermal loads, it can drop to **1–2 years**. The sign that it's time to intervene is a progressive worsening of temperatures under the same load and environment: if the PC that was at 75 °C in rendering two years ago is now at 88 °C, and dust has already been removed, the paste is the main suspect.

On the GPU. This is where replacement yields the most spectacular results, and it's also the most delicate. Graphics cards are assembled in the factory with varying quality pastes and, above all, are subject to very violent thermal cycles. On a 3–5 year old GPU, it's not uncommon to gain **10–20 °C** with a repaste, and even more on the hotspot (the hottest point of the die, measured separately on recent cards: if the delta between average core temperature and hotspot exceeds 20–25 °C, it's a strong indication of degraded paste or uneven mounting).

The procedure requires: disassembling the backplate and plate, disconnecting the fan cables, **photographing everything before disassembling** (position and thickness of thermal pads!), cleaning the GPU and base, applying the new paste (**PTM7950 is the ideal choice here**, precisely because of its resistance to pump-out), **replacing thermal pads only if necessary and with identical thickness** — and this is the trap: a 1.5 mm pad replaced with a 2.0 mm one prevents the plate

from resting on the die, and the GPU goes into severe throttling. Tighten the screws **crosswise and progressively**, following the numbering often printed on the backplate.

Regarding warranty, the same considerations apply as for the GPU waterblock in section 8.4: opening a card under warranty is a risk that must be consciously evaluated.

8.6.6 Recommendations by tier (thermal paste)

Entry / normal use: Arctic MX-4. Economical, excellent, simple, safe. End of story.

Mid / for those who want the best without risks: Noctua NT-H2 or Arctic MX-6.

High-end / GPU and laptops: PTM7950, for long-term stability. **Extreme overclocking, bare die, delidded CPU:** liquid metal, with all due precautions and full awareness of the risks.

8.7 Fans on radiators (and coolers)

8.7.1 Airflow vs. static pressure

A fan is defined by a **characteristic curve** that links airflow to pressure: at zero pressure (no obstruction), airflow is maximum; as the obstruction increases and the required pressure rises, airflow decreases, up to the point of **maximum static pressure** (mm H₂O), where the fan spins but no longer moves air.

Fans are designed by prioritizing one end of the curve or the other:

- An **airflow fan** has wide, spaced, few blades and a small motor: it moves a lot of air in free space, but "drowns" in front of an obstacle. This is the right fan for a free front panel or for unobstructed rear extraction.
- A **static pressure fan** has numerous, closely spaced blades with a more aggressive profile and a larger hub: it moves less free air but pushes much better through resistance. **This is the mandatory fan for a radiator or a dense fin stack.**

Mounting an airflow fan on a 38 mm radiator with dense fins means throwing away a significant part of the radiator's performance. This is a common mistake, often made for aesthetic reasons (the "beautiful" RGB fan is almost always an airflow fan).

In modern practice, many quality fans are hybrid and work well in both regimes (Arctic P12/P14, Noctua NF-A12x25, Phanteks T30, be quiet! Silent Wings): the most reliable way to choose is to read the **static pressure specifications** (a value above 2.0–2.5 mm H₂O is indicative of a fan suitable for radiators) and reviews with measurements at the same noise level.

8.7.2 Push, pull, push-pull

These are the three ways to arrange fans relative to the radiator.

Push: the fans are in front of the radiator relative to the flow direction, and **push** air into it. This is the most common configuration and the most efficient for a single fan, because the air reaches the radiator already under pressure.

Pull: the fans are behind the radiator and **pull** air through it. Performance-wise, it is slightly inferior to push, because the fan also draws from the edges and the flow is less uniform. However, it has a practical advantage: **the radiator remains accessible and visible** and dust settles on the outer face of the radiator instead of on the fans. In some cases, it is the only configuration that physically fits.

Push-pull: fans on both sides, pushing from one side and pulling from the other. The benefit is real but modest: typically 2–5 °C on a thick radiator with dense fins, less than 2 °C on a thin radiator. The cost is double in fans, double in potential noise (although, for equal dissipation, all fans can be kept at lower RPMs, which *reduces* overall noise), and double in occupied thickness — a 38 mm radiator in push-pull with 25 mm fans occupies **88 mm**, which doesn't fit in many cases.

Operational advice: push-pull is worthwhile on thick radiators (38 mm and above), on radiators with high fin density, and when the goal is silence (because four fans at 800 RPM make less noise than two at 1,400 RPM for the same dissipation). On a standard 27 mm radiator, in a normal system, the gain does not justify the expense.

8.7.3 Orientation: understanding which way a fan blows

A trivial question, and one that's constantly answered incorrectly. There are two rules, and they are infallible: 1. **Air enters from the side with the sticker/hub with the logo... no.** Attention: air enters from the *open* side (where you only see the blades and the frame) and exits from the side where you see the **support spokes** (the four or five arms holding the motor) and the back of the hub. In other words: **the fan blows towards the side where you see the structural "grille."** 2. Practically all fans have **two small arrows stamped on the frame** (one indicates

the direction of flow, the other the direction of rotation). Look at those and trust them.

If you make a mistake, the symptom is clear: high temperatures and an airflow that, when you put your hand near it, doesn't go where it should.

8.7.4 Control: PWM, DC, and curves

Fans connect to the motherboard with **3-pin (DC)** or **4-pin (PWM)** connectors. In **DC** (Direct Current) control, the motherboard varies the supply voltage (from 12 V downwards) to slow down the fan; below a certain voltage (typically 4–5 V), the fan doesn't start at all. In **PWM** (Pulse Width Modulation) control, the power supply remains at 12 V, and a fourth wire carries a signal that rapidly switches the motor on and off, varying the **duty cycle** (the percentage of time it's on). PWM allows for finer control and lower minimum speeds (often 200–300 RPM), and is the correct choice for a silent system.

The **fan curve** is set in the BIOS/UEFI or via software: it associates a speed with each temperature. Two concrete tips that are worth more than much expensive hardware:

- ****Linking case fan curves to CPU temperature is a common mistake****, because the CPU has very violent boost transients (from 40 to 80 °C in half a second, then down) that make the fans "pump" annoyingly. It's better to link them to a slower sensor (the motherboard temperature, or the liquid temperature if the AIO exposes it) or, if not possible, set a generous ****hysteresis / step-up and step-down time**** (3–5 seconds) to dampen oscillations.
- ****The AIO pump should not be treated like a fan****: it connects to the AIO_PUMP connector and follows the manufacturer's instructions, which in most cases is "100% constant" or a very flat curve.

8.8 Operational summary — cooling decision checklist

To be used in order, before any purchase.

1. Quantify the actual thermal load. ☐ Look for the actual power consumption under heavy load of your CPU in independent reviews (not the declared TDP: look for MTP for Intel, PPT for AMD, or better yet, measurements in Cinebench or Prime95). ☐ Add a 15–20% margin if you plan to use PBO / overclock / unlocked power limits.

2. Decide on the technology. ☐ Below ~100 W → inexpensive single tower. Done.
☐ 100–200 W → inexpensive dual tower (Thermalright Peerless/Phantom, Arctic Freezer 36). This is the default choice for the vast majority of users. ☐ 200–280 W → high-end dual tower **or** AIO 280/360. Choose air if you prioritize reliability, cost, zero maintenance. Choose liquid if you have tall RAM, want a clean aesthetic, or if your case doesn't accommodate large towers. ☐ Over 280 W → AIO 360/420. Below this threshold, you will leave performance on the table. ☐ Custom loop → only if you also want to cool the GPU, seek absolute silence, or do it as a hobby. It is never the economically rational choice.

3. Check compatibility (all three, without skipping any). ☐ **Socket:** Does the mounting kit include your socket (LGA1700 / LGA1851 / AM4 / AM5)? Verified on the cooler manufacturer's website, not the seller's product page. ☐ **Height (air):** Is the cooler's height less than the case's "max CPU cooler height," with at least 3–5 mm of clearance? ☐ **RAM (air):** Is the height of your memory modules compatible with the declared clearance? If you already have a large tower, did you choose low-profile RAM? ☐ **Radiator (liquid):** Does the case support that length *in that position*? Does the thickness (radiator + fans) fit within the declared clearance, without touching the VRM or RAM? ☐ No conflict with the first PCIe slot or M.2 heatsinks?

4. Plan the installation. ☐ (Liquid) The radiator is on the **top**, or on the **front with the tubes at the bottom. Never** on the bottom. **Never** front with tubes at the top. ☐ (Liquid) The pump is **not the highest point in the loop**. ☐ (Liquid) The pump is connected to the AIO_PUMP connector with zero speed alarm active. ☐ The fans on the radiator are **static pressure** ($\geq \sim 2$ mm H₂O), not pure airflow fans. ☐ Fan direction is verified with the arrows on the frame. ☐ (Intel LGA1700) Have you considered a **contact frame** to compensate for ILM deformation? It's worth 3–8 °C for 15–20 €.

5. Thermal paste. ☐ **Have you removed the protective film** from the base of the new cooler? (Yes, check again.) ☐ If reusing a cooler: both surfaces cleaned with **99% isopropyl alcohol** and a lint-free cloth. ☐ Quantity: a pea in the center (AM4/AM5) or a line/X (LGA1700/1851). Little, not a lot. ☐ Screws tightened **in a cross pattern, progressively**, never one at a time all the way. ☐ Cooler placed **only once**: if you lift it, clean and reapply. ☐ Paste chosen: MX-4 (default), NT-H2 (premium), PTM7950 (GPU/laptop, longevity), liquid metal (only if you know exactly what you are doing).

6. Post-installation check. □ Stress test of at least 15 minutes (Cinebench in loop, Prime95, OCCT) monitoring **simultaneously** temperature, sustained frequency, and power consumption. □ Temperatures within the ranges of the table in 8.1.4 — remembering that 90–95 °C on a Ryzen 9 under full load is not a fault, it is the designed behavior. □ No gurgling noise from the pump after the first few days of break-in. □ Fan curve set with adequate hysteresis, to avoid acoustic "pumping."

7. Scheduled maintenance. □ Dust removal from fin stack / radiator every 6–12 months. □ CPU repaste every 3–5 years (or sooner, if temperatures progressively worsen under the same load). □ GPU repaste after 3–5 years, considering PTM7950 and photographing pads before disassembly. □ (AIO) Plan for replacement after 5–7 years. □ (Custom loop) Liquid change and flush every 12 months. Leak test mandatory after each intervention.

Next chapter: the power supply and power distribution — that is, the only component whose failure can take out all the others.

Chapter 9 - PC Assembly, Step by Step

Introduction: why this chapter is different from all the others

Up to this point, the book has worked in the abstract. We have studied the CPU as an architecture, RAM as a hierarchy of latencies, the GPU as a parallel computing engine, the power supply as an energy converter. They were objects of knowledge. From this chapter onwards, they become physical objects: they have weight, they have edges, they have pins that bend, they have connectors that only fit one way and that melt if you insert them incorrectly. The difference between knowing what an LGA socket is and being able to close a retention lever without trembling is the same as between reading a flight manual and flying.

It must be said immediately, to relieve anxiety: **assembling a PC is mechanically easy**. It does not require soldering, it does not require multimeter measurements, it does not require experience. It is a game of interlocking parts designed specifically so that you cannot make too many mistakes, with polarized connectors (i.e., shaped to fit only one way) and alignment notches everywhere. An attentive adult following the instructions will finish the job in two or three hours the first time, and in forty minutes the fifth time.

But — and here's the point — mechanical ease coexists with **a few irreversible and costly actions**. A bent socket pin can mean a 300-euro motherboard to throw away. A 5V ARGB connector inserted into a 12V header destroys the LED strip and, sometimes, the controller on the motherboard. A 12V-2×6 power cable not fully inserted can melt the GPU connector under load. There isn't a long tail of medium errors: there's a huge set of harmless errors, and a tiny handful of lethal errors. This entire chapter is built around that distinction. Each step will tell you: what you need, how to do it physically, **how much force to apply** (it's the variable no one ever explains to you), what not to do for any reason, and how to check that you've done it correctly before moving on.

A final warning about the data: the socket names, connector models, and chipset acronyms mentioned here are those in circulation at the time of writing. They are the part of the book that ages fastest. The **physical procedures** — levers, notches, torque settings, order of steps — are, however, extraordinarily stable: anyone who assembled a PC in 2015 will recognize every gesture of a 2026 assembly. When data is volatile, I will explicitly indicate it.

0. Preparation: ninety percent of the work is done before opening the boxes

0.1 The tools: few, but the right ones

The tool you really need is **one**: a **Phillips PH2** screwdriver, magnetic, with a shaft at least 10-12 centimeters long.

Let's clarify the acronym, because it's the first example of how this job works.

"Phillips" is the name of the screw head type — the cross with slightly flared sides — and **PH2** is its size. PH0 and PH1 are smaller, PH3 is larger. Practically all screws in a modern PC (motherboard screws, power supply screws, panel screws, PCIe bracket screws, fan screws) use the PH2 head. The only exceptions are the micro-screws for M.2 SSDs, which require a PH1 or even a PH0, and which generally come with their own screwdriver in the motherboard box.

The reason for **magnetic** deserves a line, because it's the point that beginners are most scared of: no, a magnetized screwdriver **does not damage** PC components. The fear arises from a wrong analogy with old mechanical hard drives and floppy disks, and in any case, the field of a magnetized screwdriver is very weak and only extends a few millimeters. On the contrary, magnetism saves your day: screws stick to the tip and don't fall inside the case, into that unreachable space under the motherboard from which they can only be retrieved by turning everything upside down and shaking. A lost screw in a closed case that is then powered on is, this is true, a potential source of short circuit.

Around the screwdriver, little else is needed:

Cable ties. You'll need about ten. Disposable nylon ones are perfectly fine (to be cut with wire cutters, never torn); if you plan to frequently work inside the PC, **Velcro cable ties** are preferable, as they are reusable and often included with the power supply or case.

A light. The back of a case, the header area on the bottom edge of the motherboard, the inside of a hard drive bay: these are all dark places. An adjustable desk lamp or a headlamp transforms a fumbling operation into a clear one. Using your phone's light, held with your left hand while your right hand tries to insert a three-millimeter F_PANEL connector, is a recipe for frustration.

A large, flat, stable, non-textile surface. A clear dining table is perfect. You need space for the case lying down, for the motherboard box (which will serve as your

workbench, we'll get to that), for components awaiting installation, and for empty boxes, which should be kept: they store accessories, are useful for potential returns, and the motherboard box is the ideal surface for pre-assembly.

Wire cutters or scissors for the cable ties, and optionally a **container for screws** — a bowl, an egg carton, a jar lid. PC screws come in at least five different types, and mixing them up is a slow way to lose twenty minutes.

What you **don't** need, despite what you might be told: pliers (if you need pliers, you're doing something wrong), thermal paste if the cooler already has it applied, a professional anti-static mat, a tester.

0.2 Static Electricity (ESD): The Right Fear, in the Right Measure

ESD stands for *ElectroStatic Discharge*. It is the phenomenon where two materials rubbing against each other exchange electrons, and one of them becomes charged; when that charged body touches a conductor at a different potential, the charge discharges suddenly, with a potential difference that can easily exceed **one thousand volts** and reach tens of thousands. It's the spark you get when touching a car door handle in winter, and it's the same phenomenon that makes you "crack" when you take off a wool sweater in the dark.

The delicate point is this: the threshold at which **you** perceive a shock is around 2,000-3,000 volts, but modern transistors, with gate insulators only a few nanometers thick, can be damaged by discharges of **a few hundred volts**. This means there is a wide range where you can destroy a component **without noticing anything**. Worse: ESD damage is often *latent*. It doesn't kill the chip immediately; it weakens it. The PC works, perhaps for weeks, and then starts giving random errors, inexplicable crashes, data corruption. It is the most difficult type of fault to diagnose.

That said, it must also be stated that the odds are in your favor: modern components have ESD protection diodes on their inputs, and millions of people assemble PCs every year without precautions and without consequences. The rational strategy is not paranoia; it is **routine**. A few simple rules, always the same:

1. **Work on a hard surface, not on carpets, rugs, or sofas.** Textiles are the primary source of static charges. If the floor is carpeted, work on a table and do not drag your feet.
2. **Discharge yourself before touching components** by touching a large, freshly painted metal surface — typically the **PC case chassis** —

and repeat this action every time you have stood up, walked, or taken off or put on a sweatshirt. 3. If you want the technically correct solution, use an **anti-static wrist strap**: a conductive strap on your wrist, connected by a cable (which contains a 1 MΩ resistor for your safety) to a ground. The most convenient ground is the PC case chassis. It's good to know something almost no one explains: the wrist strap **does not "discharge" components**; it serves to keep **you and the case at the same potential**, so that there is never a voltage jump between your hand and the board. The PC does not need to be plugged into the power outlet (in fact, it should not be, we'll discuss that in a moment). Equipotentiality is sufficient. 4. **Keep components in their anti-static bags** (the gray, metallic ones that form a Faraday cage) until you use them. And never place a motherboard *on top of* its anti-static bag: the outer surface is slightly conductive and can create unwanted paths. Place it **on top of the cardboard box**, which is the ideal insulator. 5. **Handle boards by their edges**. Never by the gold contacts, never by the chips.

A note that debunks a common myth: **do not** leave the PC "plugged in with the switch off to have grounding." This was advice from the nineties, when ATX power supplies did not always have live voltages. Today, a power supply connected to the mains keeps the **5 V standby (+5VSB)** line alive even when the PC is off: if you work inside the case with the power supply connected, there are parts under voltage. The modern rule is: **power cable unplugged from the wall, always, when your hands are inside the case**.

0.3 The final compatibility check: the last exit before the highway

Opening a box means, in many stores, giving up the right to return "for second thoughts" (the right of withdrawal remains, but an opened and assembled product can be contested). So, **before breaking the first seal**, do a final check. It's tedious and will save you one out of five times.

Take the motherboard box — it's the hub of all compatibilities — and check these six points, one by one:

#	What to check	How	Symptom if wrong
1	Socket	The socket printed on the mobo box (e.g., AM5, LGA1851, LGA1700) must match the CPU's	The CPU does not physically fit

		socket <i>exactly</i>	
2	CPU BIOS support	On the mobo manufacturer's website, "CPU Support List" section, find the <i>exact model</i> of your CPU and read the minimum BIOS version required	The CPU is compatible with the socket but the PC does not POST: a BIOS update is needed
3	RAM: type and QVL	DDR4 and DDR5 are not interchangeable (different notch position, different voltage, different pinout). Also check the QVL (<i>Qualified Vendor List</i> , the list of tested kits)	RAM that doesn't fit, or fits but doesn't hold the declared frequency
4	Cooler clearance	Maximum cooler height allowed by the case (in mm) $\geq$ your cooler's height. For AIOs: does the case support a radiator of that length <i>in that position</i> ?	The side panel does not close
5	GPU length	Maximum GPU length allowed by the case $\geq$ your GPU's length (note: power cables take up another 3-4 cm if they exit sideways)	The GPU does not fit, or hits the hard drive bays
6	PSU: watts and connectors	Sufficient power (with margin) and	You cannot power everything

		physical presence of required connectors: how many 8-pin PCIe does the GPU need? Is a 12V-2×6 needed? How many EPS does the mobo need?	
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Point 2 deserves its own paragraph, as it's the most insidious trap in modern PC building. A socket has a lifespan spanning multiple CPU generations: AM5, for example, hosts CPUs released years apart. A motherboard manufactured in 2023 and left in storage might have firmware that simply **doesn't recognize** a CPU released in 2025: the socket matches, the CPU installs, but the PC won't boot because the BIOS doesn't know how to initialize it. The solution exists and is excellent, but it must be planned **in advance**: it's called **BIOS Flashback** (commercial names: *BIOS Flashback*, *Q-Flash Plus*, *Flash BIOS Button*), a feature found on mid-range and high-end motherboards that allows you to update the BIOS **without a CPU, RAM, or GPU installed**, simply by connecting the power supply, inserting a FAT32-formatted USB drive with the BIOS file renamed as required, and pressing a small button on the back. If your CPU is new and your motherboard might be old stock, **verify that the board has Flashback**: it's the difference between an hour of waiting and a week lost trying to borrow an old CPU.

0.4 The Strategy: "Bench" Pre-assembly

Here's the advice that distinguishes someone who has built ten PCs from someone building their first, and that no illustrated guide emphasizes enough: **do not install the motherboard in the case first**.

The correct method is **bench pre-assembly** (also known as *bench build* or *breadboarding*). You place the motherboard **on top of its cardboard box** — not on the anti-static bag, as mentioned — and on that comfortable, well-lit, easily accessible surface, you install: **CPU, RAM, M.2 SSD, and (if air-cooled) the cooler**. Only then do you take the assembled block and lower it into the case.

There are three solid reasons for this.

The first is **ergonomic**: installing a CPU inside a case means working vertically, with poor lighting, and the risk of the component slipping. Installing it on a table means working horizontally, with gravity assisting instead of hindering, allowing you to clearly see the pins and notches. The same applies to RAM, which needs to be pushed firmly: doing so with the motherboard already screwed into the case means flexing the PCB (the *Printed Circuit Board*, the fiberglass support on which the board is built) against its mounting points; doing it on a rigid surface means the force is absorbed by the table.

The second is **diagnostic**, and it's the most important. Pre-assembly allows for **out-of-case POST testing**: install the CPU, cooler, a single stick of RAM, the GPU, and the power supply, connect a monitor, and try to power it on by momentarily shorting the two Power Switch pins with the tip of a screwdriver (or using the integrated power button, if the board has one). If the BIOS screen appears, you are certain that the four most expensive and difficult-to-replace components **are working and compatible with each other**. If the PC doesn't boot, you have a bare system with five cables, where isolating the problem is trivial. Compare this to the alternative scenario: you install everything in the case, with forty cables, fans, hard drives, and it doesn't boot. Now the culprit could be anything, and to investigate, you have to disassemble everything. The POST test takes ten minutes and repays them tenfold in the worst-case scenario.

The third is **practical**: some cases make it impossible to mount the backplate of certain coolers once the motherboard is screwed in, if the rear cutout of the *motherboard tray* is not aligned with the socket. Better to find out beforehand.

The only exception to the rule is the **radiator of an AIO** (*All-In-One*, the closed and pre-assembled liquid cooling system). The waterblock is mounted on the CPU on the bench, but the radiator, which is connected to it by short and rigid tubes, must be mounted **after** the motherboard is in the case. We will discuss this in detail in points 4 and 8.

1. Installing the CPU on the Motherboard

It's the first and most feared step. Let's do it right.

1.1 The Two Philosophies: LGA and PGA

Historically, there have been two ways to connect a CPU to a motherboard.

PGA (Pin Grid Array): the pins — hundreds of thin metal needles — are **on the CPU**, and the socket on the motherboard is a matrix of holes. This is the scheme used by AMD up to and including the AM4 socket. The risk is entirely on the processor: if you drop it or place it incorrectly, you bend its pins, and the pins of a CPU can (with patience, a magnifying glass, and the tip of a razor blade) even be **straightened**.

LGA (Land Grid Array): the pins are **on the motherboard socket** — a thousand or more elastic, very thin needles, mounted on a plate — and the CPU only has flat, gold-plated contact pads (*lands*, precisely). This is the scheme used by Intel for twenty years and, since 2022, also by AMD with the **AM5** socket. The risk has shifted: the CPU is almost indestructible, but the motherboard socket is **extremely fragile**. A bent LGA pin is a few tens of microns wide; straightening it requires a microscope, a steady hand, and incredible luck, and in most cases, the motherboard needs to be replaced — and warranties **do not cover** mechanical damage to the socket.

Operational consequence, valid for **all** current platforms (Intel LGA and AMD AM5):

NEVER, for any reason, touch the socket pins. Not with your fingers, not with a screwdriver, not with a brush, not with a cloth. If something falls into the socket, do not "brush it away": blow gently, or use compressed air from a distance.

1.2 The Procedure, Step by Step

The socket is covered by a protective **plastic cap** (usually black). **Do not remove it by hand**. On modern sockets, it detaches by itself when you close the lever with the CPU inside, or it should only be removed after opening the mechanism. Tearing it off prematurely is one of the most common ways to damage the pins. (Keep it: if one day you need to send the motherboard for RMA — *Return Merchandise Authorization*, the warranty return procedure — it must be reattached, and some manufacturers refuse returns without the cap.)

Step 1 — Open the retention mechanism. On **AM5** and **Intel LGA**, the principle is the same: a side metal lever, held by a hook. Press the lever **down and slightly sideways** to release it from the hook, then lift it all the way up (about 90-100 degrees). You will see the retention plate (the *load plate*) above the socket rise. The

lever offers considerable resistance at this stage: this is normal, it is a preloaded spring.

Step 2 — Pick up the CPU. Pick it up **by the side edges**, with your thumb and index finger, like you would a CD. Never by the faces.

Step 3 — Align. This is the crucial step, and it must be done by **looking**, not by fumbling. There are two redundant alignment systems:

- A ****golden triangle**** printed on one corner of the CPU package must match a ****small triangle**** (or an arrow, or a chamfered corner) printed on one corner of the socket. On AM5, the triangle is clearly visible; on Intel LGA, you'll find it silk-screened on the frame.
- ****Notches**** (half-moons) on the short sides of the package fit into corresponding ****tabs**** on the socket. They are intentionally asymmetrical: if the CPU is rotated, it won't sit flat.

Step 4 — Place, don't press. This is the phrase to remember: **the CPU is placed, not inserted**. Keep it horizontal, bring it over the socket, and lower it **straight down from above**, without tilting it, without letting it slide sideways onto the pins. It must **drop into its housing with a one-millimeter movement** and stay there firmly. The force to apply in this step is **zero**. If you have to press, it's wrong: lift it straight up and try again.

How to check if you've done it correctly: with a finger, give the CPU a very light tap on the side. It should have a micro-play (a fraction of a millimeter) and return to position. If it doesn't move *at all*, it might be stuck crooked; if it wobbles noticeably, it's not in its seat. Then look at the profile **from above and from the side**: the CPU must be perfectly parallel to the socket plane, with no raised corner.

Step 5 — Close. Lower the load plate and bring the lever back down, re-engaging it under the hook. **Here, the resistance is strong, and this scares everyone**. The force required to close a modern LGA socket is on the order of **50-70 newtons** (the equivalent of five to seven kilograms of thrust): it serves to press the CPU against over a thousand elastic pins, ensuring reliable contact on each one. **It is normal to feel significant resistance and even a small creak**. What is *not* normal is a **hard stop**: if the lever does not descend with progressive and firm pressure, **stop**, reopen, and recheck the alignment. The resistance is elastic and continuous; the obstacle is hard and sudden. Learning to distinguish these two sensations is, literally, the key skill for this step.

The plastic cap, on Intel sockets, will pop off by itself upon closing. On AM5, it should be removed during the process according to the manual.

1.3 Common Errors

Bent pin. This almost always results from two actions: sliding the CPU onto the socket instead of placing it, or forcing the lever with the CPU crooked. If, after an error, the PC doesn't start, or starts but doesn't see half of the RAM (the memory channel pins pass through the socket!), suspect a bent pin. Inspect it with a grazing flashlight and a magnifying glass: the pins must form perfectly regular rows, like a combed field of wheat. A row that "wobbles" indicates a misplaced pin.

CPU mounted crooked ("at 90 degrees"). Impossible on a healthy platform, because the CPU is not square in its anchors — but it is possible by forcing it. If you even have a doubt, reopen and look at the small triangle.

Touching the golden surface (IHS). The **IHS** (*Integrated Heat Spreader*, the metal lid you see on top of the CPU) must be kept clean: fingerprints leave grease, which is a poor thermal conductor. If you touch it, clean it with a microfiber cloth and isopropyl alcohol.

Do not keep the socket cover. See above: it can void the warranty.

2. Installing RAM

This seems like the most trivial step, but it is actually the number one cause of "my new PC won't turn on."

2.1 Why not all slots are equal: dual channel

System memory communicates with the CPU through the **IMC** (Integrated Memory Controller, the memory controller that now resides **inside** the CPU, no longer in the chipset). The IMC of modern consumer CPUs is **dual channel**: it exposes two independent channels to the RAM, each with its own bus. If you populate only one channel, you halve the available bandwidth. The effect is dramatic where bandwidth matters — in **iGPUs** (integrated GPUs, the graphics integrated into the CPU, which uses system RAM as video memory) it can mean **double the frames**; in gaming with a dedicated GPU, we are still talking about double-digit percentages in the worst cases.

Now: a standard ATX motherboard has **four** DIMM slots, but there are **two** channels. Each channel has two slots. The question becomes: if I have two sticks, where do I put them?

The answer, almost universally, is: **in slots 2 and 4 counting from the CPU**, i.e., those labeled **A2 and B2** (or **DIMM_A2 / DIMM_B2**, or on some DDR5 boards with different nomenclature). These are the two slots **furthest** from the CPU among those available, and it's counterintuitive: one would think that closer is better.

The reason is **signal integrity**. The traces that carry the memory bus from the CPU to the slots are very high-frequency lines; an empty slot left "before" a populated one creates a **stub**, an unterminated section of the line that generates signal reflections. The *daisy-chain* topology used by the vast majority of consumer boards is designed and optimized for the signal to be cleaner when the **terminal slots** are populated. On AM5 and DDR5 platforms in general, where frequencies are very high and the controller is already under stress, this is not an academic detail: installing two sticks in the wrong slots can mean **not being able to start the EXPO/XMP profile**, or not POSTing at all.

The golden rule: open your motherboard manual to the "Memory Configuration" page and read the table. Thirty seconds stand between you and certainty. The table tells you exactly, for 1, 2, and 4 modules, which slots to populate. Don't guess, don't rely on memory: some models (rare, but they exist) require A1/B1.

A note worth the price of the chapter: **two sticks are better than four**. With four modules, the IMC has to drive twice the electrical loads, and the maximum stable frequency **drops**, often significantly (on DDR5, a kit that supports 6000 MT/s with 2 modules may struggle above 4800 with 4 modules). If you need 32 GB, buy **2×16**, not 4×8. And most importantly: **do not mix different kits**. Two 2×16 kits bought at different times, even if identical on the label, may have chips from different manufacturers and not work together at the nominal frequency. A kit is tested *as a whole*: if one day you want 64 GB, buy a new 2×32 kit.

2.2 The procedure

Step 1 — Open the clips. The clips are the plastic levers at the ends of the slot. Some boards have clips on **both** sides, while others (increasingly common, to make room for the GPU) have a clip on only **one side**, with the other end fixed.

Look at and open the ones that are present, pushing them outwards until they snap open.

Step 2 — Align the notch. The DIMM module has a **notch** (an indentation) on the edge of the contacts, **off-center**: it's not in the middle. The slot has a corresponding tab. Rotate the module until the notch aligns. **This notch is why you cannot install DDR4 on a DDR5 board**: it's in a different position specifically to make errors physically impossible (the two generations have incompatible voltages and pinouts, and forced insertion would destroy something).

Step 3 — Press. This is the other point where you need to overcome fear. Place the module in the slot, ensure it is **perfectly vertical and aligned**, then press **with your thumbs on the two upper ends of the module, one at a time or together**, with **firm and progressive force**. The necessary force is **considerable**: we're talking about a push comparable to what you would use to close a full suitcase. It is not a delicate gesture.

The module slides down, and the clips — pushed by the shape of the PCB — **close by themselves with an audible and distinct CLICK**. If the clips are on both sides, you should hear **two clicks**. Hence the expression you'll hear everywhere: *"press until the double click."*

How to check if you've done it correctly: look at the module **from the side, at the height of the slot**, and check that the **golden line of contacts has completely disappeared** inside the slot, **uniformly along its entire length**. Then try gently pulling the module upwards: it should not move a millimeter. Finally, check that the clips are **vertical and closed**, engaged in the side notches of the module.

2.3 The error that affects everyone: the half-inserted module

It is statistically **the number one cause** of "new PC that does nothing": the module is inserted, it seems to be in, the clip on one side has closed, but **on the other side, it has remained raised by a millimeter**. The electrical contact is partial or absent, the POST fails, the PC turns on (the fans spin!) but gives no video signal, and the **DRAM** diagnostic LED remains lit.

Why does this happen so often? Because the necessary force is greater than what instinct suggests, and because pressing with only one thumb in the center causes the module to tend to get stuck on one side and remain high on the other. **Press at the ends. Listen for the clicks. Look at the contacts.**

3. M.2 SSD Installation

3.1 What it is and which slot to choose

An **M.2 SSD** is a solid-state drive on a small rectangular circuit board (typically 22 × 80 mm, hence the designation **2280**) that plugs directly into a slot on the motherboard, without cables. In almost all modern cases, we are talking about **NVMe** (*Non-Volatile Memory Express*, the protocol specifically designed for flash memory) units that communicate over the **PCIe** (*Peripheral Component Interconnect Express*, the serial lane-based bus that connects CPU and peripherals) bus.

And here lies the important choice: **not all M.2 slots on a motherboard are equal**.

A modern motherboard typically has 2, 3, or 4 M.2 slots. Of these, **only one** — almost always the one **closest to the CPU**, between the socket and the first PCIe x16 slot — is connected **directly to the CPU's PCIe lanes**. The others go through the **chipset**, the chip that acts as an expansion hub and is connected to the CPU via a shared link (DMI on Intel, a dedicated PCIe link on AMD).

There are two practical consequences. The first is **bandwidth**: the slot from the CPU generally offers the latest PCIe generation and full x4 (four lanes); the slots from the chipset may be an older generation, and in any case, they share the chipset ↔ CPU link bandwidth with everything else (USB, SATA, network, expansion cards). The second is **latency**: going through the chipset adds a hop.

Rule: *Your primary SSD — the one with the operating system — goes into the **primary M.2 slot, the one connected to the CPU**. The manual explicitly indicates this, often calling it M2_1, M2A_CPU, or similar, and always specifies which lanes it uses.*

There's a second reason to read the manual: **resource sharing**. On many boards, populating a certain M.2 slot **disables** SATA ports, or reduces the secondary PCIe x16 slot from x4 to x2. These are normal compromises (lanes are a finite resource), but they should be known beforehand, not discovered after installing three drives that don't show up.

3.2 The procedure

Step 1 — Remove the motherboard's M.2 heatsink. Modern motherboards cover M.2 slots with a metal plate (sometimes a true finned heatsink). It's unscrewed with one or two small screws (PH1). Remove it and set it aside with its screw.

Step 2 — Remove the film from the thermal pad. This is the step that **almost everyone forgets**, and it's worth emphasizing: under the heatsink there is a **thermal pad** (a cushion of conductive rubbery material), and on top of the pad there is a protective **transparent plastic film. It must be removed.** If you leave it on, you've inserted an insulator between the chip and the heatsink, and your SSD will experience *thermal throttling* (automatic performance reduction due to excessive temperature) under load. The film is often invisible to a casual glance: run a fingernail over a corner to find its edge.

Step 3 — Adjust the standoff. The M.2 slot accepts different lengths (2242, 2260, 2280, 22110: the last two digits are the length in millimeters). The mounting point must be moved to the hole corresponding to the length of your SSD — almost always 2280, which is usually the default. Many recent boards have replaced the screw with a **snap-on clip** (commercial names: *M.2 Q-Latch*, *EZ Latch*), which is rotated or pressed without tools: much more convenient.

Step 4 — Insert at 30 degrees. The SSD does **not** go in straight. It should be presented to the slot at an **angle of about 20-30 degrees**, pushed in until the contacts are fully inserted (the notch on the contacts, like for RAM, ensures the correct orientation), and **then lowered** flat. The force is modest: you'll feel the module slide and stop. When you release it, the SSD will lift itself a few degrees — this is normal, it's the spring of the contacts.

Step 5 — Secure. Hold the SSD flat with one finger and screw in the micro-screw (or close the clip). **Do not overtighten:** you are tightening an M2 screw on a thin PCB. "Finger-tight plus a whisper" is more than enough; tightening more will strip the thread or crack the SSD.

Step 6 — Reattach the heatsink (with the film removed!) and screw it in.

How to check: The SSD should be flat, with no play. But the real check comes at the first boot, in the BIOS: if the drive does not appear in the list of storage devices, power off and recheck the insertion.

4. Cooler installation

Here is the most delicate and most misunderstood part of the entire assembly: thermal paste. Let's first clarify the principle, then the two procedures.

4.1 The principle: why thermal paste exists (and should ALWAYS be applied)

The CPU's IHS and the cooler's base (the *cold plate*) appear perfectly smooth. They are not. Under a microscope, both surfaces are mountain ranges: roughness of a few microns, plus any macroscopic deformations (many IHS are slightly concave or convex). If you place them one on top of the other, actual metal-to-metal contact occurs only on a **small fraction of the surface** — the rest are microscopic air pockets. And air is one of the **most effective thermal insulators** you know (it's why down comforters and double glazing work).

Thermal paste (*TIM*, *Thermal Interface Material*) serves exclusively to **fill those air pockets**. It is not meant to "conduct better than metal" — in fact, paste conducts heat **much worse** than copper. It serves to be better **than the air it replaces**. From this immediately follows the golden rule of application:

Thermal paste should be applied in the minimum amount sufficient to fill the gaps. More is not better: it's worse, because a thick layer of paste is a thick layer of mediocre material that moves the two metals apart.

And, equally important:

Thermal paste should ALWAYS be applied between the CPU and the cold plate. Always. Air or liquid makes no difference. This is a confusion I hear constantly: people believe that in liquid systems "the liquid takes care of it." No. The liquid works inside the waterblock; the interface between the CPU's IHS and the bottom of the waterblock is **identical** to that of an air cooler, and has exactly the same need for paste.

How much, and how. The most reliable and universal method is the **central pea**: a drop in the center of the IHS, with a diameter of **a grain of rice / a small pea** (4-5 mm). The mounting pressure spreads it radially and uniformly by itself, expelling air towards the edges — which is exactly what you want. Applying by hand with a spatula seems more "professional" but easily introduces air bubbles; the diagonal X and the five dots are acceptable variations, useful on very elongated IHSs (like the rectangular ones of certain HEDT platforms or LGA1700,

where some prefer a **thin line** along the major axis). For details on conductivity, liquid metal pastes, pump-out, and longevity, refer to the cooling chapter; here, the focus is on the action itself.

Two final warnings: **do not spread with your finger** (skin oil), and **do not assemble and disassemble "to check"**. If you lift the heatsink after placing it, the only correct way to proceed is to **clean everything with isopropyl alcohol and reapply from scratch**: the film has broken and incorporated air.

Last point: **many heatsinks come with pre-applied paste** on the cold plate (a gray layer, often with a silkscreened pattern). In that case, **do not apply more paste**. And, above all, **check if there is a plastic film on top**: we'll get to this, because it's the most catastrophic mistake in the entire chapter.

4.2 CASE A — Air Cooler

Step 1 — The backplate. Most quality air coolers do not use the motherboard's stock retention system, but their own: a metal or plastic plate (the **backplate**) that must be positioned on the **back** of the motherboard, behind the socket, and serves to distribute the strong mounting pressure over a wide area, preventing the PCB from flexing. This is exactly why pre-assembly on a test bench is convenient: the board is free, you flip it, put on the backplate, then flip it back. On AMD AM4/AM5 platforms, the backplate **is often already present** on the motherboard and must be **reused** (sometimes only the *retention module*, the black plastic frame around the socket, needs to be removed). Read the *cooler's* manual, which will tell you which path to follow.

Step 2 — The standoffs. Above the motherboard, corresponding to the four holes around the socket, four threaded posts that protrude from the backplate are screwed in. They should be tightened **by hand, until snug, without forcing**. They determine the height — and thus the pressure — of the cooler.

Step 3 — The paste. As above: pea-sized amount in the center of the clean IHS. (If the cooler already has factory paste, skip this step.)

Step 4 — Placing the cooler. Lower it **straight down, without dragging**, aligning the brackets over the standoffs. Do not "land" it at an angle.

Step 5 — Tighten in a cross pattern, incrementally. This is the technical step. The screws (usually four, sometimes two long bars) **should not be tightened one at a time all the way**. Tighten **diagonally** — 1, 3, 2, 4 — and in **partial turns**: two

turns on the first, two on the third (the opposite), two on the second, two on the fourth, and repeat. The reason is that the cooler must descend **parallel** to the IHS: if you tighten screw 1 all the way before touching 3, you tilt the cold plate, and the paste is squeezed on one side and not the other, resulting in poor contact (and, in the worst cases, PCB flexing). Tighten **until the screws bottom out and start to spin freely** or until the spring is fully compressed. Most modern kits have **captive screws with preloaded springs**, designed to make it *impossible* to overtighten: when the screw no longer advances, you are done. **Do not apply extra torque "just in case."**

Step 6 — Connect the fan to CPU_FAN. The 4-pin connector from the cooler goes into the header labeled **CPU_FAN** (not CPU_OPT, not SYS_FAN). Reason: the motherboard **monitors that header**, and many BIOSes **refuse to boot the system** (or show a "CPU Fan Error") if they don't detect a rotation signal there. The connector has a guiding tab on one side: it only goes in one way, and if you have a cooler with two fans, the second cable goes into **CPU_OPT**.

4.3 CASE B — AIO (liquid cooling)

An **AIO** is a pre-assembled closed loop: a **waterblock** with an integrated **pump** that mounts on the CPU, two **tubes**, a **radiator** with its fans. It is not filled, bled, or opened.

The order of the steps here is **different**, and getting it wrong means working in impossible conditions:

Step 1 — Mount the socket brackets on the waterblock, *first of all*, on the table. Each AIO includes different bracket kits for AMD and Intel: choose the right ones, and slide/screw them onto the block.

Step 2 — Backplate and standoffs on the motherboard, as for air cooling.

Step 3 — THE FILM. Stop. Look at the **cold plate** of the AIO, the copper or nickel plate on the bottom of the waterblock. In the vast majority of cases, the manufacturer has applied thermal paste at the factory **and covered it with a transparent or colored plastic film**.

***REMOVE THE FILM.** This is not a detail: it is the most expensive and most common mistake in this chapter. A cold plate mounted with the film on it means **zero thermal transfer**: the CPU reaches its safety temperature (**T-junction max**, typically 95-100 °C) in **a few seconds** after power-on, goes into*

extreme throttling, and then shuts down. The PC "shuts down by itself after ten seconds" is almost always this. It doesn't damage the CPU (the protections work), but it costs you an afternoon.

If the cold plate is **bare** (no pre-applied paste), then apply the pea-sized amount yourself on the CPU's IHS, as for air cooling.

Step 4 — Secure the waterblock. Place it straight and tighten the nuts/screws **crosswise and incrementally**, exactly as for air cooling, until snug.

Step 5 — The radiator: NOT NOW. The radiator remains hanging from its tubes, resting on the table. **It should be mounted in the case after installing the motherboard** (point 8). Attempting to mount the radiator and motherboard together, outside the case, means having to lower a rigid, heavy object constrained by two short tubes into the case: it's the best way to tear something. In the meantime, **gently rest the radiator** and do not bend the tubes at sharp angles.

Step 6 — Electrical connections. An AIO generally has two (or three) cables:

- the ****pump**** cable, which goes into a dedicated header: ****AIO_PUMP**** if it exists, otherwise ****CPU_FAN**** (the pump header always provides 100% voltage, without a regulation curve: that's what you want, a pump should always run at maximum or at a fixed speed);
- the cable (or cables) for the ****radiator fans****, which go into ****CPU_FAN**** and/or ****CPU_OPT****, or — if the kit has a ****splitter**** or a hub — into a single header. If you connect the pump to AIO_PUMP and the fans to CPU_FAN, the BIOS is happy and you can build a sensible curve.
- optionally an ****ARGB/USB**** cable for lighting and monitoring (be careful with ARGB: see point 10).

5. Case preparation

The motherboard + CPU + RAM + M.2 + air cooler block is ready. Before lowering it in, prepare the nest.

Step 1 — Open everything. Remove **both** side panels (the glass/metal one and the rear one, behind the *motherboard tray*: this is where you'll route cables) and, if possible, the front and top panels, to access the fan brackets. Put the panel screws in a separate container.

Step 2 — Check the standoffs. This step is often overlooked and is potentially **destructive**. The case's **standoffs** are small threaded hexagonal pillars, screwed into the motherboard tray, on which the motherboard will rest. They have **two functions**: to lift the board a few millimeters from the metal, to prevent the solder joints on the back from touching the frame and **short-circuiting**; and to define the anchoring points.

Cases come with standoffs **pre-mounted for the ATX format**. But there are three formats (and we'll list them, because the choice is yours):

Format	Typical Dimensions	Typical Mounting Holes	PCIe Slots	RAM Slots
ATX	305 × 244 mm	9	up to 7	4
Micro-ATX (mATX)	244 × 244 mm	6-8	up to 4	2-4
Mini-ITX	170 × 170 mm	4	1	2

***The critical check:** Place (without screwing) the motherboard in the case and check that **under every hole on the board there is a standoff**, and — **most importantly** — **that there is NO standoff where the board does NOT have a hole**. An "orphan" standoff touching the back of the motherboard is a **short circuit**: the PC won't start on the first boot, and in the worst case, the board will be damaged. This is a classic mistake made by those who install a Micro-ATX board in a case designed for ATX. Excess standoffs should be unscrewed (by hand or with a 5 mm socket wrench, often included).*

Many cases have a central **raised, non-threaded standoff**: it serves as support and for centering, not for screws. It's perfectly fine for it to be there.

Step 3 — The I/O shield. The **I/O shield** is the rectangular metal plate that covers the rear ports of the motherboard. On today's mid-range and high-end motherboards, it is **integrated** (already mounted on the board, often with a plastic cover over the VRMs): in that case, you don't need to do anything and can skip this step. If your board has a **separate** I/O shield — a metal plate in the box — it must be installed **now**, because you won't be able to do it later.

It is installed **from the inside of the case outwards**: place it in the rectangular opening at the back, check the **orientation** (the port labels must be readable from

the outside, and the shape must match your board's ports: look at the motherboard while you do this) and press on the **four corners** until it clicks into place with a metallic sound. Considerable **force** is required, and the metal sheet has sharp edges: be careful with your fingers. An I/O shield installed backward is one of the most bitter memories of PC building, because you only realize it after the motherboard is already screwed in. A small warning: some I/O shields have **elastic metal tabs** that protrude inwards. They must press **on top** of the ports (they act as grounding), not **inside** them: when you lower the motherboard, check that no tab has entered a USB or Ethernet socket.

Step 4 — Pre-route front cables. The front panel cables (power switch, USB, audio) and the cables for the pre-installed case fans are **already there**, dangling. Before installing the motherboard, route them through the **nearest grommets** to their destination, so they are already positioned correctly. Doing it later means having to thread your hand between the motherboard and the chassis.

6. Installing the PSU (Power Supply Unit)

The **PSU** (Power Supply Unit) is almost always located at the **bottom** in modern cases, in a separate compartment (shroud), and this is a good thing: it draws cool air from below and expels it directly outside, without heating up the rest of the system.

6.1 Fan Orientation: The Question Everyone Gets Wrong

The power supply fan can face **downwards** or **upwards (into the case)**. The rule is simple but needs to be understood:

- If the case has ****tall feet and a filtered grille under the power supply**** (practically all modern cases), the fan should face ****downwards****. The power supply draws ****cool air from outside****, uses it to cool itself, and expels it out the back. This is a completely independent circuit from the case's airflow: excellent.
- If the case rests directly on the floor without airflow, or the bottom is ****closed**** (old cases, desktop cases with the power supply at the top), then the fan should face ****inwards****: it will draw hot air from the case. This is suboptimal, but it's better than not getting any air at all. A fan facing a closed bottom will ****suffocate**** the power supply, and an overheating power supply will age quickly or trigger thermal protection.

If you place it on carpet, the feet are not enough: put a board under the case.

6.2 The Procedure

Step 1 — Connect modular cables BEFORE inserting the power supply. If your PSU is **modular** (detachable cables) or **semi-modular** (fixed 24-pin and EPS, the rest detachable), this is the time to attach the cables to the **power supply side**, while you have it in your hand and the ports are accessible. Once it's screwed into the bottom of the case, behind the shroud, those ports become inconvenient. Which cables do you need? Take inventory: **1× ATX 24-pin** (always), **1 or 2× EPS 8-pin** (for the CPU; check how many your motherboard requires), **PCIe** cables for the GPU (how many 8-pin? or a **12V-2×6?**), **1× SATA power** if you have 2.5" SSDs or hard drives, and any cables for fans/hubs.

CRITICAL WARNING, and this is not an exaggeration: modular cables are NOT interchangeable between different power supplies, even from the same manufacturer. The PSU-side connector may be physically identical but have a completely different pinout. Using a cable from one PSU on another is one of the very few ways to instantly destroy all connected components, sending 12V to a line expecting 3.3V or putting ground where there should be voltage. Only use the cables that came in the box with that power supply. If you've lost them, they can be purchased from the manufacturer, for that specific model.

Step 2 — Insert the PSU into its bay (sometimes from the rear, sometimes from the inside; some cases have a removable bracket that screws to the PSU first and then to the case).

Step 3 — Screw in the four screws on the back of the case. These are coarse-thread screws (UNC 6-32): do not force them.

Step 4 — Route the cables behind the motherboard tray, towards their destinations. Do not connect them yet: the motherboard goes in first.

7. Installing the Motherboard in the Case

Step 1 — Lower the board. Hold it by the edges (or by the CPU cooler, if it's an air cooler, which is now attached) and lower it into the case with a slightly **oblique motion**: first the rear ports towards the I/O shield, then the board flat.

Step 2 — Align with the I/O shield. The board should be **pushed towards the back of the case** until the ports protrude through the I/O shield holes. This is when you check that no metal tabs have ended up inside a port. When the board is in position, the **holes for its mounting points align perfectly with the standoffs**: if a hole is misaligned by two millimeters, do not force it — the board is not pushed back enough, or a standoff is in the wrong place.

Step 3 — Screw it in. Place **the first screw in a corner, without tightening it**, just to hold the position. Then the others. Then go back and tighten all of them, **working in a cross pattern** (never in a circle) to avoid putting tension on the PCB.

How much force? Little. The screw should **just make contact**: as soon as you feel metal touching metal, **stop**. Tightening a motherboard screw "with force" means deforming the PCB around the hole and, in the long run, cracking the copper traces. A properly screwed-in motherboard **does not flex** if you press a finger on it, but the screws are not as tight as wheel lug nuts.

Check: count the screws. There should be as many as there are standoffs. A standoff without a screw is an unsecured support point, and when you press the RAM or GPU, the board will flex there.

8. Installing the AIO Radiator (if present)

Now that the motherboard is in, the radiator finds its home.

8.1 Where to put it, and the rule of air in the pump

The two sensible positions in a standard ATX case are:

Top exhaust (at the top, expelling air) — the recommended choice. The radiator sits on the roof of the case and pushes hot air upwards/out. Advantages: it doesn't heat up the rest of the system; the GPU receives fresh air from the front; hot air goes where physics already wants to send it. Disadvantage: the liquid in the radiator receives air already slightly warmed by the components, so CPU temperatures are a few degrees higher compared to the front intake configuration.

Front intake (at the front, drawing in air) — maximum performance for the CPU. The radiator sits at the front and draws fresh air from the outside. The CPU gets the best possible temperatures, but **all the air entering the case has been**

preheated by the radiator, and the GPU, VRMs (Voltage Regulator Module, the CPU power stages), and SSDs pay the price.

However, there's a constraint that **overrides everything**, and it's hydraulic. The AIO **pump** must be **at the lowest point of the circuit**, or at least **never at the highest point**. The reason: in every AIO, over time, a small amount of air accumulates (permeation through the tubes, residual factory air). That air **floats** and goes to the highest point of the circuit. If the highest point is the **radiator**, the air collects there and does no harm. If the highest point is the **pump**, the air enters the impeller, the pump spins "dry" — and starts making an unmistakable noise, a cyclic gurgle/hum — and **loses efficiency, eventually seizing**.

Practical rule: the radiator tubes must exit from the BOTTOM SIDE of the radiator (or at least the waterblock must be below the level of the top part of the radiator). In a top configuration, mount the radiator with the tubes towards the back/bottom. In a front configuration, mount the radiator high, with the tubes at the bottom, never with the tubes at the top above the pump.

A position to **avoid**: the radiator mounted **at the top with the tubes pointing upwards** and the pump higher than the radiator itself — a typical situation in some very compact cases. And **rear** mounting (120 mm at the back) is only suitable for small AIOs.

8.2 The procedure

Step 1 — Mount the fans on the radiator, deciding the direction of the airflow (fans have arrows stamped on the housing: one indicates the air direction, the other the rotation). In top exhaust, the fans push air **through the radiator outwards**; they can be underneath (to push) or on top (to pull) — underneath is more common for space reasons.

Step 2 — Screw the radiator+fan assembly to the case. The screws that go through the radiator are **long and dangerous**: if they are too long, the tip will **puncture the fins and the liquid channel**, and the AIO is dead. Use **only the screws provided in the kit**, and distinguish between "short" ones (radiator screwed directly to the case) and "long" ones (radiator + fans together). When in doubt, screw **by hand, counting the turns**, and stop as soon as they meet resistance.

Step 3 — Connect the fans (to CPU_FAN/CPU_OPT or the kit's hub) and ensure that the tubes do not press against the RAM or the VRM heatsink.

9. GPU Installation

9.1 Which slot, and why it's not trivial

The motherboard has one, two, sometimes three **PCIe x16** slots — physically identical, long. They seem interchangeable. **They are not.**

The GPU goes into the **first PCIe x16 slot, the one closest to the CPU**. And here's why, which is the explanation missing in nine out of ten guides.

The **PCIe** bus operates on **lanes**: each lane is a point-to-point differential pair. A modern consumer CPU exposes a **fixed and limited** number of PCIe lanes — typically **20-28** usable — of which **16 are reserved for the graphics connection** and 4 (or 8) for the primary NVMe SSD. The rest of the connectivity (USB ports, SATA, network, secondary expansion slots) is managed by the **chipset**, which connects to the CPU via a single shared link.

This means that, on a consumer motherboard:

- the **first x16 slot** is **electrically x16** and is wired **directly to the CPU**;
- subsequent x16 slots are **physically x16** (the length is the same), but electrically they are **x4** or **x1**, and often pass through the **chipset**.

Putting a modern GPU into an x4 slot from the chipset means **strangling it**: bandwidth is drastically reduced and latency increases, with performance losses that can reach double-digit percentages in games (and much more in compute loads that continuously transfer data between RAM and VRAM). This is not a detail for purists: it's the kind of mistake that makes you buy an 800 euro GPU and get the performance of a 500 euro one.

The motherboard manual always states this, with notations like **"PCIEX16_1: x16 mode (CPU)"** and **"PCIEX16_2: x4 mode (Chipset)"**. Read it.

Exceptions, which exist and are legitimate:

- **AIO radiator or cooler clearance.** On some compact cases, the first slot is too close to something. In that case, an x8 slot from the CPU (if available, on boards that support x8/x8 splitting) is better than a GPU that doesn't fit.

- ****Workstation/HEDT**** (*High-End Desktop*, e.g., Threadripper) boards: here the CPU lanes are 48, 64 or more, and all slots are x16 electrical. The rule does not apply.
- ****Airflow.**** Some prefer the second slot to give a thick GPU some breathing room. It's a compromise: a few degrees less, halved bandwidth. Almost never worth it.

9.2 The procedure

Step 1 — Remove the rear brackets. Count how many **expansion slots** your GPU occupies (modern GPUs are 2, 2.5, or 3 slots thick) and remove the corresponding **metal brackets** from the back of the case, unscrewing them (**do not bend them until they break**: in cheap cases they are tear-off, and once broken they don't go back). Remove the **correct** ones: if the GPU goes into the first x16 slot and occupies 3 slots, you need to remove the corresponding bracket and the two below it.

Step 2 — Open the slot clip. At the end of the PCIe slot, towards the inside of the motherboard, there is a plastic **retention clip**. It must be lowered/pushed aside **before** inserting the GPU. This is a detail that is often missed, and forcing the GPU with the clip closed can break the slot.

Step 3 — Insert. Align the GPU's **gold connector** with the slot (the notch on the connector, as always, ensures the correct orientation) and the **bracket** with the holes on the back of the case. Press **straight down, with uniform force on both ends** of the card (one above the connector, one on the free end). The GPU slides in and the clip **snaps into place with a CLICK**. Verification: look at the connector from the side — the gold contacts must be **completely hidden** in the slot, and the card must be **parallel** to the motherboard.

Step 4 — Screw the bracket to the case (one or two screws). Do not overtighten.

9.3 GPU power: the paragraph that can save your house

GPUs consume more than the PCIe slot can provide (the slot provides a maximum of 75 W), so they draw power directly from the power supply. There are two families of connectors.

Classic family: PCIe 6+2 pin (commonly "8-pin"). Each 8-pin connector is certified for **150 W**; the 6-pin for 75 W. A GPU can have one, two, or three of these.

The daisy-chain rule. Many power supplies provide PCIe cables with **two connectors on the same cable** (one at the end and one "mid-cable," called a piggyback or daisy-chain). These are convenient but **share the same conductors**. On a low-end GPU (only one power input, or two inputs but modest power consumption), this is perfectly fine. On a powerful GPU — typically **250 W and up**, or whenever the GPU requires **two or more connectors** — **use separate cables, one for each GPU input, starting from different power supply outlets**. A single cable carrying 300 W on conductors designed for 150 W will heat up, degrade, and in documented cases, **melt**.

New family: 12V-2×6 (evolution of 12VHPWR). This is the high-density connector introduced with the **PCIe 5.0 / ATX 3.0** standard: twelve power pins (six for 12V, six for ground) plus four signal pins (sideband), capable of carrying up to **600 W** from a single cable. The "2×6" is the revision that shortened the signal pins compared to the original 12VHPWR: if the connector isn't fully seated, the shorter pins lose sense contact first, so the power supply cuts delivery to 0 W instead of still allowing up to 150 W through a loose connection — a safety/detection fix, not a mechanical barrier to partial insertion.

SAFETY WARNING, the most serious in this book: the 12V-2×6 connector must be inserted ALL THE WAY IN, until the clip clicks, and must be visually verified.

The reason is physical and unforgiving. A connector inserted **partially** (even by a single millimeter) or **bent at too sharp an angle** immediately after exiting has **high contact resistance** on some pins. The power dissipated in a contact is $P = R \times I^2$: with currents in the order of 40-50 amperes, even a few milliohms of excess resistance produce tens of watts concentrated on a small piece of plastic and brass. The documented result, with abundant photos online, is the **melting of the connector** — on the GPU side, PSU side, or both — damaging the video card.

The rules: (1) insert **until it clicks**, verifying that the **golden area** of the terminals is **no longer visible**; (2) **do not bend the cable** within **at least 35 mm** of the connector (this is the minimum distance specified by the standard)

— if the side panel of the case presses on the cable forcing an immediate sharp bend, you have a design problem and need an angled cable or a certified 90° adapter; (3) use the **native power supply cable** if it is ATX 3.x, or the adapter provided with the GPU, never generic adapters of dubious origin; (4) **recheck** the connector after closing the case and after any movement of the PC.

Final step — The anti-sag support. Modern GPUs can weigh **up to 2 kilograms** and are supported by two points: the PCIe slot and the bracket. The consequence is **sag**: the card tilts downwards, permanently flexing the slot and the GPU's PCB. If you look at the mounted GPU **from the side** and see that the free end visibly drops compared to the slot, add a **support** (an adjustable stand resting on the bottom of the case, or a retractable bracket: many GPUs and cases include one). This is not just aesthetics: a heavy GPU left in flexion for two years can develop micro-cracks in the solder joints.

10. Complete Cabling, One by One

We're at the part that scares people because "there are fifty cables." In reality, there are nine types, and each has only one place to go. Let's proceed in order, from top to bottom.

10.1 ATX 24-pin (motherboard power)

The largest connector, **24-pin** (often in two separable pieces, 20+4). It goes in the only place it can go: the long connector on the **right edge** of the motherboard. It is polarized (the pins have different shapes) and has a **side clip** that must **snap into place**. **Force**: considerable. The 24-pin requires a firm push, and beginners often leave it halfway. Push until the clip clicks into place. Verification: try to pull it; it should not come out.

10.2 EPS 8-pin (CPU power) — the most confusing cable of all

The **EPS** is an **8-pin** connector that powers the CPU's VRMs, and it is located **top left on the motherboard**, near the socket, often labeled **CPU_PWR1**, **ATX_12V**, or **EPS12V**.

The risk: The EPS 8-pin and PCIe 8-pin look identical. They have the same number of pins, the same general shape, the same color (black). But the pinout is different (the 12 V and ground positions do not match) and the shape of the individual pins is different: some are square, others are beveled, precisely to prevent errors. Never force an 8-pin connector. If it doesn't slide

*in smoothly, you have the wrong cable. A PCIe forced into an EPS socket can **destroy the motherboard**.*

***How to always distinguish them:** Power supply cables are **labeled** — look for **"CPU"** or **"EPS"** on the connector itself, versus **"PCIe"** or **"VGA"**. If the labels are missing, look at the separation: PCIe is **6+2** (it splits into a 6-block and a 2-block); EPS is **4+4** (it splits in half). This is the foolproof method.*

High-end motherboards have **two** EPS connectors (8+8, or 8+4). With consumer CPUs, **only one is almost always sufficient** for normal operation; the second is for extreme overclocking and very high currents. If you have two and the power supply has the cables, connect both: it doesn't hurt.

Accessibility warning: The EPS is in the most awkward corner of the case, often already covered by the cooler. **Connect it before** mounting the top radiator, or even before installing the motherboard if the case is cramped.

10.3 PCIe / 12V-2×6 (GPU power)

Already covered in section 9.3. I'll summarize the rule: **no daisy-chain on powerful GPUs; 12V-2×6 all the way, without tight bends within 35 mm.**

10.4 SATA power and SATA data

If you have a 2.5" SSD or a mechanical hard drive, you need **two** cables: one **SATA power** (flat L-shaped connector, wide, coming from the power supply, with multiple plugs on the same cable — here daisy-chaining is perfectly fine, power consumption is minimal) and one **SATA data** (small, L-shaped, going from the drive to a **SATA** port on the motherboard). Both have a clip. The SATA data cable is often included with the motherboard. If you only have M.2 drives, **you don't need any of this.**

10.5 Front panel (F_PANEL) — the pin puzzle

This is the most hated set: **four or five thin cables, with two-pin connectors**, coming from the front of the case, which go into a block of **9-10 pins** on the **bottom right** edge of the motherboard, labeled **F_PANEL**, **PANEL1**, **JFP1**, or similar.

The cables are:

Cable	Function	Polarity
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POWER SW (or PWR_SW)	The power button: closes a momentary contact	Indifferent (it's a switch)
RESET SW	The reset button	Indifferent
POWER LED (sometimes in two pieces, + and –)	The power LED	Matters: the positive pole (colored wire) goes on the + pin
HDD LED (or H.D.D. LED)	The disk activity LED	Matters
SPEAKER (if present)	The buzzer that emits diagnostic beeps	Indifferent in practice

The key point: there is no universal standard for pin order. Each motherboard manufacturer uses a slightly different layout. **The only source of truth is the diagram in the motherboard manual**, which shows the block pin by pin. Open it, keep it open, and connect them one by one with the help of a flashlight. This is the only part of the assembly where you cannot rely on intuition.

Good news: **making a mistake here won't break anything.** If you reverse an LED, that LED won't light up (and you just need to flip the connector). If you swap power and reset, the power button will perform a reset. No permanent damage. Some motherboards (especially ASUS) include a **Q-Connector** adapter: a block where you conveniently insert the cables **outside** the case, and then the whole thing plugs into the board. If it's there, use it.

If your case has a power button but you can't figure out which one is POWER SW: remember that, to turn on a PC, you just need to **short-circuit the two POWER SW pins for an instant** with the tip of a screwdriver. This is exactly what the button does. Useful for bench POST testing.

10.6 Front Panel Data Headers

USB 3.0 / USB 3.2 Gen 1 header: a **blue, wide, 19/20-pin** connector, with a guide notch. It goes into the corresponding header on the motherboard. **Caution:** it's rigid and the pins are thin; it only goes in one way but must be inserted straight. This is the connector that is most often **damaged by bending pins**, because it's on the edge of the board and awkward. Take your time.

USB-C header (USB 3.2 Gen 2 / Gen 2×2): a small, rectangular, "E-key" type connector, often labeled **USB 20G** or **USB-C**. It easily goes in one way. If your

motherboard **does not have** this header, the front USB-C port on the case **will not work**: this is a compatibility check to perform during purchase.

USB 2.0 header: 9 pins (one missing acts as a key). Used for the control cable of many AIOs, for RGB hubs, for some card readers.

HD Audio (or AAFP / F_AUDIO): a 9-pin connector, very similar to USB 2.0 (**but NOT the same thing**: look at the label, "HD AUDIO"). It goes into the audio header, which is almost always in the **bottom left corner** of the motherboard, as far as possible from digital circuits (to reduce interference). This is the cable for the front headphone/microphone jacks.

10.7 Fans: PWM vs DC

Each fan has a **3 or 4-pin** connector, with a guide tab that ensures its orientation, and plugs into one of the **SYS_FAN / CHA_FAN** headers on the motherboard.

The difference between 3 and 4 pins is conceptually important:

	3 pin (DC)	4 pin (PWM)
Pin	GND, 12V, Sense (tachometer)	GND, 12V, Sense, PWM
How speed is regulated	By varying the voltage (from 12V downwards)	By keeping 12V fixed and sending a variable duty cycle control signal
Low-speed regulation	Imprecise; below a certain voltage, the fan stops and does not restart	Precise and stable even at low RPMs
Compatibility	A 3-pin fan works on a 4-pin header, but only if the header is set to DC mode in the BIOS	A 4-pin fan on a 3-pin header will always run at 100%

Practical consequence: if you have 3-pin fans that are always running at full speed and making noise, go into the BIOS, find the fan section (Q-Fan, Smart Fan, Fan Control), and set that header to **DC** instead of **PWM/Auto**. This is the solution to a problem that frustrates many people.

If you have more fans than available headers, use a **splitter** (a cable that splits; the tachometric signal comes from only one fan) or a **PWM hub** powered directly from the PSU with a SATA power connector (mandatory if you connect more than

3-4 fans: a motherboard header typically provides **1 A / 12 W**, and exceeding this can burn out the control circuit).

10.8 ARGB and RGB: the hardware-saving paragraph

This is where things really get destroyed, and the cause is that the two connectors look similar.

	RGB 12 V (4 pin)	ARGB 5 V (3 pin)
Voltage	12 volts	5 volts
Pin	4 (12V, G, R, B)	3 (5V, Data, empty , GND) — it has an empty space!
What it can do	The entire strip changes color together	Each LED is individually addressable ("rainbow" effects)
Label on the mobo	RGB_HEADER, JRGB, 12V G R B	ARGB_HEADER, JRAINBOW, ADD_HEADER, 5V D G

NEVER CONNECT a 5V ARGB device to a 12V RGB header. The connectors are physically similar, and on some headers (which have 4 pins), it is possible to accidentally insert a 3-pin connector. The result is that the device receives more than double the voltage it is designed for: the LED strip or fan burns out instantly, and with a certain probability, it takes the motherboard's LED controller with it, which is not repairable.

How to protect yourself: the 3-pin ARGB header has an unmistakable characteristic: there are **4 positions but one pin is MISSING** (there's a hole). The ARGB connector has a corresponding blanking hole. Count the pins, **read the silkscreen on the board** (it always says "12V" or "5V" next to the header), and if in doubt, **do not connect anything and consult the manual page**. This is a case where thirty seconds of reading are worth a hundred euros of components.

10.9 Cable management

It's not (just) aesthetics. A bundle of cables running through the case in front of the cooler or GPU **obstructs airflow**, raises temperatures, and makes any future maintenance impossible.

There are three principles: 1. **Anything that can go behind the motherboard tray, goes behind.** Modern cases have a dedicated rear compartment and rubber grommets at strategic points. The cable exits the grommet closest to its destination, travels a few centimeters in view, and connects. 2. **Group and secure with zip ties**, without tightening them so much that they crush the cables. Secure them to the case's anchor points (eyelets, slots). 3. **Do not obstruct fan intake or cooler exhaust.** No cable should pass *in front* of the front fan or over the CPU cooler.

If the rear panel doesn't close, you have too much stuff behind it: redistribute, don't force it.

11. First Boot

11.1 Pre-Power-On Checklist

Before pressing the button, stop. Five minutes here are worth an hour later. Review each point:

1. **Is the CPU socket lever closed and latched?** 2. **Is the cooler securely fastened?** Try to move it: it should not rotate or wobble. (And: **did you remove the film from the cold plate?**) 3. **Is the cooler's fan/pump connected to CPU_FAN / AIO_PUMP?** 4. **Is the RAM in the correct slots (A2/B2) and are ALL clips closed?** 5. **Is the GPU in the first x16 slot, is the clip snapped shut, is the bracket screwed in?** 6. **Are all power cables connected?** 24-pin, EPS 8-pin (don't forget it: it's the most common mistake by far), PCIe/12V-2×6 for the GPU, SATA power. 7. **Is the 12V-2×6 inserted ALL THE WAY?** Look at it. 8. **Are there no screws, zip ties, or tools inside the case?** Turn the case upside down and shake it gently: if you hear rattling, search for them. 9. **No orphaned standoffs under the motherboard?** 10. **Is the switch on the back of the power supply set to "O"?** Keep it off until you've connected the power cable.

11.2 Connecting the Monitor: To the GPU, Not the Motherboard

It seems obvious and **one out of three people** make this mistake.

If you have a **dedicated GPU**, the monitor cable (DisplayPort or HDMI) goes into the **GPU's ports**, which are the **horizontal ones, at the bottom, on the video card's bracket**. The video ports **on the motherboard** (vertical, at the top, in the

rear I/O cluster) belong to the CPU's **iGPU** and, by default, **are disabled when a dedicated GPU is present**.

Symptom of the error: the PC turns on, fans spin, LEDs light up, but **the monitor says "No Signal"**. Unjustified panic. Move the cable.

(Corollary: if your CPU **does not have** integrated graphics — many AMD X-series models and some Intel "F" models — the motherboard's video ports never work, and a dedicated GPU is **absolutely necessary**.)

11.3 The POST and How to Read Diagnostics

Turn it on: PSU switch to "I", then the case's power button.

The **POST** (*Power-On Self-Test*) is the self-diagnostic sequence that the firmware performs before handing control to the operating system. It initializes, in order: **CPU → memory → PCIe/graphics devices → boot devices**. If something fails, the POST **stops there**, and tells you where.

How it tells you depends on the motherboard:

Diagnostic LEDs (EZ Debug LED / Q-LED). Four LEDs on the edge of the motherboard, labeled **CPU**, **DRAM**, **VGA**, **BOOT**. During POST, they light up in sequence and turn off as each phase passes. **The LED that remains lit indicates the phase where the system got stuck:**

LED lit solid	Meaning	What to check, in order
CPU	CPU has not been initialized	EPS 8-pin connected? Then: CPU properly seated? BIOS updated (CPU too new)? Socket pins intact?
DRAM	Memory is not responding	RAM inserted until double click? Correct slots (A2/B2)? Try only one module , in A2. Clear CMOS.
VGA	Graphics not detected	GPU inserted until click? PCIe cables connected to GPU? Try the other slot.
BOOT	All OK, but no bootable disk found	Normal if you haven't installed the OS yet!

		Otherwise: SSD detected in BIOS? Installation USB stick connected?
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A note of realism that avoids many panic attacks: on **first power-on** with DDR5 RAM, the system can take **from 30 seconds to several minutes** to POST, with the DRAM LED lit and a black screen, and perhaps one or two automatic reboots. This is **memory training**: the controller is calibrating dozens of electrical parameters for each module, and stores the result. **This is normal, and only happens the first time (and after every RAM change or clear CMOS)**. Do not turn everything off after twenty seconds: **wait at least five minutes** before declaring failure.

Q-Code / POST code display. High-end motherboards have a small **two-digit hexadecimal** display that shows the code of the current phase. The code **it gets stuck on** identifies the problem with surgical precision; the complete table is **in the appendix of the manual**. Some recurring codes (vary between manufacturers — always check your manual): codes in the **0d/19/53/55** series typically indicate memory problems; **d6/d7** indicate absence of a graphics device; **A0/A2/AA** indicate that POST was successful and the system is proceeding to boot. If the display shows a solid **00** or **FF**, generally the CPU is not executing anything: suspect CPU power or unrecognized CPU.

Beep code (buzzer). If you have connected the **speaker**, a **single, short beep** means "POST passed, all good". Long/repeated sequences indicate errors (typically: repeated beeps = RAM; one long and two short = graphics). It's worth connecting it: it costs nothing and gives you information.

11.4 If it doesn't start: the systematic procedure

The golden rule of diagnostics: **change one variable at a time, and start with the simplest**. The temptation to dismantle everything and reassemble randomly is strong and only leads to confusion.

Level 0 — The absurdities (really, check them). Is the power supply switch on? Is the cable plugged into the outlet? Does the outlet work? Is the power button connector (POWER SW) on the correct pins? (If the PC shows **no** signs of life — not even an LED, not even a fan twitch — the culprit is almost always here, or it's the power supply.)

Level 1 — Reseating. Turn off, unplug, **hold the power button for 10 seconds** (discharges capacitors). Then:

- **RAM:** Remove all modules and reinsert **only one**, into slot **A2** (the second from the CPU). Test. If it doesn't start, try the same module in **B2**. If it doesn't start, try **the other module** in A2. This four-attempt matrix distinguishes a faulty module from a faulty slot from an insertion error, and resolves the vast majority of cases.
- **GPU:** Reseat, and recheck power cables.
- **EPS 8-pin:** Recheck it. Seriously. It's the most common error.

Level 2 — Clear CMOS. The **CMOS** is the memory (powered by the CR2033 coin cell battery) where the BIOS stores settings. An impossible setting (typically: an overly aggressive XMP/EXPO profile that your CPU can't handle) can prevent POST. *Clear CMOS* resets everything to factory defaults. This can be done in three ways, in order of convenience: 1. Press the **"Clear CMOS" or "BIOS Reset" button** on the rear panel (mid/high-end motherboards have this); 2. Short-circuit the two **"CLR_CMOS" / "JBAT1" pins** on the motherboard for 5-10 seconds (with a screwdriver or the included jumper), **with the PC off and unplugged**; 3. **Remove the CR2032 battery** for a couple of minutes, again with the PC off and unplugged. After a clear CMOS, all settings (XMP, boot order, fan curves) return to default and must be reconfigured. And memory training restarts from scratch: **be patient on the subsequent first boot.**

Level 3 — Breadboarding. If nothing has worked, it's back to the bench. **Remove the motherboard from the case** (yes, everything) and reassemble it bare on its cardboard box, with **the bare minimum**: CPU, cooler, **one** RAM module, GPU, power supply, monitor. No hard drives, no case fans, no front USB, no LED strips. Power on by short-circuiting the POWER SW pins. If it **starts this way**, the culprit was in the case: almost always an **orphan standoff** causing a short, or a poorly connected front panel cable, or a faulty peripheral. Reassemble **one piece at a time**, testing each time. If it **doesn't start this way**, the culprit is among the five remaining components, and at that point, spare components are needed for substitutions (the part where a friend with a PC becomes invaluable).

12. Initial Configuration

The PC POSTs and displays something. You're not done: you've finished the **mechanical part**. Now comes the part where the computer becomes *yours*, and where you reclaim performance you've already paid for but, at factory settings, **are not using**.

12.1 Entering the BIOS/UEFI

On first power-on, repeatedly press **DEL** — or **F2** on some motherboards — as soon as the PC starts. You will enter the **UEFI** (*Unified Extensible Firmware Interface*, the modern successor to BIOS; the two terms are now used synonymously), the firmware configuration interface.

First check: is everything there? The initial screen (*EZ Mode*) shows you a summary. Verify:

- The **CPU** is recognized with the correct name;
- The **total amount of RAM** matches what you have installed (if you have 32 GB and it only sees 16, one module is not making contact: power off and re-seat);
- All **SSDs/hard drives** are listed;
- The **fan speed** is non-zero for the CPU;
- The **CPU temperature** is reasonable (at idle, below 50-60 °C on a well-assembled system; if you see 80-90 °C at rest, **power off immediately**: the cooler is not making contact, and the first thing to check is **the film**).

12.2 Activating XMP / EXPO: five seconds, ten percent performance

This is the single action with the best effort-to-benefit ratio in the entire process.

RAM modules, according to **JEDEC** specifications (the body that standardizes semiconductors), always start at a **conservative base frequency** — much lower than what you paid for. The 6000 MT/s kit you bought, once installed, runs at 4800 MT/s with relaxed timings. **The performance you paid for is there, but it's off.**

To turn it on, there is a profile stored in the module's **SPD** chip, which the BIOS can load with a click:

- **XMP** (**Extreme Memory Profile**) is the name of the profile on **Intel** platforms;
- **EXPO** (**EXtended Profiles for Overclocking**) is the **AMD** equivalent;
- some boards generically call it **D.O.C.P.** or **A-XMP**.

They are conceptually the same thing: a predefined set of frequency, timings, and voltages, tested by the memory manufacturer. **Find the option (it's prominently displayed in EZ Mode) and activate it.** The PC will restart and perform memory training again (again: **patience, it can take minutes**).

Technically it's an overclock, so: if after activation the system is unstable (crashes, blue screens, no POST), it's not a fault — it's your specific IMC that can't handle that profile. Solutions: try the **secondary** XMP/EXPO profile (often more conservative), or manually reduce the frequency by one step. And if it no longer POSTs: **clear CMOS**, and start over.

12.3 Updating the BIOS

If the PC is working, it is **advisable** to update the BIOS to the latest stable version: revisions after a platform's launch substantially improve memory compatibility and stability, fix bugs, and sometimes increase performance.

The correct procedure (commercial names vary: *EZ Flash*, *M-Flash*, *Q-Flash*): 1. On the **website of your motherboard manufacturer**, on the page **for your exact model** (pay attention to variants: a "B650 Gaming X" is not a "B650 Gaming X AX"), download the latest stable BIOS file; 2. Copy it to a **USB stick formatted in FAT32**, in the root directory; 3. Enter the BIOS, open the update utility **integrated into the BIOS itself** (never, ever use update utilities from within Windows: they are the historical cause of "bricked" BIOSes), select the file; 4. **DO NOT POWER OFF, DO NOT TOUCH ANYTHING** for the entire duration. A power interruption during the flash can render the motherboard unusable (unless it has a dual BIOS or Flashback).

After the update, settings will be reset: **re-do XMP/EXPO and fan curves**.

12.4 Fan Curves and Resizable BAR

Fan curves. In the dedicated BIOS section (Q-Fan Control, Smart Fan, Fan Xpert), you can define a **curve** for each header that links the speed (in % or RPM) to a reference temperature. The default is almost always too aggressive (noisy) or too sluggish. A sensible curve for the CPU: silent up to ~50 °C, gradual increase between 50 and 75 °C, maximum beyond 80 °C. Important: set a **response time** (*step up/down time, hysteresis*) of a few seconds, to prevent the fan from "pumping" up and down with every momentary temperature spike — the phenomenon that makes a PC annoying even when it's quiet on average. And remember: header on **DC** for 3-pin fans, **PWM** for 4-pin fans.

Resizable BAR (Re-Size BAR / Smart Access Memory on AMD). This is a PCIe feature that allows the CPU to access the **entire VRAM** of the GPU at once, instead of 256 MB windows at a time. The benefit in games ranges from none to **over 10%** depending on the title. It must be enabled **in the BIOS** (often two settings are needed: **Above 4G Decoding** to *Enabled* and **Re-Size BAR Support** to *Auto/Enabled*). It's free: enable it.

12.5 Operating system and driver installation

Media preparation: On another PC, download the official media creation tool (for Windows, the *Media Creation Tool* from the Microsoft website; for Linux, the distribution's ISO written to a USB stick with a tool like Rufus or balenaEtcher). You need a USB stick of at least 8 GB, which will be **formatted**.

Boot: Insert the USB stick into a **rear USB port** (more reliable than front ports during installation), restart, and select the USB stick from the **Boot Menu** (usually **F8**, **F11**, or **F12** — the board tells you at startup) or set it as the first boot device in the BIOS.

During installation, when it asks where to install, select your **NVMe SSD** (identifiable by size). If there are pre-existing partitions, delete them and use the unallocated space.

After installation, driver order matters: 1. **Chipset drivers** — first, always. Download them from the **motherboard** website (or directly from AMD/Intel, often more up-to-date). They teach the operating system how to communicate with the platform: power management, core scheduling, controllers. 2. **GPU drivers** — directly from the **NVIDIA / AMD / Intel** website, **not** from the disc included in the box (it's months old). 3. **Network, audio, and other drivers** — Windows Update installs many of them automatically; for the rest, check the motherboard website.

12.6 Testing: checking temperatures and stability

Don't assume a PC is good just because it turns on. A PC that turns on can still have a poorly mounted cooler, a wrong fan curve, or an unstable memory profile. Testing is the proof.

The tools (all free):

- ****HWiNFO64**** (in **Sensors only** mode): the window of truth. Shows temperatures, frequencies, voltages, and power consumption of each

component, with **minimum/maximum/average** values. Keep it open during every test.

- **Cinebench** (R23 or later versions): a benchmark that loads **all CPU cores to 100%** for 10 minutes. It's the most immediate test: it gives you a score (comparable online with identical CPUs to yours, to understand if yours is performing as it should) and brings the CPU to its realistic peak temperature.
- **OCCT** or **Prime95**: more severe stress tests. Useful for verifying the stability of an XMP/EXPO profile (OCCT's **Memory** test, for an hour, is the standard validation) and the PSU's endurance.
- **FurMark** or **3DMark**: GPU load.
- **MemTest86** (booted from a USB stick, outside the operating system): the definitive RAM test. A complete pass, without errors, certifies that your memory is fine.
- **CrystalDiskMark** or **CrystalDiskInfo**: SSD speed and health.

What to look for, and what is normal:

Component	At idle	Under full load	Alarm
CPU	~30-50 °C	Up to 85-95 °C is normal on many modern CPUs, which are designed to operate at their thermal limit and regulate themselves	If it reaches the limit (95-100 °C) within a few seconds of starting the test → cooler poorly mounted / film not removed
GPU	~30-45 °C	65-80 °C typical (VRAM/hotspot can be higher)	> 85-90 °C sustained, or constant 100% fan speed
NVMe SSD	35-50 °C	Up to ~70 °C in sustained writing	Visible throttling (speed drops) → heatsink/film
Stability	—	No crashes, no errors, no reboots	Even a single error in MemTest or a crash in OCCT = real problem, almost always XMP/EXPO

An important clarification, because it generates unnecessary anxiety: **a modern CPU that hits 90 °C under Cinebench is not "faulty" nor "poorly mounted"**. Today's CPUs use opportunistic boost algorithms that push frequency and voltage **until they saturate the available thermal budget**: if the cooler is good, the CPU runs faster and still remains hot. What matters is not the absolute number, but **how long it takes to get there** (seconds = bad, minutes = normal), **if it stays there without dropping**, and **what frequency it maintains**. A 10-minute test that ends at 88 °C with full frequencies is a healthy system. A test that reaches 100 °C in five seconds and drops to base frequencies is a system with a mounting problem.

Operational Summary — the assembly checklist

Keep this open next to your table. Do not move to the next step until the previous one is verified.

BEFORE OPENING THE BOXES

- [] **PH2 magnetic** screwdriver, zip ties, light, large non-textile surface
- [] Final compatibility check: **socket** / **BIOS** that supports the CPU (is **Flashback** available?) / **correct DDR RAM** / **cooler clearance** / **GPU length** / **PSU: wattage + connectors**
- [] Power cable **unplugged from the wall** at every stage of the work
- [] Discharge yourself by touching the case chassis (or anti-static wristband) and repeat after each break

BENCH PRE-ASSEMBLY (mobo on its BOX, not on the anti-static bag)

- [] **CPU:** lever open → align triangle → **placed**, not pressed (force = zero) → lever closed (force = high, elastic resistance **yes**, hard stop **no**)
- [] **RAM:** slots **A2/B2** (**read the manual**) → align notch → press at ends until **double click** → golden contacts **disappeared**
- [] **M.2:** **primary (CPU)** slot → **THERMAL PAD FILM REMOVED** → inserted at 30° → screw without forcing
- [] **Cooler:** backplate + standoff → **THERMAL PASTE** always (pea-sized dot in center) → **AIO: REMOVE FILM** from cold plate → tighten **crosswise, incrementally**, until snug → fan/pump on **CPU_FAN / AIO_PUMP**
- [] **(Recommended)** **Bench POST test** before putting everything in the case

CASE

- [] Standoffs: **one under each hole**, **no orphaned standoffs** (short circuit risk)
- [] **I/O shield** mounted in the correct orientation (if not integrated)
- [] Front cables pre-routed through grommets

PSU

- [] Fan **facing down** if there's a grilled filter, facing inwards if the bottom is closed
- [] **Only original cables for THAT PSU model** (never cables from other PSUs!)
- [] Modular cables connected **before** inserting it into the case

MOBO AND RADIATOR

- [] Mobo aligned with I/O shield, screwed in **crosswise**, **without overtightening**
- [] AIO Radiator: **top exhaust** (or front intake) with **tubes at the bottom** — **the pump should never be the highest point**
- [] Only use screws from the kit (screws that are too long will **puncture the radiator**)

GPU

- [] **First x16 slot** (the only electrically x16 from the CPU) — check manual
- [] Brackets removed, clip open, inserted until **click**, screwed in
- [] **NO daisy-chaining** on PCIe cables for powerful GPUs
- [] **12V-2×6 inserted ALL THE WAY IN**, no bends within **35 mm** — **recheck with case closed**
- [] Anti-sag support if the card sags

CABLING

- [] **24-pin** (clip clicks) — **EPS 8-pin** (top left: **it's "4+4"**, PCIe is **"6+2"**) — **PCIe/12V-2×6** — **SATA power**
- [] **F_PANEL** according to the **manual's diagram** (wrong connection won't break anything, but it won't start)
- [] Headers **USB 3.0** (fragile pins!), **USB-C**, **USB 2.0**, **HD Audio** (do not confuse with USB 2.0)

- [] Fans: **CPU_FAN** mandatory; 3 pin = **DC**, 4 pin = **PWM** (set in BIOS)
- [] **ARGB 5V** (3 pin, with a gap) $\neq$ **RGB 12V** (4 pin). NEVER mix them: everything will burn.
- [] Cables behind the tray, zip ties, **no cables in front of fans or heatsink**

FIRST BOOT

- [] Pre-power-on checklist (lost screws? tools inside? EPS connected?)
- [] **Monitor connected to the GPU, NOT the motherboard**
- [] POST: interpret **CPU / DRAM / VGA / BOOT LEDs** or **Q-Code**
- [] **DDR5**: the first POST may take minutes (memory training). WAIT.
- [] If it doesn't start: **one variable at a time** $\rightarrow$ re-seat RAM $\rightarrow$ **only one module in A2** $\rightarrow$ **clear CMOS** $\rightarrow$ **breadboarding**

CONFIGURATION

- [] BIOS: everything detected? (CPU, **correct total RAM**, drives, fans, **healthy temperatures**)
- [] **Enable XMP / EXPO** (it's performance you've already paid for)
- [] Update BIOS (**from internal utility, never from Windows**) $\rightarrow$ re-do XMP and fans
- [] **Resizable BAR** + **Above 4G Decoding** = Enabled
- [] Fan curves with hysteresis; correct DC/PWM headers
- [] OS from USB $\rightarrow$ **chipset drivers FIRST**, then **GPU drivers from manufacturer's website**
- [] Testing: **Cinebench + HWiNFO** (CPU temperatures), **OCCT/MemTest86** (XMP stability), **FurMark/3DMark** (GPU)
- [] High peak temperatures $\neq$ fault. **Peak temperatures reached in a FEW SECONDS** = poorly mounted cooler.

Note on data: the socket designations, chipsets, and connector models mentioned reflect the market status at the time of writing and are the most volatile part of this chapter; the physical procedures, forces, and assembly orders, however, have been substantially stable for over a decade. Always check your specific motherboard manual: it is the only authoritative source for slot, header, and pin layout.

Chapter 10 - CPUs for AI Workstation and HEDT

Introduction: why this chapter exists

In previous chapters, we built, piece by piece, the mental model of the consumer PC: a CPU at the center, a GPU attached to the main slot, two RAM modules, one or two NVMe SSDs, a power supply sized with a reasonable margin. In that world, the dominant question is "how many frames per second do I get?" or "how long does it take to compile?", and the answer almost always revolves around clock frequency and core count.

When the goal becomes an **AI/ML workstation** — that is, a machine that trains neural networks, performs fine-tuning of language models, does local inference on quantized LLMs, processes video streams with detection models, or simply serves as a test bench for pipelines that will then run in datacenters — that mental model no longer holds. Not because it's wrong, but because **the question changes**. The question is no longer "how fast is the CPU," but "how much material can I simultaneously move between memory, drives, network, and accelerators, without anything waiting."

This is the central point of the entire chapter, and it's worth stating brutally: **in an AI workstation, the CPU's main purpose is to not be a bottleneck for everything else**. Heavy computation is done by GPUs. The CPU must ensure three things: (1) enough communication lanes (PCIe lanes) for GPUs and SSDs to communicate at full speed, (2) enough memory bandwidth and enough RAM for data to reach GPUs smoothly, (3) enough cores so that data preparation (image decoding, augmentation, tokenization) doesn't leave GPUs idle.

From these three needs arises an entire category of products — **HEDT and workstation/server** — that the consumer market does not know and that has completely different rules, nomenclature, prices, and pitfalls. HEDT stands for *High-End DeskTop*: it is a historical segment, created for those who needed more than a desktop but did not want (or could not) manage a server. Today, HEDT essentially means AMD Threadripper and, in part, Intel Xeon W.

The chapter proceeds as follows: first, we understand *why* a consumer CPU is not enough (and, equally important, *when it is perfectly sufficient*), then we delve into Threadripper platforms, then server platforms (EPYC and Xeon), then we explore PCIe as an infrastructure, then workstation motherboards, then sizing based on

workload, and finally, typical errors. We conclude with an operational summary, which is effectively a purchase checklist.

A methodological note before we begin. **Numerical data in this sector ages very quickly**: prices, availability, generations, and even the compatibility matrix between CPU and chipset change with every firmware update. Whenever data is volatile, I will explicitly mark it with the tag *[volatile data]*. Architectures and principles, however, remain valid for much longer: these are what are truly worth learning.

1. Why a Consumer CPU Is Not Enough

1.1 The Shift in Perspective: The CPU as a Dispatch Node

In a consumer PC, the CPU is the protagonist: it executes application code, and everything else revolves around it. In an AI workstation, the protagonist is the accelerator (the GPU), and the CPU takes on what we might call an *infrastructural* role. It is the traffic dispatch node: it reads data from disk, decodes it, transforms it, packages it, and sends it to the GPU; then it collects the results, writes them, synchronizes them across multiple GPUs, and sends them over the network.

If even one of these steps is bottlenecked, the GPU — which costs as much as the rest of the system combined — remains idle. The critical parameter in an AI workstation is called **GPU utilization**: the percentage of time the accelerator cores are actually computing. A well-designed workstation keeps GPUs above 90–95% utilization during training. An unbalanced workstation — consumer CPU with four GPUs attached to reduced lanes, dataset on a single SSD, little RAM — can easily drop to 40–50%, which means **half of the hardware's economic value is wasted**.

Let's look, one by one, at the four resources that consumer systems fail to provide.

1.2 PCIe Lanes: The True Currency of the Workstation

PCIe stands for *Peripheral Component Interconnect Express*: it is the bus (i.e., the connection system) through which the CPU communicates with high-speed peripherals — graphics cards, NVMe SSDs, network cards, dedicated accelerators, storage controllers.

PCIe is a **serial and point-to-point** bus: it is not a shared road where everyone crowds (like the old PCI or ISA), but a set of dedicated connections. The

elementary unit is called a **lane**. A lane is, physically, a pair of differential connections: one for transmitting, one for receiving. This means PCIe is **full-duplex**: it can transmit and receive simultaneously at full nominal speed in both directions.

Lanes are aggregated: a slot can be x1, x4, x8, x16. The number indicates how many lanes are wired to that slot. More lanes mean more bandwidth: bandwidth grows linearly with the number of lanes.

The second variable is the **generation** (or *revision*) of PCIe, which determines the speed of each lane. Here is the overview, which is worth memorizing because we will use it constantly:

Generation	Indicative Year	Bandwidth per Lane (per direction)	x4 Bandwidth	x8 Bandwidth	x16 Bandwidth
PCIe 3.0	2010	~0.985 GB/s	~3.94 GB/s	~7.88 GB/s	~15.75 GB/s
PCIe 4.0	2017	~1.97 GB/s	~7.88 GB/s	~15.75 GB/s	~31.5 GB/s
PCIe 5.0	2019	~3.94 GB/s	~15.75 GB/s	~31.5 GB/s	~63 GB/s
PCIe 6.0	2022 (spec)	~7.88 GB/s	~31.5 GB/s	~63 GB/s	~126 GB/s

To be read as follows: **each generation doubles the bandwidth per lane**, for the same number of lanes. Consequently, a PCIe 5.0 x8 slot has the same bandwidth as a PCIe 4.0 x16 slot. This equivalence is very important and will come up repeatedly.

[volatile data] PCIe 6.0 exists as a specification and is being adopted in the datacenter world, but on workstation platforms available today, the reference standard is PCIe 5.0. PCIe 7.0 has already been ratified as a specification but does not yet have real products on the workstation market.

Why Lanes Are the Real Bottleneck

A CPU has a **finite** number of PCIe lanes, because each lane requires physical pins on the package, transceivers, dedicated silicon, and power consumption. It is an expensive resource, and manufacturers ration it to segment the market.

Let's count what a serious AI workstation needs:

- ****One x16 PCIe 5.0 GPU****: 16 lanes. If there are four GPUs, ****64 lanes**** are needed just for the accelerators.
- ****NVMe SSDs****: Each NVMe (*Non-Volatile Memory express*, the protocol by which SSDs communicate directly on the PCIe bus) typically occupies 4 lanes. An AI workstation often has 3–4: one for the operating system, one or two for the "hot" dataset, one for checkpoints. That's ****12–16 lanes****.
- ****Network card****: A 10 GbE NIC (*Network Interface Card*) generally occupies 4 PCIe 3.0/4.0 lanes; a 25/100 GbE one occupies 8–16 lanes. Let's say ****4–16 lanes****.
- ****HBA/RAID controller**** for mass storage (datasets of tens of TB): ****8 lanes****.
- ****Capture cards, dedicated accelerators, acquisition cards****: variable.

Realistically summing up for a 4-GPU workstation: $64 + 16 + 8 + 8 = \mathbf{96 \text{ lanes}}$. A consumer CPU has about twenty. The gap is a factor of four or five, and no software optimization can bridge it.

How Many Lanes Do Consumer CPUs Really Have

Here, one must pay attention to a detail that confuses many buyers: the lanes **of the CPU** and the lanes **of the chipset** are not the same thing.

The **chipset** (in AMD, it's also called *PCH*, *Platform Controller Hub*, originally an Intel term) is a separate chip on the motherboard that expands connectivity. But the chipset is connected to the CPU through a link that is itself limited — typically x4 PCIe 4.0. Everything connected to the chipset (secondary slots, some NVMe ports, USB, SATA, integrated LAN) **shares that single x4 link**. It's like having a villa with ten rooms but only one meter-wide corridor to reach them all.

For this reason, when designing a workstation, **only the CPU's native lanes matter**, those directly connected. Chipset lanes are fine for slow peripherals (audio, USB, gigabit network, archive SSDs) and should be avoided for high-performance GPUs and NVMe drives.

Here is the consumer overview [*indicative data, verify for specific models*]:

- ****AMD AM5 (Ryzen 7000/8000/9000)****: 28 total PCIe lanes from the CPU, of which ****24 are actually usable**** (16 for the GPU + 4 for one NVMe + 4 for a second NVMe), plus 4 lanes dedicated to the chipset connection. Versions with powerful integrated graphics ("G" series) have even fewer.
- ****Intel LGA1700 (12th–14th gen)****: ****20 lanes**** from the CPU (16 PCIe 5.0 for the GPU + 4 PCIe 4.0 for an NVMe), plus the DMI connection to the chipset.

- ****Intel LGA1851 (Core Ultra 200 series "Arrow Lake")****: ****24 lanes**** from the CPU (16 PCIe 5.0 + 4 PCIe 5.0 + 4 PCIe 4.0), plus DMI.

[volatile data] These numbers change with each new platform; the principle (consumer = ~20–28 lanes, and no more) has been stable for over a decade.

1.3 Comparative table: lanes by platform segment

Platform	Segment	PCIe Lanes from CPU	Generation	Notes
Intel LGA1700 (Core 12th/13th/14th gen)	Consumer	20	5.0 (16) + 4.0 (4)	Single GPU at x16
Intel LGA1851 (Core Ultra 200S)	Consumer	24	5.0 (20) + 4.0 (4)	One GPU + 2 NVMe
AMD AM5 (Ryzen 7000/9000)	Consumer	24 usable (+4 chipset)	5.0	One GPU + 2 NVMe
AMD Threadripper 7000/9000 (TRX50)	HEDT	48 PCIe 5.0 (+ additional 4.0 lanes; AMD states up to ~92 platform lanes)	5.0 / 4.0	2 GPUs at full x16 + NVMe
AMD Threadripper PRO 7000WX/9000WX (WRX90)	Workstation	128 PCIe 5.0	5.0	Up to 7 x16 slots
Intel Xeon W-2400/2500 (W790)	Entry Workstation	64	5.0	2 GPUs at x16 + storage
Intel Xeon W-3400/3500 (W790)	Workstation	112	5.0	4+ GPUs
AMD EPYC 9004/9005 (SP5)	Server	128 per socket	5.0	160 total lanes in dual-socket
Intel Xeon 6 (6700P/6900P)	Server	88 / 96 per socket	5.0	Depends on SKU

[volatile data: always check the datasheet for the precise SKU, as variants with different counts exist within the same family]

Practical reading of this table: the jump from consumer to HEDT **doubles or triples** the lanes; the jump from HEDT to professional workstation **doubles them again**. And this is why — not for the number of cores — one moves up in platform.

1.4 Memory channels: why RAM bandwidth matters

The **memory controller** is the circuit that manages communication between the CPU and RAM. For about fifteen years, it has been integrated into the CPU itself, and is called the **IMC** (*Integrated Memory Controller*). Previously, it was on the chipset (the infamous "northbridge"), and its integration into the CPU was one of the major performance breakthroughs of the 2000s.

The IMC communicates with the RAM through one or more **channels**. A channel is an independent path to the memory modules: more channels mean more parallel paths, and thus more aggregated bandwidth.

Let's do the math. A DDR5 (*Double Data Rate 5*, the fifth generation of SDRAM memory) module at 5600 MT/s (*Mega Transfers per second*, i.e., millions of transfers per second) provides approximately **44.8 GB/s per channel** (5600×8 bytes). From this:

Configuration	Channels	Theoretical Bandwidth (DDR5-5600)	Platforms
Dual channel	2	~90 GB/s	Consumer AM5, LGA1700/1851
Quad channel	4	~180 GB/s	Threadripper non-PRO (TRX50), Xeon W-2400/2500
Octa channel	8	~358 GB/s	Threadripper PRO (WRX90), Xeon W-3400/3500
12-channel	12	~537 GB/s	AMD EPYC 9004/9005, Intel Xeon 6 (6900P)

[note: officially supported frequencies vary; server platforms typically use registered DDR5 at 4800–6400 MT/s depending on the number of modules per channel. Volatile data.]

Why RAM Bandwidth Matters in an AI Workstation

There are three reasons, all concrete.

First: data loading. During training, the flow is as follows: raw data (JPEG images, audio files, texts) is read from disk, decoded into RAM, transformed (resizing, normalization, augmentation), packaged into tensors, and finally copied to the GPU's memory via PCIe. All these intermediate steps reside in system RAM. With four GPUs requesting, say, 3–4 GB/s of data each, and considering that each byte is touched multiple times (reading, decoding, transforming, copying to *pinned memory*), RAM is traversed at tens of GB/s. A dual channel at a theoretical 90 GB/s — which in practice translates to 60–70 GB/s real-world — becomes a bottleneck.

Second: CPU inference. When running an LLM (*Large Language Model*) directly on the CPU — with `llama.cpp`, with ONNX Runtime backend, with oneDNN — the **token-by-token generation** phase is almost entirely *memory-bound*, meaning limited by memory bandwidth, not computational power. The reason is that to generate each single token, **all model weights** must be read from RAM. A 70 billion parameter model quantized to 4 bits occupies about 40 GB. To generate 10 tokens per second, 400 GB/s must be read. On a dual-channel platform with 70 GB/s real-world, this results in about 1.5–2 tokens/s: unusable for interactive use. On a 12-channel platform with 400+ GB/s, this results in 8–10 tokens/s: slow but usable. **This is why there is a real niche of people who buy used EPYC for LLM inference on CPU: not for the cores, but for the bandwidth.**

Third: classic preprocessing. Pandas/polars pipelines on tabular datasets of tens of GB, join operations, groupby, feature engineering: all operations that move mountains of bytes and scale with memory bandwidth more than with clock frequency.

1.5 RAM Capacity: The Real Limits and Registered Modules

Beyond bandwidth, there's **capacity**. And here the consumer limit is clear.

A consumer platform has **four DIMM slots** (on two channels: two modules per channel). The practical limit depends on the maximum capacity of available UDIMM modules:

[volatile data] On AM5 and LGA1851, with 48 GB UDIMM modules, up to 192 GB is achieved; with the arrival of 64 GB modules and related AGESA/BIOS updates, 256 GB is reached. Beyond that, it's not possible, and already with 4 modules, operating frequencies drop because the IMC struggles to drive two modules per channel at high speed.

This is the critical point: **on a consumer platform, filling all four slots degrades RAM frequency**. A DDR5-6000 kit with two modules runs at 6000; the same four modules often drop to 3600–4400 MT/s. You pay for capacity in exchange for bandwidth — exactly the compromise you don't want to make in a workstation.

UDIMM, RDIMM, LRDIMM

To overcome this limit, the professional world uses different modules.

- ****UDIMM**** (*Unbuffered DIMM*): This is the consumer module. Address and command signals from the IMC arrive ****directly**** at all memory chips on the module. Simple, inexpensive, low latency—but the electrical load on the controller increases with the number of chips, which limits how many modules can be placed per channel and at what frequency.
- ****RDIMM**** (*Registered DIMM*): There is an additional chip on the module, the ****RCD**** (*Registering Clock Driver*), which acts as a buffer for address and command signals. The IMC drives only one load (the register) instead of dozens of chips. Cost: one additional clock cycle of latency. Benefit: much larger modules can be installed (64, 96, 128 GB and more) and more modules per channel while maintaining high frequencies. ****Threadripper PRO, EPYC, and Xeon W/Scalable platforms require RDIMM: they do not accept UDIMM.****
- ****LRDIMM**** (*Load-Reduced DIMM*): This buffers data lines as well, not just addresses and commands. It allows for extreme capacities (up to 256 GB per module) at the cost of additional latency. Used in servers with enormous memory requirements.
- ****MRDIMM**** (*Multiplexed Rank DIMM*): The recent evolution, which multiplexes two ranks to double the effective bandwidth. Supported on the latest server platforms. **[volatile data, evolving ecosystem]**

Capacity Limits per Platform

Platform	Typical DIMM Slots	Memory Type	Indicative Max Capacity
Consumer AM5 / LGA1851	4	UDIMM (optional unofficial ECC)	192–256 GB <i>[volatile]</i>

Threadripper 7000/9000 (TRX50)	4	RDIMM ECC (4 channels)	~1 TB
Threadripper PRO (WRX90)	8	RDIMM ECC (8 channels)	~2 TB
Xeon W-3400/3500 (W790)	8	RDIMM ECC (8 channels)	~2 TB
EPYC 9004/9005 (SP5)	12 or 24	RDIMM ECC (12 channels)	6 TB and beyond per socket

[volatile data: maximum capacities increase with each new RDIMM module on the market. Check the motherboard's QVL — Qualified Vendor List.]

The jump is an order of magnitude. And it's truly needed: loading an entire image dataset into RAM to eliminate disk I/O, holding a huge graph in memory, performing CPU inference on a 405-billion-parameter model, running simulations —these are all cases where 256 GB is insufficient.

1.6 ECC: What it is and why it's non-negotiable in professional environments

ECC stands for *Error-Correcting Code*.

The problem it solves is physical. A DRAM memory cell stores a bit as an electrical charge in a microscopic capacitor. That charge can be altered: by a cosmic ray (a high-energy neutron passing through silicon), by alpha particles emitted from radioactive impurities in package materials, by electromagnetic interference, or by thermal or power instabilities. The result is a **soft error**: a bit that flips, from 0 to 1 or vice versa, without anything being physically broken.

The frequency is low but not negligible. Field studies (the most cited being Google's on tens of thousands of servers) indicate orders of magnitude of **thousands of correctable errors per gigabyte per year** under real conditions. On a desktop with 32 GB accessed eight hours a day, an occasional error: annoying, maybe a crash, you restart. On a workstation with **512 GB running 24/7 for three weeks** during training, the probability of at least one bit flip becomes **high**.

How ECC works

The classic implementation is called **SECCDED**: *Single Error Correction, Double Error Detection*. For every 64 bits of useful data, 8 bits of control code (an extended Hamming code) are added, meaning the module has extra chips: an ECC module typically has 9 or 18 chips instead of 8 or 16. When the CPU reads those 72

bits, it recalculates the code: if a single bit is wrong, the system **understands which one** and corrects it on the fly, transparently, logging the event. If two bits are wrong, the system **detects** the error but cannot correct it: it generates a machine exception and generally halts the system, because continuing would mean working with corrupted data.

This is the key point: **without ECC, a bit flip is silent**. The computer doesn't know the data is wrong. It just keeps going.

Why it's essential in AI/ML

Let's imagine the concrete consequences of a silent bit flip in an AI workstation:

- During a 72-hour training session, a bit flips in the exponent of a floating-point number representing a weight. The weight changes from 0.003 to 3×10^{30} . The gradient explodes, the loss becomes NaN (*Not a Number*), and the problem is noticed — in the best case — hours later. ****Three days of GPU wasted.****
- Worse: the bit flip occurs at a point that doesn't cause anything to explode, but subtly corrupts a batch of data or a weight. The model still converges, but it is ****subtly wrong****, and no one will ever notice. The scientific result or product is compromised in a non-reproducible and non-diagnosable way.
- A bit flip in a checkpoint written to disk corrupts the file, and it's only noticed weeks later when trying to resume training.

In a professional context — publishable research, models going into production, calculations on which decisions are made — **silent non-reproducibility is unacceptable**. ECC costs 10–15% more for modules and about one percentage point in performance. It's the cheapest insurance for the system.

Beware of two common confusions

Confusion 1: DDR5's "on-die ECC" is not system ECC. All DDR5 modules, even the cheapest gaming ones, have a form of error correction *internal to the chip* (on-die ECC), introduced because cell densities have become so high that internal errors would otherwise be unsustainable. But this correction only protects the cell; **it does not protect the path between the chip and the CPU**, does not report errors to the operating system, and provides no end-to-end guarantee. Seeing "DDR5 with on-die ECC" on the packaging of a gaming kit and thinking you have an ECC machine is a very common and very costly mistake.

Confusion 2: ECC support on a consumer platform. Some AM5 CPUs technically support UDIMM ECC modules, and some motherboards (especially ASRock and

some ASUS Pro) expose it in the BIOS. This is real support but **not officially validated** for professional use, with limited module availability and no guarantees. It's not a shortcut to avoid the workstation platform: it's a "better than nothing" for a homelab.

1.7 What if consumer is enough?

Before proceeding, it's honest to say something that the rest of the chapter might make you forget: **for many real-world use cases, a high-end consumer platform is the right choice.**

If the project is: a single GPU (even an RTX 5090 or an RTX PRO 6000), 128 GB of RAM, two NVMe drives, fine-tuning with LoRA, local inference, prototype development — then a Ryzen 9 9950X or a Core Ultra 9 285K on a consumer platform **does exactly what's needed**, costs a quarter, has higher single-thread frequencies (useful for the non-parallelized Python code), and doesn't require exotic cooling.

Spending 3,000 euros on a Threadripper PRO CPU + motherboard to drive a single GPU is, as we'll see in the errors section, the most frequent and most expensive mistake in this sector. **You upgrade platforms when a precise boundary is crossed, not for prestige.** That boundary is: **two or more GPUs at x16 electrical**, or the need for more than 256 GB of RAM, or the requirement for certified ECC.

2. AMD Threadripper: The Complete Overview

2.1 What Threadripper is and where it fits

Threadripper is AMD's line launched in 2017 to fill the gap between Ryzen (consumer) and EPYC (server). Technically, it's not a separate project: it's **repackaged EPYC silicon** on a dedicated package and socket, with higher frequencies, fewer memory channels (in non-PRO versions), and a motherboard ecosystem designed for desktop rather than rack.

The positioning is precise:

- ****Ryzen**** (AM5): 1 GPU, 2 RAM channels, ~24 lanes. Maximum frequencies, minimum latencies, low price.
- ****Threadripper**** (sTR5/TRX50): 2 GPUs at x16, 4 RDIMM RAM channels, 48 PCIe 5.0 lanes. The "extreme desktop."

- ****Threadripper PRO**** (sTR5/WRX90): 4–7 GPUs, 8 RDIMM RAM channels, 128 PCIe 5.0 lanes. The true workstation.
- ****EPYC**** (SP5): 12 channels, 128 lanes per socket, dual socket, but lower frequencies and no "desktop" support.

It's important to understand immediately: **PRO and non-PRO share the same physical socket (sTR5) but not the same platforms.** This is the number one source of confusion.

2.2 Threadripper non-PRO: The 7000X and 9000X Series

The Zen 4-based generation (codename *Storm Peak*, launched in late 2023) includes:

Model	Cores / Threads	Indicative Max Boost	L3 Cache	TDP
Threadripper 7960X	24 / 48	~5.3 GHz	128 MB	350 W
Threadripper 7970X	32 / 64	~5.3 GHz	128 MB	350 W
Threadripper 7980X	64 / 128	~5.1 GHz	256 MB	350 W

The next generation, based on **Zen 5** (codename *Shimada Peak*, 9000 series, available from 2025), maintains the same structure:

Model	Cores / Threads	Notes
Threadripper 9960X	24 / 48	Direct successor to 7960X
Threadripper 9970X	32 / 64	Successor to 7970X
Threadripper 9980X	64 / 128	Successor to 7980X

[volatile data: frequencies, prices, and availability change; the core/lane structure is stable]

Platform characteristics of non-PROs on TRX50 chipset:

- ****Memory****: ****4-channel**** DDR5 RDIMM ECC, 4 DIMM slots. Important note: even non-PRO Threadrippers ****require registered RDIMM****, not consumer UDIMM. You cannot reuse a gaming kit.
- ****PCIe****: ****48 PCIe 5.0 lanes**** from the CPU usable on the TRX50 platform, plus additional PCIe 4.0 lanes from the chipset and CPU. AMD commercially

advertises a total platform "up to 92 lanes" by summing everything (CPU + chipset, mixed generations). *[Caution: this is a marketing number. What matters for GPUs are the 48 native PCIe 5.0 lanes.]*

- **TDP**: **350 W** (*Thermal Design Power*, the thermal design power that the cooling system must be able to dissipate under sustained conditions; actual peak consumption may exceed it).

What it's used for: two GPUs at full x16 PCIe 5.0, plus 2–3 NVMe at x4, plus a NIC. This is the typical configuration for a serious individual researcher's AI workstation, or a small team.

2.3 Threadripper PRO: The 7000WX and 9000WX Series

This is a different category. The PRO line is AMD's professional workstation, and the difference is not in the core count (which also increases) but **in the platform**.

Model	Cores / Threads	L3 Cache	TDP
Threadripper PRO 7955WX	16 / 32	64 MB	350 W
Threadripper PRO 7965WX	24 / 48	128 MB	350 W
Threadripper PRO 7975WX	32 / 64	128 MB	350 W
Threadripper PRO 7985WX	64 / 128	256 MB	350 W
Threadripper PRO 7995WX	96 / 192	384 MB	350 W

The Zen 5 generation (9000WX series) replicates the structure, with the **9995WX** at 96 cores at the top, flanked by 9985WX (64c), 9975WX (32c), 9965WX (24c), 9955WX (16c). *[volatile data: check updated SKU list]*

PRO platform features on **WRX90** chipset:

- **Memory**: **8-channel** DDR5 RDIMM ECC, 8 DIMM slots, up to ~2 TB.
- **PCIe**: **128 native PCIe 5.0 lanes**. This is the number that justifies the entire existence of the line.
- **Professional features**: AMD PRO Manageability (enterprise-level remote management), AMD PRO Security (Memory Guard, memory encryption), ISV (*Independent Software Vendor*: certification that professional software such

as CAD, simulators, and rendering suites are tested on that platform) guarantees and validation.

2.4 The Comparison That Really Matters

Let's summarize, because this is **the most important table in the entire chapter**:

Feature	Threadripper (non-PRO) on TRX50	Threadripper PRO on WRX90
Socket	sTR5	sTR5 (same physical socket)
Chipset	TRX50	WRX90
Native PCIe 5.0 Lanes	48	128
RAM Channels	4	8
DIMM Slots	4	8
Max RAM	~1 TB	~2 TB
RAM Type	RDIMM ECC	RDIMM ECC
Max Cores	64	96
GPUs at x16 electrical	2	up to 7
Remote Management (IPMI/BMC)	rarely	typical
Platform Cost (CPU+MB)	High	Very High

A subtle point that causes a lot of confusion: **a Threadripper PRO CPU can be mounted on a TRX50 motherboard**, because the socket is the same. But in that case, it **operates with the limitations of the TRX50**: 4 memory channels instead of 8, and a reduced number of exposed lanes (TRX50 boards do not have the wiring for 128 lanes). The opposite is **not true**: a non-PRO CPU **does not work** on a WRX90 motherboard. WRX90 only accepts PRO.

This opens up an interesting practical scenario: **buying a PRO CPU on a TRX50 board** only makes sense if you want access to a higher core count (e.g., the 96-core 7995WX) without needing 128 lanes. It's a niche case — but it exists, for example, for rendering or massive compilation workloads with a single GPU.

2.5 Decoding the nomenclature

Let's take **7995WX** and break it down:

- ****7****: generation. The 7000 series corresponds to Zen 4; the 9000 series to Zen 5. ***(Note: AMD skipped the 8000 series on Threadripper, aligning with Ryzen numbering.)***

- ****99****: tier within the generation. Higher = more cores. 95 → 96 cores, 85 → 64 cores, 75 → 32 cores, 65 → 24 cores, 55 → 16 cores. The two-digit logic is "position in the stack," not a mathematical formula.
- ****5****: segment/revision digit, practically constant in the line.
- ****WX****: ****the suffix is the most important thing of all****. ***WX*** = ***WorkstationeXtreme***, i.e., ****PRO****: 8 channels, 128 lanes, WRX90.
- ****X**** (without W): non-PRO. 4 channels, 48 lanes, TRX50.

So: **7970X** ≠ **7975WX**. The first has 32 cores, 4 channels, 48 lanes. The second has 32 cores, 8 channels, 128 lanes. **Same cores, completely different platforms, very different price.** Those who don't know the meaning of "WX" are highly likely to buy the wrong product.

2.6 sTR5 Socket, TRX50 and WRX90 Chipsets

The **sTR5** socket is an LGA (*Land Grid Array*: the pins are on the socket, not on the CPU) with **4844 pins**. It's physically enormous — a rectangle of about 76 × 58 mm — and this has practical consequences:

- ****Mounting****: the CPU is inserted into an orange plastic **carrier** that slides into a metal rail; the lid closes with ****three Torx screws**** to be tightened ****in numbered order (1, 2, 3)**** and with ****controlled torque****. AMD provides a torque screwdriver in the box. This is not a whim: tightening out of order or with incorrect torque on such a large surface can cause missed contacts or permanent damage to the package.
- ****Cooling****: the IHS (**Integrated Heat Spreader**, the metal lid of the CPU) is much larger than consumer ones. ****An AM5 or LGA1700 cooler will not cover the surface**** and does not have the correct mounting. Specific sTR5 coolers are required.

The **chipset**, as we have seen, is the chip on the motherboard that expands connectivity. The choice between TRX50 and WRX90 **is the real architectural decision**, and it must be made **before** choosing the CPU:

Question	If the answer is...	Platform
How many GPUs at electrical x16?	1 or 2	TRX50
How many GPUs at electrical x16?	3 or more	WRX90
Do you need more than ~500 GB of RAM?	Yes	WRX90

Do you need maximum memory bandwidth (CPU inference)?	Yes	WRX90 (8 channels)
Do you need remote IPMI/BMC management?	Yes	WRX90 (or server board)
Is budget the main constraint?	Yes	TRX50

2.7 Why Threadripper PRO is **the** platform for AI workstations

There is only one reason, and it lies in the 128 lanes. Let's count on an **ASUS Pro WS WRX90E-SAGE SE** board, which is the reference in this category:

- ****7 PCIe 5.0 x16 slots**** — all ****true electrical x16****, all simultaneously. $7 \times 16 = \mathbf{**112 \text{ lanes**}}$.
- Approximately 16 lanes remain for NVMe (often 4 M.2 slots, some via bifurcation) and for the rest.

Seven full x16 slots mean you can install **four dual-slot GPUs with adequate space between them** (using slots 1, 3, 5, 7), or **seven single-slot cards**. No other desktop-derived platform offers this. It is literally the reason why multi-GPU workstations exist under the desk instead of in racks.

2.8 Brand and Price Range Comparison (Threadripper)

[All prices are indicative and volatile; use them only as orders of magnitude.]

Configuration	Indicative CPU	Motherboard	CPU+MB Cost Range
HEDT entry	TR 7960X / 9960X (24c)	TRX50 (ASUS TRX50-SAGE, ASRock TRX50 WS)	~2,000–2,800 €
HEDT high	TR 7970X / 9970X (32c)	TRX50	~3,000–4,000 €
Workstation entry PRO	TR PRO 7955WX (16c)	WRX90 (ASUS WRX90E-SAGE SE)	~3,500–4,500 €
Workstation medium PRO	TR PRO 7975WX (32c)	WRX90	~5,500–7,000 €
Workstation top PRO	TR PRO 7995WX (96c)	WRX90	~12,000–15,000 €

The point to note: **the 16-core 7955WX is the "budget" PRO CPU**, and for many AI workstations, **it is the right choice**, because you buy it for the lanes and memory channels, not for the cores. Sixteen Zen 4 cores are more than enough to handle data loading for four GPUs. Paying three times as much for 96 cores that will remain at 15% utilization is wasted money — unless the workload is also CPU-intensive (rendering, simulation, compilation).

3. AMD EPYC and Intel Xeon: When to Move to Server

3.1 AMD EPYC (Genoa, Bergamo, Turin)

EPYC is AMD's server line. Recent generations, on **SP5** socket (LGA6096):

- **EPYC 9004 "Genoa"** (Zen 4, 2022): up to **96 cores** per socket.
- **EPYC 9004 "Bergamo"** (Zen 4c, density-optimized cores): up to **128 cores**.
- **EPYC "Genoa-X"**: with **3D V-Cache**, L3 cache up to over 1 GB. Useful for cache-sensitive workloads (CFD simulation, EDA).
- **EPYC 9005 "Turin"** (Zen 5, 2024): up to **128 Zen 5 cores**, or **192 cores** in Zen 5c variants.

SP5 platform features:

- **128 PCIe 5.0 lanes per socket.** In a **dual-socket** configuration, some lanes are used for inter-processor interconnection (the **Infinity Fabric** inter-socket link), so the total usable is not 256 but typically **160 lanes** (configurable: more lanes can be sacrificed for a wider inter-socket link, or fewer lanes for more free ones).
- **12 DDR5 RDIMM channels per socket** → in dual socket, **24 channels**, with theoretical aggregated bandwidth over **1 TB/s**. This is a number no desktop platform can approach.
- **Huge RAM capacity**: several TB per socket.
- **Lower frequencies** compared to Threadripper (typically 2.0–3.5 GHz base, boost up to ~4.x GHz only on low core count SKUs), because the thermal budget is divided among many more cores and because servers optimize for efficiency, not peak single-thread.

The EPYC compromise in a word: you gain in parallelism, bandwidth, and connectivity; you lose in single-thread performance. And this matters, because

a lot of orchestration Python code (the training loop, PyTorch's DataLoader before it distributes to workers, non-vectorized preprocessing code) is **single-thread**. An EPYC workstation clocked at 2.4 GHz can be *slower* than a Ryzen 9 in all the "glue" parts of the code.

3.2 Intel Xeon W and Xeon Scalable

Intel divides the professional world into two:

Xeon W is the workstation line, conceptually equivalent to Threadripper PRO. On LGA4677 socket and W790 chipset:

- **Xeon W-2400** (Sapphire Rapids, up to 24 cores): **4 channels** DDR5, **64 PCIe 5.0 lanes**.
- **Xeon W-3400** (up to 56 cores): **8 channels** DDR5, **112 PCIe 5.0 lanes**.
- **Xeon W-2500 / W-3500**: the generation refresh, with slightly higher core counts (up to ~60 cores on the W-3500 line) and the same platform features.
[volatile data: check current SKU list]

Xeon Scalable / Xeon 6 is the server line, equivalent to EPYC:

- **Xeon 6 "Granite Rapids"** (P-core, high performance per core): 6900P series up to 128 cores, **12 channels** DDR5/MRDIMM, ~96 PCIe 5.0 lanes; 6700P series with 8 channels and ~88 lanes.
- **Xeon 6 "Sierra Forest"** (E-core, optimized for density and efficiency): very high core counts (up to 288 E-cores), but **E-cores without AVX-512 and without AMX**, thus less suitable for CPU inference.

[volatile data: Intel's roadmap is rapidly evolving]

Intel's Countermove: AMX

There's an Intel ace up its sleeve worth knowing about, because it's specifically relevant to AI: **AMX (Advanced Matrix Extensions)**, introduced with Sapphire Rapids. It's an instruction set that adds two-dimensional **"tile" registers** and a dedicated matrix multiplication unit to the CPU. In practice, it puts a small tensor core inside each CPU core.

The result is that CPU inference of quantized models (INT8, BF16) on Xeon with AMX can be **3 to 8 times faster** than on the same CPU without AMX, at the same clock and core count, when the software supports it (oneDNN, OpenVINO, PyTorch with Intel backend). **This is the strongest argument in favor of Intel in**

the AI on CPU domain, and should be kept in mind by those doing CPU-only or hybrid inference.

3.3 Why AMD Dominates in Core Count and Lanes Today

It's useful to understand the structural *why*, not just observe the fact.

The main reason is **architectural**, and it's called **chiplet**. Since 2017, AMD has been building its server CPUs by assembling multiple small dies:

- ****CCDs**** (*Core Complex Dies*): small dies that contain only the cores and cache, manufactured on the most advanced production node available (5 nm, then 4 nm, then 3 nm).
- An ****IOD**** (*I/O Die*): a larger die that contains the memory controllers, PCIe controllers, and interconnect infrastructure, manufactured on a cheaper, more mature node (where I/O transistors don't benefit from extreme miniaturization anyway).

The advantage is economic and statistical. **Production yield** (the percentage of functional dies on a wafer) drops exponentially with die area, because the larger it is, the more likely a random defect is to strike it. By producing **eight or twelve small dies** instead of **one giant monolithic die**, AMD achieves much higher yields, can select the best dies, and can scale the number of cores simply by **adding CCDs to the package**.

Intel used monolithic dies for years, and when it wanted to increase core count, it encountered increasingly worse yields and rising costs. It too has moved to a tile-based approach (with Sapphire Rapids and successors), but with a delay of several years, exacerbated by known difficulties in transitioning to advanced production nodes (the famous stagnation at 10 nm).

The same applies to PCIe lanes: the PCIe logic is in the IOD, a large and "cheap" die, and AMD could afford to put a lot of it there. 128 native lanes versus Intel's ~96 is not a coincidence: it's a direct consequence of the packaging choice.

Honest Conclusion: AMD currently has a structural advantage in cores, lanes, and memory channels. Intel maintains advantages in specific integrated accelerators (AMX), optimized software ecosystems (OpenVINO, oneAPI), certain low-latency SKUs, and sometimes in the price of used platforms. *[Volatile situation: the competitive advantage between the two can change rapidly.]*

3.4 Server or HEDT? The Decision for a Researcher or Small Team

Here are the concrete criteria, beyond marketing.

Choose HEDT (Threadripper / Threadripper PRO) if you:

- Work under a desk, in an office, and need ****silence**** (a 2U server with 15,000 rpm fans is unsustainable in a work environment).
- Want ****high frequencies**** because part of the workload is single-threaded.
- Have ****1–4 GPUs****.
- Want to be able to ****assemble everything yourself**** with retail components.
- Need ISV certifications for professional software.

Choose server (EPYC / Xeon Scalable) if you:

- Need ****more than 4 GPUs**** or dual-socket.
- Need ****extreme memory bandwidth**** (CPU inference on huge models, HPC).
- Already have a ****rack****, with three-phase power, cooling, and isolated noise.
- Need ****multi-terabyte RAM****.
- Want to ****maximize price/performance ratio on the used market****: decommissioned datacenter servers (EPYC Rome/Milan, first and second-generation Xeon Scalable) can be found at fractions of their original price, and for those who need lanes and RAM more than clock speed, they can be excellent deals. This is a path taken by many homelabbers and small labs.
[The used server market is extremely volatile in pricing.]

A practical observation worth its weight in gold: if you need *a lot* of power *occasionally*, the economically sound answer is often **not to buy anything** and rent cloud instances (or on-demand GPUs from specialized providers). A local workstation is justified when usage is **continuous** (months of training, daily use), when **data cannot leave** the organization (privacy, GDPR, industrial secret), or when a rapid **iterative cycle** matters more than peak power.

4. PCIe in Workstations: In-depth

4.1 Review and Extension

We have already defined lanes and generations. Now let's add three distinctions that can be ignored in the consumer world but are decisive in the workstation world.

Distinction 1: physical slot vs. electrical slot. A slot can be **physically** x16 long (i.e., the connector has the length to accommodate an x16 card) but only 4 or 8 lanes are **electrically** wired. This is very common, and motherboards indicate it in their documentation with notations like "PCIe 5.0 x16 (x8 mode)" or "x16 slot, x4 electrical". A GPU inserted into a physical x16 but electrical x4 slot **will work**, but with a quarter of the bandwidth.

Golden Rule: in the motherboard manual, **always** look for the "PCIe lane distribution table" or "block diagram". It is the only document that tells the truth. The commercial description on the website does not.

Distinction 2: shared lanes. On many motherboards, populating a certain slot **disables or reduces** another slot or an M.2 slot. The manual contains a table like "if PCIe_2 slot is used, M.2_3 slot is disabled." On workstation platforms with 128 native lanes, these shared configurations are much less frequent — and this is precisely one of the benefits you pay for.

Distinction 3: downward negotiation. PCIe automatically negotiates both width (number of active lanes) and speed (generation). A disturbed signal, a low-quality riser, or a dirty contact can cause the link to drop to x8 instead of x16, or to Gen3 instead of Gen5, **without any visible error**: the system works, just slower. On Linux, this can be checked with `lspci -vv` looking for `LnkCap`` (capability) and `LnkSta`` (current status) lines; on NVIDIA, also with `nvidia-smi -q | grep -i pcie``. **This check should always be performed after every assembly.**

4.2 Why 2–4 GPUs need true x16 electrical lanes

Here we need to be precise, because there is a lot of misinformation circulating in both directions.

When x16 is NOT necessary. If the workload is: a single GPU, the dataset fits in VRAM or is loaded once, and training is purely GPU-bound — then PCIe bandwidth matters little. Known tests on a single GPU show differences of 2–5% between x8 and x16. In this case, spending on a workstation platform just to have x16 is irrational.

When x16 is indispensable. The picture changes completely in three scenarios:

Scenario A: multi-GPU distributed training. When using *data parallelism* (each GPU has a copy of the model and processes a different batch), at each training step the GPUs must **synchronize gradients**. The operation is called **all-reduce**: each

GPU must receive the sum of gradients from all others. The data volume is equal to the model size, **at every single step**. For a 1 billion parameter model in FP16, that's 2 GB per GPU per step. With 4 GPUs and one step every 100 ms, **tens of GB/s are continuously moved across the PCIe bus**. At x8 Gen4 (15 GB/s) this becomes **the dominant bottleneck**: the GPU computes in 60 ms and then waits 150 ms for the gradients to arrive. Scaling efficiency plummets: four GPUs perform like two and a half.

Scenario B: model parallelism / pipeline parallelism. When the model is too large for one GPU and is split between multiple GPUs, **activations** must pass from one GPU to another at each forward and backward pass. Traffic is continuous and sensitive to latency. Here, PCIe bandwidth is even more critical.

Scenario C: offloading and streaming. Techniques like ZeRO-Offload or offloading part of the weights to system RAM (used when VRAM is insufficient) continuously transfer weights back and forth over PCIe. Bandwidth is everything.

Conclusion: if you have **one** GPU, x8 is perfectly fine. If you have **two or more** and use them together, every lane counts.

4.3 PCIe switches (PLX / Broadcom): what they are and when to use them

A **PCIe switch** is a chip that does what a network switch does for Ethernet: it takes a certain number of incoming lanes (*upstream*, towards the CPU) and expands them into a larger number of outgoing lanes (*downstream*, towards peripherals), dynamically routing packets.

The most well-known chips are those originally produced by **PLX Technology**, a company later acquired by **Avago/Broadcom**. Hence, by habit, "PLX" is used to refer to any PCIe switch. The historical family is **PEX 87xx/88xx** (Gen3/Gen4); recent Gen5 generations are marketed by Broadcom under names like **Atlas 3 / PEX 89xxx**.

How it works in practice. Imagine a switch with 16 upstream lanes to the CPU and 32 downstream lanes to two GPUs (x16 each). The two GPUs each see an x16 link and are happy. But towards the CPU, there is only one shared x16.

The real, and often misunderstood, benefit: the switch **does not create bandwidth out of thin air**. If both GPUs want to communicate with the CPU simultaneously at full speed, they contend for the 16 upstream lanes. **The benefit is that the two GPUs can communicate with each other, through the switch, at**

full x16 speed, without ever disturbing the CPU. This local **peer-to-peer** communication is exactly what is needed in multi-GPU training. A switch transforms four GPUs connected at x4 each into a cluster where the GPUs exchange gradients at full bandwidth among themselves, using the uplink to the CPU only to load data.

When a switch makes sense:

- On consumer or entry-workstation motherboards, to run 4 GPUs with an insufficient number of CPU lanes.
- On server motherboards designed for GPU density (the classic 8-GPU baseboards).
- On expansion cards (backplanes) that expand an x16 slot into multiple slots.

When it does NOT make sense:

- If you already have 128 native lanes (Threadripper PRO, EPYC). ****Native lanes are always better****: lower latency, no contention point, no cost, no additional power consumption.
- PCIe switches are expensive (hundreds of dollars per chip), consume 10–25 W, and add latency. It's technology to be used out of necessity, not by choice.

4.4 Bifurcation: splitting an x16 slot

Bifurcation is the ability of the CPU and BIOS to **split a single x16 slot into multiple independent links**: typically x8+x8, x8+x4+x4, or **x4+x4+x4+x4**.

It's important to understand that **bifurcation is passive**. There is no additional chip: the CPU is simply instructed to treat those 16 lanes as four separate 4-lane links. The CPU's PCIe chip natively supports this configuration; the BIOS exposes the option (usually in the "PCIe Bifurcation" or "PCIe Slot Configuration" menu).

The main application in AI workstations: quad M.2 cards. These are passive PCIe x16 expansion cards, without any controller on board, that expose four M.2 slots. Each NVMe SSD uses 4 lanes, and the total is exactly 16. They cost a few tens of euros (the PCB, connectors, and a heatsink) and allow you to add **four full-speed NVMe SSDs** using a single slot.

Common examples: ASUS Hyper M.2 X16, Gigabyte AORUS Gen4 AIC, ASRock Hyper Quad M.2. Some workstation motherboards already include dedicated connectors (SlimSAS, MCIO) that expose lanes in a similar way.

Beware of the classic trap: if the BIOS **does not support bifurcation** on that slot, a passive quad M.2 card will show **only one SSD** (the first one), or none at all. This is not a defect of the card: it is the platform that cannot split the lanes. There are also **active** quad M.2 cards, which contain a PCIe switch and work everywhere — but cost 5–10 times as much. **Before buying a passive quad M.2 card, check the motherboard manual to ensure the slot supports x4x4x4x4 mode.**

Threadripper/PRO and workstation platforms almost always support it; consumer platforms often do not, or only on one slot.

4.5 P2P (peer-to-peer) over PCIe vs NVLink

P2P (peer-to-peer) is the ability of two GPUs to **exchange data directly**, without routing data through system RAM. Without P2P, a copy from GPU0 to GPU1 requires: GPU0 → system RAM → GPU1. With P2P: GPU0 → GPU1, directly on the PCIe bus. The savings are in bandwidth (one copy instead of two), latency, and CPU utilization.

This capability is the basis of collective communication libraries like **NCCL** (*NVIDIA Collective Communications Library*), which is what PyTorch and every distributed training framework use under the hood for all-reduce of gradients.

The problem: NVIDIA has **disabled P2P on GeForce GPUs** starting from recent generations (from RTX 30 onwards, support has been progressively removed; on RTX 40 and 50, P2P via official drivers **is not available**). This is a commercial segmentation choice, not a technical one: the silicon could do it. The consequence is that on a workstation with four RTX 5090s, NCCL is forced to use the path through system RAM, with significant costs in bandwidth and latency.

(A well-known community curiosity: there is a modified driver, developed by the tinygrad team, that re-enables P2P on RTX 4090s. It works, but it is unsupported, fragile with respect to updates, and unsuitable for a production environment.)

NVLink is NVIDIA's proprietary alternative: a **dedicated** interconnect, separate from PCIe, with much higher bandwidth (on the order of hundreds of GB/s per link, compared to 63 GB/s for an x16 Gen5) and much lower latency. Historically available via a physical "bridge" between two cards on Quadro/RTX A-series, and via NVSwitch in server configurations (SXM).

The current situation, to be well understood [volatile data]: NVIDIA has **removed the NVLink connector from professional PCIe format cards** starting with the Ada Lovelace generation (RTX 6000 Ada and later do not have the

bridge). Today, "true" NVLink is reserved for GPUs in **SXM format** (modules mounted directly on the baseboard in server systems like DGX/HGX) and rack-scale platforms.

Practical consequence, and this is the message that brings this section into the CPU chapter: **on a workstation with PCIe format GPUs — which is what the vast majority of us will build — inter-GPU communication goes through PCIe. Period.** There is no NVLink to save a configuration with insufficient lanes. And this is precisely why the CPU's PCIe lanes are not a specification detail: **they are the cluster's interconnect.** It is the ultimate technical reason why Threadripper PRO exists.

(We will delve into NVLink, NVSwitch, SXM formats, and multi-GPU topologies in the chapter dedicated to GPUs.)

5. Choosing the Workstation Motherboard

The motherboard, in this segment, is not an accessory: it is **the document that defines what the machine will be able to do.** Here are the criteria, in order of importance.

5.1 Number of Physical AND Electrical x16 Slots

Already covered, but it bears repeating because it is the number one criterion. Look in the manual (not on the commercial website) for the **block diagram** and verify how many lanes are actually wired to each slot and in which generation. A board with "5 PCIe x16 slots" that actually has two electrical x16, one x8, and two x4 is a different product from one with five true x16 slots.

5.2 Slot Spacing (the Most Underestimated Criterion)

This is, statistically, **the most frequent mistake by far** for those building their first multi-GPU workstation.

Modern high-end GPUs occupy **2, 2.5, or even 3 slots in thickness** due to oversized coolers. A custom RTX 5090 can be a 3.5-slot card. If the motherboard has x16 slots spaced by **two slots**, and the GPU occupies **three**, **the second GPU simply won't physically fit.** This is not a matter of performance: it's a matter of geometry.

And even when they do fit, there are two further problems:

- ****Thermal Suffocation****: two adjacent GPUs with open-fan coolers will have the intake of the second one a millimeter from the backplate of the first. Temperatures rise by 15–25 °C, and the GPUs go into **thermal throttling** (automatic frequency reduction to limit temperature). Performance plummets, and noise explodes.
- ****Weight and Flex****: a modern GPU weighs 2 kg. Four GPUs mounted cantilevered will flex the motherboard and the GPUs' own PCBs. Support brackets are needed.

How to solve this: 1. Choose motherboards designed for multiple GPUs: reference WRX90 boards have **7 slots with calculated spacing** specifically to accommodate 4 dual/triple-slot cards in alternating slots (1, 3, 5, 7). 2. Prefer **blower-style** GPUs (turbine, with air expelled out of the case) or **2-slot** format — these are the formats of professional cards (RTX PRO/A-series are designed to be stacked). 3. Use quality **PCIe risers** (certified for the correct generation: a Gen3 riser on a Gen5 link causes silent degradation) to move GPUs to ventilated positions, with a **mining/open-frame case** or a large workstation case.

Practical rule: before buying, **measure**. Take the thickness of the GPU (in millimeters, not "in slots") and the distance between the slots on the motherboard. A standard PCIe slot has a **20.32 mm** (0.8 inch) pitch.

5.3 RDIMM Channels and Slots

Verify **how many DIMM slots** there are and **how they are organized into channels**. A WRX90 board with 8 DIMM slots on 8 channels gives one module per channel: maximum frequency, ideal configuration. Some boards have more slots per channel (2 DIMMs per channel, "2DPC"): they allow for more capacity but often at reduced frequency.

Critical point that ruins many builds: on 8-channel platforms, **populating only 4 slots halves memory bandwidth**. If you buy a WRX90 and put in 4x 32GB modules instead of 8x 16GB modules, you have the same capacity (128GB) but **half the bandwidth**. You paid for 8 channels and are only using 4. **Always fill all channels**. This single rule is perhaps the most valuable advice in the entire chapter, because the error is invisible: the system works, everything seems fine, and it simply runs at half speed.

5.4 Network: 10G Ethernet and beyond

High-end workstation motherboards typically integrate **dual 10 GbE** (10 gigabits per second). Why it matters:

- ****Datasets on NAS/network storage****: A 10 TB dataset won't fit on a local SSD. It resides on a storage server. With 1 GbE (~110 MB/s real), loading is impossible. With 10 GbE (~1.1 GB/s real), it becomes feasible.
- ****Distributed training across machines****: If you have two workstations and want to distribute training, the network becomes the interconnect, and it's already very slow compared to PCIe. 10 GbE is the minimum; for serious work, 25/100 GbE or ****InfiniBand**** (a very low-latency networking technology dominant in HPC, which supports ****RDMA**** — **Remote Direct Memory Access** — i.e., the ability to write directly into another machine's memory without involving the CPU) is needed.

If the motherboard doesn't have integrated 10 GbE, remember that a 10G NIC occupies one PCIe slot and 4 lanes. It must be counted in the lane budget.

5.5 IPMI / BMC: remote management

BMC stands for *Baseboard Management Controller*: it is an **autonomous microcontroller** soldered onto the motherboard (the most common is the ASPEED AST2500/AST2600), with its own CPU, its own RAM, its own firmware, and **its own Ethernet port**. It works **even when the machine is off**, as long as the power supply is connected to the electrical grid.

IPMI (*Intelligent Platform Management Interface*) is the standard protocol for communication.

What it allows you to do, from a browser, from anywhere in the world:

- ****Turn on, turn off, and restart**** the machine.
- View the ****console video**** as if you were sitting in front of the monitor — ****including the BIOS and POST**** (**Power-On Self-Test**, the diagnostic sequence at startup). This function is called ****KVM over IP**** (**Keyboard, Video, Mouse over IP**).
- ****Mount a remote ISO image**** as if it were a USB stick inserted into the machine: you can therefore ****reinstall the operating system remotely****.
- Read ****all sensors****: temperatures, voltages, fan speeds, power consumption.
- Consult ****hardware logs**** (SEL, **System Event Log**), including ****corrected ECC events****.

Why it's crucial in an AI workstation: because an AI workstation is a machine you leave on for days, often in another room, in the basement, or in a small rack. When the training gets stuck at three in the morning and the machine doesn't respond via SSH, without BMC you have to get up, go there physically, and connect a monitor. With BMC, you open the browser and see the kernel panic screen. **The first time it saves you, the BMC has paid for itself.**

BMC is almost always present on **server** boards (Supermicro, ASRock Rack, Gigabyte, Tyan) and on some high-end workstations (some WRX90). It is not present on consumer boards or most "desktop" TRX50 boards.

5.6 Reference Brands and Models

[Models and availability volatile]

Brand	Positioning	Typical Models	Notes
ASUS Pro WS	"Desktop-friendly" workstation	Pro WS WRX90E-SAGE SE, Pro WS TRX50-SAGE WIFI, Pro WS W790E-SAGE SE	Industry benchmark for multi-GPU workstations. The WRX90E-SAGE SE with 7 full x16 slots is effectively the standard.
Supermicro	Server and server-grade workstations	M13SWA-TF (sTR5), H13/X13 series	Server quality and reliability, BMC/IPMI as standard, but spartan BIOS and support oriented towards businesses, not individuals.
ASRock Rack	Servers, excellent value for money	WRX90D8-2T, TRX50D8-2L2T	Server form factor (SSI-EEB or proprietary), IPMI, often more affordable. Pay attention to case form factor.
Gigabyte (Server)	Servers and barebones	MZ73/MZ33 (EPYC), TRX50 AERO D	Good range, from complete barebones to

			single boards.
Tyan	Servers, HPC niche	Various	Less common in retail in Europe.

Practical advice: If you are building a workstation to keep in the office and want a "desktop" experience (usable BIOS, quiet fans, standard ATX/E-ATX form factor, consumer RMA support), go for **ASUS Pro WS**. If you are building something that will live in a rack or a technical room and want BMC and 24/7 reliability, go for **Supermicro** or **ASRock Rack**, but be prepared for a more "industrial" experience.

5.7 Power Supply: Multiple EPS and Server-Grade PSUs

An aspect that first-time builders systematically underestimate.

The EPS12V connector (often simply called "EPS" or "CPU power") is the **8-pin** connector that supplies 12 volts to the CPU's **VRM** section. **VRM** stands for *Voltage Regulator Module*: it is the circuit on the motherboard that converts the 12 V from the power supply into the much lower voltage (approximately 1.0–1.3 V) and very high current (hundreds of amperes) that the CPU requires. The VRM is made of **phases** (parallel conversion stages): more phases mean more current can be delivered with less thermal stress on each.

A Threadripper CPU with a 350 W TDP, during transients, can draw well over 400 W. With a VRM efficiency of around 90%, the power supply must provide more than 450 W **on the CPU rail alone**. A single 8-pin EPS connector is specified for approximately **300 W. This is not enough**. This is why sTR5 motherboards have **two (or even three) 8-pin EPS connectors**, and they **must all be connected**. Connecting only one will, at best, result in power-on with warnings and limited performance; at worst, instability under load or overheating of the connectors.

The overall power budget of a multi-GPU workstation:

Component	Indicative Consumption
Threadripper PRO CPU (350 W TDP)	400–450 W peak
High-end GPUs (e.g., 450–600 W TDP each)	$4 \times 500 = \mathbf{2,000\ W}$
RAM (8 RDIMM modules)	~50 W
NVMe (4)	~40 W
Fans, pumps, motherboard	~100 W
Total Peak	~2,600 W

With a 20% margin for transients (modern GPUs have instantaneous peaks — *transient spikes* — that can exceed nominal consumption by 50–100% for a few milliseconds, and a power supply that doesn't absorb them will trigger OCP/OPP protection and shut down the machine), this reaches **3,000 W or more**.

Here we encounter two obstacles:

Wall 1: The electrical outlet. In Italy, a standard household outlet on a 16 A, 230 V circuit theoretically provides ~3,680 W, but household circuits and standard supply contracts (3 kW or 4.5 kW) cannot handle a 3 kW machine **plus everything else in the house**. A dedicated circuit is needed, and often an increase in contractual power. In 110 V countries (USA), the problem is even more severe: a 15 A, 120 V outlet provides 1,800 W theoretical, ~1,440 W in continuous use — **which is why 4-GPU workstations in the USA almost always require two separate circuits or a 240 V outlet**.

Wall 2: The power supply. Consumer ATX power supplies typically reach 1,600–2,000 W [*volatile*]. To go beyond this, the options are:

- ****Dual ATX power supply**** synchronized with an "add2psu" (an adapter that turns on the second PSU when the first one turns on, by bridging the PS_ON signal). It works, is common in mining and homelab environments, but must be done carefully: ****never power the same card from two different PSUs**** (for example: PCIe slot from PSU 1 and additional GPU connectors from PSU 2), because the grounds can have different potentials and damage the hardware.
- ****Server-grade power supplies**** (Delta, Murata, HP/Dell "Common Slot" power supplies) with breakout boards, often 240 V input, which can reach 2,400–3,000 W in a single module. This is the professional route, but requires attention to noise (the fans of these power supplies are designed for a rack, not an office).
- ****Reduce consumption via software****: with `nvidia-smi -pl <watt>` you can set a power limit per GPU. Reducing a GPU from 500 W to 350 W (–30%) typically costs only 5–10% in performance, because the power/performance curve is highly non-linear at the high end. ****On a 4-GPU workstation, this single command line can reduce consumption from 2,000 to 1,400 W practically for free.**** This is a standard technique in serious labs, not a poor man's compromise.

5.8 sTR5 socket cooling

A note that deserves its own space, because it's a beginner's mistake with immediate consequences.

The IHS of an sTR5 CPU is **much larger** than that of an AM5 CPU. A consumer cooler, even the best one, has a base designed to cover ~40×40 mm. On sTR5, it would leave much of the surface uncovered, and – even worse – with Threadripper, the dies (the CCDs) are **distributed across the entire package area**, not concentrated in the center. A cooler that only covers the center leaves the peripheral CCDs to cook.

Specific solutions are needed for sTR5:

- **Air cooling**: Noctua NH-U14S TR5-SP6 (and variants), IceGiant ProSiphon. These are coolers with enormous bases and a large number of distributed heatpipes.
- **AIO liquid cooling** (*All-In-One*, the closed pump+radiator system): SilverStone XE360-TR5, Arctic Liquid Freezer for TR5 *[volatile availability]*. **Crucial**: the *cold plate* (the cold plate in contact with the CPU) must be **full-coverage size** for sTR5. An AIO with a consumer cold plate, even mounted with adapters, **will not work**.
- **Custom loop**: with a dedicated sTR5 waterblock.

And, I repeat because it's important: mounting requires the **torque screwdriver** supplied with the CPU and **tightening in the numbered order** stamped on the socket. It is not a formality.

6. How to size the CPU for the AI workload

Let's get to the practical question: **which CPU should I actually buy?** The answer depends on *what* the machine does.

6.1 GPU-bound Training: The CPU is for Data Loading

In typical computer vision training or models with large datasets, the cycle is:

1. A **worker** (a separate process, managed in PyTorch by `DataLoader` with `num\_workers`) reads a file from disk.
2. It **decodes** it (a JPEG needs to be decompressed: this is a CPU-intensive operation, and for a high-resolution image, it can take milliseconds).
3. It applies **transformations** (resize, crop, flip, color jitter, normalization): more CPU work.
4. It packages the result into a tensor and

copies it to **pinned memory** (non-pageable, "locked" memory, from which the GPU's DMA can read directly). 5. The tensor is transferred to the GPU via PCIe.

If the workers don't keep up, the GPU waits. This is **the** most common bottleneck in unbalanced workstations and is easily diagnosable: ``nvidia-smi dmon`` showing GPU utilization fluctuating between 30% and 100% instead of staying flat at 95% is the classic symptom.

The rule of thumb, shared in the community and consistent with reference system configurations (NVIDIA DGX systems historically have about 8 physical cores per GPU):

4–8 physical cores per GPU, depending on the intensity of the preprocessing pipeline.

- **Light pipeline** (already preprocessed data, binary tensors, `.npy`` or Arrow/WebDataset format, no decoding): **2–4 cores per GPU** are sufficient.
- **Medium pipeline** (standard JPEGs, classic augmentations): **4–6 cores per GPU**.
- **Heavy pipeline** (very high-resolution images, video, complex augmentations, audio decoding): **8+ cores per GPU**, or — a better solution — **move preprocessing to the GPU** with libraries like NVIDIA DALI (which performs JPEG decoding in hardware, on the GPU's own NVDEC/nvJPEG decoders) or with accelerated video decoding.

Important corollary: if your problem is data loading, the *best* answer is often not to buy more cores, but to **preprocess the dataset once** and save it in a ready-to-use format (tensors, TFRecord, WebDataset shard). It's free and shifts the bottleneck to the disk, where an NVMe handles it effortlessly.

6.2 When Cores Really Matter

There are workloads where the CPU **is** the computation, not just support:

CPU Inference. As discussed, this is dominated by memory bandwidth during the generation phase. But in the **prefill** phase (processing the initial prompt, which is a large matrix multiplication), cores matter a lot, and vector instructions (**AVX-512, AMX** on Intel) make an enormous difference. A CPU with AMX can perform prefill in a fraction of the time. **If CPU inference is central to your workflow, seriously consider Intel Xeon with AMX or AMD platforms with many memory channels.**

Classic Preprocessing and Feature Engineering. Pandas/Polars/DuckDB pipelines on datasets of tens or hundreds of GB. Here, cores scale linearly (if the code is parallelized), and memory bandwidth matters greatly. This is the use case where a 32 or 64-core Threadripper truly shines.

Tokenization. Tokenizing a corpus of hundreds of GB of text to train a language model is a **massively parallel and CPU-bound** operation. With Hugging Face's `tokenizers` library (written in Rust, which releases the GIL), 64 cores can do in one hour what 8 cores do in eight hours.

Compilation. If you work on CUDA code, Triton kernels, compiling PyTorch from source, or compiling models with TensorRT: `make -j64` on a Threadripper is a revelation. Compilation is perhaps the workload that scales best with cores.

Simulation and classical processing. CFD, scientific computing, 3D rendering to generate synthetic datasets, combinatorial optimization: all CPU-bound.

6.3 Typical Configurations by Budget Range

Here is the operational core. *[Indicative, volatile prices, referring to the European market. Always verify.]*

ENTRY Tier — Single-GPU Workstation (~€2,500–€5,000)

The most important message: here the consumer WINS.

Component	Choice
CPU	AMD Ryzen 9 9950X (16c) or Intel Core Ultra 9 285K
Motherboard	High-end X870E / Z890
RAM	64–128 GB DDR5 (2 modules to avoid frequency degradation; 4 modules if capacity is needed)
GPU	1 × RTX 5090 (32 GB) or RTX PRO 4500/5000
Storage	2 × NVMe Gen4/Gen5 (1 TB system + 2–4 TB dataset)
PSU	1,000–1,200 W
Notes	No certified ECC. If ECC is needed, the only alternative is to upgrade the platform.

Who it's for: those doing fine-tuning with LoRA/QLoRA, local inference, prototyping, computer vision on medium datasets, courses, and individual research. **This covers 70–80% of real-world cases.** Do not upgrade the platform if you don't have a reason.

MID Tier — 2 GPU Workstation (~€7,000–€12,000)

Component	Choice
CPU	AMD Threadripper 7960X / 9960X (24 cores)
Motherboard	TRX50 (ASUS Pro WS TRX50-SAGE WIFI, ASRock TRX50 WS)
RAM	128–256 GB DDR5 RDIMM ECC, 4 modules on 4 channels
GPU	2 × RTX 5090 or 2 × RTX PRO 6000 (at full x16 PCIe 5.0)
Storage	2–4 × NVMe Gen5
PSU	1,600 W+ (or dual PSU)
Cooling	AIO 360 mm specific sTR5

Who it's for: the serious researcher, small teams, those who need ECC and two GPUs that truly communicate. **24 cores are more than enough:** 12 per GPU, well above the rule of thumb.

HIGH-END Tier — 4+ GPU Workstation (~€20,000–€50,000)

Component	Choice
CPU	AMD Threadripper PRO 7955WX / 7965WX / 9955WX (16–24 cores) — or 7975WX/7995WX if the workload is also CPU-bound
Motherboard	WRX90 (ASUS Pro WS WRX90E-SAGE SE, ASRock Rack WRX90D8-2T)
RAM	256 GB–1 TB DDR5 RDIMM ECC, 8 modules on 8 channels — fill all channels!
GPU	4 × RTX PRO 6000 Blackwell (96 GB each) or 4 × RTX 5090
Storage	4+ × NVMe Gen5 (one for the system, an array for datasets)
Network	Integrated 10/25 GbE or dedicated NIC
PSU	2 × 1,600 W, or server-grade 3,000 W PSU on a dedicated circuit

Case	Large workstation chassis or open-frame
Notes	Check slot spacing against the thickness of the chosen GPUs.

Pay attention to the central point: in the high-end segment, **the PRO CPU is purchased for its 128 lanes and 8 channels, not for its cores.** A 16-core 7955WX costs a fraction of the 7995WX and provides **the same 128 lanes and 8 channels**. Sixteen cores for four GPUs = 4 cores per GPU: at the low end of the rule of thumb, but sufficient with a well-optimized pipeline (preprocessed dataset, DALI). If preprocessing is heavy, move up to the 7965WX (24c) or 7975WX (32c). The 7995WX (96c) only makes sense **if the workload is genuinely CPU-intensive** (rendering, simulation, massive compilation) — otherwise, it's ~€9,500–10,000+ of inactive silicon.

EXTREME Tier — Server (€50,000+)

Here, we move from workstations to servers: **EPYC 9004/9005** or **Xeon 6**, dual socket, 8 GPUs on the baseboard, rack, three-phase power, direct liquid cooling. This is a different category, with different suppliers (Supermicro, Dell, HPE, Lenovo, Gigabyte) and different purchasing logic (you buy a system, you don't assemble it). It deserves a chapter of its own.

6.4 The three-line sizing rule

If I had to condense all of section 6:

1. **Count the GPUs.** 1 → consumer. 2 → Threadripper. 3+ → Threadripper PRO. 8+ → server. 2. **Count the necessary lanes.** GPU × 16 + NVMe × 4 + NIC. If the total exceeds the CPU's lanes, upgrade the platform. 3. **Count the cores.** 4–8 per GPU. If the result is less than the cores of the CPU you need for the lanes, **do not pay for extra cores:** get the cheapest SKU for that platform.

7. Common mistakes

We collect them here, in order of frequency and severity, because each one costs real money.

Mistake 1 — Buying Threadripper PRO for a single GPU

The mistake. "I want a serious AI workstation, so I'll get a Threadripper PRO." Then only one GPU is installed.

Why it's a mistake. The PRO platform costs €3,000–€5,000 more (CPU + motherboard + RDIMM RAM) and provides 128 lanes, of which you'll use 20. In addition, the single-thread frequencies of a Threadripper are **lower** than those of a Ryzen 9 of the same generation, and your orchestration Python code — which is single-thread — will run **slower**. You spent more to go slower in the part you use every day.

What to do instead. Get a Ryzen 9 9950X or a Core Ultra 9, invest the difference in **a better GPU or more RAM**. A single RTX PRO 6000 with 96 GB of VRAM allows you to train models that four 32 GB RTX 5090s cannot contain without complex partitioning techniques.

Mistake 2 — Ignoring slot spacing

The mistake. Buying four triple-slot custom GPUs and a motherboard with slots spaced by two.

Why it's a mistake. The GPUs **won't fit**. Or they'll fit but thermally choke each other and throttle. The cost is: returning the GPUs (if possible), or buying risers and a different case, or living with 20–30% reduced performance.

What to do instead. Measure in millimeters before buying. Check the slot layout in the motherboard manual. Prefer **blower** or professional 2-slot GPUs for dense configurations. The standard pitch of a PCIe slot is 20.32 mm.

Mistake 3 — Wrong RAM

This mistake has three variants, all common.

Variant A: UDIMM on an RDIMM platform. Buying a nice DDR5 gaming kit only to find out that the TRX50/WRX90 board **doesn't even POST**. Threadripper (even non-PRO), Threadripper PRO, EPYC, and Xeon W **require ECC RDIMMs**. It's not an option, it's a requirement.

Variant B: Not filling all channels. Buying 4x 64 GB modules (256 GB total) for an 8-channel WRX90. It works, you have the capacity you wanted — **and half the memory bandwidth**. You paid for an 8-channel platform and are only using 4. **Always buy a number of modules equal to the number of channels** (or a multiple).

Variant C: No ECC where it's needed. Training for three days on a machine without ECC and never understanding why the loss occasionally goes to NaN.

Error 4 — Underestimating power supply

The error. Connecting only one EPS connector on a board that has two. Buying a 1,000 W power supply for a machine that consumes 2,000 W. Ignoring transient GPU peaks. Plugging everything into a household outlet on a 3 kW contract.

Why it's an error. The symptoms are the worst possible: **random shutdowns under load**, occurring after hours of training, with no logs, no reproducible pattern. Weeks are spent blaming software, drivers, the kernel. It's the power supply going into protection.

What to do instead. Balance the power budget with a 25–30% margin. Connect **all** EPS connectors. Seriously consider **power limiting** GPUs (`nvidia-smi -pl`): the cost/benefit ratio is unbeatable. Check the electrical circuit of the room.

Error 5 — Underestimating sTR5 cooling

The error. Using an AM5 cooler with an adapter, or an AIO with a consumer cold plate.

Why it's an error. The sTR5 package is huge and the CCDs are distributed. A small cold plate cools the center and leaves the peripheral dies at throttling temperatures. Result: a €5,000 CPU running at 60% of its frequency.

What to do instead. **sTR5 specific** cooler, with a full-coverage base. Tighten with a **torque screwdriver**, in the **numbered order**.

Error 6 — Trusting the commercial description of slots

The error. Buying a motherboard because it "has 4 PCIe x16 slots" and finding out that two are x4 electrical and connected to the chipset.

What to do instead. Download the **PDF manual** and look for the **block diagram** and the **PCIe lane distribution table**. If the manufacturer doesn't publish them, it's a red flag.

Error 7 — Overestimating core needs, underestimating RAM needs

The error. Spending on a 96-core and putting in 128 GB of RAM.

Why it's an error. The 96 cores will remain at 10% during training. The 128 GB will run out quickly, because the rule of thumb is **system RAM $\geq 1.5\text{--}2\times$ total VRAM** (to hold the dataset in cache, worker buffers, pinned memory, and any offloading). With 4x 48 GB GPUs (192 GB VRAM), 128 GB of RAM is **insufficient**: at least 384 GB are needed.

What to do instead. Cheaper CPU SKU of the right platform + more RAM.

Error 8 — Buying hardware when the cloud was needed

The error. Spending €30,000 on a workstation that will be used at full load two weeks a year.

What to do instead. Make an honest assessment of the hours/GPUs needed per year and compare it with the cost of cloud instances or on-demand GPU providers. A local workstation is justified for **continuous use, sensitive data that cannot leave the premises, or rapid daily iteration**. Not for prestige.

Operational Summary: AI Workstation CPU Decision Checklist

To be used **in this order**. Each step depends on the previous one.

PHASE 1 — Define the Workload (before looking at a single product)

- ☐ ****How many GPUs**** will I install, today and realistically within 2 years? (This is the determining question.)
- ☐ Is the workload ****GPU-bound**** (training/inference on GPU) or is there a significant ****CPU-bound**** component (heavy preprocessing, tokenization, compilation, simulation, CPU inference)?
- ☐ Will the GPUs need to ****communicate with each other**** (distributed training) or will they work independently (e.g., parallel inference on different requests)? If they communicate, lanes are critical; if not, much less so.
- ☐ How much ****total VRAM**** will I have? → System RAM should be ****1.5–2 times**** that amount.
- ☐ Is the ****dataset**** on local disk or on the network? → If on the network, 10 GbE or more is needed.
- ☐ Do I really need local hardware, or is the ****cloud**** more economical for my usage profile?

PHASE 2 — Calculate Necessary PCIe Lanes

- ☐ GPU × 16 = _____ lanes
- ☐ NVMe × 4 = _____ lanes
- ☐ NIC (4 for 10G, 8–16 for 25/100G) = _____ lanes
- ☐ Other cards (HBA, capture, accelerators) = _____ lanes
- ☐ ****TOTAL lanes needed = _____****

PHASE 3 — Choose the Platform Based on the Total

Lanes Needed	GPUs	Platform
≤ 24	1	Consumer (Ryzen 9 / Core Ultra) — and it's perfectly fine
25–48	2	Threadripper non-PRO on TRX50
49–128	3–7	Threadripper PRO on WRX90
> 128	8+	EPYC / Xeon Scalable, server

- ☐ Is **certified ECC** needed? → If yes, Threadripper minimum (RDIMM ECC mandatory).
- ☐ Is **more than 256 GB of RAM** needed? → If yes, Threadripper minimum.
- ☐ Is **maximum memory bandwidth** needed (CPU inference on large LLMs)? → 8 channels (WRX90) or 12 channels (EPYC).
- ☐ Is **remote management** needed? → Motherboard with BMC/IPMI.

PHASE 4 — Choose the CPU SKU within the Platform

- ☐ Calculate necessary cores: **4–8 physical cores per GPU**.
- ☐ Add cores if the workload has a real CPU-bound component (tokenization, compilation, simulation, heavy preprocessing not transferable to GPU).
- ☐ **If the result is less than the cores of the cheapest SKU on that platform, take the cheapest SKU.** Don't pay for cores that will remain idle. (Example: 4 GPUs + light pipeline → **7955WX/9955WX** with 16 cores on WRX90, not the 7995WX.)
- ☐ Check the **suffix**: **WX = PRO** (8 channels, 128 lanes). **X without W = non-PRO** (4 channels, 48 lanes). **Do not confuse them.**
- ☐ If CPU inference is central → consider **Intel Xeon with AMX**.

PHASE 5 — Choose the Motherboard

- ☐ Download the **PDF manual** and find the **block diagram** / **PCIe lane distribution table**. Do not trust the product page.
- ☐ Count the **ELECTRICAL x16** slots (not just physical) and their PCIe generation.
- ☐ **Measure the slot spacing** (standard pitch: 20.32 mm per slot) and compare it with the **thickness in mm** of the GPUs you will buy.

- [] Verify the number of ****DIMM slots**** and their channel organization. Confirm that you will buy ****one module per channel****.
- [] Verify ****bifurcation**** support (x4x4x4x4) if you plan to use passive quad M.2 cards.
- [] Verify the presence of integrated ****10 GbE**** (or account for lanes for a NIC).
- [] Verify the presence of ****BMC/IPMI**** if the machine will not be within reach.
- [] Count the required ****EPS 8-pin**** connectors and verify that your power supply has all of them.
- [] Verify the ****form factor**** (E-ATX, SSI-EEB, proprietary) and compatibility with the chosen case.

PHASE 6 — Memory

- [] Type: ****RDIMM ECC**** on all Threadripper/PRO/EPYC/Xeon W platforms. No exceptions.
- [] Number of modules = ****number of channels**** (4 on TRX50, 8 on WRX90, 12 on EPYC). ****Do not leave channels empty.****
- [] Total capacity $\geq$ ****1.5–2 × total VRAM****.
- [] Check the motherboard's ****QVL**** (Qualified Vendor List) before buying modules.

PHASE 7 — Power and cooling

- [] Power budget: CPU + (GPU × number) + RAM + storage + fans, ****+25–30% margin for transients****.
- [] Verify that the ****room's electrical circuit**** can handle it (in Italy: dedicated contract and circuit for machines above ~2 kW).
- [] Consider ****power limiting**** for GPUs (`nvidia-smi -pl`): –30% consumption for –5/10% performance.
- [] Connect ****all**** EPS connectors on the motherboard.
- [] ****sTR5-specific cooler**** with full-coverage base. ****Never use AM5/LGA1700 adapters.****
- [] Mount the CPU with the ****supplied torque screwdriver****, in the ****numbered order**** stamped on the socket.

PHASE 8 — Post-assembly verification (never skip it)

- [] `lspci -vv` → compare ****LnkCap**** (capability) with ****LnkSta**** (actual state) for each GPU and NVMe. If a link is negotiated at x8 instead of x16, or at Gen3

instead of Gen5, **there is a hardware problem to solve now**, not in three months.

- [] ``nvidia-smi -q | grep -i pcie`` → verify GPU link width and generation.
- [] Verify in BIOS/OS that **ECC is active and reports events** (on Linux: ``edac-util`, `dmidecode -t memory``, the BMC's SEL log).
- [] Run a **combined stress test** (CPU + all GPUs at 100% simultaneously) for at least an hour and monitor temperatures, clocks (to verify absence of throttling), and power stability.
- [] Verify **real memory bandwidth** with a benchmark (STREAM, Intel's ``mlc``): if it's half of what's expected, you've populated the channels incorrectly.
- [] Run a **real training** and observe **GPU utilization**: if it's not stably above 90%, you have a bottleneck to diagnose (almost always: data loading, or PCIe link negotiated downwards).

A final note on data volatility: prices, availability, CPU generations, and compatibility matrices change rapidly in this industry. All cited models, prices, and SKU counts should be verified against official manufacturer documentation at the time of purchase. Architectural principles—lanes as currency, channels as bandwidth, ECC as insurance, spacing as a physical constraint—are stable, however, and will remain valid even as product names change.

In the next chapter, we will discuss GPUs for AI: architectures, VRAM, tensor cores, NVLink and NVSwitch, SXM versus PCIe form factors, and how to choose the right accelerator for your workload.

Chapter 11 - Nvidia GPUs for AI: From Workstation to Datacenter

Preliminary methodological note

This chapter deals with an industry where data ages faster than a book can be printed. All information reported is current as of mid-2026, but three categories of data should be considered **volatile by nature** and always verified at the source before making a purchasing decision:

1. **Prices.** Datacenter GPUs do not have a stable public price list: they are purchased through OEMs (Original Equipment Manufacturers, i.e., server manufacturers like Dell, Supermicro, HPE, Lenovo) or system integrators, with prices depending on volume, contract, and historical moment. The figures you will find here are orders of magnitude documented by industry press, not price lists. 2. **Availability.** In 2024–2026, demand consistently outstripped supply: an "announced" product can have a real waiting time of 6–12 months. 3. **The latest generation.** Nvidia has adopted an annual architectural cadence. What is "the latest" at the time of writing will be second-to-last within twelve months.

I will explicitly mark with **[VOLATILE DATA]** every point where this issue is particularly acute.

1. The complete Nvidia hierarchy

1.1 Why there isn't "the Nvidia GPU," but three separate worlds

Those new to artificial intelligence hardware almost always make the same conceptual mistake: they think there is a continuous scale of products, from the cheap graphics card to the expensive graphics card, and that the only difference is "how fast it is." This is not the case. Nvidia produces three product lines that share the basic architecture but are **different products, with different markets, different licensing agreements, and different physical constraints**. Confusing them leads to incorrect purchases costing tens of thousands of euros.

First, let's define the term **GPU**: *Graphics Processing Unit*. It originated in the 1990s as an accelerator for 3D rendering in video games, meaning for a very specific task: applying the same mathematical operation (a matrix multiplication, a coordinate transformation, a lighting calculation) to millions of independent elements, all at once. This is the definition of **massive parallel computing**. The

historically decisive fact is that training neural networks requires exactly the same thing: multiplying enormous matrices, repeated billions of times. The GPU was not "adapted" for AI; it happened to be in the right place at the right time, and Nvidia had the foresight to build a software ecosystem on top of it (CUDA, which we will discuss in section 7) ten years before it was truly needed.

The three families are:

GeForce (consumer). Gaming cards: RTX 4090, RTX 5090, RTX 5080, and their smaller siblings. Sold retail, installed in desktop PCs, with air coolers and fans. The acronym **RTX** = *Ray Tracing Texel eXtreme*, introduced with the Turing generation (2018) to indicate the presence of hardware units dedicated to ray tracing. For AI purposes, ray tracing is irrelevant; what matters is that these same cards also have **Tensor Cores** (units dedicated to low-precision matrix multiplication), and this is why a GeForce is a perfectly valid AI machine — within very precise limits.

RTX PRO / professional workstation (formerly Quadro). The line historically called *Quadro*, then renamed first "RTX A" (Ampere: A4000, A5000, A6000), then "RTX Ada" (Ada Lovelace: RTX 5000 Ada, RTX 6000 Ada), and from 2025 **RTX PRO Blackwell** (RTX PRO 4000/5000/6000 Blackwell). These cards are intended for professional workstations: CAD, rendering, simulation, medical visualization, and increasingly, local AI development. They cost 3–5 times a GeForce with equivalent silicon.

Datacenter (formerly Tesla). A100, H100, H200, B100, B200, GB200, and the new Rubin generation. They do not have video outputs (in most cases), do not have fans in the SXM variant, and cannot be bought on Amazon. They are server components, designed to run at 100% load for years.

1.2 The seven differences that justify three lines

Let's look at them one by one, because each has practical consequences.

(1) VRAM. *Video RAM*: the memory mounted on the card, physically separate from system RAM. It is **the number one constraint** in AI. An RTX 5090 has 32 GB; an RTX PRO 6000 Blackwell has 96 GB; an H200 has 141 GB; a B200 has 192 GB. The difference is not "faster/slower": it is **you can run the model or not**. A model that doesn't fit into VRAM simply won't start, or will run at ridiculous speeds by relying on system RAM.

(2) Memory type. Consumer cards use **GDDR** (*Graphics Double Data Rate*, currently GDDR7): memory chips soldered around the GPU on the PCB. Datacenter cards use **HBM** (*High Bandwidth Memory*): stacks of memory dies vertically stacked and connected to the GPU via a silicon *interposer*, with a very wide bus (thousands of bits versus hundreds for GDDR). HBM costs enormously more but offers 3–5 times higher bandwidth. We will see in section 8 why bandwidth is almost as important as capacity.

(3) ECC. Error Correcting Code: memory capable of detecting and correcting single-bit errors. In a gaming PC, a bit flip due to a cosmic ray or a marginal memory cell produces a graphical artifact for one frame, and no one notices. In a three-week training run on 1,000 GPUs, it produces a NaN (*Not a Number*) that corrupts gradients and wastes days of computation. Datacenter and RTX PRO cards have ECC; GeForce cards do not (or have partial and uncertified forms).

(4) Certified Drivers. Nvidia distributes *Game Ready* drivers for GeForce cards (optimized for newly released games, updated every two weeks) and **enterprise/production branch** drivers for professional and datacenter cards: rarely updated, tested for months, guaranteed stable with specific certified software, and with long-term support. If your cluster needs to run for two years without surprises, this difference is worth money.

(5) Form factor and cooling. An RTX 5090 is three slots wide and has three fans that blow air into the case. Putting four of them in a server is impossible: they would suffocate each other. Datacenter cards come in two formats: **PCIe** (a double-slot card, *passive*, meaning without its own fans: it relies on the very strong airflow generated by the server chassis fans) and **SXM** (a proprietary module, without a PCIe connector, that screws directly onto a dedicated motherboard and allows for much higher TDP and interconnections). We will return to SXM in section 3.

(6) Datacenter Use License. This is the point that almost no one knows about and that causes the worst problems. The **EULA** (*End User License Agreement*) for Nvidia GeForce drivers explicitly prohibits *datacenter deployment* of GeForce cards, with a historical exception for blockchain mining and for academic/research use. In practice: **you cannot legally build a commercial cloud service based on RTX 4090/5090**. You can use them in your company, on your workstation, for your internal product; you cannot rent them to third parties as computing capacity, nor install them in a datacenter as service infrastructure.

It is a contractual clause, not a technical block: the card works perfectly fine. But if you are a startup that wants to sell inference, this is a real legal risk. **[VOLATILE DATA: the EULA text has changed multiple times; always verify it as of the date of purchase.]**

(7) NVLink. The high-speed interconnection between GPUs. Present on datacenter cards, **absent on the entire consumer RTX 40 and RTX 50 line** and on most recent RTX PRO cards. This single product choice is what prevents building the economic equivalent of a datacenter node with consumer cards. It is the subject of section 4, and is probably the most important concept in the entire chapter.

1.3 How to read the nomenclatures

The three lines have three completely different naming logics, and this in itself is a source of confusion.

Line	Schema	Example	How to read
GeForce	RTX + [generation][tier]	RTX 5090	"50" = 50 series (Blackwell consumer); "90" = highest tier. 80 = high, 70 = mid, 60 = entry
Workstation	RTX PRO + [tier number] + [architecture]	RTX PRO 6000 Blackwell	6000 = top-tier; 5000/4000/2000 = lower tiers. The architecture is in the name to distinguish it from previous eponymous models (RTX 6000 Ada ≠ RTX PRO 6000 Blackwell!)
Datacenter	[Architecture letter] + [number]	H100, B200	A = Ampere, H = Hopper, B = Blackwell, R = Rubin. The number increases with positioning. Suffixes: GB200 = with Grace CPU; NVL = NVLink

			configuration; the number after NVL indicates the domain size
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Classic nomenclature trap: there is an "RTX 6000 Ada" and an "RTX PRO 6000 Blackwell". These are two different cards, two generations apart, with 48 GB and 96 GB respectively. They are often confused in reseller listings. Always read the architecture, never just the number.

2. Architectures: What truly changes between generations

2.1 The concept of "architecture" and "manufacturing process"

Two terms must be clearly separated.

Architecture is the logical design of the chip: how many computing units, how they are organized, what types of operations they can perform, how the cache hierarchy is structured, how memory is connected. Nvidia names each architecture after a scientist: Pascal (2016), Volta (2017), Turing (2018), Ampere (2020), Hopper (2022), Ada Lovelace (2022, consumer/pro only), Blackwell (2024), Rubin (2026).

The **manufacturing process** (or *node*) is the foundry technology with which the chip is physically manufactured: "4 nanometers", "3 nanometers". Caution: these numbers **no longer correspond to any real physical dimension** for at least a decade. They are commercial names. TSMC's "4N" used for Hopper and Blackwell is a derivative of the 5 nm node optimized for Nvidia. What truly matters is **transistor density** and energy efficiency per operation. A more advanced node allows more transistors to be placed in the same area and consumes less per operation — which is why the industry continues to pay exorbitant amounts to access it.

2.2 Ampere (2020) — A100

Ampere introduced, on the datacenter front, three things we now take for granted.

The first is the **third-generation Tensor Core with TF32 support**. A **Tensor Core** is a computing unit that doesn't perform one multiplication at a time, but executes

a small matrix-matrix multiplication with accumulation (typically 4x4 or similar) in one go, which is the elementary operation of every neural network. **TF32** (*TensorFloat-32*) is a clever numerical format: it has the same dynamic range (the 8 exponent bits) as classic FP32 but only 10 mantissa bits like FP16. Result: models written in FP32 ran faster **without modifying a single line of code**, because the Tensor Core silently converted. It was a huge adoption accelerator.

The second is **HBM2e memory** with bandwidth around 2 TB/s in the 80 GB version — in comparison, a consumer GPU of those years was under 1 TB/s.

The third is **MIG** (*Multi-Instance GPU*): the ability to divide a single A100 into up to seven hardware-isolated logical GPUs, each with its own slice of memory, cache, and compute units. This is useful in the cloud, where a customer performing inference on a small model doesn't have to occupy (and pay for) an entire GPU.

The A100 exists in 40 GB and 80 GB, PCIe and SXM4. **[VOLATILE DATA]** In 2026, it is technically obsolete but commercially very active in the used market and on low-cost clouds, at prices that are a fraction of the original. For many inference workloads and for fine-tuning medium models, it is still the option with the best price/performance ratio on the secondary market.

2.3 Hopper (2022) — H100 and H200

Hopper is the generation that coincided with the explosion of LLMs (*Large Language Models*) and is, in fact, the workhorse on which most famous models have been trained.

Key innovations:

The Transformer Engine and the FP8 format. The Transformer Engine is a hardware+software block that dynamically analyzes, layer by layer, which tensors can be represented in **FP8** (8-bit floating point) without degrading model quality, and converts them on the fly while keeping the rest in FP16/BF16. Going from 16 to 8 bits **doubles theoretical throughput and halves the memory occupied by tensors**. It is the single most important leap of this generation.

HBM3 memory (80 GB, approximately 3.35 TB/s on SXM5) and, with the H200 revision, **HBM3e**: 141 GB and approximately 4.8 TB/s. Note: H200 **does not change the compute chip**. It is the same silicon as H100 with larger and faster memory. This is a pattern that Nvidia will repeat, and it is crucial to understand: **a higher number does not imply a more powerful chip, it often only implies more**

memory. For LLM inference workloads — which are *memory-bound*, meaning limited by memory bandwidth and not by compute power — H200 is still significantly superior to H100, even with the same TFLOPS.

Fourth-generation NVLink at 900 GB/s per GPU and third-generation NVSwitch (see section 4).

Confidential Computing: the ability to perform computations on encrypted data in a hardware enclave, relevant for healthcare and finance.

2.4 Blackwell (2024–2025) — the architectural leap

Blackwell deserves extended discussion because it introduces a paradigm shift in the physical construction of the chip.

The problem: the reticle limit. In photolithography, the circuit image is projected onto the wafer through a mask. The maximum area that the machinery can expose in one go is called a **reticle**, and it is approximately 800 mm². A chip cannot be larger than that. Hopper was already practically at the limit (814 mm²). Nvidia therefore had two paths: wait for a better node (slow, uncertain) or **stop making single chips**.

The solution: dual-die. Blackwell is composed of **two separate dies**, each close to the reticle limit, mounted side-by-side on the same package and connected by an interconnect called **NV-HBI** (*NVIDIA High Bandwidth Interface*), which offers approximately **10 TB/s** of bandwidth between the two dies. This is an enormous figure: ten terabytes per second. The consequence is that, from the perspective of software, drivers, and programmers, **the two dies appear as a single coherent GPU**: a single memory space, a single CUDA device. It's not disguised multi-GPU; it's a single chip, stitched together in silicon. This is the true technical content of the "Blackwell leap." In total, approximately 208 billion transistors on TSMC 4NP node.

FP4 and the second-generation Transformer Engine. If FP8 halved the bits compared to FP16, Blackwell halves them again with **FP4** (specifically the proprietary **NVFP4** format, which uses block-level scaling factors to contain precision loss). Four bits per number means **sixteen possible values in total**. This seems absurd, and for training, it almost always is; but for **inference**, where the model is already trained and only needs to be executed, research has shown that with the right quantization techniques, the quality degradation is minimal.

The gain is twofold: doubling of throughput and halving of memory occupied by weights.

The "sub-generational": B100, B200, and GB200. Here, a confusing point needs to be clarified. B100 and B200 **use the exact same silicon**. The difference is the configuration: TDP (*Thermal Design Power*, the thermal power that the cooling system must dissipate, used as a proxy for consumption) and frequencies.

- **B100**: ~700 W. Designed to be a *drop-in replacement* in racks designed for H100, which were already sized for 700 W. No infrastructure modifications, superior performance. It had a short commercial life and a marginal role: Nvidia pushed everyone towards B200.
- **B200**: ~1000–1200 W depending on the configuration. Same chip, unlocked. Significantly higher performance, but requires new racks and cooling.
- **GB200**: **is not a GPU**. It is a *superchip*: **one Grace CPU + two B200 GPUs** on a single module, connected by **NVLink-C2C** (*Chip-to-Chip*) at 900 GB/s. It is not a simple "commercial package": the CPU-GPU connection via NVLink-C2C, instead of via PCIe, is about seven times faster and is **cache-coherent**, meaning that the CPU and GPU see the same memory space without explicit copies. This changes how code is written: weights or KV-cache can be *offloaded* to the CPU's LPDDR memory (hundreds of GB) without paying the devastating price of a PCIe transfer.

Blackwell Ultra (B300 / GB300). As H200 was to H100, B300 is to B200: same architecture, increased HBM3e memory (up to ~288 GB per package), and optimizations for low-precision inference, with approximately 208 billion transistors. This is the "mid-generation" covering 2025–2026.

2.5 Rubin (2026) — the current generation

[VOLATILE DATA — this section describes a product in ramp-up: verify everything.]

Rubin is the architecture succeeding Blackwell. Nvidia announced its full production in early 2026, with availability to partners and cloud providers in the second half of the year. The key points, as publicly known in mid-2026:

- **Rubin GPU (often referred to as R100/R200)**: TSMC 3nm node, **dual-die** design like Blackwell but with approximately **336 billion transistors** (~1.6× Blackwell).

- **HBM4 Memory**: approximately **288 GB** per package with a declared bandwidth of up to approximately **22 TB/s** — almost triple that of Blackwell. The leap primarily comes from doubling the bus width per stack.
- **Compute**: in the order of **50 PFLOPS in NVFP4** per package (compared to ~20 for Blackwell).
- **Vera CPU**: replaces Grace. Custom ARM CPU with 88 "Olympus" cores, up to 1.5 TB of LPDDR5X, connected to GPUs via NVLink-C2C at 1.8 TB/s.
- **NVLink 6**: approximately 3.6 TB/s per GPU, double the Blackwell generation.
- **System**: the reference rack is the **VR200 NVL72** (72 GPU packages), with power consumption in the order of 190–230 kW per rack — compared to ~120–130 kW for the GB200 NVL72. Liquid cooling is **mandatory**: air-cooled configurations do not exist.

The declared roadmap continues with **Rubin Ultra** (2027, "Kyber" rack at ~600 kW) and **Feynman** (2028). The strategic message is clear and must be understood by anyone planning infrastructure: **Nvidia has adopted an annual cadence where not only the chip but the entire physical enclosure changes** — power, cooling, network. Anyone building a data center today is building it for hardware that, within two cycles, will no longer fit.

2.6 Generational summary table

	Ampere (A100)	Hopper (H100)	Hopper (H200)	Blackwell 1 (B200)	Blackwell 1 Ultra (B300)	Rubin (R100)
Year	2020	2022	2023–24	2024–25	2025–26	2026 (ramp)
Node	TSMC 7N	TSMC 4N	TSMC 4N	TSMC 4NP	TSMC 4NP	TSMC 3nm
Die	single	single	single	dual-die (NV-HBI)	dual-die	dual-die
Transistors	54 bln	80 bln	80 bln	208 bln	~208 bln	~336 bln
Memory	HBM2e 40/80 GB	HBM3 80 GB	HBM3e 141 GB	HBM3e 192 GB	HBM3e ~288 GB	HBM4 ~288 GB
Bandwidth	~2.0 TB/s	~3.35 TB/s	~4.8 TB/s	~8 TB/s	~8 TB/s	~22 TB/s
Minimum Precision	FP16/BF16	FP8	FP8	FP4 (NVFP4)	FP4	FP4/FP6

NVLink	4th gen, 600 GB/s	4th gen, 900 GB/s	900 GB/s	5th gen, 1.8 TB/s	1.8 TB/s	6th gen, 3.6 TB/s
TDP (SXM)	400 W	700 W	700 W	~1000– 1200 W	~1400 W	~1800– 2300 W

All values are indicative and refer to top-tier SXM variants. PCIe versions always have lower TDP and NVLink bandwidth.

3. Datacenter cards, one by one

3.1 Premise: SXM versus PCIe

Before looking at individual products, it's important to understand this distinction, as it's the first question any salesperson will ask you.

PCIe (*Peripheral Component Interconnect Express*) is the standard bus by which any card connects to the motherboard of any computer. A datacenter GPU in PCIe format looks like a very long, dual-slot video card, **without fans**: it is *passive*, and relies on the strong airflow that the server fans push through the chassis.

Advantages: it mounts in a standard server, costs less, and can be bought one at a time. Disadvantages: **TDP is limited** (typically 300–600 W, because the connector and slot cannot handle more) and, critically, **interconnection between GPUs is limited** — at most a single NVLink bridge between two cards, or nothing at all on recent generations.

SXM is a proprietary Nvidia format. The module does not have a PCIe connector: it bolts onto a dedicated **baseboard** (typically HGX, the 4 or 8 GPU board that Nvidia sells to OEMs). The baseboard provides power (up to 700–1400 W per module) and NVLink lines to the NVSwitches. Advantages: much higher TDP (thus higher frequencies, thus more performance from the same silicon), and **full-bandwidth NVLink between all GPUs in the node**. Disadvantages: it can only be purchased as a complete 4 or 8 GPU system, it is very expensive, and it requires a specially designed server.

The practical rule: if you are doing distributed *training* on multiple GPUs, you need SXM. If you are doing *inference* with models that fit on a single GPU, PCIe is more than enough — and much cheaper.

3.2 A100 (40 GB / 80 GB)

- **VRAM:** 40 GB or 80 GB HBM2e
- **Bandwidth:** ~1.55 TB/s (40 GB) / ~2.0 TB/s (80 GB SXM)
- **TDP:** 250–300 W (PCIe) / 400 W (SXM4)
- **Form factor:** PCIe and SXM4
- **Indicative performance:** ~312 TFLOPS in FP16 with sparsity, ~19.5 TFLOPS FP32
- **Price [VOLATILE DATA]:** long out of production; on the used and refurbished market, they can be found at a fraction of the original price (which was around \$10,000–\$15,000 for the 80 GB). On clouds, it is among the cheapest options available.
- **Use cases:** inference of models up to ~30–70 billion quantized parameters, fine-tuning, academic research, classic HPC workloads (simulation, CFD). It does **not** have FP8: for modern LLM inference, which relies on FP8/FP4, this is a serious architectural limitation.
- **Who should buy it:** those with a tight budget, a known workload, and the expertise to manage used hardware. It is the "safe used option" in the industry.

3.3 H100

- **VRAM:** 80 GB HBM3 (there is also an H100 NVL with 94 GB, a dual-board variant)
- **Bandwidth:** ~3.35 TB/s (SXM5) / ~2.0 TB/s (PCIe)
- **TDP:** 700 W (SXM5) / 350 W (PCIe)
- **Performance:** ~1,979 TFLOPS FP8 with sparsity (SXM5). The PCIe version performs **about 20–30% less** due to halved TDP and lower bandwidth.
- **Price [VOLATILE DATA]:** the publicly reported order of magnitude is \$25,000–\$40,000 per card; an HGX H100 8-GPU node is in the \$250,000–\$400,000 range.
- **Use cases:** it was *the* GPU of the LLM boom. Training models up to tens of billions of parameters (in clusters), high-throughput inference, serious fine-tuning.

Common mistake: buying the H100 PCIe thinking it's "a slightly cheaper H100". It's not: it's a significantly slower GPU, with limited or no NVLink, which can perform half as well in a multi-GPU workload. If your workload is multi-GPU, the PCIe is almost always a false economy.

3.4 H200

- **VRAM:** 141 GB HBM3e
- **Bandwidth:** ~4.8 TB/s
- **TDP:** 700 W (SXM5), same as its predecessor
- **Compute chip:** identical to H100. No increase in TFLOPS.
- **Price [VOLATILE DATA]:** above the H100, in the order of \$30,000–\$40,000.
- **Use cases:** LLM inference. Here the gain is dramatic — often 1.5–2x compared to H100 on the same model — not because it computes more, but because autoregressive inference is limited by memory bandwidth. Each generated token requires re-reading all model weights from VRAM. More bandwidth = more tokens per second, period.

This is the moment to state something clearly that will recur throughout the chapter: **in LLM inference, memory bandwidth matters more than TFLOPS**. Anyone comparing GPUs by only looking at TFLOPS is looking at the wrong metric.

3.5 B100, B200, GB200

- **B100:** ~192 GB HBM3e, ~8 TB/s, 700 W. Compatible with H100 infrastructure. Transitional product, not widely adopted.
- **B200:** ~192 GB HBM3e, ~8 TB/s, ~1000–1200 W. Approximately 20 PFLOPS in FP4 dense. This is the true Blackwell datacenter GPU, available in 8-GPU HGX B200 baseboards.
- **GB200:** 1 Grace CPU + 2 B200 GPUs on a superchip. ~384 GB of HBM3e (2×192) plus up to 480 GB of LPDDR5X on the Grace. Module power consumption around 2700 W. **Requires liquid cooling.**
- **Price [VOLATILE DATA]:** a single B200 is in the range of \$30,000–\$40,000; a complete GB200 NVL72 system has been estimated by the specialized press at around \$3 million.

3.6 GB200 NVL72: when the compute unit becomes the rack

This is the most important concept in this section, and it marks the end of an era.

Up until Hopper, the compute unit was **the node**: a server with 8 GPUs connected by NVLink, and the nodes themselves connected by network (InfiniBand). The bandwidth gap between "inside the node" (900 GB/s) and "outside the node" (50–100 GB/s) was an order of magnitude, and the entire theory of distributed training was built around this disparity.

The **GB200 NVL72** rewrites the rule. It is a rack that contains **36 GB200 superchips**, meaning **72 B200 GPUs + 36 Grace CPUs, all connected by a single NVLink domain** through 9 NVSwitch trays. The 72 GPUs see a shared memory space of approximately 13.5 TB of HBM3e, with an aggregated NVLink bandwidth in the order of 130 TB/s. From a software perspective, **the entire rack behaves like a gigantic GPU**.

Why is this important? Because a model with hundreds of billions or trillions of parameters, which previously had to be split across many nodes paying the network penalty at each layer, now lives within a single, ultra-high-bandwidth domain. For **MoE** (*Mixture of Experts*) models (architectures where each token activates only a subset of "experts" and thus generate a lot of routing traffic between GPUs), the gain is enormous.

The price to pay is physical: ~120–130 kW per rack, mandatory liquid cooling, and an electrical infrastructure that most existing datacenters in the world **simply do not have**. A traditional rack consumes 5–15 kW.

3.7 Grace: why ARM in the datacenter

Grace is the CPU designed by Nvidia on the **ARM** architecture (specifically Neoverse V2 cores), with 72 cores, **LPDDR5X** memory soldered on the package (up to 480 GB, ~500 GB/s bandwidth) and — this is the key point — an **NVLink-C2C** connection to the GPU at 900 GB/s, **cache-coherent**.

Why ARM and not x86? Three reasons, in order of real importance:

1. **The connection.** Nvidia couldn't put NVLink-C2C into an Intel or AMD processor: it doesn't control those designs. To have a CPU–GPU path seven times faster than PCIe, it had to make its own CPU. ARM gave it an architecture license to build upon. **The Grace CPU exists primarily to be the right socket next to the GPU**, not because it is a revolutionary CPU in itself.
2. **Power efficiency.** In a 130 kW rack, every watt spent by the CPU is a watt taken away from the GPUs. ARM cores with LPDDR memory (low-power memory, like in phones, soldered and not expandable) offer an excellent performance-per-watt ratio for the role they need to play here: feeding data to the GPUs, managing I/O, orchestrating.
3. **Vertical control.** Nvidia wants to sell the complete system, not just a component. Grace is part of this strategy.

With Rubin, Grace is replaced by **Vera**, with entirely custom ARM cores.

Practical implication for the developer: a Grace system is an **aarch64** machine, not **x86_64**. Your Docker container, your compiled Python wheel, your proprietary binary must exist for ARM. In the mainstream AI ecosystem (PyTorch, CUDA, Nvidia's NGC containers) this is now resolved. In legacy enterprise software, often not. **It's the first check to make before buying.**

4. NVLink and Interconnections

4.1 Why PCIe is not enough

Imagine training a model on 8 GPUs with **data parallelism**: each GPU receives a different slice of the data, calculates its gradients (the derivatives of the loss function with respect to each parameter), and then all of them must **average the gradients among themselves** before updating the weights. This operation is called **all-reduce** and must be performed **at every single training step**, i.e., tens of thousands of times.

How much data is exchanged? As much as the model's parameters, multiplied by the byte size of each. For a 7 billion parameter model in BF16, that's 14 GB, and all-reduce moves about twice that.

Now let's do the math. **PCIe 5.0 x16** offers about **64 GB/s** per direction. **NVLink 4** offers **900 GB/s**. The ratio is 14 to 1. In a real training scenario, this means that with PCIe, GPUs spend a huge fraction of the time **idle, waiting for data from their companions**. Actual utilization plummets, and eight GPUs perform like three or four.

Then there's **latency**, which is a distinct problem from bandwidth: PCIe goes through the CPU's *root complex*, with protocol overhead. NVLink is a direct point-to-point connection between GPUs, with an order of magnitude lower latency. In **tensor-parallel** models (where a single layer is split across multiple GPUs and each GPU must exchange activations **within the forward pass**, dozens of times per token), latency is more critical than bandwidth. With PCIe, tensor parallelism is simply **impractical**.

4.2 NVLink: Generations

NVLink is Nvidia's proprietary interconnect introduced in 2016 with Pascal. It is not a bus: it is a set of **point-to-point serial links** (called *bricks* or *links*) that can

be aggregated. Each generation has increased the speed per link and the number of links per GPU.

Generation	Year	Architecture	Bidirectional Bandwidth per GPU
NVLink 1	2016	Pascal (P100)	160 GB/s
NVLink 2	2017	Volta (V100)	300 GB/s
NVLink 3	2020	Ampere (A100)	600 GB/s
NVLink 4	2022	Hopper (H100/H200)	900 GB/s
NVLink 5	2024	Blackwell (B200/GB200)	1,800 GB/s (1.8 TB/s)
NVLink 6	2026	Rubin	~3,600 GB/s (3.6 TB/s) [VOLATILE DATA]

NVSwitch. Connecting 8 GPUs "all-to-all" directly would require 28 physical links, and it doesn't scale. The NVSwitch is a dedicated **switch chip**: each GPU connects to the switch, and the switch routes the traffic. The result is that **each GPU talks to every other GPU at full bandwidth**, simultaneously, without bottlenecks. In the GB200 NVL72, NVSwitches move out of the server and become dedicated trays in the rack, extending the NVLink domain to 72 GPUs. This is exactly the leap that made "the rack as a compute unit" possible.

4.3 The removal of NVLink from consumer and professional cards

This paragraph is probably **the most important in the chapter for anyone building a workstation.**

- RTX 3090 (Ampere) had NVLink: two cards could be paired with a bridge.
- ****RTX 4090 (Ada) and RTX 5090 (Blackwell consumer) do NOT have NVLink.****
No bridge, no option.
- Even in the professional line, support has drastically reduced: the RTX 6000 Ada did not have NVLink; the RTX PRO Blackwell cards don't either.

The concrete implications:

1. **You cannot sum VRAM.** Two RTX 5090s do not make a 64 GB GPU. They make two separate 32 GB GPUs. A model that doesn't fit into 32 GB doesn't fit, period — unless you manually split it over PCIe, with the penalties discussed above. 2.

Tensor parallelism is out of the question. You can do *pipeline parallelism* (put layers 1–20 on GPU A and 21–40 on GPU B, with only one exchange at the boundary): this also works on PCIe because the traffic is much lower. But the speed will be mediocre. 3. **Consumer multi-GPU is mainly for independent workloads:** four different jobs on four GPUs, four models in inference, four experiments in parallel. This works perfectly. Distributed training of a single large model, no.

It's not a technical limitation: it's a **commercial decision**. Nvidia doesn't want people building cheap training nodes with gaming cards. Knowing this will prevent you from spending €10,000 on a workstation that won't do what you hope.

4.4 Beyond the node: InfiniBand and Ethernet/RoCE

When you go beyond the GPUs of a single node (or a single NVL72 rack), you need to move to the **network**.

InfiniBand is a networking technology designed for HPC (*High Performance Computing*), acquired by Nvidia with Mellanox. Its distinctive feature is **RDMA** (*Remote Direct Memory Access*): a GPU can write directly into the memory of a GPU on another server **without involving the CPUs or operating systems**. The result is very low latency (~1 microsecond) and reliable congestion control. Current generations are NDR (400 Gb/s) and XDR (800 Gb/s). It is the classic choice for large training clusters.

Ethernet with RoCE (*RDMA over Converged Ethernet*) brings RDMA to standard Ethernet. Advantages: lower cost, utilizes existing company expertise and equipment, avoids vendor lock-in. Disadvantages: Ethernet is inherently *lossy* (can drop packets under congestion), and AI requires configuring lossless networking mechanisms (PFC, ECN) which are notoriously difficult to fine-tune. Nvidia sells its own AI-optimized Ethernet platform (**Spectrum-X**) precisely to bridge this gap.

Rule of thumb: below ~64 GPUs, the network is rarely the main bottleneck, and Ethernet is fine. Above thousands of GPUs, the quality of the interconnection determines the scalability of the entire investment, and every percentage point of efficiency is worth millions.

5. Workstations and Pre-built Systems

5.1 The DGX Line

DGX is Nvidia's line of "turnkey" systems: complete servers, designed, assembled, and supported by Nvidia, with pre-installed software stack. You're not buying GPUs: you're buying a **guaranteed working system**, with enterprise support, and saving months of integration effort.

- ****DGX H100:**** 8× H100 SXM5 (640 GB HBM3), 2 Intel Xeon CPUs, ~10 kW. ****[VOLATILE DATA]**** Historically reported list price around \$300,000–\$400,000.
- ****DGX B200:**** 8× B200 (1,440 GB HBM3e), air-cooled (the last DGX to allow this), ~14.3 kW. Order of magnitude: \$500,000.
- ****DGX GB200 / GB200 NVL72:**** the complete rack described in section 3.6. Millions of dollars.
- ****DGX SuperPOD:**** the aggregation of many DGX units into a turnkey supercomputer, with networking, storage, and management software. We're talking tens of millions.

Who is the DGX line for? Organizations where the cost of engineering time outweighs the hardware premium: funded research labs, Fortune 500 companies, cloud providers. If you're calculating how much you'd save by buying components separately, **you are not the DGX customer** — and that's perfectly fine.

5.2 DGX Spark (formerly Project DIGITS)

This product deserves attention because it's Nvidia's first serious attempt to bring datacenter architecture to the desktop at a non-absurd price, and because it is **profoundly misunderstood**.

What it is: a box the size of a small Mac Mini, based on the **GB10 Grace Blackwell** superchip: an ARM CPU (20 cores, in collaboration with MediaTek) and a Blackwell GPU on a single package, with **128 GB of unified LPDDR5X memory** shared between CPU and GPU. Approximately **1 PFLOP of FP4 compute**. Power consumption around 240 W: powered by a standard household outlet. **[VOLATILE DATA]** Launch price around \$3,000–\$4,000.

What it's really for. The extraordinary thing is the **memory capacity**: 128 GB accessible to the GPU means being able to load models that would never fit on any consumer GPU. Nvidia talks about models up to ~200 billion parameters (quantized) on a single unit, and up to ~400 billion by connecting two. For a

developer who needs to **prototype, experiment, test pipelines, perform local inference on large models**, it is an unparalleled object at that price.

The limit, and it must be said with brutal clarity: memory bandwidth. The GB10's LPDDR5X offers in the order of **270 GB/s**. Compare that to the ~1,800 GB/s of an RTX 5090 or the 4,800 GB/s of an H200. It is **6–18 times slower**. And we have already established that **LLM inference is limited by memory bandwidth**. Practical consequence: on the DGX Spark, a large model **fits**, but generates tokens slowly. It is not a production machine, it is not a throughput benchmark machine.

The correct positioning is this: the DGX Spark is a **development machine**. It is used to write and validate code that will then run on a real DGX or in the cloud, with the **exact same software stack and the same ARM+Blackwell architecture**. The value is in the fidelity of the environment, not in speed. Anyone who buys it expecting a fast inference GPU will be disappointed, and the fault lies with ambiguous marketing, not the product.

5.3 DGX Station (GB300)

[VOLATILE DATA] The new-generation DGX Station is a desktop workstation (tower format) based on the **GB300 Grace Blackwell Ultra** superchip, with approximately **784 GB of total coherent memory** (HBM3e on the GPU + LPDDR5X on the CPU), integrated high-speed ConnectX NIC, and standard wall outlet power. It is the missing link between the DGX Spark and the rack: a **true** datacenter machine, with true HBM bandwidth, in a format that sits next to a desk. Expected price: in the order of tens of thousands of dollars, far from the Spark and well below a rack.

5.4 The alternative: building your own workstation

Does it make sense? Let's make an honest comparison.

Configurat ion	Total VRAM	Bandwidth per GPU	NVLink	Indicative Cost [VOLATIL E]	Suitable for
2× RTX 5090	2×32 GB (not summable)	~1.8 TB/s	✗	~€7,000– 9,000	Inference of medium models, LoRA, Stable Diffusion, parallel

					jobs
4× RTX 5090	4×32 GB	~1.8 TB/s	✗	~€15,000–20,000	As above ×2; power and cooling become a serious problem
2× RTX PRO 6000 Blackwell	2×96 GB	~1.8 TB/s	✗	~€20,000–25,000	Large model inference; serious fine-tuning; legitimate commercial use
DGX Spark	128 GB unified	~0.27 TB/s	—	~€4,000	Development, prototyping, huge but slow models
1× H100 PCIe (used)	80 GB	~2.0 TB/s	limited	~€20,000–25,000	Professional inference, legitimate datacenter use
HGX 8× H100 Node	640 GB	~3.35 TB/s	✓ 900 GB/s	~€250,000–400,000	True distributed training

Fundamental note on domestic power. Four RTX 5090s alone draw 2,300 W peak, plus CPU, fans, and power supply losses: easily exceeding 3,000 W. A standard Italian domestic socket provides 3,000–3,300 W **for the entire apartment**. This is not a detail: it is a constraint that physically closes the discussion. Beyond two high-end cards, a dedicated line and serious consideration for heat extraction from a room where you are dumping the thermal equivalent of three space heaters are needed.

6. Consumer GPUs for AI: when they are enough

6.1 RTX 4090 and RTX 5090

RTX 4090 (Ada, 2022): 24 GB GDDR6X, ~1,008 GB/s bandwidth, 450 W. **RTX 5090** (Blackwell, 2025): 32 GB GDDR7, ~1,792 GB/s bandwidth, 575 W. **[VOLATILE DATA: MSRP ~ \$2,000, street price historically much higher.]**

The leap of the 5090 is significant for AI **especially for bandwidth** (+78%) and for native FP4 support, not so much for the 32 GB — which is still little.

What you actually do with it, concretely:

- ****Inference of quantized LLMs.**** A 7–14 billion parameter model in 4-bit occupies 4–9 GB and flies. A quantized 32B barely fits into 32 GB. A 70B in 4-bit requires ~40 GB: ****it doesn't fit****, and this is the number one frustration for 5090 owners.
- ****LoRA / QLoRA fine-tuning.**** ****LoRA**** (*Low-Rank Adaptation*) is a technique that freezes the weights of the original model and trains only small additional low-rank matrices, reducing trainable parameters by orders of magnitude. ****QLoRA**** adds 4-bit quantization of the base model. Together, they allow fine-tuning of 7–13B models on a single consumer card. This is the use case where a 4090/5090 truly shines, and it's why there's an entire ecosystem of fine-tuned models built by people with a single gaming card.
- ****Image/video generation**** (Stable Diffusion, Flux, and similar): excellent. These models are compute-bound rather than memory-bound and fit comfortably in 24–32 GB.
- ****Computer vision, YOLO, ASR, embedding, training small models from scratch:**** perfect.

The wall: 32 GB. There's no way around it. No NVLink. No expansion. When the model doesn't fit, the card — however powerful — is useless for that task.

6.2 RTX PRO 6000 Blackwell

[VOLATILE DATA] The RTX PRO 6000 Blackwell offers **96 GB of GDDR7 with ECC**, bandwidth around 1.8 TB/s, TDP of 600 W in the workstation variant (there is also a 300 W *Max-Q* version and a passive *Server Edition* for racks). Indicative price: \$8,000–\$10,000.

When does it justify the price? The calculation is straightforward. It costs about 4–5 times an RTX 5090 and offers **3 times the VRAM** for the same bandwidth. If you only considered TFLOPS per euro, it would be a bad deal. But:

1. **96 GB allows models to fit that 32 GB cannot.** A 70B in 4-bit fits comfortably. The leap is not quantitative, it is **binary**: you can or you cannot. 2. **It has ECC**, and for fine-tuning that lasts days, it matters. 3. **It does not have the EULA restriction on the datacenter**: you can use it in a commercial service. 4. **In a workstation, two RTX PRO 6000 = 192 GB of VRAM** (in separate cards, without NVLink, but with practicable pipeline parallelism) at a cost of about €20,000, compared to much higher figures for equivalent datacenter hardware in capacity.

It is, today, **the rational choice for the vast majority of small to medium-sized companies that want to do AI seriously locally**. It is not cheap; it is the least expensive among those that truly solve the problem.

6.3 The three limits to always keep in mind

1. **No NVLink**. GPUs do not sum memory. Let's repeat this because it is the most expensive mistake in the industry. 2. **VRAM is a wall, not a ladder**. There is no gradual degradation: there is "it works" and "OutOfMemoryError". 3. **The datacenter license**. If your business model involves selling access to computation, GeForce cards are out of the question.

7. The software moat: CUDA

7.1 What it is and why it is Nvidia's real product

CUDA (*Compute Unified Device Architecture*) is the parallel computing platform introduced by Nvidia in 2007. At the time, it seemed like a strange gamble: why invest heavily to allow researchers to use video cards for general-purpose mathematics, when the market was video games? The answer came fifteen years later.

CUDA is a language (an extension of C/C++), a compiler, a runtime, a programming model. But **the real moat is not CUDA itself: it's what has grown on top of it**.

- ****cuDNN**** (*CUDA Deep Neural Network library*): Hand-optimized kernels, for each GPU architecture, for convolutions, normalizations, activation functions. Every deep learning framework in the world relies on it.

- **cuBLAS**: Linear algebra (matrix multiplications) at the maximum performance obtainable from the silicon.
- **NCCL** (*NVIDIA Collective Communications Library*): Multi-GPU communication primitives (all-reduce, all-gather, broadcast) optimized for NVLink and InfiniBand. Without NCCL, distributed training would have to be written by hand.
- **TensorRT** and **TensorRT-LLM**: Inference compilers that take a trained model and transform it into an optimized engine for a specific GPU (kernel fusion, quantization, automatic selection of the best algorithms). The gain compared to naive execution is often 2–5×.
- **Triton Inference Server**, **NeMo**, **NGC** containers: The entire production stack.

7.2 What this means in practice when choosing hardware

It means that when you compare an Nvidia GPU with an alternative (AMD Instinct MI300X/MI355X, Intel Gaudi, Google TPU, startup accelerators) **you are not comparing two chips. You are comparing two ecosystems.** And the comparison must be made honestly on three levels:

Level 1 — Does it work? On Nvidia: `pip install torch`, the code runs, done. On alternatives: it depends on the maturity of the stack (AMD's ROCm has improved enormously but remains more fragile), the version, the model.

Level 2 — Does it work fast? Low-level optimization (fused kernels, optimized attention like FlashAttention, quantization) comes out first and better on CUDA. A new model on Nvidia runs optimized from day one; on alternatives, often months later, or never.

Level 3 — Does it work when it breaks? When a training crashes at three in the morning, the probability that someone has already seen that error and solved it on Stack Overflow, GitHub, or a forum is enormously higher on CUDA. **This is the real moat**, and it's made of people, not silicon.

The practical consequence for purchasing: the total cost of ownership of a cheaper alternative includes the time of your engineers. If an engineer costs €400/day and the alternative makes you lose twenty days of integration, you've burned €8,000 — often more than the savings. Alternatives make sense under two conditions: **at large scale** (where the unit saving multiplies by thousands of units and you can afford a dedicated team) or **for a narrow and stable workload** (a

single model, in inference, which you validate once and never touch again). In between, Nvidia wins for reasons that have nothing to do with TFLOPS.

8. How to evaluate a GPU for AI: explicit criteria

8.1 Criterion #1: VRAM

Always, in any case, before anything else. A super-fast GPU with too little memory is useless; a slow GPU with enough memory still gets the job done.

The calculation must be made by summing **three items**:

(a) Model weights. Number of parameters $\times$ bytes per parameter:

- FP32 (32 bit) = 4 bytes $\rightarrow$ a 7B model occupies 28 GB
- FP16/BF16 (16 bit) = 2 bytes $\rightarrow$ 14 GB
- FP8 (8 bit) = 1 byte $\rightarrow$ 7 GB
- INT4/FP4 (4 bit) = 0.5 bytes $\rightarrow$ 3.5 GB

(b) Optimizer state, if you are training. With Adam (the standard optimizer), you need **two additional states per parameter** (momentum and variance) plus gradients. Rule of thumb: **mixed precision training costs approximately 16–20 bytes per parameter**, which is **8–10 times** the inference-only cost in FP16. A 7B model that occupies 14 GB for inference requires **over 100 GB** for full fine-tuning. This is why LoRA exists.

(c) Activations and KV-cache. Activations are the intermediate results of each layer and grow **linearly with batch size**. The **KV-cache** (*Key-Value cache*) is the memory that, during text generation, stores already computed attention keys and values to avoid recomputing them for each token: it grows **linearly with context length and the number of concurrent requests**. In an inference service with long contexts and many users, **the KV-cache can exceed the size of the model itself**. This is the item everyone forgets to calculate, and it's what makes production servers explode.

8.2 Criterion #2: Memory Bandwidth

As repeatedly stated: **autoregressive LLM inference is memory-bound**. To generate **every single token**, the GPU must read **all model weights** from VRAM. The theoretical upper limit of tokens per second is therefore, as a first approximation:

$$\text{tokens/s} \approx \text{memory bandwidth (GB/s)} \div \text{model size in memory (GB)}$$

Example: a 14 GB model on a card with 1,800 GB/s yields a ceiling of approximately 128 tokens/s (in practice, 60–80% of this value is achieved). The same card with double the bandwidth would yield double the tokens. **TFLOPS do not appear in this formula.** This is why H200 beats H100 despite having the same chip.

Note: the formula applies to **generation** (the *decode* phase). The **prefill** phase (processing the input prompt, which is parallelizable across all tokens together) is instead **compute-bound**: there, TFLOPS matter a lot. A workload with very long prompts and short responses has a completely different profile from a conversational chatbot.

8.3 Criterion #3: Numerical Precision

This needs to be well understood, because manufacturers declare TFLOPS in the precision that makes them look best.

Format	Bits	Structure	Typical Use
FP64	64	1 sign, 11 exponent, 52 mantissa	Scientific simulation. Irrelevant for AI — but datacenter GPUs have it and consumer GPUs do not
FP32	32	1+8+23	Historical "full" precision for deep learning
TF32	19 used (in 32-bit container)	1+8+10	Nvidia format: FP32 range, FP16 mantissa. Transparent acceleration
FP16	16	1+5+10	Classic mixed precision. Restricted dynamic range → risk of overflow/underflow, requires loss scaling

BF16	16	1+8+7	<i>Brain Float</i> : same range as FP32, less mantissa. More robust than FP16 in training , now the de facto standard
FP8	8	E4M3 or E5M2	Inference and (with caution) training on Hopper/Blackwell. Doubles throughput and halves memory
FP4 / NVFP4	4	block scaling	Inference only, on Blackwell and later. Doubles again. Requires accurate quantization

The golden rule when reading spec sheets: a TFLOPS number without specified precision **means nothing**. And if you read "with sparsity," know that this number is **double** the real number under normal conditions: it assumes a 2:4 sparsity structure in weights that most real models do not have. Nvidia always states this, in a small note. **Divide by two**.

8.4 Criterion #4: Interconnect

If the workload is **multi-GPU on a single model**, interconnect is not a detail: it's the difference between working and not working. Review section 4. If the workload is **multi-GPU with independent jobs**, interconnect is irrelevant and you can ignore it completely.

8.5 Criterion #5: TDP, Power, and Cooling

TDP is the thermal design power. It must be multiplied by the number of cards and added to the rest of the system; then a 20–30% margin must be added for transient peaks (modern GPUs have instantaneous peaks much higher than the nominal TDP, and undersized power supplies shut down abruptly). Then the problem of **where that heat goes** must be solved: a workstation with two 5090s under load heats a room in a way that no one who hasn't experienced it can

imagine. And finally, the **cost of energy** must be calculated: $2 \text{ kW} \times 8 \text{ hours/day} \times 250 \text{ days} \times \text{€}0.30/\text{kWh} \approx \text{€}1,200/\text{year}$ just for electricity, not counting air conditioning.

8.6 Criterion #6: Cost per GB of VRAM

A crude but useful metric for an initial comparison. [ALL VOLATILE DATA]

Card	VRAM	Indicative Price	€/GB
RTX 5090	32 GB	~€2,500	~€78/GB
RTX PRO 6000 Blackwell	96 GB	~€9,000	~€94/GB
DGX Spark	128 GB	~€4,000	~€31/GB (but 6× lower bandwidth!)
H100 PCIe	80 GB	~€25,000	~€310/GB
H200 SXM	141 GB	~€35,000	~€250/GB

This table says two things. First: **the DGX Spark is unbeatable for euro/GB, and terrible for euro/(GB/s)** — confirming it's a development machine, not a production one. Second: **datacenter cards cost 3–4 times more per GB than professional cards**, and you're paying for ECC, HBM, NVLink, licensing, and support. If you don't use any of those four things, **you're paying for nothing**.

8.7 Criterion #7: Buy or Rent?

This question should be asked before any other, and the answer depends on only one number: **the utilization factor**.

A cloud GPU costs (order of magnitude, [VOLATILE]) \$2–4/hour for an H100. In a year of continuous use, that's ~ \$25,000 — which is **the purchase price of the card**. Hence:

- ****Usage below 30–40%:**** Rent. Period. Buying hardware that sits idle is burning capital.
- ****Usage above 60–70%, for at least 18–24 months:**** Buying is convenient, provided you can manage the infrastructure (and the cost includes space, electricity, cooling, network, maintenance, and a system administrator's time).
- ****Unpredictable peaks:**** Hybrid — local hardware for baseline, cloud for peaks.
- ****Sensitive data / regulatory constraints (healthcare, defense, strict GDPR):**** Local hardware may be mandatory regardless of economic convenience.

A very common mistake in small businesses: buying a €20,000 workstation for a project that runs two weeks a month. It would have cost €800 in the cloud.

8.8 Decision table: workload → recommended GPU

Workload	Minimum VRAM	Recommendation	Economical alternative
Learning, courses, tutorials	8–12 GB	RTX 4060 Ti 16GB / RTX 5070	Free Google Colab
Stable Diffusion / image generation	12–24 GB	RTX 4090 / 5090	RTX 4070 Ti Super 16GB
LLM inference 7–14B quantized	8–16 GB	RTX 4090 / 5090	Mac with Apple Silicon (unified memory)
LLM inference 30–70B quantized	40–80 GB	RTX PRO 6000 Blackwell (96GB)	2× RTX 5090 with pipeline parallelism; DGX Spark (slow)
LoRA/QLoRA fine-tuning on 7–13B	24–32 GB	RTX 5090	Used RTX 4090
LoRA fine-tuning on 70B	80+ GB	RTX PRO 6000 / H100	Cloud by the hour
Full fine-tuning 7B	100+ GB	H100 / H200 (or multi-GPU)	Cloud, always
Production inference, high throughput	depends	H200 (bandwidth is everything)	H100 if budget is tight
Training a large model from scratch	thousands of GB	HGX B200 / GB200 NVL72	Cloud. Always and only cloud, unless you are a hyperscaler
Development and prototyping on huge models	128+ GB	DGX Spark	Cloud by the hour
Computer vision, YOLO, ASR, embedding	8–24 GB	RTX 4090 / 5090	Any recent RTX 12–16 GB
Commercial inference service for third parties	depends	Any datacenter or RTX PRO	None: GeForce are excluded by the EULA

9. Typical buyer mistakes

We list them here because they are the most costly and recur with impressive regularity.

- 1. Buying two GPUs thinking you can sum the VRAM.** It doesn't happen. Without NVLink (and consumer cards don't have it), two 32 GB cards do not make 64 GB for a single model.
- 2. Looking at TFLOPS instead of VRAM and bandwidth.** TFLOPS are the number marketing wants you to look at. For LLM inference, they are almost irrelevant.
- 3. Not realizing that the stated number is "with sparsity."** Halve it.
- 4. Confusing H100 PCIe and H100 SXM.** Same name, very different performance.
- 5. Forgetting KV-cache in sizing.** The model fits, the service goes into production, and at the tenth concurrent user, the server explodes.
- 6. Buying hardware for 15% utilization.** The cloud existed.
- 7. Ignoring the EULA on the datacenter.** Building a commercial product on GeForce is a legal risk that no serious investor will accept during due diligence.
- 8. Underestimating power and heat.** The 4-GPU workstation project almost always dies at the electricity meter, not on the budget.
- 9. Buying the DGX Spark expecting speed.** It's a development machine. Bandwidth is 270 GB/s. Reread section 5.2.
- 10. Not considering the ARM architecture.** If you buy a Grace/Vera system, verify that your stack exists for aarch64 **before** signing.
- 11. Buying "the latest generation" on principle.** If your workload is inference on medium models, a used A100 or a second-hand H100 will give you a price/performance ratio that no new card can approach.

Operational Summary — Decision Checklist for Choosing an AI GPU

Follow the order. Do not skip steps: each one can close the decision on its own.

Step 1 — Define the workload, in writing.

- ☐ Training, fine-tuning, or inference only?
- ☐ Which model, how many parameters, at what precision?
- ☐ What context length and how many concurrent requests at peak?
- ☐ Is it a one-off project or an ongoing service?

Step 2 — Calculate the necessary VRAM.

- ☐ Weights = parameters $\times$ bytes/parameter (4 / 2 / 1 / 0.5)
- ☐ If training: $\times$ 8–10 for optimizer and gradients (or use LoRA/QLoRA)
- ☐ Add activations ($\propto$ batch size) and KV-cache ($\propto$ context $\times$ concurrency)
- ☐ Add a **20% margin**
- ☐ $\rightarrow$ This number is the **eliminary filter**. Anything below it is excluded.

Step 3 — Decide whether to buy or rent.

- ☐ Expected usage below 40% $\rightarrow$ **cloud**, end of discussion
- ☐ Privacy/regulatory constraints prohibiting cloud $\rightarrow$ **on-premises**, at any cost
- ☐ Usage above 60% for 18+ months $\rightarrow$ evaluate purchase, including electricity, space, cooling, and man-hours in TCO

Step 4 — Check if you need multi-GPU on a single model.

- ☐ If **yes** (tensor parallelism, distributed training) $\rightarrow$ you need **NVLink** $\rightarrow$ you are in the SXM/datacenter world. Consumer cards are excluded.
- ☐ If **no** (independent jobs, one model per GPU) $\rightarrow$ you can stick to PCIe and consumer/pro cards, saving an order of magnitude.

Step 5 — Check the licensing constraint.

- ☐ Will the computation be sold or rented to third parties? $\rightarrow$ **GeForce excluded**. RTX PRO or datacenter is needed.
- ☐ Internal use or research? $\rightarrow$ GeForce allowed.

Step 6 — Check bandwidth, not just capacity.

- ☐ Is the workload generative inference? $\rightarrow$ **memory bandwidth** is your performance predictor. Estimate: tokens/s $\approx$ bandwidth $\div$ model size.
- ☐ Is the workload long prefill or training? $\rightarrow$ then **TFLOPS** (at the right precision!) really matter.

Step 7 — Check system compatibility.

- [] Power supply: total TDP + 30% margin
- [] Sufficient CPU/chipset PCIe lanes for the number of cards (x16 or at least x8 per card)
- [] Physical space in the case and actual airflow
- [] Electrical line: above 2 kW continuous, in Italy a serious consideration of the meter is needed
- [] CPU architecture: if Grace/Vera, does your software exist for **aarch64**?

Step 8 — Choose by price range.

- [] **Entry (< €3,000):** RTX 5070 Ti / 5080, or used RTX 4090. For learning, computer vision, small LLMs. Alternatively: **no hardware and €500 of cloud.**
- [] **Mid (€3,000–12,000):** RTX 5090 (one or two), or DGX Spark if the constraint is capacity and not speed. Covers 90% of the real needs of a professional or an SME.
- [] **High-end (€15,000–30,000):** RTX PRO 6000 Blackwell (one or two, for 96/192 GB), or an H100/H200 PCIe if datacenter license and HBM bandwidth are needed. This is the range where an SME can do serious AI in-house.
- [] **Datacenter (€100,000 +):** HGX H200/B200 node. At this point, you have a dedicated team and a purchasing department, and this chapter only serves to ask the right questions to your supplier.

Step 9 — Before signing.

- [] Have you checked prices **today**, and not on this page?
- [] Have you checked actual delivery times (not the declared ones)?
- [] Have you tested your exact workload on that GPU **in the cloud**, for an hour, before buying it? This is the single piece of advice with the best value/cost ratio in the entire chapter: **€20 of cloud can save you €20,000 of error.**

Final note on data volatility: architectures, prices, and availability in this sector change on a quarterly basis. The architectures, sizing principles, and selection criteria described in this chapter remain valid; specific numbers must be re-verified as of the decision date.

Chapter 12 - AMD Instinct GPUs for AI and the Nvidia vs. AMD Comparison in the Datacenter

Note on Data. *This chapter touches upon the most volatile hardware segment in the entire IT industry. Architectures succeed each other annually, prices have no public list and are negotiated contract by contract, and roadmaps are rewritten every six months. All reported numbers are updated to mid-2026, and where the data is inherently unstable (prices, availability, generations not yet shipped in volume), I explicitly mark it with the phrase **[VOLATILE DATA]**. At the time of writing: MI300X and MI325X are mature, widely available products; the MI350 series (MI350X/MI355X) has been in production since mid-2025; and the MI400 series with the "Helios" rack-scale platform was unveiled at CES in January 2026, with volumes expected in the second half of 2026. On the Nvidia side, Blackwell (B200/GB200) and Blackwell Ultra (B300/GB300) are in the field, while Vera Rubin (VR200) has entered production with volume shipments expected in the second half of 2026. The reasoning method, however—how to read a datasheet, how to evaluate an accelerator, what mistakes to avoid—will remain valid much longer than the numbers.*

1. The AMD Instinct Line

1.1 What it is, in elementary terms

When we talk about "GPU" (Graphics Processing Unit), our minds immediately go to the video card plugged into the PC's PCIe slot, with fans and an HDMI output. AMD Instincts are not that. They are accelerator processors that share architectural DNA with video cards—thousands of parallel computing units executing the same instruction on many different data—but have shed everything needed to draw pixels: they have no video outputs, no rasterization units, and (in recent generations) no ray tracing engines. They are matrix multiplication machines, and nothing else.

The Instinct line was launched in 2016 as AMD's answer to Nvidia's Tesla line (later simply rebranded "Data Center GPU"). For many years, it was a weak response: honest hardware, unusable software, irrelevant market share. The turning point came in 2023-2024, when the global scarcity of accelerators for generative AI suddenly made the second vendor interesting—and AMD found

itself, for the first time, with a product (MI300X) that, on a crucial metric, onboard memory, beat the world champion.

Definition of two acronyms we will use continuously. *HBM* (High Bandwidth Memory) is DRAM stacked vertically in towers of multiple dies, soldered on the same substrate as the processor and connected to it with very wide buses (thousands of traces). It is very expensive, consumes a lot of power, but offers an order of magnitude higher bandwidth than the GDDR in gaming cards. *TDP* (Thermal Design Power) is the power in watts that the cooling system must be able to dissipate continuously; AMD in the datacenter more often uses the term *TBP* (Total Board Power), which includes the entire board, memory included. In the rest of the chapter, I will treat them as practical synonyms, because on these products they almost always coincide.

1.2 Positioning: Why AMD Exists in This Market

The positioning of Instinct can be explained with a single sentence: **no rational buyer wants a single supplier for the largest expenditure item in their budget.** In 2024-2026, an AI training cluster costs more than the real estate that houses it, and Nvidia holds a market share that, depending on how Google's TPUs are counted, ranges between 80% and 90%. Hyperscalers (Microsoft, Meta, Oracle, Amazon), frontier labs, and governments have an enormous strategic interest in the existence of a credible second supplier: not necessarily to buy in bulk, but to **have negotiating leverage** and to avoid being held hostage by a price list and delivery schedule decided by third parties.

AMD is playing exactly this game, and it's playing it on three levers:

1. **More memory per GPU.** This is the main lever, the one on which AMD has built its entire narrative since 2023. If a model with 70 billion parameters fits into a single GPU instead of two, the service architecture simplifies, the communication cost between the two GPUs disappears, and the cost per token served plummets.
2. **Lower price per unit of memory and per FLOP.** AMD cannot afford Nvidia's premium, because the software is inferior: it has to discount. This discount has a precise name — *ecosystem discount* — and we will analyze it in section 4.
3. **Open standards.** Where Nvidia has NVLink (proprietary), AMD pushes UALink (consortium); where Nvidia has InfiniBand, AMD pushes Ultra Ethernet; where Nvidia has CUDA (proprietary), AMD has ROCm (open source). This is only seemingly an ideological choice: it is the classic strategy of the challenger seeking to commoditize the ground on which the leader has built its moat.

1.3 CDNA vs. RDNA: Why Two Separate Architectures

Until 2018, AMD had only one GPU architecture, GCN (Graphics Core Next), used for both gaming and computation. It was a compromise and, like all compromises, it didn't excel anywhere: it brought graphics-dedicated silicon to the datacenter that no one used, and it brought an emphasis on double-precision computing to gaming that no video game ever required.

In 2019, AMD made the structural choice that still defines its portfolio today: **bifurcating the architecture.**

- **RDNA** (Radeon DNA) is the *consumer/graphics* line. Optimized for latency, real-time rendering, ray tracing, and power consumption within the 350-600 W of a desktop card. It uses GDDR memory. This is what you find in the Radeon RX 7000/9000 series.
- **CDNA** (Compute DNA) is the *datacenter/compute* line. Optimized for pure throughput, filled with matrix units (the *Matrix Cores*, the functional equivalent of Nvidia's Tensor Cores), hungry for memory bandwidth, powered by HBM, designed to fit into a server chassis cooled by 40 °C inlet air or liquid.

Why it's the right choice. A silicon die is a finite area budget. Every square millimeter spent on rasterization units is a square millimeter not spent on Matrix Cores. For a customer buying 8,192 accelerators to train a language model, every transistor dedicated to graphics is burned money and wasted watts. Nvidia does exactly the same thing, less explicitly: the GB100 die of Blackwell datacenter and the GB202 die of the GeForce RTX 5090 are completely different chips, with the same commercial label "Blackwell".

Practical consequence for the reader. A Radeon RX 7900 XTX and an Instinct MI300X are not "the same thing in two sizes." They are distant relatives. Code optimized for one is not automatically fast on the other, low-level functionalities differ (RDNA has 32-thread wavefronts, CDNA has 64), and even ROCm support follows different schedules. **Typical buyer's mistake: thinking "I'll try ROCm on my Radeon and then scale up to Instinct."** The code can be ported, but the performance and optimized kernels cannot.

1.4 How to read Instinct nomenclature

The acronym is simpler than Nvidia's. Let's take `MI355X`:

Element	Meaning
`MI`	"Machine Intelligence." Family brand,

	unchanged since 2016.
`3` (first digit)	Generation/series. MI100 → CDNA1, MI200 → CDNA2, MI300 → CDNA3, MI350 → CDNA4, MI400 → CDNA5.
`5` (second digit)	Position in the series. MI300 is the progenitor, MI325 and MI350 are the incremental "refreshes" (typically more memory or faster memory).
`5` (third digit)	Product variant within the same family (MI350X vs MI355X: same silicon, different thermal envelope and cooling).
`X` (suffix)	High-end variant, typically GPU-only , for AI/HPC workloads.
`A` (suffix)	APU variant: x86 CPU cores and GPU cores with unified memory coexist on the same package (e.g., MI300A).

The suffix is the most confusing part: `X` = pure GPU accelerator; `A` = hybrid CPU+GPU. In the MI400 series, AMD introduced a third dimension of specialization, no longer just thermal but **functional** (see §2.6): MI430X for high-precision HPC, MI440X/MI455X for low-precision AI.

2. The cards, one by one

Before going into detail, three definitions that will be used in each card description.

OAM (OCP Accelerator Module) is the standardized physical format from the Open Compute Project for datacenter accelerators: a rectangular module, without its own heatsink, without cable power connectors, which bolts onto a baseboard (UBB, Universal Baseboard) typically capable of housing eight of them. It is the open equivalent of Nvidia's proprietary SXM format. An OAM module **cannot be bought and installed in a PC**: it is bought inside a server, period.

PCIe (Peripheral Component Interconnect Express) is the standard system bus. PCIe versions of datacenter GPUs exist, in the form of a dual-slot add-in card, but they are always depowered versions (less watts, less bandwidth between GPUs) and serve niche cases, not serious clusters.

FLOPS (Floating Point Operations Per Second) is the raw computation metric. It should always be read in conjunction with: (a) **precision** (FP64, FP32, FP16, BF16, FP8, FP6, FP4 — fewer bits, more operations per second, less numerical accuracy); (b) whether the number is **dense** or **sparse** (with structured sparsity, manufacturers double the declared number: two "PFLOPS" numbers can differ by 2× just because of this, and it is the most common marketing trick in the industry).

2.1 MI250X (CDNA2, 2021) — the historical chapter

Description and function. The MI250X is the card that gave AMD its first real symbolic victory in supercomputing. CDNA2 architecture, manufactured at 6 nm by TSMC, adopts an **MCM** (Multi-Chip Module) design: two complete GPU dies — called GCD, Graphics Compute Die — placed side-by-side on the same package and connected by Infinity Fabric.

Specification	Value
Architecture	CDNA2, TSMC 6 nm
Configuration	2 GCD × 110 CU = 220 Compute Units
Memory	128 GB HBM2e
Bandwidth	~3.2 TB/s
Vector/Matrix FP64	~47.9 / ~95.7 TFLOPS
Matrix FP16	~383 TFLOPS
TBP	500-560 W
Form Factor	OAM

The architectural weakness, and it's didactically important. The two GCDs **were not seen by the software as a single GPU**: the operating system saw two distinct devices. Programmers had to explicitly manage the partitioning of work between the two halves. This is exactly the problem AMD will solve with MI300X, where the chiplet design becomes transparent to the programmer. Remember this distinction: *visible chiplet vs transparent chiplet* is one of the most important conceptual differences between CDNA2 and CDNA3.

Historical use case: Frontier. The Frontier supercomputer at Oak Ridge National Laboratory (USA), operational since 2022, was the first machine in the world to officially exceed exaFLOPS in double precision (10^{18} operations per second in FP64). It is built with over 37,000 MI250X paired with EPYC CPUs. Frontier is proof that AMD was already capable of making first-class HPC hardware *before* the AI wave: AMD's problem has never been silicon, it has always been software (\$4).

Indicative price. [VOLATILE DATA] Out of production for the new market; modules can be found on the used/decommissioning market for a few thousand euros. **Do not buy it.** 128 GB of HBM2e seems like a lot, but without FP8 support and with a software stack of that generation, it's a dead end for modern AI.

2.2 MI300X (CDNA3, late 2023) — the card that changed the conversation

This is the card that made AMD relevant in AI, and it deserves the longest discussion.

The 3.5D chiplet design. The MI300X is, from a packaging engineering perspective, one of the most complex chips ever produced. It is not a monolithic die: it is a tower.

- At the base are **4 IODs** (I/O Die, manufactured at 6 nm): they contain the memory controllers, shared cache (256 MB of "Infinity Cache"), and the internal interconnect fabric.
- Above the IODs, soldered with **hybrid bonding** technology (direct copper-to-copper connection, without microbumps, which allows much better connection density and latencies), are **8 XCDs** (Accelerator Complex Die, at 5 nm), for a total of **304 active Compute Units**.
- Around them, on an interposer, **8 HBM3 stacks** for a total of **192 GB**.
- All this for approximately **153 billion transistors**.

The keyword is "3.5D": vertical stacking (3D, XCDs above IODs) combined with horizontal side-by-side placement on an interposer (2.5D, HBM stacks next to the logic complex).

Why it matters. Unlike the MI250X, here the chiplets are **transparent to the software**: PyTorch sees *one* GPU with 192 GB. The cost of this transparency is paid in hardware, by the internal Infinity Fabric and the shared cache.

Specification	MI300X
Architecture	CDNA3 (5 nm XCD + 6 nm IOD)
Compute Units	304
Memory	192 GB HBM3
Bandwidth	5.3 TB/s
FP16/BF16 (dense)	~1.3 PFLOPS
FP8 (dense)	~2.6 PFLOPS
Matrix FP64	~163 TFLOPS
TBP	750 W
Form Factor	OAM (8-GPU platform); a niche MI300X

	PCIe also exists
--	------------------

The VRAM Advantage on H100: What it Really Means. Nvidia's H100 SXM, its direct contemporary, has 80 GB of HBM3 with 3.35 TB/s. The MI300X has **192 GB with 5.3 TB/s**: 2.4x the capacity, 1.6x the bandwidth. Translated into practice:

- A 70 billion parameter model in FP16 occupies ~140 GB just for the weights. On H100, **two** GPUs are needed (with all the complications of **tensor parallelism**, i.e., splitting matrices between GPUs, which introduces communication at each layer). On MI300X, **one** is enough, and ~50 GB remains for the KV cache (the key-value cache, which is the memory of the already processed context, growing linearly with context length × number of concurrent requests).
- An 8-MI300X platform offers **1.5 TB of aggregated HBM** compared to 640 GB of an 8-H100 node. For inference of enormous models (a DeepSeek or a Llama 405B in FP8), this is the difference between "it fits in one node" and "I have to distribute it across two nodes via the network," which is a significant jump in complexity and latency.
- **LLM inference during generation is memory-bound, not compute-bound.** This sentence should be memorized. To generate a token, the GPU must read **all** the model weights from memory. The bottleneck is memory bandwidth, not compute units, which remain largely idle. It follows that **HBM bandwidth predicts generation speed better than peak TFLOPS**. And this is precisely the area where AMD is competitive.

Indicative Price. [VOLATILE DATA] AMD does not publish price lists. Market estimates for the MI300X in volume have been in the order of **\$10,000-\$15,000 per module** compared to \$25,000-\$40,000 asked for an H100 SXM at the peak of scarcity. On the cloud, the hourly rental of an MI300X has been **15-40% below** that of a comparable H100. Consider this an order of magnitude, not a price.

Use Cases. High-volume LLM inference; fine-tuning of medium-to-large models; mixed HPC (strong FP64). Less suitable for: research that depends on third-party custom CUDA kernels (\$4).

2.3 MI300A (the APU) and El Capitan

The MI300A is the most conceptually interesting and commercially more niche variant. Take the MI300X, remove two XCDs (leaving 6 XCDs, 228 CUs) and in their

place put **three CCD chiplets with 24 Zen 4 cores** (the same CPU as EPYC). Add **128 GB of HBM3 coherently shared between CPU and GPU**.

What "coherent unified memory" means and why it's a huge deal. In the classic model, the CPU has its RAM and the GPU has its VRAM. To compute on the GPU, data must be *copied* from RAM to VRAM via the PCIe bus, the computation performed, and then copied back. In HPC workloads with many alternating CPU/GPU phases, this copying dominates execution time and is the most common source of slow code. In the MI300A, **there is no copying**: CPU and GPU see the same physical addresses on the same HBM, with coherence maintained by hardware. An entire class of optimizations (and bugs) is eliminated.

TBP: ~550 W nominal, up to 760 W in boosted configuration. **Form factor:** socket/OAM in dedicated nodes.

El Capitan. The supercomputer at Lawrence Livermore National Laboratory, operational from late 2024, built on MI300A, has become the fastest system in the world on the TOP500 list, exceeding 1.7 exaFLOPS FP64 (**[VOLATILE DATA]**: rankings change every six months). It is the second demonstration, after Frontier, that in "true" scientific computing, AMD is not the second vendor: it is *the* vendor.

Typical error. Looking at the MI300A and thinking "128 GB unified, perfect for AI." No: for pure AI, the MI300X with 192 GB of GPU-only memory is almost always the better choice. The MI300A excels in mixed CPU/GPU HPC codes (fluid dynamics, nuclear simulation, weather), where memory coherence is more valuable than capacity.

2.4 MI325X (CDNA3 refresh, late 2024/2025)

This is not a new architecture: it's **the same CDNA3 silicon as the MI300X with better memory**. It's a classic "mid-cycle refresh," and should be understood as such.

Specification	MI325X	(for comparison) MI300X
Architecture	CDNA3	CDNA3
Memory	256 GB HBM3e	192 GB HBM3
Bandwidth	~6.0 TB/s	5.3 TB/s
Compute (FP8/FP16)	unchanged	—
TBP	1,000 W	750 W

Positioning against H200. The H200 is, similarly, the H100 with HBM3e: **141 GB at 4.8 TB/s**, 700 W. The comparison is therefore 256 GB vs 141 GB and 6.0 vs 4.8 TB/s: AMD maintains the capacity advantage (1.8×) and adds a bandwidth advantage (1.25×), paying for it with 300 W more per module.

The takeaway. The MI300X → MI325X jump is **+33% memory and +13% bandwidth at +33% power, with zero compute gain**. This is an upgrade that makes sense *only* if your bottleneck is memory — that is, again, if you are doing LLM inference. If you are training, the MI325X gives you almost nothing more than the MI300X. **Typical error: buying the refresh because the number is bigger, without having profiled your workload.**

2.5 MI350X / MI355X (CDNA4, mid-2025)

Here AMD truly changes architecture. CDNA4 on a 3 nm process, and most importantly: **native support for FP6 and FP4**.

What are FP8, FP6, FP4 and why they are the news. These are very low-precision floating-point number formats: 8, 6, and 4 bits per number respectively, compared to 16 for FP16/BF16. Fewer bits simultaneously mean three things: (1) the model occupies less memory (a 70B model in FP4 weighs ~35 GB instead of 140); (2) fewer bytes to read from memory per generated token, thus **faster inference on the memory-bound path**; (3) matrix units complete more operations per cycle. The cost is a loss of numerical accuracy, which can be mitigated with sophisticated quantization techniques (block scales, calibration, "microscaling" formats like MXFP4/MXFP6). The point is that from 2025 onwards, **the currency of inference is FP4**, which is why both AMD (CDNA4) and Nvidia (Blackwell) have built dedicated silicon.

Specification	MI350X	MI355X
Architecture	CDNA4, TSMC 3 nm	CDNA4, TSMC 3 nm
Memory	288 GB HBM3e	288 GB HBM3e
Bandwidth	~8 TB/s	~8 TB/s
FP8 (dense)	~5 PFLOPS	~5 PFLOPS
FP4/FP6 (dense)	~9-10 PFLOPS	~10 PFLOPS
FP64	~72 TFLOPS	~79 TFLOPS
TBP	1,000 W (air)	1,400 W (liquid)
Cooling	air-cooled	liquid-cooled only

The MI350X/MI355X difference is, essentially, cooling. Same silicon: the MI350X fits into a 1,000W air-cooled envelope and slides into existing data centers; the MI355X pushes to 1,400W but **requires liquid cooling** and thus an equipped data center. Those with cold plates get the MI355X and gain frequencies and throughput; those with only air get the MI350X. **This is the first time in this book that the *facility* constraint — not the chip — determines which product you can buy.** From now on, it will always be this way.

Declared comparison with Blackwell. AMD positions MI355X against Nvidia's B200/B300. The MI355X's 288GB of HBM3e exactly matches the B300's (Blackwell Ultra) 288GB, and bandwidth is equivalent (~8 TB/s). The capacity advantage AMD had against H100 **is gone**: Nvidia has responded. On FP4 FLOPS, the B300 claims ~15 PFLOPS dense versus the MI355X's ~10: Nvidia is ahead on raw compute. **[VOLATILE DATA]** In independent MLPerf Inference benchmarks published in spring 2026, the MI355X achieved AMD's best result ever, placing within a few percentage points of the B200 on server inference workloads — a result that would have been unthinkable two years prior.

Indicative price. [VOLATILE DATA] Order of magnitude \$25,000-\$35,000 per module in volume, typically sold only as an 8-GPU platform. A complete 8-MI355X node with EPYC CPUs, network, and chassis is on the scale of **hundreds of thousands of euros**.

2.6 The MI400 Series and the Helios Platform (2026)

[ENTIRELY VOLATILE SECTION — data from CES 2026 announcements, product not yet shipped in volume at time of writing.]

This is the most ambitious generational leap ever attempted by AMD, and it changes the very nature of the product: **AMD stops selling GPUs and starts selling racks.**

The MI400 series (CDNA 5, TSMC 2 nm) for the first time features *functionally* different variants:

Model	Target	Distinctive Feature
MI455X	Large-scale training + inference	The flagship. ~320 billion transistors, 432 GB HBM4 , ~19.6 TB/s, ~40 PFLOPS FP4 / ~20 PFLOPS FP8. Optimized

		for low precision (FP4/FP8/BF16).
MI440X	Enterprise on-premise	Designed for the "normal" 8-GPU server to be placed in an existing enterprise rack, not for rack-scale.
MI430X	HPC and "sovereign AI"	Retains full FP32/FP64. This is the variant for scientific computing and for states that want data within their own borders. Intended for supercomputers like Discovery (Oak Ridge) and Alice Recoque (France).

The logic of specialization must be well understood, because it is a general trend in the industry: by separating the AI variant (which does not need FP64) from the HPC variant (which desperately needs it), AMD **eliminates redundant logic from the die** and improves its power efficiency and cost. Nvidia is doing the same thing in a different way (with the Rubin CPX specialized in *prefill*). The era of the "generalist GPU that does everything well" is ending.

Helios. This is the rack. It is not a server: it is a double-width, integrated, liquid-cooled cabinet that AMD sells as an indivisible computing unit:

- **72 MI455X accelerators** + **EPYC "Venice"** (Zen 6) CPU + 800G Pensando "Vulcano" NIC (Network Interface Card);
- **31 TB** of aggregated HBM4, with **1.4 PB/s** of total memory bandwidth;
- **~2.9 exaFLOPS FP4** for inference and **~1.4 exaFLOPS FP8** for training per rack;
- power consumption and cooling requirements necessitate a new-generation datacenter (on the order of **hundreds of kW per rack**).

Why AMD had to build a rack. Because Nvidia did it first (GB200 NVL72) and shifted the battlefield. When 72 GPUs are wired into a single coherent memory domain, the customer no longer compares "GPU versus GPU": they compare "rack

versus rack," and whoever doesn't have a rack isn't in the race. The roadmap continues with the **MI500 series on CDNA 6 and HBM4E announced for 2027**.

3. The AMD Interconnect

3.1 The Problem, Before the Solution

A single GPU cannot train a frontier model. Thousands are needed, and they must continuously exchange data: in every distributed training step, at the end of gradient computation, all GPUs must sum their gradients with those of all others (an operation called *all-reduce*). If this exchange is slow, the GPUs remain idle waiting, and you are paying €40,000 per card to let them rest.

Hence the fundamental distinction, which applies to both manufacturers:

- ****Scale-up**** ("vertical" interconnect, within the node or within the rack): few devices, very low latency, enormous bandwidth, coherently shared memory. This is the domain of NVLink and Infinity Fabric.
- ****Scale-out**** ("horizontal" interconnect, between nodes and between racks): many devices, using a real network (InfiniBand or Ethernet). Bandwidth is an order of magnitude lower, latency is higher.

The universal design rule: **keep the most intense traffic within the scale-up domain and exit to scale-out as little as possible.**

3.2 Infinity Fabric and the 8-GPU OAM Platform

Infinity Fabric (IF) is AMD's proprietary interconnect fabric, born in 2017 with Zen to connect CPU chiplets and then extended to everything: chiplet-to-chiplet within the package, CPU-to-GPU on the socket, GPU-to-GPU on the baseboard.

In the standard Instinct platform — **8 OAM modules on a UBB** — the 8 GPUs are connected to each other by Infinity Fabric links in an **all-to-all** topology: each GPU has a direct connection to each of the other seven. This is architecturally elegant and has a practical advantage: **a dedicated switch chip is not needed**. Nvidia, to achieve the same result, uses **NVSwitch**, additional switching chips.

- ****The AMD advantage:**** less silicon, less cost, less power consumption, simple topology.
- ****The AMD limitation:**** an all-to-all does not scale. It works perfectly with 8 GPUs. With 72, the number of direct connections explodes ($n \cdot (n-1) / 2$) and

becomes impossible. **It is precisely for this reason that Nvidia, with NVSwitch, was able to build the GB200 NVL72 — 72 GPUs in a single NVLink domain — while AMD remained stuck at 8 for an entire generation.** If you remember only one sentence from this chapter, remember this: *Nvidia's structural advantage in 2024-2025 was not the GPU, it was the switch.*

3.3 NVLink: What AMD Must Beat

NVLink is Nvidia's proprietary interconnect. In the Blackwell generation, it offers ~1.8 TB/s bidirectional per GPU; in the Rubin generation, it increases to ~3.6 TB/s. Combined with NVSwitch, it creates a **coherent memory domain across 72 GPUs**: from a programmer's perspective, the NVL72 rack behaves like a giant GPU with shared memory. It's a deep technological moat, and Nvidia defends it: NVLink is closed, not licensed to third parties.

3.4 UALink: the open anti-NVLink standard

UALink (Ultra Accelerator Link) is the industry's collective response. It's a consortium bringing together AMD, Intel, Google, Meta, Microsoft, Broadcom, Cisco, HPE, and others, with a stated goal: **to define an open standard for scale-accelerator-to-accelerator interconnection**, so that a rack can contain accelerators from different vendors and switches from different vendors, as is already the case with Ethernet in networking.

The real situation in 2026, without rhetoric. MI400 accelerators will be the first to support UALink alongside Infinity Fabric. But an interconnect standard without **switch silicon** is a specification, not a product: for UALink to truly work, switch chips are needed, expected from vendors like Astera Labs, Broadcom, Enfabrica, and others. Until they arrive in volume, the first Helios systems will use **UALink-over-Ethernet** — a functional solution but not what UALink was designed for.

Translation for the buyer: UALink is a credible promise with uncertain timing. Do not base a purchasing plan on it today.

Ultra Ethernet is the sibling on the scale-out side: a parallel consortium that extends Ethernet with the latency and reliability features required by AI clusters, to replace InfiniBand (Nvidia proprietary after the acquisition of Mellanox). Here, AMD's position is stronger, because Ethernet already exists and network cards (Pensando Pollara 400G, Vulcano 800G) are real products.

4. The weak point: software

If you've read the previous sections thinking "but then AMD is competitive, why isn't anyone buying it?", the answer is all here.

4.1 CUDA: understanding the moat before understanding the bridge

CUDA (Compute Unified Device Architecture) is not "Nvidia's driver." It's a complete stack accumulated over **nearly twenty years**: a language, a compiler, a runtime, and above all a **pyramid of libraries** (cuBLAS for linear algebra, cuDNN for neural networks, NCCL for multi-GPU communication, TensorRT-LLM for optimized inference, CUTLASS for matrix kernels) on which *all* the world's AI software rests. And above that: millions of lines of research code, tutorials, Stack Overflow answers, doctoral theses, pre-packaged containers, and an entire generation of engineers who learned to program this way.

The moat is not technical. It is **social and cumulative**. And that's why it cannot be filled with a faster chip.

4.2 ROCm: what it is and where it really stands

ROCm (Radeon Open Compute platform) is AMD's answer: an open-source stack, functionally equivalent to CUDA. As of 2026, the current line is **7.x** (ROCm 7 introduced stated improvements of over 3-4× on inference and training compared to ROCm 6.0), with regular releases and first-class Linux support.

Where ROCm works well, today — and this has changed a lot in the last 18 months:

- **PyTorch**: Official, upstream support with ready-to-use builds. In the vast majority of cases, a PyTorch model running on CUDA runs on ROCm with zero code changes: `torch.cuda` continues to be called that way even on AMD (it's an alias, for compatibility).
- **vLLM and SGLang**: The two most widely used LLM inference engines in production have official and mature ROCm support. AMD has invested heavily here, with optimized proprietary kernels (the **AITER** library) that bring multiple throughput improvements in attention backends compared to generic paths. In 2026, ROCm entered vLLM's official CI (Continuous Integration): every commit is tested on AMD silicon, meaning regressions are caught before reaching the user. This is a huge change in status, from being "a community porting" to being "a first-class platform."

- **Standard LLM Inference**: On PyTorch + vLLM workloads without custom kernels, independent measurements place ROCm on MI300X/MI355X **around 90-95% of Nvidia's counterpart throughput** — and with large batches, the gap narrows further. **[VOLATILE DATA]**
- **llama.cpp / Ollama**: They work, even on consumer cards.

Where ROCm still suffers, and it's honest to say:

- **Third-party custom CUDA kernels**. This is **the** problem, the one that causes migrations to fail. A lot of research code — and not a few production libraries — ships hand-written CUDA kernels, sometimes with PTX assembly or Nvidia-specific intrinsics. That code **does not port automatically**. You find yourself choosing between: looking for a compatible fork, rewriting the kernel, or waiting for the community to do it. If your project depends on a niche library with custom kernels, **migration to AMD can get stuck on a single `.cu` file**.
- **TensorRT-LLM and the Nvidia optimization ecosystem** do not have a complete ROCm equivalent. At very small batches (low-latency inference, one user at a time), where TensorRT-LLM performs best, Nvidia maintains a significant throughput advantage.
- **FlashAttention**: ROCm ports exist and work, but tend to lag behind the CUDA version by weeks or months. If you need the latest features for training, expect friction.
- **Documentation and edge cases**. Greatly improved, but the density of ready answers on the Internet remains an order of magnitude lower. When you have an obscure CUDA error, you Google it and find three people who already solved it in 2022. On ROCm, often, you find an open issue on GitHub. **This translates into man-hours, and man-hours have a cost.**
- **Windows and macOS**: For serious ML work, **Linux only**. Ubuntu LTS or RHEL. No qualifiers.

4.3 HIP: The Bridge

HIP (Heterogeneous-Compute Interface for Portability) is the compatibility layer. Syntactically it's almost a clone of CUDA: `cudaMalloc` becomes `hipMalloc`, `__global__` remains `__global__`, the concepts (grid, block, thread, shared memory) are the same. AMD provides **hipify**, an automatic source-to-source translation tool that converts CUDA code to HIP code.

How well it works, realistically:

- For "idiomatic" CUDA code that uses standard APIs: ****automatic conversion succeeds in the vast majority of cases.****
- For code that uses Nvidia-specific architectural intrinsics, warp shuffle with assumptions about warp size (32 on Nvidia, 64 on CDNA!), inline PTX assembly, or the latest Tensor Core APIs: ****conversion fails, or worse, succeeds but produces correct yet extremely slow code.****

The subtle, and crucial, point: **HIP guarantees portability, not performance.** A mechanically translated kernel will run. A *fast* kernel on CDNA needs to be rewritten with CDNA in mind. Anyone selling migration to AMD as "just run hipify" is selling you an illusion.

4.4 The "ecosystem discount": why AMD hardware costs less

Now we can close the economic loop. Why does an MI300X with 192 GB cost less than an H100 with 80 GB?

Because the price of an accelerator is not the price of the silicon: it's the price of the silicon plus the value of the ecosystem, minus the cost of risk. AMD must compensate, in price, for everything the customer will have to spend extra:

- engineering hours to port and optimize code;
- risk of getting stuck on an unsupported dependency;
- risk of delivery, drivers, regressions;
- less liquid used market and lower residual value;
- smaller talent pool (finding an engineer who can optimize HIP kernels is harder than finding one who can optimize CUDA, and costs more).

This discount is real and should only be taken if you have the organization to do so. A 25% discount on hardware is an excellent deal for a hyperscaler with a team of 40 system engineers optimizing kernels. It's a bad deal for an SME with two data scientists, because those two will spend three months fighting with drivers, and those three months cost more than the discount. **Rule: the ecosystem discount is worth as much as your internal capacity to absorb it.**

5. Nvidia vs. AMD in the datacenter: do they beat or not beat?

Let's make an honest, criterion-by-criterion comparison, without bias.

5.1 VRAM per GPU — ****AMD wins**, historically**

On this, AMD has had a continuous structural advantage. 192 GB vs. 80 (MI300X vs. H100), 256 vs. 141 (MI325X vs. H200). **However, beware:** with Blackwell Ultra (B300, 288 GB), Nvidia has **matched** the MI355X. The advantage reopens with MI455X (432 GB HBM4) vs. Rubin VR200 (288 GB HBM4). **The historical pattern is clear: AMD opens the lead, Nvidia closes it in the next generation.** Don't take it for granted that it will last.

5.2 Memory bandwidth — ****substantially on par, with generational head-to-head****

5.3 vs. 3.35 TB/s (MI300X vs. H100: AMD wins). 8 vs. 8 (MI355X vs. B300: tied). 19.6 vs. 22 TB/s (MI455X vs. VR200: **Nvidia wins**, which with HBM4 at over 11 Gbps per pin has pushed the frequency higher). This is the most important metric for inference, and it is the most contested front.

5.3 Training performance — ****Nvidia wins, and it's not close****

Not so much for the silicon, but for: (a) higher peak FLOPS in low precision; (b) NCCL and the entire collective communication stack, mature and optimized for years; (c) the 72-GPU NVLink domain, which for training enormous models is an architectural, not incremental, advantage; (d) FlashAttention, Transformer Engine, Megatron/NeMo and the entire large-scale training toolkit, which originated on CUDA. **If you need to train a frontier model from scratch, in 2026 buy Nvidia.** This is the conclusion, however uncomfortable.

5.4 Inference Performance — ****AMD is competitive, and in some scenarios wins****

Because inference during generation is memory-bound, and memory is AMD's territory. More VRAM per GPU eliminates entire axes of parallelism, simplifies deployment, and serves a large model on fewer GPUs. On PyTorch + vLLM/SGLang, with large batches, the gap narrows to single-digit percentages. **If your business is serving tokens in volume, AMD deserves serious consideration.**

5.5 TCO and Price — ****AMD wins on purchase price; TCO depends on you****

TCO (Total Cost of Ownership) is the sum of: purchase + energy + cooling + space + personnel + software + risk cost. AMD clearly wins on the first item. It typically loses on energy (higher TDPs for the same generation) and personnel (\$4.4). The verdict **depends entirely on the scale and maturity of your team.**

5.6 Software — ****Nvidia wins, clearly****

See §4. There's nothing else to add. It's why this chapter doesn't end with "buy AMD."

5.7 Availability — ****the wild card****

In times of extreme scarcity, the question isn't "which is better," it's "what can I get." An MI355X delivered in eight weeks beats a B300 delivered in ten months, regardless of benchmarks. **[VOLATILE and extremely variable DATA.]** This is, historically, the primary reason AMD has sold Instinct.

5.8 Who buys AMD, and why

1. **Hyperscalers.** Microsoft, Meta, Oracle. They have the teams to absorb the software cost, buy in volumes that justify dedicated optimization, and have an existential strategic interest in not depending on a single vendor. They buy AMD **also to have something to show Nvidia during negotiations.** It's negotiating leverage, even more than technology. 2. **Inference farms.** Companies whose product is serving tokens (API providers, cloud inference engines). Memory-bound load, standardized software stack (vLLM), extreme sensitivity to cost per token. The ideal profile for AMD. 3. **HPC and the public sector.** Frontier, El Capitan, European exascale systems. AMD is the champion here, not the challenger — strong FP64, unified memory, and no lock-in. 4. **"Sovereign AI."** States that want their own infrastructure, often with an explicit political mandate in favor of open standards and against monopoly.

Who does **not** buy AMD: the startup with six engineers and eighteen months of runway, who cannot afford to lose three weeks on a kernel. And rightly so.

5.9 Final Comparative Table

[ALL DATA IN THIS TABLE IS VOLATILE.] FLOPS in *dense* values, not sparse. Indicative market prices, not list prices.

	H100 SXM	H200 SXM	B200	B300 / GB300	VR200 (Rubin)	MI300X	MI325X	MI355X	MI455X
Manufacturer	Nvidia	Nvidia	Nvidia	Nvidia	Nvidia	AMD	AMD	AMD	AMD
Architecture	Hopper	Hopper	Blackwell	Blackwell	Rubin	CDNA 3	CDNA 3	CDNA 4	CDNA 5

e				Ultra					
Year	2022	2023	2024-25	2025-26	H2 2026	2023	2024-25	2025	H2 2026
Memory	80 GB HBM3	141 GB HBM3e	180-192 GB HBM3e	288 GB HBM3e	288 GB HBM4	192 GB HBM3	256 GB HBM3e	288 GB HBM3e	432 GB HBM4
Bandwidth	3.35 TB/s	4.8 TB/s	8 TB/s	8 TB/s	~22 TB/s	5.3 TB/s	6.0 TB/s	8 TB/s	~19.6 TB/s
FP8 (dense)	~2 PF	~2 PF	~4.5 PF	~5 PF	~35 PF (NVFP4 train)	~2.6 PF	~2.6 PF	~5 PF	~20 PF
FP4 (dense)	—	—	~9 PF	~15 PF	~50 PF	—	—	~10 PF	~40 PF
FP64	34 TF	34 TF	~40 TF	reduced	reduced	163 TF	163 TF	79 TF	(MI430X)
TDP/TBP	700 W	700 W	1,000 W	~1,400 W	~1,800-2,300 W	750 W	1,000 W	1,400 W	~2,300+ W
Cooling	air/liquid	air/liquid	air/liquid	liquid	liquid only	air	air	liquid	liquid only
Scale-up	NVLink 4	NVLink 4	NVLink 5	NVLink 5 (NVL72)	NVLink 6 (NVL72)	Infinity Fabric (8 GPUs)	IF (8)	IF (8)	IF + UALink
Rack-scale	HGX 8×	HGX 8×	GB200 NVL72	GB300 NVL72	VR200 NVL72	—	—	—	Helios (72)
Indicative Price	\$25-30k	\$30-35k	\$30-40k	\$40k+	n.d.	\$10-15k	\$15-25k	\$25-35k	~\$30k (estimates)
Software	CUDA ★★	★★★	★★★	★★★	★★★	ROCm	★★★☆☆	★★★	★★★

	★★ ★	★	★	★	★	★★ ★☆ ☆	☆	☆	☆
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How to read this table. Look at the last three rows. Price and software tell the whole story: AMD offers more memory per dollar, Nvidia offers less friction per dollar. The choice is ultimately a bet on **which of the two resources—money or your engineers' time—is scarcer in your organization.**

6. Final Decision Guide for an AI Build (Workstation Part)

Let's conclude by descending from the clouds of the datacenter to the desktop, because that's where 99% of this book's readers will have to make a real decision. Price ranges are **[VOLATILE DATA]** and in 2026 have been further distorted by memory scarcity: consider them orders of magnitude.

6.1 The Zero Question: Buy or Rent?

Before picking a tier, ask this first. **The cloud is convenient when usage is low, variable, or uncertain. Buying is convenient when usage is high, constant, and predictable.**

The calculation is simple: take the hourly cost of the equivalent cloud instance (**[VOLATILE DATA]** — approximately, in 2026, a few euros per hour for an H100/H200-class GPU on specialized providers, more on large generalist clouds), multiply it by the hours you will *actually* use per month, and compare it with (hardware price / 36 months) + electricity + your own administration time.

The typical result is that **the break-even point sits around 30-50% continuous usage.** If your GPU would stay on and loaded less than 8 hours a day, the cloud almost always wins. If you keep it at 90% for two years, buying clearly wins.

But there are three **non-economic** reasons to buy that are often decisive: **data privacy** (if customer data can't leave your premises — often the case in Italy under GDPR — the public cloud is precluded or extremely complicated); **latency and availability** (your GPU is yours, always, without queues or quotas); and **learning** (assembling, configuring, breaking, and repairing a system teaches things renting an instance never will).

6.2 Tier A — under €3,000 : the high-end consumer PC

Who you are: a developer, a researcher, a student. You want to fine-tune small models, perform local inference, develop and prototype, and maybe play games in the evening.

What you buy: a single high-end consumer GPU (RTX 5090 with 32 GB, or a used RTX 4090 with 24 GB), a consumer CPU (Ryzen 7/9 or Core i7/i9), **64-128 GB of system RAM** — rule of thumb: system RAM $\geq 2 \times$ total VRAM, because it's needed for loading, quantization, and preprocessing — and a 2 TB NVMe.

What you can do: inference of models up to ~30 B in 4-bit quantization; **LoRA** (Low-Rank Adaptation: only a small set of additional matrices are trained instead of all weights, drastically reducing memory requirements) fine-tuning on models up to 7-13 B; all the computer vision you want; development of pipelines that will then run elsewhere.

What you CANNOT do: train from scratch; full fine-tuning of large models; serve in production with real concurrency.

Common mistake: buying two 16 GB consumer GPUs and expecting to get 32 GB. **VRAM does not stack** if the model is not designed to be partitioned. For LLM inference, a single 24 GB GPU almost always beats two 12 GB GPUs. **Maximize the VRAM of a single chip, not the number of chips.**

6.3 Tier B — €5,000-€15,000: The Threadripper Workstation with 2 GPUs

Who you are: A consultant, a small team, a pre-seed startup. You need a machine that can handle serious workloads and run for a week straight.

What you buy: A **Threadripper** (not a Ryzen) or a Xeon W. Why? For **PCIe lanes** (the "lanes" of the bus: each GPU ideally wants x16 lanes; a consumer CPU offers ~24-28 usable lanes in total, meaning with two GPUs you drop to x8 each and the bottleneck shifts to the bus when GPUs need to communicate). A Threadripper offers 48-92, a Threadripper PRO up to 128. Then: 2 GPUs (2× RTX 5090, or 1-2 RTX PRO Blackwell 48-96 GB if the budget allows), 128-256 GB of ECC RAM, a 1,600 W power supply, a full-tower case with serious airflow.

Common mistake: installing two *open-air* consumer GPUs (with open fans, i.e., the vast majority of gaming models) adjacent to each other. **See §6.5: this is the most expensive and most common mistake in this entire tier.**

6.4 Tier C — €15,000-€50,000: The Professional Workstation

Who you are: A company doing serious AI but without a datacenter (yet).

What you buy: Threadripper PRO or single-socket EPYC, and **2-4 RTX PRO 6000 Blackwell (96 GB each)**. Four of these give you **384 GB of VRAM on a single machine**, with blower cooling specifically designed for side-by-side installation, ECC, certified drivers, and — a non-trivial detail — a license that permits datacenter use, which GeForce cards **do not have** (Nvidia's license terms prohibit the use of GeForce cards in datacenters, with limited exceptions: if you provide services to clients, check with a lawyer).

The specialized alternative: Desktop systems based on superchips (DGX Spark, DGX Station, and OEM derivatives), which offer large amounts of **coherent unified memory** (up to hundreds of GB) on a single ARM+GPU chip. They are excellent for **loading huge models** that would never fit into a discrete GPU, but their memory bandwidth is much lower than HBM: these are machines for *development and prototyping on large models*, not for high-throughput production. **[VOLATILE DATA]** Choose them if your constraint is "the model doesn't fit," not if it's "the model is slow."

6.5 Tier D — over €50,000 or at scale: Datacenters, and the Real Question

Here the question is no longer "which GPU," it's **"buy or rent."**

The calculation you need to make, and almost no one does. Take the cost of an 8-GPU datacenter node (order: €250,000-€400,000). Add: high-speed network, storage, racks, UPS, cooling, and — the forgotten item — **energy**. Eight 1,000 W GPUs plus the rest of the node make ~10 kW continuous; at €0.20/kWh, that's ~€17,500/year **[VOLATILE DATA, and in Italy the industrial energy price is higher than the European average]**. Add depreciation over 3-4 years (because the generational cadence is annual and residual value plummets) and a person to administer it.

Then compare it with cloud rental of the same capacity.

The rule of thumb, which generally applies:

- ****Cloud is convenient**** if usage is ****less than 50-60%****, if the load is spiky, if you are still exploring what hardware you need, if you need to scale from 8 to 512 GPUs for two weeks and then back to 8, or if you don't have system

personnel. The cloud sells you ****optionality****, and optionality in a market that changes every twelve months is worth a lot.

- ****Purchasing is convenient**** if usage consistently exceeds ****70-80% 24/7**** (at which point the break-even on cloud typically occurs between 12 and 24 months), if you have non-negotiable data residency requirements (healthcare, defense, personal data), if you already have the facility, or if the load is predictable and constant.
- ****The middle ground, often the best:**** buy a workstation for daily development (which is 90% of man-hours) and rent the cluster for training sessions (which are 90% of GPU-hours).

6.6 Compatibility Checklist for a Multi-GPU Workstation

This is the section to reread before clicking "buy." Each point below corresponds to a mistake I've seen made.

1. PCIe Lanes — count, don't hope. Each GPU wants x16. Sum the lanes required by all GPUs **plus** those of the NVMe drives (4 lanes each) **plus** those of the network card. Compare this with what the CPU *actually* offers (not what the chipset "exposes," which are often shared upstream). Check the motherboard manual for the **slot division scheme**: almost always, populating the second x16 slot will downgrade the first to x8. For single-GPU inference, x8 is not dramatic; for multi-GPU training with continuous gradient exchange, it is.

2. Physical Slot Spacing. Modern GPUs occupy 3-4 slots in thickness. A motherboard with two x16 slots "spaced out" on paper may actually leave **no air gap** between the two cards. Measure. If there is no space, solutions with PCIe *risers* (extensions) and vertical mounts exist, but they are a source of signal instability at PCIe 5.0: buy them certified for the correct generation, never generic ones.

3. Power Supply Unit (PSU) — and transient calculation. Sum the TDPs and add **at least 30-40% margin**. Not for general prudence, but because modern GPUs have **transient peaks** (microsecond spikes) that can reach double the nominal TDP and trigger the overcurrent protection (OCP) of an undersized power supply — resulting in an unexpected reboot, usually in the middle of a thirty-hour training session. Two RTX 5090s (2 × 575 W) plus a Threadripper (350 W) plus the rest require **1,600 W minimum**. Above ~1,600-2,000 W, you enter the territory of **dual PSUs**: two power supplies synchronized with an "add2psu" or a pre-configured server motherboard. Non-negotiable rule: **a single GPU must never**

be powered by two different PSUs (potential differences on ground references → damage). One GPU, one power supply. And, for the connectors: **12VHPWR/12V-2x6 must be inserted all the way, with a click, and must not be bent within 35 mm of the connector**. Cases of connector melting are, in almost all instances, cases of incomplete insertion.

4. Cooling — blower vs. open-air, the €3,000 mistake. This is the most important and least known difference in the entire section.

- An ****open-air**** GPU (the 2-3 fan design of almost all gaming cards) draws air from below and ****blows it sideways into the case****. Designed for a single GPU in a well-ventilated case.
- A ****blower**** GPU (radial fan) draws air and ****expels it entirely out of the case through the rear bracket****. Designed to be placed alongside others.

If you install **two adjacent open-air GPUs**, the upper card draws in the hot air expelled by the lower one. The result is not a failure: it's **thermal throttling**, i.e., the automatic reduction of frequencies to avoid exceeding temperature limits. The upper card will run stably 15-25 °C hotter and lose a significant fraction of performance, **silently and permanently**. You will have paid for two GPUs and get one and a half, without any warning.

Solutions, in order of preference:

- buy ****professional GPUs with blower coolers**** (RTX PROs are designed exactly for this — it's one of the reasons they cost more, and why in multi-GPU setups they are often more economical for the same performance achieved);
- ****liquid cooling**** with dedicated blocks (block on each GPU, generous radiators): effective, expensive, and introduces a hydraulic point of failure;
- ****physically space out**** the cards with risers, in a very large case, with dedicated exhaust fans;
- ****limit power**** (`nvidia-smi -pl`): counterintuitive, but limiting a GPU to 80% of its TDP typically costs only 5-10% in performance and drastically reduces heat. In multi-GPU setups, it's almost always the rational choice.

5. The domestic electrical circuit — the 3 kW limit in Italy. And here we come to the most concrete and most ignored constraint, which no American website will ever mention.

The standard domestic electricity contract in Italy is **3 kW** (with a typical tolerance of 10%, so ~3.3 kW before the limiter trips). A Schuko socket on a 16 A, 230 V circuit nominally handles ~3.6 kW.

Do the math: a workstation with **two RTX 5090s** under full load realistically draws **1,400-1,800 W from the socket** (the upstream draw is higher than the downstream output because the power supply has an 88-92% efficiency). Add the monitor, and you're at ~1,900 W. **That leaves ~1,100 W for everything else in the house.** The electric oven demands 2,000. The washing machine during heating, 2,000. The air conditioner, 1,000-1,500. The hairdryer, 1,800.

Operational conclusion: with a 3 kW contract, a mid-range multi-GPU workstation will trip your circuit breaker every time someone in the house turns something on. And it won't trip at noon: it will trip at three in the morning, in the middle of a fine-tuning, because the water heater turned on.

What to do, in order:

- ****Increase the contractual power**** to 4.5 or 6 kW. This is done by requesting it from your provider, it entails an adjustment to the fixed charges on your bill, and it makes sense starting from band B.
- ****Check the circuit****: an 1,800 W workstation should be on a ****dedicated circuit****, not shared with the kitchen. If your system has a single 16 A outlet circuit for half the apartment, call an electrician.
- ****Consider an uninterruptible power supply (UPS)**** with adequate **real** power: be aware that UPS units are sold in VA (volt-amperes), not watts. A "1,500 VA" UPS typically delivers ~900-1,000 real W. Size it based on watts.
- ****Above ~3 kW for the machine, the workstation stops being an appliance****: an industrial line is needed, and at that point, you have implicitly answered the "buy or cloud" question.

6. Thermal load in the room. Last point, and the most underestimated: **all electrical power absorbed by a computer is converted into heat.** Not "almost all": all. An 1,800 W workstation is, from a thermodynamic perspective, **an 1,800 W heater constantly on.** In a 15 m² room without air conditioning, in summer, this is physically unsustainable — for the machine and for you. Factor in an air conditioner, and remember that it also consumes from your 3 kW contract.

7. Operational Summary — Decision Checklist

A) Choosing between Nvidia and AMD in the datacenter

1. **Is my workload training or inference?** → *Frontier training* → **Nvidia**, no discussion. → *Volume inference* → AMD is in the running. Proceed. 2. **Does my stack depend on custom CUDA kernels, TensorRT-LLM, or niche libraries?** → Yes → **Nvidia**. The AMD discount won't compensate for porting. → *No, I'm on standard PyTorch + vLLM/SGLang* → AMD is a serious choice. Proceed. 3. **Do I have system engineers capable of independently solving a driver issue or a slow kernel?** → *No* → **Nvidia**. The ecosystem discount is only realized with internal expertise. → *Yes* → Proceed. 4. **Is my bottleneck GPU memory capacity?** → *Yes* (the model doesn't fit, or barely fits) → **this is the ideal use case for AMD**. 5. **Have I benchmarked MY model, with MY context length and MY batch?** → *No* → **Stop and do it**. No number in this chapter replaces a measurement on your workload. Almost all clouds offer MI300X/MI355X by the hour: **spend €200 on rental before spending €200,000 on hardware**. 6. **Have I verified actual availability and delivery times?** → The best GPU is the one that arrives. 7. **Have I considered negotiating leverage?** → An AMD quote in hand changes the Nvidia quote. This is, for many companies, the main reason for this entire chapter.

B) Compatibility Checklist for a Multi-GPU Workstation

- [] **PCie Lanes**: sum the required lanes (GPU + NVMe + network) and compare them with the *actual* lanes of the CPU. Read the slot allocation diagram in the motherboard manual.
- [] **Physical Spacing**: verify that there is air between the GPUs. Measured, not estimated.
- [] **Coolers**: **blower** (or liquid) if GPUs are adjacent. Never two open-air coolers stuck together.
- [] **PSU**: sum of TDPs + 30-40% margin. 12VHPWR cables fully inserted, not bent near the connector.
- [] **Dual PSU** (if >1,600 W): one GPU = one power supply. Never straddle.
- [] **System RAM** ≥ 2× total VRAM. ECC if the machine needs to run for days.
- [] **Power limit** set to 80% in multi-GPU: you lose 5-10%, gain stability and silence.
- [] **Electricity contract**: 3 kW is **not enough** beyond band A. Request an increase to 4.5-6 kW.
- [] **Dedicated circuit** for the workstation, verified by an electrician.
- [] **UPS** sized in **watts**, not in VA.

- [] **Room air conditioning**: the machine is a heater. Treat it as such.
- [] **Operating system**: Linux. Ubuntu LTS. Both on Nvidia and (especially) on AMD.
- [] **Exit plan**: if in 18 months you will need 4× this power, is this machine an investment or a dead end? If it's a dead end, consider the cloud.

End of chapter.